

4 Cod in the North Sea, West of Scotland, eastern English Channel, and Skagerrak (Northern Shelf)

cod.27.46a7d20 – Gadus morhua in Subarea 4, divisions 6.a and 7.d, and Subdivision 20

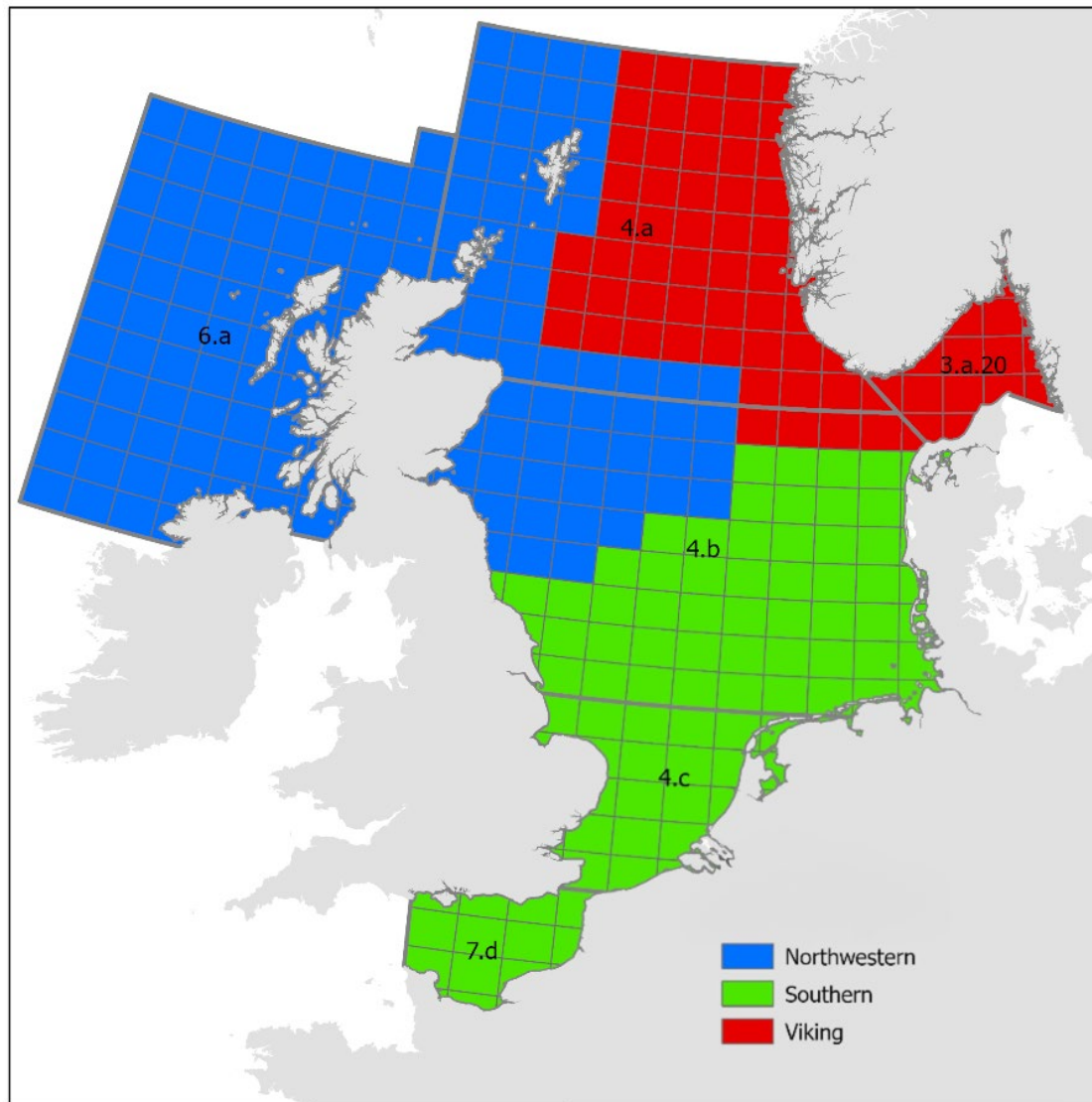
4.1 General

This assessment relates to substocks of cod in the North Sea (Subarea 4), West of Scotland (Division 6.a), the Skagerrak (Subdivision 20), and the eastern English Channel (Division 7.d). This assessment is presented as an update from last year.

A stock annex records more detail and references information on the stock definition, ecosystem aspects and the fisheries. This report section records only recent developments and new information presented to WGNSSK.

4.1.1 Stock definition

The Northern Shelf cod stock consists of reproductively isolated populations of Viking and Dogger cod, with the Dogger cod displaying phenotypic spatial structure either side of a 50 m bathocline in the central North Sea. Based on this, the multistock assessment adopts a three substock hypothesis, with substocks defined as Northwestern, Viking, and Southern (see Figure below), with different rates of sexual maturity, reproduction, and growth. A comprehensive summary of available information on stock definition can be found in ICES WKNSCodID (2020) and ICES WK6aCodID (2022).



Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Assumed distribution of substocks at spawning time.

4.1.2 Ecosystem aspects

Both the North Sea and the West of Scotland are characterized by episodic changes in productivity of key components of the ecosystem. Phytoplankton, zooplankton, demersal and pelagic fish have all exhibited such cycles in variability. Managers should expect long-term change and ensure that management plans have the potential to respond to new circumstances. For example, a regime shift occurred in the region in the mid-1980s, and evidence suggests another from around 1998, a time since which cod recruitment has been low. A summary of available information on ecosystem aspects is presented in the Stock Annex.

4.1.3 Fisheries

4.1.3.1 North Sea: Subarea 4, Division 7.d, and Subdivision 20

Cod are caught by virtually all the demersal gears in Subarea 4, Subdivision 20 (Skagerrak) and Division 7.d, including beam trawls, otter trawls, seine nets, gillnets, trammel nets, and lines. Most of these gears take a mixture of species. In some of them, cod is considered a bycatch (for

example in beam trawls targeting flatfish), and in others, the fisheries are directed mainly towards cod and other gadoids such as haddock and saithe (for example, in large-meshed otter trawls and some fixed gear fisheries). The main gears landing cod in the North Sea area are primarily otter trawlers with >100 mm mesh (TR1), mainly operated by Scotland and Denmark, but also GN1 (mainly Denmark and Norway), TR2, and beam trawlers with >120 mm mesh (BT1). Cod are also an important target for marine recreational fisheries (see section 4.2.1.4).

4.1.3.2 West of Scotland: Division 6.a

The demersal fisheries in Division 6.a are predominantly conducted by otter trawlers (typically using 120 mm mesh) fishing for haddock, anglerfish, and whiting, with bycatches of cod, saithe, megrim, lemon sole, ling, and skate spp. There is a substantial trawl fishery for *Nephrops* in more inshore waters which uses a smaller mesh size (< 100 mm) and also has a bycatch of small cod, the majority of which are discarded. Fishing in the area is conducted mainly by vessels from Scotland, France, Ireland, Norway, and Spain with Scottish vessels taking the majority of cod catch. Typically, large trawl gear vessels targeting finfish are responsible for around 90% of cod catches in Division 6.a, the *Nephrops* fleet take approximately 4% and the remainder are taken by other gears, including longliners and gillnets.

4.1.3.3 Technical Conservation Measures

The recovery plan for cod (EC 1342/2008) triggered considerable improvements in selectivity and cod avoidance through incentives that were linked to the fishing effort regime and through national measures, such as the Scottish Conservation Credits Scheme. The Conservation Credits scheme was suspended on 20 November 2016 and the fishing effort regime discontinued in 2017 (EC 2094/2016). Further details of these measures are presented in the Stock Annex.

The expansion of the closed-circuit TV (CCTV) and FDF programmes in 2010–2016 in Scotland, Denmark, Germany, England, and the Netherlands is expected to have contributed to a reduction of cod mortality. The cod specific FDF scheme terminated at the end of 2016. Further details are presented in the Stock Annex.

Temporary and permanent spatial closures to protect cod have been a feature of cod recovery measures since the early 2000s. A seasonal closure (covering the main spawning period: February–April) in the Clyde was initially implemented as a temporary emergency EU measure to protect spawning cod. Since then, this measure has been implemented through a Scottish Statutory Instrument (SSI) and remains in place for 2024 and 2025. Since 2020, seasonal closures have been implemented to reduce fishing pressure on cod in the North Sea (EC 2021/92, EC 2022/109, EC 2023/194 and EC 2024/257). Details of other spatial closures that have been implemented for cod are presented in the Stock Annex.

A summary of information on cod fisheries in the North Sea and West of Scotland including past and current technical measures used for the management of cod is presented in the Stock Annex.

4.1.3.4 Pre-assessment meeting with industry representatives

Following the recommendations from ICES WKRRCOD (2022), a pre-assessment meeting was held with industry representatives 10th April 2025. Industry representatives reported good cod availability throughout the assessment area, and that fishers were better at avoiding cod during spawning. UK representatives also noted the development of a new recreational UK fishery.

4.1.4 Management

Management of Northern Shelf cod is by TAC and technical measures. The agreed TACs for Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20 (Skagerrak) over the last ten years were as follows (Table 4.1a):

TAC (000t)	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
20 (Skagerrak)	4.2	4.8	5.7	8.0	4.2	2.1	1.9	1.9	3.1	2.8	2.8
2.a + 4	29.2	33.7	39.2	43.2	29.4	14.7	13.2	13.2	21.7	19.9	19.9
6.a	0	0	0	0	1.7	1.3	1.3	1.3	1.2	1.1	1.1
7.d	1.7	2.0	2.1	1.7	1.7	0.9	0.8	0.8	1.3	1.2	1.2

For 2015–2016, Council Regulations (EC) N°2015/104 and N°2016/72 allocated different amounts of Kw*days by Member State and area to different effort groups of vessels depending on gear and mesh size as stipulated by Council Regulation (EC) N°43/2009. The effort regime has now been discontinued, and the TACs for 2017–2025 are given in Council Regulations (EC) N°2017/127, N°2018/120, N°2019/124, N°2020/123, N°2021/1239, N°2022/109, N°2023/194, N°2024/257 and N°2025/202 respectively. From 2012 to 2018, the TAC for cod in Division 6.a was set to zero with allowance for a bycatch of cod to be landed provided that it does not comprise more than 1.5% of the live weight of the total catch retained on board per fishing trip. From 2015, this provision was not allowed for catches subject to the landing obligation. Since 2020, and in addition to TACs, remedial measures have been agreed by the EU, UK, and Norway to reduce the fishing pressure on cod in the North Sea. Details of these remedial measures are given in EC 2021/92, EC 2022/109, EC 2023/194, EC 2024/257, and EC 2025/202.

The EU landing obligation was implemented from 1 January 2017 for several gears, including TR1, BT1 and fixed gears. The EU landing obligation was fully implemented in the North Sea and Skagerrak from 1 January 2018 and in the eastern English Channel and West of Scotland from 1 January 2019, although a few exemptions exist. In 2019, a bycatch TAC of 1735 t was set for Division 6.a to allow mixed fisheries to continue. In 2020, this TAC reduced to 1279 t and remained at this level until 2023, although the quota share changed (as agreed under the EU-UK Trade and Cooperation Agreement). In 2024 and 2025 the TAC reduced to 1114 t.

Demersal fisheries in the area are mixed fisheries, with many stocks exploited together in various combinations in the various fisheries. In these cases, management advice must consider both the state of individual stocks and their simultaneous exploitation in demersal fisheries. Stocks in the poorest condition, particularly those which suffer from reduced reproductive capacity, become the overriding concern for the management of mixed fisheries, where these stocks are exploited either as a targeted species or as bycatch.

4.1.4.1 Cod recovery and management plans

A Cod Recovery Plan which detailed the process of setting TACs for cod in the North Sea, Skagerrak, eastern English Channel and to the West of Scotland was in place until 2008 (EC 423/2004). In December 2008, the European Commission and Norway agreed on a new cod management plan that aimed to be consistent with the precautionary approach and was intended to achieve sustainable fisheries and high yield. In addition to the EU–Norway agreement, the EU implemented effort restrictions, reducing KW-days available to EU vessels in the main métiers catching cod in direct proportion to reductions in fishing mortality until the long-term phase of the plan was reached (EC 1342/2008). A historical evaluation of the effectiveness of these plans (ICES WKROUNDMP, 2011; Kraak *et al.*, 2013) concluded that the plans had not controlled F as envisaged.

In December 2012, the cod management plan was amended. Details are given in EC 1243/2012.

In November 2016, the cod management plan was amended to discontinue the effort regime set out in EC 1342/2008 as it became an obstacle to the implementation of the landing obligation. Details of the amended cod management plan are given in EC 2016/2094.

In July 2018, the European Union agreed to a multiannual management plan for demersal fisheries in the North Sea (MAP). However, the plan was not adopted by Norway and the UK. Details of the plan are given in EC 2018/973.

In March 2019, the European Union adopted a multiannual management plan for stocks fished in Western Waters and adjacent waters. Cod in Division 6.a are considered a bycatch species within this regulation. Details of the plan are given in EC 2019/472.

In 2023, it was decided to assess the previous North Sea and West of Scotland cod stocks as a combined Northern Shelf cod stock consisting of three substocks (Northwestern, Southern and Viking). Advice for substocks of Northern Shelf cod is given according to the ICES MSY approach and precautionary considerations for substocks in concurrent fisheries.

4.2 Data available

4.2.1 Catch

Landings data from human consumption fisheries for recent years as officially reported to ICES together with those estimated by the WG are given for each area separately and combined in Table 4.1a.

The total catch estimate for 2024 is 32 338 tonnes, and above the TAC of 325 028 tonnes for 2024. The catch is split as follows for the separate areas (tonnes):

	TAC	Landings	Discards*	BMS**
20–Skagerrak	2846	2777	244	0
4	19910	22657	5364	2
6.a	1114	1007	277	0
7.d	1158	9.93	0	0
Total	25028	26451	5886	2

* Discards include BMS landings.

**BMS landings uploaded to InterCatch.

No discards or age data are available for substocks of Northern Shelf cod. Catch-at-age data are therefore prepared separately for cod in the North Sea, eastern English Channel and Skagerrak (27.47d20) and for cod to the West of Scotland (27.6a) according to the procedures established for single-stock assessments prior to 2023. Catch numbers for the total Northern Shelf cod stock are then taken as the sum of numbers-at-age from the North Sea and West of Scotland components and catch weights as the weighted average of the annual catch mean weights (weighted by the catch numbers-at-age). The sampling coverage for discard ratio, and landings and discards age compositions for 2024 is provided in Table 4.1b, along with the contributions to total landings, discards and BMS from each area prior to raising. Raising procedures for the North Sea and West of Scotland components are described in the following sections.

Figure 4.1a plots reported landings and estimated discards (including BMS landings) used in the assessment. Landings in numbers-at-age are given in Table 4.2a. These data do not include industrial fishery bycatches landed for reduction purposes prior to 2002 (values from 2002 onwards were entered into InterCatch for all relevant nations except Norway, and were included in the raising, although the numbers were very small). Discard numbers-at-age (including BMS landings from 2016) are shown in Table 4.2b. The proportions of the estimated numbers discarded for ages 1–4 and the proportion of the estimated total discards by weight and number are shown in Figure 4.1b. Total catch numbers-at-age are shown in Table 4.2c. Reported landings, estimated discards (including BMS landings from 2016) and total catch (sum of landings and discards), given as SOPs in tonnage, are shown in Table 4.3.

4.2.1.1 North Sea (27.47d20) catch-at-age data

Prior to the use of InterCatch for discard estimation, discard numbers-at-age were estimated for areas 4 and 7.d by applying the Scottish discard ogives to the international landings-at-age, and were based on observer sampling estimates for Subdivision 20–Skagerrak. Discard raising for 2002–2024 was performed in InterCatch, with the different nations providing information by area, quarter, and métier. Sampling for discards and age compositions was poor in area 7.d in 2002–2003, and this necessitated combining areas 4 and 7.d in those years. The provision of discard information has vastly improved since the reform of the EU's data collection framework in 2008 (see <http://datacollection.jrc.ec.europa.eu/>) and covered 72% of the landings in 2024. All nations apart from Norway now provide discard information and age compositions were provided by all nations except Belgium for 2024 data.

Norwegian discarding is illegal, so although this nation has accounted for 7–15% of cod landings over the period 2002–2024 (InterCatch data), it does not provide discard estimates. Nevertheless, the agreed procedure applied in InterCatch is that discards raising should include Norway (i.e. Norway will be allocated discards associated with landings in reported métiers). Furthermore, tagging and genetic studies have indicated that Norwegian coastal cod are different from North Sea cod and do not generally move into areas occupied by North Sea cod. Therefore, Norwegian coastal cod data have been removed from North Sea cod data by uploading only North Sea cod data into InterCatch for 2002 onwards, and by adjusting catches prior to 2002 to reflect the removal of Norwegian coastal cod data (an annual multiplicative adjustment of no more than 2.5% was made using Norwegian coastal cod data (see ICES WKNSEA, 2015, for more details).

For cod in 4, 20–Skagerrak and 7.d, ICES first raised concerns about the misreporting and non-reporting of landings in the early 1990s, particularly when TACs became intentionally restrictive for management purposes. The WG believes that underreporting of landings may have been significant in 1998 because of the abundance in the population of the relatively strong 1996 year-class as 2-year-olds. The landed weight and input numbers-at-age data for 1998 were adjusted to include an estimated 3000 tonnes of underreported catch. The 1998 catch estimates remain unchanged in the present assessment (apart from the adjustment for Norwegian coastal cod). The UK Buyers and Sellers legislation, introduced towards the end of 2005, is expected to have improved the accuracy of reported cod landings.

Since the WG has no basis to judge the overall extent of underreported catch over time, it has no alternative but to use its best estimates of landings, which in general are in line with the officially reported landings.

InterCatch

InterCatch was used for estimation of landings, discards (including BMS landings) and total catch-at-age and mean weight at age in 2024. Data co-ordinators from each nation were tasked to input data into InterCatch, disaggregated to quarter and métier. Allocations of discard ratios and age compositions for unsampled strata were then performed to obtain the data required for

the assessment. The approach used for discard ratio allocations was to do it by area (20, 4 and 7.d), giving three broad categories. Annual discards were first matched to quarterly landings. Then, within each of these three categories, ignoring country and season, where métiers had some samples, these were pooled and allocated to unsampled records within that métier. At the end of this process, any remaining métiers were allocated an all samples pooled discard ratio for the given category.

The landings and discards imported in weight or raised for 2024 are as follows (tonnes):

Catch Category	Raised or Imported	CATON	Percentage
BMS landing	Imported	1.8	100
Discards	Imported	4278	76
Discards	Raised	1329	24
Landings	Imported	25444	100
Logbook Registered Discard	Imported	0	NA

A similar approach was used for allocating age compositions, except that there were six broad categories because discards (including BMS landings) were treated separately to landings. However, age compositions for Division 7.d had to be allocated from métiers in Subarea 4 as there was no landings or discards age sampling in 7.d in 2024.

The landings and discards imported in weight or raised, with age distribution sampled or estimated for 2024 are as follows (tonnes):

Catch Category	Raised or Imported	Sampled or Estimated	CATON	Percentage
Logbook Registered Discard	Imported	Estimated	0	NA
Landings	Imported	Sampled	21771	86
Landings	Imported	Estimated	3673	14
Discards	Imported	Sampled	4250	76
Discards	Raised	Estimated	1329	24
Discards	Imported	Estimated	28	0
BMS landing	Imported	Sampled	0	NA
BMS landing	Imported	Estimated	2	100

The landings, discards (including BMS landings) and total catch-at-age and mean weights at age in 2021–2023 were re-estimated following resubmission of data from UK-England for 2021–2023 and from the Netherlands for 2023. This resulted in only minor changes to the total landings and discards and had a negligible effect in the stock assessment.

	2021	2022	2023
Discards			
Resubmitted (t)	3812	6191	7540

	2021	2022	2023
Original (t)	3766	6221	7386
Difference (t)	46	-29	154
Difference (%)	1.2	-0.5	2.1
Landings			
Resubmitted (t)	14590	15208	23838
Original (t)	14589	15205	23819
Difference (t)	1	3	19
Difference (%)	0.0	0.0	0.1

4.2.1.2 West of Scotland (27.6a) catch-at-age data

National estimates of landings and where available, discards and associated age compositions are available in InterCatch from 2003 onwards, and prior to 2003 from historical assessment WGCSE files. Age composition data for landings and discards are typically provided by UK (Scotland) and Ireland for the main métiers over the time-series, and France have provided discard estimates for 2009 onwards but no landings or discards age compositions.

Reported landings of cod in Division 6a are considered to have been significantly impacted by area misreporting since the mid-2000s (i.e. cod which are caught in Division 6a are incorrectly reported as being taken from the North Sea and elsewhere, resulting in the reported landings being an underestimate of actual landings from Division 6a). At the 2020 benchmark an approach was agreed which utilized VMS data and associated daily reported landings records to estimate an amount of area-misreported landings (ICES WKDEM, 2020). Catch data used in the previous West of Scotland assessment have been adjusted in the submission to InterCatch to account for area-misreporting i.e. the area-misreported quantity is added to the reported landings for Division 6a, and an equivalent quantity has been subtracted from the North Sea landings in most years. Late availability of estimated area-misreported landings between 2019 and 2021 meant that only landings for the West of Scotland cod stock were adjusted in these years. This is likely to result in a small amount of double accounting (< 2% of total landings in 2019 and 2020, and < 0.5% in 2021) when summing InterCatch data for 6a cod and N Sea cod for these years.

Since the adoption of the multistock assessment approach for Northern Shelf cod in 2023 (ICES WKBCOD 2023), area misreported landings are no longer relevant as commercial landings data from the North Sea and West of Scotland are combined prior to being input into the assessment. As such, no adjustment for area-misreporting was made for 2022–2024, nor will be made in future assessments of Northern Shelf cod.

Between the mid-1990s and mid 2000s there was believed to be significant underreporting of landings from Division 6.a due to restrictive TACs. Given the lack of estimates of underreporting, previous assessments of cod in this area have excluded total catches for this period (1995–2006) and estimated them within the assessment model.

InterCatch

The catch estimation process in InterCatch involves two stages: (i) allocating discard ratios to fleets for which only landings have been imported and (ii) age composition allocation by catch

category (for unsampled catches). Age samples are allocated for landings and discards separately. Below Minimum Size landings are combined with discards for the purpose of age composition estimation.

To estimate discard ratios, discards are first automatically matched to landings by country, area, métier and season (year or quarter) in InterCatch. The resulting discard-landings ratios are then used to estimate discards for landings from fleets without discard estimates. Due to the mix of both quarterly and annual data submitted for each year, strata for allocating discard rates were independent of season. The strata by year are as follows: large mesh demersal target fleets were allocated a discard-landings ratio on the basis of the weighted average of all available ratios from large mesh demersal target fleets (weighted average of Scottish, Irish and French when available); small mesh fleets were allocated discard ratio on the basis of all available ratios from small mesh fleets (usually only Scottish *Nephrops* fleet); longline fleets are allocated discard proportions from other longline fleets (and when not available are allocated zero discard rate as observed discard rates appear very low compared with other fleets); all other fleets are given a weighted average of all available discard proportions. For 2024 data, the usual allocation scheme needed to be changed because there were no Scottish landings samples in 2024, all landings allocations in 6.a are therefore based on Irish age compositions. Discard allocation schemes were not impacted and remained as described above.

The allocation of age compositions to un-sampled landings and discards follows the same stratification as described for the allocation of discard ratios. The exception being the longline fleets which were included in the ‘other fleets’ category as there were no age composition data provided.

The landings and discards imported in weight or raised, with age distribution sampled or estimated for 2024 are as follows (tonnes):

Catch Category	Raised or Imported	Sampled or Estimated	CATON	Percentage
Logbook Registered Discard	Imported	Estimated	0	NA
Landings	Imported	Sampled	123	12
Landings	Imported	Estimated	884	88
Discards	Imported	Sampled	236	85
Discards	Raised	Estimated	41	15
Discards	Imported	Estimated	0.066	0
BMS landing	Imported	Estimated	0	NA

4.2.1.3 Spatially and seasonally disaggregated landings

Landings data disaggregated by ICES rectangle are used to calculate the proportion of landings by substock for Northern Shelf cod. Landings data from quarter 1 are used to determine the relative substock proportions over time as it is assumed that the substocks do not mix in quarter 1. Spatially disaggregated landings data are available from 1995 onwards, with data available from eleven ICES countries (Belgium, Denmark, England (UK), France, Germany, Ireland, Netherlands, Northern Ireland (UK), Norway, Scotland (UK) and Sweden). Four countries are missing spatial landings data at the beginning of the time-series (i.e. Belgium, France, the Netherlands and Northern Ireland), and there are also missing spatial landings for Ireland between 2011–2013. For each of these countries, data gaps are filled based on a 3-year mean of the substock

landings proportions and a 3-year mean of the quarter 1 landings proportions using the first years in the time-series with available data. These proportions are then applied to the ICES official landings data to get landings estimates for the missing years for each country. The relative quarter 1 substock proportions since 1995 are given in Figure 4.1c and for 2024 are calculated as 47% Northwestern, 51% Viking and 2% Southern. The Q1 substock proportions over time are as follows:

Year	Northwestern	Southern	Viking
1995	0.21	0.44	0.35
1996	0.23	0.47	0.29
1997	0.20	0.49	0.31
1998	0.20	0.51	0.29
1999	0.15	0.51	0.34
2000	0.17	0.45	0.38
2001	0.18	0.38	0.44
2002	0.23	0.32	0.45
2003	0.14	0.43	0.43
2004	0.11	0.42	0.47
2005	0.11	0.37	0.52
2006	0.13	0.35	0.53
2007	0.19	0.37	0.44
2008	0.14	0.45	0.41
2009	0.14	0.45	0.42
2010	0.18	0.39	0.43
2011	0.25	0.28	0.47
2012	0.21	0.25	0.54
2013	0.29	0.18	0.53
2014	0.29	0.19	0.52
2015	0.27	0.25	0.48
2016	0.27	0.14	0.58
2017	0.30	0.05	0.64
2018	0.46	0.03	0.51
2019	0.50	0.02	0.48
2020	0.51	0.03	0.46

Year	Northwestern	Southern	Viking
2021	0.41	0.06	0.53
2022	0.44	0.02	0.53
2023	0.47	0.04	0.49
2024	0.47	0.02	0.51

Disaggregated landings data are also used to calculate the proportion of total landings by quarter (Figure 4.1d), which are input into the assessment. Generally speaking, the proportion of total landings taken in Q1 and Q2 has decreased over the years, with larger proportions being taken in Q3 and Q4 in recent years. The proportion of total landings by quarter (rounded) are given below, and are calculated as 21% Q1, 23% Q2, 28% Q3 and 27% Q4 for 2024:

Year	Q1	Q2	Q3	Q4
1995	0.21	0.32	0.27	0.20
1996	0.23	0.27	0.27	0.23
1997	0.20	0.28	0.27	0.25
1998	0.22	0.31	0.25	0.22
1999	0.29	0.27	0.24	0.20
2000	0.24	0.26	0.23	0.26
2001	0.24	0.26	0.24	0.26
2002	0.22	0.27	0.25	0.26
2003	0.30	0.26	0.25	0.20
2004	0.26	0.24	0.22	0.28
2005	0.23	0.24	0.27	0.26
2006	0.23	0.23	0.30	0.24
2007	0.22	0.26	0.28	0.24
2008	0.26	0.29	0.26	0.18
2009	0.23	0.24	0.28	0.25
2010	0.22	0.26	0.25	0.27
2011	0.23	0.24	0.27	0.25
2012	0.23	0.27	0.28	0.22
2013	0.20	0.24	0.29	0.27
2014	0.16	0.24	0.31	0.29
2015	0.21	0.25	0.30	0.25

Year	Q1	Q2	Q3	Q4
2016	0.21	0.24	0.29	0.26
2017	0.23	0.23	0.26	0.27
2018	0.24	0.24	0.27	0.26
2019	0.19	0.25	0.31	0.26
2020	0.19	0.25	0.29	0.27
2021	0.18	0.26	0.28	0.28
2022	0.19	0.25	0.29	0.27
2023	0.19	0.25	0.31	0.25
2024	0.21	0.23	0.28	0.27

4.2.1.4 Recreational catches

Recreational catches were estimated for 2010–2024 from data provided by Belgium, Denmark, Germany, Sweden, Norway, the Netherlands and, but are considered provisional and not included in the assessment due to length of time-series and unknown age structure and uncertainty. Further details are provided in the Stock Annex and ICES WKBCOD (2023). Estimates of commercial and recreational removals along with the percentage of recreational removals and percentage of recreational removals derived from imputation are shown in the following table. The percentage of recreational removals derived by imputation has increased in recent years because estimates from Germany and Scotland are no longer available.

Year	Landings	Commercial removals (t)		Recreational removals (t)		
		Discards	Total	Total	% recr.	%imputed
2010	37115	13899	51014	1170	2.2	81
2011	32323	12515	44839	1277	2.8	81
2012	32363	10554	42916	1066	2.4	82
2013	30744	12073	42817	1837	4.1	72
2014	35048	13906	48954	3112	6.0	73
2015	37938	15038	52976	2164	3.9	75
2016	38754	13702	52456	1991	3.7	13
2017	38355	12202	50557	1645	3.2	11
2018	41147	10170	51317	1594	3.0	2
2019	33995	3677	37672	1325	3.4	9
2020	20185	4905	25090	1479	5.6	10
2021	15754	4405	20159	1137	5.3	14

Year	Landings	Commercial removals (t)		Recreational removals (t)		
		Discards	Total	Total	% recr.	%imputed
2022	16150	6761	22911	1146	4.8	35
2023	24634	7783	32418	1090	3.3	36
2024	26232	6167	32399	831	2.5	37
Mean	30716	9850	40566	1524	3.7	42

4.2.2 Weight-at-age

Mean weights-at-age for landings, discards (including BMS landings from 2016) and catch are taken as a weighted average of the annual catch mean weights from the North Sea and West of Scotland (weighted by the catch numbers-at-age) and are given in Tables 4.4a–c. Long-term trends in mean catch weights-at-age by catch component for ages 1–7+ are plotted in Figure 4.2a, which indicates an overall decline from around 2010 for ages 3 and above. Ages 1 and 2 show little absolute variation over the long term.

Stock mean weights are derived by substock from the NS-IBTS, SWC-IBTS and SCOWCGFS survey data in Q1, together covering the North Sea, West of Scotland and Skagerrak. A bootstrap procedure is used to derive stock mean weights from the observed length distribution and estimated length-weight relationships, accounting for length-stratified sampling. Differences have been noted in the historic period due to the random allocation of CA records with missing haul IDs; however, a sensitivity analysis showed this to have no impact on stock weights in the recent period and minimal effect in the stock assessment (see Annex 8 in ICES WGNSSK, 2023). Stock mean weights used as input to the assessment are given in Tables 4.5a and plotted in Figure 4.2b. Mean stock weights have been variable across the time-series for the Viking substock and have declined in most recent years for ages 2–5 (Figure 4.2b). There is more of a declining trend in mean stock weights for the Northwest and Southern substocks.

4.2.3 Maturity and natural mortality

Annually varying maturity ogives are estimated for each substock based on the Q1 survey data (NS-IBTS, SWC-IBTS and SCOWCGFS) from 1983. Data are weighted following the procedures described in ICES WKMOG (2008) and maturity ogives produced by fitting a GLMM to the weighted data. A sensitivity analysis shows some impact of randomly allocating CA records with missing haul IDs on the maturity ogives but minimal influence in the assessment (see Annex 8 in ICES WGNSSK, 2023). Furthermore, there are concerns about the quality of maturity records prior to 1991, although this does not impact the advice. The time-varying maturity ogives used as input to the assessment are given in Tables 4.5b and illustrated in Figure 4.2c. Time-varying ogives are stable for the Northwestern substock. A slight declining trend is observed for the Viking substock whereas a strong increasing trend for maturity ogives is observed for the southern component (Figure 4.2c)

Table 4.5c and Figure 4.2d show estimates of M based on multispecies considerations adopted for the assessment. Estimates of natural mortality are derived from multispecies analyses updated by the Working Group on Multispecies Stock Assessment Methods (WGSAM) every three years in so-called “key runs” to account for improved knowledge of predation on cod by other species (mainly seals, harbour porpoises and gurnards) and cannibalism. The last update occurred in 2023 but included an ad-hoc inflation of the M s for cod to account for migrations to the

West of Scotland, given that the distribution of the Northern Shelf cod stock is larger than the SMS model domain covering only the North Sea. This ad-hoc adjustment follows the methodology of the 2021–2022 assessments of the former North Sea cod stock but is inconsistent with the new stock definition and multi-stock assessment adopted since 2023, which explicitly includes the West of Scotland. The Ms from the 2023 key run were therefore considered inappropriate for Northern Shelf cod. In 2024, WGSAM provided an alternative model run without the M adjustment for use in the Northern Shelf cod assessment. This alternative run is almost identical to the official key run for everything except cod and provides Ms that benefit from improved knowledge of predation whilst retaining consistency with the multi-stock assessment and assumptions. Natural mortality estimates from the North Sea key run used as input to the assessment are given in Table 4.5c and Figure 4.2d and are assumed to apply to all three substocks of Northern Shelf cod. There is an increased level of natural mortality from 2000 onwards across all age classes (except for age 7 with remains constant), compared with the earlier part of the time-series (1983–1999).

4.2.4 Catch, effort and research vessel data

Only survey and combined commercial landings and discard information are analysed within the assessment presented.

Seven survey series are available for use within this assessment:

- North Sea International Bottom Trawl Survey in Q1 (NS-IBTS-Q1): covering the North Sea and Skagerrak from 1976. This multi-vessel survey uses a fixed station design of at least two tows per rectangle with the GOV trawl.
- Scottish West Coast Bottom Trawl Survey in Q1 (SWC-IBTS-Q1): covering the West of Scotland for the period 1985–2010. This survey used a fixed station design with the GOV trawl.
- Scottish West Coast Groundfish Survey in Q1 (SCOWCGFS-Q1): covering the West of Scotland from 2011. This survey series uses a more robust GOV groundgear (compared to the SWC-IBTS) and a random stratified design.
- North Sea International Bottom Trawl Survey in Q3 (NS-IBTS-Q3): covering the North Sea and Skagerrak from 1991. This multi-vessel survey uses a fixed station design of at least two tows per rectangle with the GOV trawl.
- Scottish West Coast Bottom Trawl Survey in Q4 (SWC-IBTS-Q4): covering the West of Scotland for the period 1996–2009. This survey used a fixed station design with the GOV trawl.
- Scottish West Coast Groundfish Survey in Q4 (SCOWCGFS-Q4): covering the West of Scotland from 2011. This survey series uses a more robust GOV groundgear (compared to the SWC-IBTS) and a random stratified design.
- Irish Groundfish Survey in Q4 (IE-IGFS-Q4): covering the southern part of Division 6.a from 2003. This survey uses a random stratified design and the GOV trawl.

Standardized age-based survey indices for Northern Shelf cod are calculated based on GAMs and Delta-distributions. The general methodology is described in Berg and Kristensen (2012) and Berg *et al.* (2014) and was run with R version 4.1.2 in the Windows Subsystem for Linux (WSL) based on the DATRAS¹ and surveyIndex packages. The Delta-GAM is fit to the survey data for Quarter 1 (NS-IBTS-Q1, SWC-IBTS-Q1 and SCOWCGFS-Q1) ages 1–7+, and Quarters 3 and 4 combined (NS-IBTS-Q3, SWC-IBTS-Q4, SCOWCGFS-Q4 and IE-IGFS-Q4) ages 0–7+. The final model formulation selected for Northern Shelf cod comprises a high resolution stationary spatial

¹ <http://rforge.net/DATRAS/>

model with low resolution yearly independent deviations and includes ship, year, depth, time of day and haul-duration effects. The Q3+Q4 model also includes a gear effect to account for use of the Aberdeen trawl between 1992–1997. More details of the method used to produce Delta-GAM indices are provided in the Stock Annex and can be found in ICES WKBCOD (2023), as well as the above-mentioned publications.

Where historic records in the DATRAS CA (age) data are missing haul IDs, random allocations are made based on year, quarter, country, ship, and either ICES rectangle or roundfish area, to avoid losing age samples. A sensitivity analysis shows these random allocations to have minimal influence on the indices (see Annex 8 in ICES WGNSSK, 2023).

Maps showing the predicted distribution of cod in recent years are presented in Figures 4.3a-b. Retrospective analyses with three peels give average Mohn's rho values of 0 and -0.01 across all ages for the Q1 and Q3+Q4 models, respectively (Figures 4.3c-d). Indices by substock are produced by summing the spatial model predictions by year, age, and substock area. This is done for Q1 based on the assumption that any mixing during Q1 is small/negligible, and for age 0 in Q3+Q4 based on the assumption that age 0 cod will not have moved far since spawning. Mixing is expected for age 1+ fish outside Q1, and the Q3+Q4 indices therefore summed across the whole area to represent a mix of the three substocks. Because the first age in the multistock SAM assessment is age 1, estimates of age 0 cod from Q3+Q4 are retained as separate recruitment indices (one for each substock) and forward shifted to 1st January to represent age 1 cod the following year. The Delta-GAM indices and associated standard deviations used in the assessment are given in Tables 4.6a-c and plotted in Figures 4.3e-i. These show increased uncertainty about the youngest and oldest ages, which is carried through to the multistock SAM assessment. The latest values from the Q3+Q4 recruitment (2024 forward shifted to 2025) and Q1 (2025) indices indicate a weak incoming 2024 year-class across all three substocks. Northern shelf cod are concentrated more in the northern parts of their stock area in recent years, with older age classes concentrated more northerly in Q3 compared with Q1.

Following the recommendations of ICES-WKRRCOD2 (2023), an exploratory run of the Q3+Q4 index including data from a localised inshore survey around Shetland (Fraser *et al.*, 2024) was presented. The index model appeared to pick up some signal around the Shetland area and resulted in some improvements to internal consistency for ages 0–2 but made little difference to the indices as they go into the assessment. This is likely because the current index model covers too large an area at too coarse a scale for localised data to have much impact. However, it was agreed that this data should be considered at a future benchmark.

4.3 Data analyses

4.3.1 Assessment audit

The assessment audit for Northern Shelf cod was completed, and no significant issues found for the assessment itself. Additional checks on the forecast will be carried out during the ICES WGMIXFISH-ADVICE meeting in 2025.

4.3.2 Exploratory survey-based analyses

Survey abundance indices are plotted in log-mean standardized form by year and cohort for each substock in the Q1 surveys in Figures 4.4a-c, together with log-abundance curves and associated negative gradients for the age range 2–4. Similar plots are shown for the Q3+Q4 surveys for all substocks combined in Figure 4.4d. The log-mean standardized curves track cohort signals well, although there is some loss of signal between the 2012 and 2013 cohorts associated with an

apparent positive year effect towards the end of 2016 (Q3+Q4) into the beginning of 2017 (Q1). This is most apparent for the Northwestern substock, with all ages except age 2 at a relatively high level in that year and is likely compounded by a single very large haul in the SCOWCGFS-Q1 survey. This positive year effect is apparent to a lesser extent for the other substocks, with all three displaying a larger recruitment in 2017. The log abundance curves showed a sharp decrease, with a substantial decrease in the negative gradient for ages 2–4, following the 2017 year-class, likely due to the reductions in catches from 2019 onwards. The negative gradients have, however, recently increased for all substocks.

Figures 4.5a-d show within-survey consistency (in cohort strength) for the Q1 substock indices and Q3+Q4 mixed index, respectively. These show generally good consistency, particularly for the Northwestern and Southern substocks where all correlations between adjacent ages are ≥ 0.7 , justifying their use for survey tuning. There is some deterioration with age for all indices, but particularly for ages 5+ in both the Viking substock in Q1 and for the mixed-stock in Q3+Q4. Figure 4.5e shows between survey correlations for the recruitment indices (age 0 in Q3+Q4 forward shifted to represent age 1 on 1st January) and age 1 from the Q1 indices, indicating strong external consistency for all substocks.

4.3.3 Exploratory catch-at-age-based analyses

4.3.3.1 Catch-at-age matrix

The total catch-at-age matrix for the mixed-stock (Table 4.2c) is expressed as numbers-at-age, and proportions-at-age, standardized over time in Figure 4.6. It clearly shows the contribution of the 2005, 2009, 2013 and 2016 year-classes to catches and indicates a relative increase in the number of older fish in the catches in recent years.

4.3.3.2 Catch curve cohort trends

The top panel of Figure 4.7 presents the log catch curve plot for the catch-at-age data for the mixed-stock (all substocks) while the bottom panel plots the negative slope of a regression fitted to the ages 2–4, the age range used as the reference for mortality trends. These gradients show a sharp increase for the 2015–2016 cohorts and, similar to equivalent plots for the survey indices, a sharp decrease for the 2017-year class, likely due to the reductions in catches from 2019 onwards. The points representing the 2019–2020 year-classes (the last two year-classes to reach age 4 in the catches) are the lowest in the time series. This opposite trend is found in the survey catch curves which show an increase in mortality in Q1 surveys for all substocks. For Q34 survey catch curves, consistent values are found for the two most recent years for all substocks.

Figure 4.8 shows cohort correlations in the catch-at-age matrix for the mixed-stock, plotted as log-numbers. These correlations show good consistency within cohorts up to the plus-group, verifying the ability of the catch-at-age data to track relative cohort strengths over the full time-series.

4.3.4 Final assessment

The final assessment used multistock SAM (multistock State Space Assessment Model; Albertsen *et al.*, 2018) run with R *stockassessment* (version 0.12.4) and *multiStockassessment* (version 0.4.0) packages in R version 4.3.2. The data used in the assessment are given in Tables 4.2 and 4.4–4.6 and include mixed substock catch-at-age data, substock-wise Q1 indices, a mixed substock Q3+Q4 index, substock-wise Q3+Q4 forward-shifted recruitment indices, substock composition proportions from Q1 landings, proportions of landed weight per quarter, and substock-wise estimates of stock weights, maturity, and natural mortality (where the same natural mortalities are used for all three substocks). The stock-wise configuration (adopted for all three substocks) and

stock complex configuration arguments are given in Tables 4.7a–b, respectively. All single-stock observations are set to NA and a fleet-to-stock key used to specify the vulnerability of each substock to a catch, survey, or auxiliary fleet in the shared observations. The fleet-to-stock key is given in Table 4.7c.

Random walk processes are used to model recruitment and fishing mortality-at-age, where the random walks for fishing mortality are correlated among ages according to an AR(1) process and are connected between substocks via scaled selectivities with flexible F_{bar} . Where fleets cover a mixture of substocks, observations are predicted as the sum of substock-wise predictions according to the weights specified in the fleet-to-stock key (Table 4.7c). Modelled catches and Q1 and Q3+Q4 survey observations are correlated across ages according to an AR(1) structure (Berg and Nielsen, 2016). Substock composition proportions and seasonal landings proportions are modelled using additive logistic normal distributions and seasonal effects are estimated by substock and quarter. Stock weights, maturity and natural mortality are modelled as Gaussian Markov Random Field (GMRF) processes with cohort- and within year correlations. The GMRF for stock weights also includes an additional correlation across years in the plus group. Model fitting diagnostics are given in Table 4.7d.

Figures 4.9a–e show summary plots of the final assessment in terms of substock population trends. Estimates of fishing mortality-at-age, substock numbers-at-age, catches-at-age, stock weights-at-age, maturity-at-age and natural mortality-at-age for each substock are given in Tables 4.8–13 respectively, while summary tables for estimates of recruitment (age 1), TSB, SSB, catches and F_{bar} (2–4) by substock are given in Tables 4.14 (along with 95% confidence bounds). Mean fishing mortality split into ages by substock is shown in Figure 4.10a and selectivity in F is shown in Figure 4.10b, while log numbers-at-age and log catches-at-age are shown for each substock in Figures 4.11–4.12. Fits of the GMRF process for stock weights-, maturity- and natural mortality- at age by substock (along with 95% confidence bounds) are shown in Figures 4.13–4.15. Although the GMRFs for natural mortality are fit to the same input data for each substock, separate random effects are estimated, resulting in slight differences in M between substocks.

One-step-ahead quantile residuals are shown in Figure 4.16 and, for most fleets, show no discernible patterns. However, for the substock composition data from the Q1 landings, there is a tendency to over-predict one of the substocks at the beginning of the time-series and under-predict towards the end. Retrospective plots for SSB, average fishing mortality, recruitment-at-age 1 and TSB are shown in Figures 4.17a–d, respectively. Mohn’s rho statistics based on five-year peels are calculated as follows:

	Northwest	South	Viking	Total
SSB	0.037	0.159	0.029	0.049
$F_{\text{bar}}(2-4)$	-0.045	-0.0097	-0.029	-0.025
R (age 1)	-0.005	0.237	-0.226	-0.028

A comparison with last year’s assessment (ICES WGNSSK, 2024) is provided for each substock in Figures 4.18a–c. Differences between the assessments are due to the addition of one year of catch, survey, and landings proportions data and slight revisions to the delta-GAM indices and GMRF processes. These changes result in a downscaling of SSB in recent years for all substocks and an upscaling of fishing mortality in recent years for all substocks.

4.4 Historic stock trends

The historic substock and fishery trends are presented in Figures 4.9–12 and Tables 4.14.

For the Southern substock, a steady decline was estimated in SSB from around 17 400 tonnes in 1983 to 2800 tonnes in 2020, followed by a small increase from 2021 to 2025. The Northwestern substock was estimated to be the largest component of the stock complex. From 1983 to 1997, SSB was estimated to fluctuate around 50 000 tonnes, followed by a large decline to 15 000 in 2005. Since then, there has been a generally increasing trend in SSB, except for the years 2017–2020 and 2023–2025. Throughout the period, the SSB of the Viking substock has been fluctuating around 13 000. Consequently, trends in the total stock complex SSB followed the trends in the Northwestern substock.

Trends in fishing mortality were similar across substocks. For all three substocks, F was steady or slightly increasing from 1983 to 1999, followed by a decline until 2014. From 2014 to 2018, F increased for all substocks, and decreased again. In most recent years, F has been increasing across all sub-stocks, with a strong increase observed for the southern substock. Throughout the period, F was estimated to be highest for the Southern substock, followed by the Northwestern substock and lowest for the Viking substock.

For all substocks, recruitment has fluctuated at a relatively low level from 1998, with the 1996 year-class being the last large year-class to contribute to the fishery. The 2025 recruitment (total of all sub-stocks) remains low following the poor incoming cohorts in 2023 and 2024. Recruitment in 2025 is the second lowest in the time-series across all sub-stocks but has shown a slight increase in Q1 2025 in the assessment.

4.5 Recruitment estimates

Recruitments in the intermediate year (2025) were sampled from normal distributions about the assessment estimates and are reported as the median of those samples. Estimates of recruitment for subsequent years were simulated from a lognormal distribution with the same median and log-variance as the recruitment estimates from the 2022–2024 year-classes, reflecting recent low levels of recruitment.

4.6 Biological reference points

Reference point estimation is performed with the EqSim software (ICES WGMG, 2013), in accordance with the ICES guidelines. Reference points were re-estimated for substocks of Northern Shelf cod at the 2024 meeting of WGNSSK following the adoption of updated natural mortalities (please see ICES (2024) for details).

Biological reference points and their technical basis are as follows:

Northwestern substock

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY $B_{trigger}$	40 823	B_{pa} in tonnes	ICES WGNSSK (2024)
	F_{MSY}	0.196	EqSim analysis based on the recruitment period 1998–2023	ICES WGNSSK (2024)
	$F_{MSY lower}$	0.121	EqSim analysis based on the recruitment period 1998–2023	ICES WGNSSK (2024)
	$F_{MSY upper}$	0.335	EqSim analysis based on the recruitment period 1998–2023	ICES WGNSSK (2024)

Framework	Reference point	Value	Technical basis	Source
Precautionary approach	B_{lim}	29 378	Estimated change point of a segmented regression fit to substock stock recruit pairs, excluding the 2024 recruitment; in tonnes	ICES WGNSSK (2024)
	B_{pa}	40 823	$B_{pa} = 1.4 \times B_{lim}$; in tonnes	ICES WGNSSK (2024)
	F_{lim}	0.666	F giving 50% probability of $SSB < B_{lim}$	ICES WGNSSK (2024)
	F_{pa}	0.543	F that provides a 95% probability for SSB to be above B_{lim} (with advice rule [AR])	ICES WGNSSK (2024)

Viking substock

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY $B_{trigger}$	13 732	B_{pa} ; in tonnes	ICES WGNSSK (2024)
	F_{MSY}	0.187	EqSim analysis based on the recruitment period 1998–2024	ICES WGNSSK (2024)
	$F_{MSY\ lower}$	0.102	EqSim analysis based on the recruitment period 1998–2024	ICES WGNSSK (2024)
	$F_{MSY\ upper}$	0.297	EqSim analysis based on the recruitment period 1998–2024	ICES WGNSSK (2024)
Precautionary approach	B_{lim}	9619	Average SSB in the lower 50 th percentile producing above average recruitment; in tonnes	ICES WGNSSK (2024)
	B_{pa}	13 732	$B_{pa} = B_{lim} \times e^{(1.645 \times \sigma)}$ where σ = multiSAM estimate of the standard deviation of $\ln(SSB)$ in the terminal year; in tonnes.	ICES WGNSSK (2024)
	F_{lim}	0.379	F giving 50% probability of $SSB < B_{lim}$	ICES WGNSSK (2024)
	F_{pa}	0.307	F that provides a 95% probability for SSB to be above B_{lim} (with advice rule [AR])	ICES WGNSSK (2024)

Southern substock

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY $B_{trigger}$	19 851	B_{pa} ; in tonnes	ICES WGNSSK (2024)
	F_{MSY}	0.231	EqSim analysis based on the recruitment period 1998–2024	ICES WGNSSK (2024)

Framework	Reference point	Value	Technical basis	Source
	$F_{MSY\ lower}$	0.144	EqSim analysis based on the recruitment period 1998–2024	ICES WGNSSK (2024)
	$F_{MSY\ upper}$	0.366	EqSim analysis based on the recruitment period 1998–2024	ICES WGNSSK (2024)
Precautionary approach	B_{lim}	14 002	Estimated change point of a segmented regression fit to substock stock recruit pairs; in tonnes	ICES WGNSSK (2024)
	B_{pa}	19 851	$B_{pa} = B_{lim} \times e^{(1.645 \times \sigma)}$ where σ = multiSAM estimate of the standard deviation of $\ln(SSB)$ in the terminal year; in tonnes	ICES WGNSSK (2024)
	F_{lim}	0.695	F giving 50% probability of $SSB < B_{lim}$	ICES WGNSSK (2024)
	F_{pa}	0.464	F that provides a 95% probability for SSB to be above B_{lim} (with advice rule [AR])	ICES WGNSSK (2024)

4.7 Short-term forecasts

Forecasting takes the form of short-term stochastic projections using the “modelforecast” function from the *multiStockassessment* R package (version 0.4.0). A total of 1000 samples are generated from the asymptotic normal approximation of survivors. These replicates are then simulated forward according to the model formulation and forecast assumptions (see table below) with the constraints on F set by the catch scenarios. The simulations are done in isolation. In each simulation, the median of the fishing mortality rate process, F, is set based on a deterministic forecast with non-linearity correction of past F and abundance, N, to hit the catch scenario target. In turn, realized F and abundance is simulated based on the model and forecast settings. Consequently, constraints set using predicted values, such as catch, will not be matched exactly by the realized catch due to process variability. From 2025, a manual back-correction procedure is applied to ensure that forecast targets are matched exactly. This procedure finds the scaling factor η needed to adjust a preliminary forecast F_0 to match a target constraint, and the back-corrected forecast obtained by re-doing the forecast with the same seed but with $F = F_0 \exp(\eta)$.

The intermediate year assumption was taken as $F_{status\ quo}$. This results in predicted catches for 2025 39% above the combined TAC in 2025 for the North Sea + West of Scotland + eastern English Channel + Skagerrak. This is in the range of the differences between the model estimated catches and the TAC in the last three years (27–92%). The combined catch (ICES landings and discards) for 2024 was 1.5% more than the combined TAC.

Forecast assumptions are as follows (note that the values that appear in Tables 4.15 are medians from the distributions that result from the stochastic forecast):

Model component/option	Setting
Method	Simulation based with 1000 replicates
Base year	Last year with catch data
Resample first year	Yes, N and F in the intermediate year are resampled from asymptotic normal distribution of the corresponding estimates

Model component/option	Setting
Maturity	Forecasted according to GMRF process
Natural mortality	Forecasted according to GMRF process
Stock weights	Forecasted according to GMRF process
Catch/landing/discard weights	Average of final three years (before intermediate year)
Landing fraction	Average of final three years (before intermediate year)
F and M before spawning	Average of final three years (before intermediate year)
Recruitment	Simulated from a lognormal with the same median and log-variance as recruitment from 2023 to 2025
Selectivity	Forecasted according to the SAM F process
F covariance	Stationary using asymptotic normal of estimates
Non-linearity correction	Yes
Intermediate year assumption	Determined by working group based on the best knowledge of the fishery at the time

Several catch scenarios were considered:

- Precautionary considerations to protect the Southern substock: F_{bar} , south corresponding to $SSB(2027) = B_{lim}$; F_{bar} , Northwest/Viking (2026) = $F(2025) \times 0.074$ (as F_{bar} , south (2026) / F_{bar} , south (2025) = 0.074)
- Without precautionary considerations: $F_{2025} \times 0.36$
- MSY approach: $F_{bar}(2026) = F_{MSY} \times \min\{1; SSB_{2026}/B_{trigger}\}$
- FMSY lower (with reduction): $F_{bar}(2026) = F_{MSY\ lower} \times \min\{1; SSB_{2026}/B_{trigger}\}$
- FMSY upper: $F_{bar}(2026) = F_{MSY\ upper}$ (presented only for substocks with $SSB_{2026} > B_{trigger}$)
- Zero catch: $F_{bar}(2026) = 0$
- F_{pa} : $F_{bar}(2026) = F_{pa}$
- $SSB(2027) = B_{lim}$: F corresponding to $SSB(2027) = B_{lim}$
- $SSB(2027) = B_{trigger} = B_{pa}$: F corresponding to $SSB(2027) = B_{trigger} = B_{pa}$
- $SSB(2027) = SSB(2026)$: F corresponding to $SSB(2027) = SSB(2026)$
- Status quo – constant F: $F_{bar}(2026) = F_{bar}(2025)$
- FMSY: $F_{bar}(2026) = F_{MSY}$ (where different from the MSY approach, i.e., for substocks where $SSB_{2026} < B_{trigger}$)
- FMSY lower: $F_{bar}(2026) = F_{MSY\ lower}$ (where different from FMSY lower (with reduction), i.e., for substocks where $SSB_{2026} < B_{trigger}$)

For the MSY approach, $F_{MSY\ lower}$ and $F_{MSY\ upper}$ scenarios, the ICES advice rule was applied manually to median SSBs rather than following the full replicate approach inherent to the “modelforecast” function, which does not incorporate rebuilding and precautionary considerations when substocks are below B_{lim} . Forecast results are given for each substock in Tables 4.15, and plots of forecasted SSB, F, recruitment and catch are given in Figures 4.19a–d. For without precautionary considerations an F multiplier is used that equates approximately to the cumulative substock advice when following the MSY approach for each substock independently and equates to a total

catch of 12 675 t. The group could not reach consensus, and two advice options are thus presented: (1) MSY approach is applied for the Southern substock and precautionary considerations to protect the Southern substock are applied for the Northwestern and Viking substocks, and (2) zero catch advice based on precautionary considerations to protect the southern substock.

The catch derived from the sum of catch advice by substock in the precautionary considerations scenario for the Northwestern and Viking substocks, consistent with the MSY approach for the Southern substock, is equal to 2812 tonnes, which is 81.9% lower than the combined advice given in 2024 (15 511 tonnes) and 88.8% lower than the combined TAC in 2025 for the North Sea + West of Scotland + eastern English Channel + Skagerrak (25 028 tonnes). This change in advice is due to:

1. Lower incoming recruitments. The recruitment assumption was changed, shortening the time-series from 1998-2024, to 2023-2025 following the lower recruitments in 2023-2025.
2. The southern substock is below B_{lim} .
3. Downscaled stock sizes compared to the forecast last year leading to a lower stock size estimate at the start of the advice year and, for the Southern substock, to a lower target fishing mortality in order to reach B_{lim} at the end of the advice year with a 50% probability.

The justification for a zero catch advice is as follows:

1. The southern substock is below B_{lim} and ICES is not able to identify any catch level that is likely to rebuild the southern substock above B_{lim} .
2. Precautionary considerations to protect the Southern substock, given the downscaling of the 2024 SSB, the continued low productivity, confirmed in survey data and age structure data, and the southern substock being below B_{lim} since 2017.

An additional forecast scenario was requested post WG meeting via ACOM. The scenario reflects an MSY approach applied to the Northwestern and Viking substocks without precautionary considerations, and the MSY approach with precautionary considerations applied to the Southern substock. Median total catches in 2026 thus must be no more than 8670 tonnes for Northwestern, 3610 for Viking and 0 tonnes for the Southern substocks. An F multiplier ($F_{2025} \times 0.35$) that equates approximately to the cumulative substock advice (12 280 tonnes) is used. As the three substocks are caught together this will result in a probability of 68% for the Southern substock of remaining below B_{lim} .

4.8 Medium-term forecasts

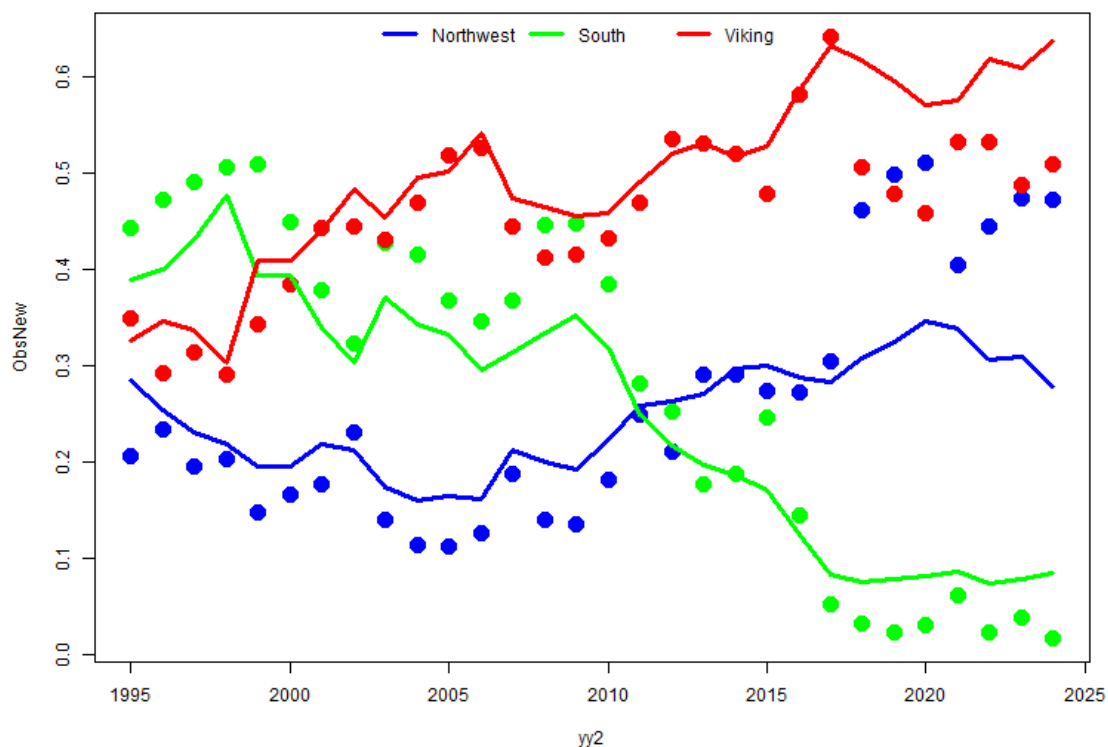
Medium-term projections are not carried out for this stock.

4.9 Quality of the assessment

The quality of the commercial landings and catch-at-age data for this stock deteriorated in the 1990s following reductions in the TAC without associated control of fishing effort. The WG considers the international landings figures for both the North Sea and West of Scotland components to have inaccuracies from the mid-1990s to the mid-2000s but no longer accounts for this within the assessment model (ICES WKNSEA, 2021).

Several historic records (from 1983–2003) in the DATRAS CA (age) data are missing haul IDs, which can preclude linking these records with the corresponding HH (hydro) records for subsequent analysis and assessment input preparation. Random haul allocations are made based on year, quarter, country, ship, and ICES rectangle or roundfish area to avoid losing age samples. A sensitivity analysis shows this to have some impact on the calculation of stock weights and maturity, mostly in the early period, but minimal influence on the indices and assessment (see Annex 8 in ICES WGNSSK, 2023). There are also concerns about the quality of maturity records prior to 1991, although this has no impact on the advice.

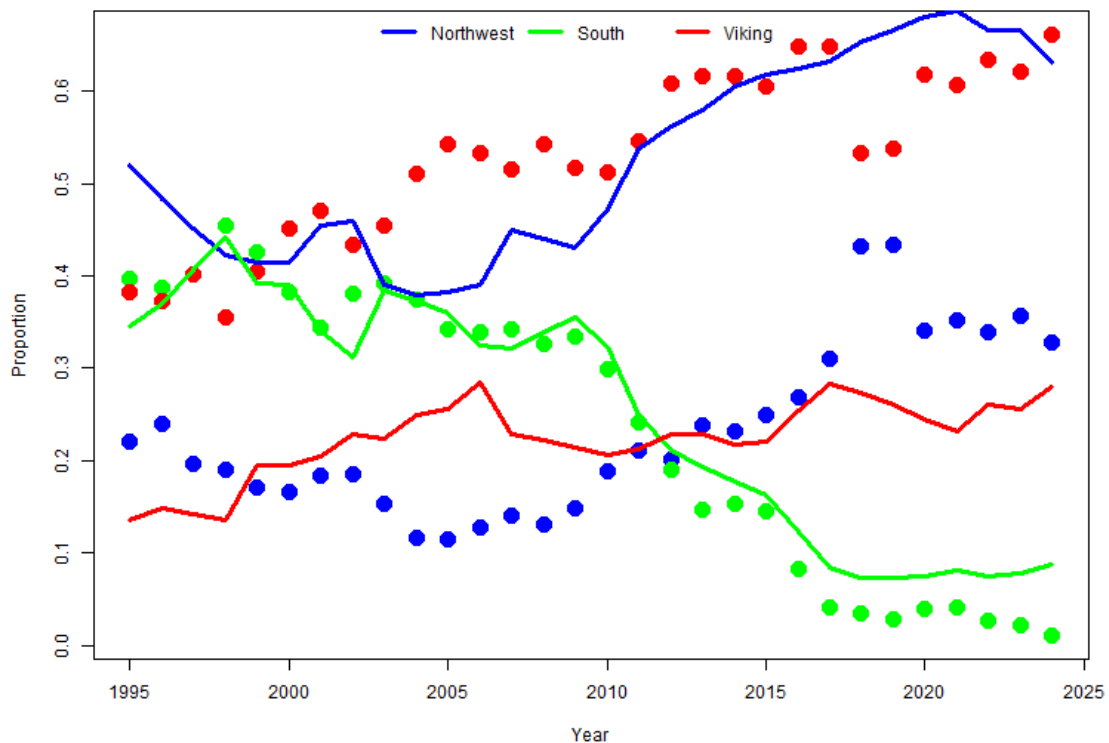
There is a lack of fit of the multistock SAM model to the substock composition data from the quarter 1 landings, which may be a consequence of assuming total stock landings fractions that are not applicable to substocks. A sensitivity run that excluded the Q1 landings proportions for substocks was performed by ICES WKBCOD (2023) and gave similar assessment results, so the benchmark concluded that this aspect of lack-of-fit was not a serious concern. Furthermore, industry representatives have warned of potential mismatches between landings and abundance due to quota and management regulations (ICES WGNSSK, 2023).



Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of modelled and observed substock landings proportions in quarter 1.

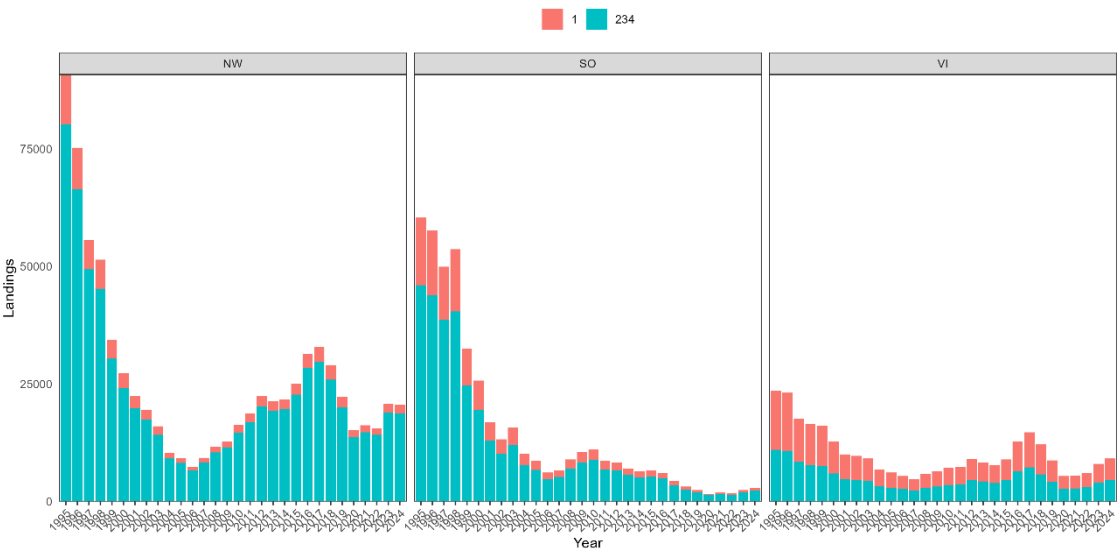
The multistock SAM assessment models the population dynamics of substocks, which are assumed to be reproductively isolated with no transfer of individuals between substocks. Aside from the assumption of geographic isolation during spawning in quarter 1, the model does not say anything about the distribution of substocks. Estimated annual landings by substock (lines) do not match the annual landings taken in the corresponding substock area (points), suggesting strong mixing during the other quarters and that catches taken in a substock area may include fish belonging to other substocks of origin. No genetic or tagging information is currently available that can be used to validate the model estimates of substock composition or substock origin.

of catches. Furthermore, it is difficult to make judgments on the quality of the assessment by substock with diagnostics that are available only for the combined stock (e.g. fits to the catch-at-age matrix and combined Q3+Q4 survey index). The IBTS Q1 survey reveals that cod aggregate near the border between the substock areas, adding uncertainty to the multistock assessment.

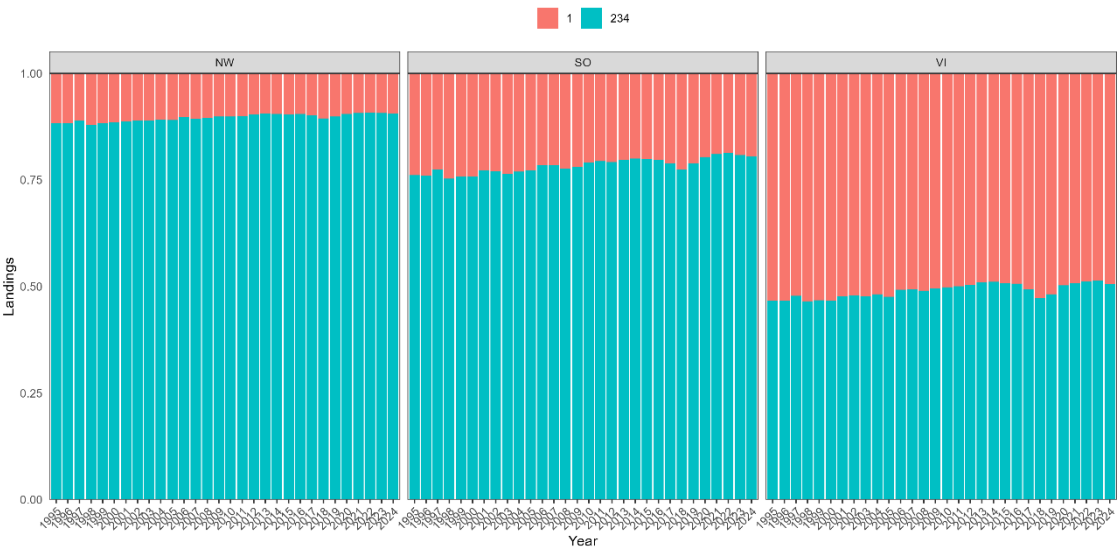


Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of annual modelled substock landings proportions derived from the multistock SAM fit to observed landings proportions by substock area. Note that annual landings proportions are not input to multistock SAM or used during model fitting.

The multistock SAM assessment estimates seasonal effects, allowing variation in fishing mortality rates by substock and quarter. Plots of absolute estimated landings and estimated landings proportions for quarter 1 against quarters 2–4 show a large proportion of the Viking substock to be landed in quarter 1 (~50%) while most of the Northwestern and Southern sub-stocks are landed in quarters 2–4 (~75–90%). No explanation for this pattern was found by the working group and no genetic information is currently available that can be used to validate these model estimates. Furthermore, while estimates of the substocks' spawning biomass are consistent with the trends apparent from the survey indices; estimates of recruitment and fishing mortality show strong correlations between substocks. This may follow from the limited data available to resolve the dynamics of the different substocks; however, the strong synchrony in recruitment may reflect similar spawning and nursery environments impacting all substocks, despite reproductive isolation (ICES WKNSCodID, 2020).



Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multi-stock SAM estimates of absolute landings in quarter 1 and quarters 2–4.



Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multi-stock SAM estimates of landings proportions in quarter 1 and quarters 2–4.

The multistock SAM assessment systematically overestimates total catches from 2011. This could be due to differences in substock catch weights that are not accounted for in the mixed substock assessment inputs and needs further exploration. Note any differences in observed catch between this table and others, e.g., Table 4.3, are due to decimal differences in catch numbers following log transformations.

Year	Observed catch	Estimated catch	% difference
2011	44836	48727	-8.0
2012	42918	50856	-15.6
2013	42811	49498	-13.5

Year	Observed catch	Estimated catch	% difference
2014	48956	48134	1.7
2015	52966	55403	-4.4
2016	52459	65655	-20.1
2017	50556	66501	-24.0
2018	51323	54568	-5.9
2019	37670	37267	1.1
2020	25093	25677	-2.3
2021	20159	30343	-33.6
2022	22914	33062	-30.7
2023	32419	40452	-19.9
2024	32119	39695	-19.1

4.10 Status of the stock

All substocks have shown an increase in SSB since 2020/2021, although this is starting to reverse for the northwestern and Viking substocks. The SSB of the Northwestern and Viking substocks is between B_{lim} and $MSY B_{trigger}$ but the SSB of the Southern substock is below B_{lim} .

Fishing mortality has declined from a peak in 2018 but is above the F_{pa} precautionary reference point, for the southern and Viking substocks and has shown an increase in the most recent years for all substocks. Fishing mortality is above F_{MSY} , the level that achieves the long-term objective of maximum yield, for all substocks.

Recruitment of 1-year old cod has varied considerably since the 1980s, but since 1998, average recruitment has been lower than any other time. The last larger recruitment observed during this period was the 2016-year class, most predominant in the Northwestern substock, while the 2025 recruitment (total of all sub-stocks) is estimated to be the second lowest in the time series.

4.11 Management considerations

Cod has been fully under the EU landing obligation since 2018 in Subarea 4 and Subdivision 20, and since 2019 in divisions 6a and 7d although there are some *de minimis* exemptions in Subarea 4 and Subdivision 20 (see Section 4.1.4). BMS landings of cod reported to ICES are currently negligible. Discarding continues based on observations from sampling programmes, with the discard rate estimated at 18% of the total catch by weight in 2024.

It is uncertain whether if and to what extent the discontinuation of the days-at-sea regulation in 2017, which was part of the cod recovery plan, had an impact on the decline of the cod substocks in the late 2010s. Although F has reduced since 2018, recent catches have not been in line with ICES advice. The SSB of the Southern substock remains below B_{lim} , with an increased risk of impaired recruitment.

The Northern Shelf cod consists of reproductively isolated populations of Viking cod and Dogger cod, with the Dogger cod displaying phenotypic spatial structure either side of a 50 m bathocline in the central North Sea (ICES WKNSCodID, 2020). Based on this, the multistock assessment adopts a three substock hypothesis, with substocks defined as Northwestern, Viking and Southern, with different rates of sexual maturity, reproduction, and growth. The multistock assessment only models the population dynamics of the substocks and does not say anything about the distribution of those substocks during Q2-4. Therefore, advice based on substocks should not be taken as advice based on substock areas.

Cod are taken by towed gears in mixed demersal fisheries, which include haddock, whiting, *Nephrops*, plaice, and sole. They are also taken in directed fisheries using fixed gears. It is important to consider both the species-specific assessments of these species for effective management, but also the broader mixed-fisheries context. This is not straightforward when stocks are managed via a series of single-species TACs that do not incorporate such mixed-stocks considerations. However, a reduction in effort on one stock may lead to a reduction or an increase in effort on another, and the implications of any change need to be considered carefully. The ICES WGMIXFISH Group monitors the consistency of the various single-species management plans under current effort schemes, to estimate the potential risks of quota over- and under-shooting for the different stocks.

Area-specific catch advice cannot be provided without additional data, including stock composition samples routinely collected from both commercial fisheries and scientific surveys. Given that the degree of mixing between substocks cannot be quantified, the advice is based on an approach to protect the weakest substock. Two advice options are presented this year as the group could not reach consensus. Both options are consistent with protection of the southern substock. Option 1) MSY approach is applied for the Southern substock and precautionary considerations to protect the Southern substock are applied for the Northwestern and Viking substocks. This involves applying a common fishing mortality reduction compared to the intermediate year across all the substocks in line with the mortality reduction for the Southern substock in the MSY approach scenario (–96%). Based on current fishing patterns, assumed stock-mixing, and single-stock assumptions, the substock catch advice would be reached simultaneously for the three substocks. However, deviations from the assumptions could result in different levels of reduction in fishing mortality for each substock. Option 2) zero catch advice based on precautionary considerations to protect the southern substock. The southern substock is below B_{lim} and ICES is not able to identify any catch level that is likely to rebuild the southern substock above B_{lim} . Management measures should thus be implemented to facilitate recovery of the vulnerable Southern substock.

4.12 Issues for future benchmarks

The stock was last benchmarked in 2023 and the primary changes consisted of updating the stock definition to combine the previous North Sea and West of Scotland cod stocks and use of a new assessment model (multistock SAM) to explicitly account for substock dynamics. Below is a list of issues that were either left unresolved from the last benchmark or have arisen during subsequent WGNSSK meetings. A scoring system has been developed to aid working groups in prioritizing stocks to be put forward for benchmark (see Annex 5 for further details). The current scoring for this stock is:

1. Assessment quality	2. Opportunity to improve	3. Benchmark preparedness	4. Perceived risks	5. Changing ecosystem and ability to include impact	6. Time since last benchmark	Total Score
3	2	2	5	0	2	3

4.12.1 Data

4.12.1.1 Stock identity

No stock composition (i.e., genetics, otoliths) or tagging information is currently available that can be used to validate the model estimates of substock composition or substock origin of catches. ICES WKBCOD (2023) recommended that genetic information be sampled on a regional basis. If it were to become available, genotype data can be included directly in the multistock SAM model. A new ICES special request initiative, WKCODSCOPE and WKCODSAMPLING will address several data needs for Northern Shelf cod, including collection of stock composition data and possible future assessment approaches, it is unknown when these data will become available.

4.12.1.2 Commercial catches

ICES WKBCOD (2023) recommended that national catch sampling programs take the new cod substock structure into account when planning future commercial catch sampling. Along with stock composition samples and improved knowledge of mixing, this could allow catch numbers and weights at age to be provided by substock.

4.12.1.3 Maturity

There are concerns about the reliability of DATRAS maturity records prior to 1990. This does not impact the advice but should, nonetheless, be investigated.

ICES WKNSEA (2015) raised concerns that accounting for the increase in maturity may give the impression that the spawning stock is in better condition than it is given the possibility of lower fecundity of younger age groups and the potential for a maternal age effect on survival, and recommended exploration of the significance of spawner age on reproductive potential.

4.12.1.4 Natural mortality

Further work should be conducted to split natural mortality to substock, either by explicitly modelling three substocks in the SMS multispecies model or by post-processing the outputs of the multispecies keyrun. Further research is required to quantify seal predation on the North-western substock.

4.12.1.5 Surveys

Catchability issues and year effects have become apparent in the survey indices, with reduced cohort consistency and lower than expected catch rates of older fish in the late 2010s. This was not completely resolved by combining the former the North Sea and West of Scotland cod stocks. Age reading issues may contribute and should be investigated.

Use of spatial GAM modelling approaches could allow the inclusion of smaller localised surveys in the indices for Northern Shelf cod, such as data from the Shetland Inshore Survey (Fraser *et al.*, 2024).

4.12.1.6 Recreational catches

Recreational catches are estimated to account for 2.2-6.0% of the total removals of this stock but are not included in the assessment due to length of time-series and unknown age structure and uncertainty (Section 4.2.1). Work on standardization of recreational inputs should be given relevance for future consideration in the assessment.

4.12.2 Assessment

It should be investigated how to improve the multistock SAM model fits to the quarter 1 landings by substock. This may involve substock specific landings fractions or catch weights at age. Furthermore, in the absence of new data, it could be explored to have Q1 and Q2–4 effects instead of seasonal effects per quarter.

ICES WKBCOD (2023) recommended to investigate if the optimal multistock SAM configuration is the same for each substock and to conduct simulation self-tests of the preferred model.

4.12.3 Forecast

Short-term forecasts in a given year tend to be more optimistic than realized values in subsequent years. Although recent catches have not been in line with ICES advice, potential biases in the forecasting procedure should be investigated.

4.13 References

- Albertsen, C.M., Nielsen, A. and Thygesen, U.H. 2018. Connecting single-stock assessment models through correlated survival. *ICES Journal of Marine Science*, 75: 235–244.
- Berg, C.W. and Kristensen, K. 2012. Spatial age-length key modelling using continuation ratio logits. *Fisheries Research*, 129:119–126.
- Berg, C.W. and Nielsen, A. 2016. Accounting for correlated observations in an age-based state-space stock assessment model. *ICES Journal of Marine Science*, 73: 1788–1797.
- Berg, C.W., Nielsen, A., Kristensen, K. 2014. Evaluation of alternative age-based methods for estimating relative abundance from survey data in relation to assessment models. *Fisheries Research*, 151: 91–99.
- Fraser, S., Thomason, L., McAllister, M., and Harlow, L. 2024. Shetland Inshore Fish Survey (SIFS) 2024: summary of catch rates, size compositions, and spatial distributions of commercial demersal fish species. UHI Shetland.
- ICES-WGMG. 2013. Report of the Working Group on Methods of Fish Stock Assessments (WGMG), 30 September - 4 October 2013, Reykjavik, Iceland. ICES CM 2013/SSGSUE:08. 130 pp.
- ICES-WGNSSK 2023. Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK). ICES Scientific Reports. 5:39. 1536 pp. <https://doi.org/10.17895/ices.pub.22643143>
- ICES. 2024. Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK). ICES Scientific Reports. To be added Report. <https://doi.org/10.17895/ices.pub.25605639.v2>
- ICES-WK6aCodID. 2022. Workshop on Stock Identification of West of Scotland Sea Cod (WK6aCodID; outputs from 2021 meeting). ICES Scientific Reports. 4:5. 24pp. <http://doi.org/10.17895/ices.pub.10031>
- ICES-WKBCOD. 2023. Benchmark workshop on Northern Shelf cod stocks (WKBCOD). ICES Scientific Reports. 5:37. 425 pp. <https://doi.org/10.17895/ices.pub.22591423>
- ICES-WKDEM. 2020. Benchmark Workshop for Demersal Species (WKDEM). ICES Scientific Reports. 2:31. 136 pp. <http://doi.org/10.17895/ices.pub.5548>
- ICES-WKMOG. 2008. Report of the Workshop on Maturity Ogive Estimation for Stock Assessment (WKMOG), 3–6 June 2008, Lisbon, Portugal. ICES CM2008/ACOM:33. 72 pp.
- ICES-WKNSCodID. 2020. Workshop on Stock Identification of North Sea Cod (WKNSCodID). ICES Scientific Reports. 2:89. 82 pp. <http://doi.org/10.17895/ices.pub.7499>.
- ICES-WKNSEA. 2015. Report of the Benchmark Workshop on North Sea Stocks (WKNSEA), 2–6 February 2015, Copenhagen, Denmark. ICES CM 2015/ACOM:32. 253 pp.

- ICES-WKNSEA. 2021. Benchmark Workshop in North Sea Stocks (WKNSEA). ICES Scientific Reports. 3:25. 756 pp. <https://doi.org/10.17895/ices.pub.7922>
- ICES-WKREBUILD2. 2023. Workshop on guidelines and methods for the design and evaluation of rebuilding plans for category 1-2 stocks (WKREBUILD2). ICES Scientific Reports. 5:112. 79 pp. <https://doi.org/10.17895/ices.pub.24763293>
- ICES-WKROUNDMP. 2011. Report of the Joint ICES-STEFC Workshop on management plan evaluations for roundfish stocks (WKROUNDMP/EWG 11-01), 28 February - 4 March 2011, ICES Headquarters, Copenhagen. 67 pp.
- ICES-WKRRCOD. 2022. Workshop on research needs and a roadmap for further research on cod in the northern shelf seas (including cod in the Celtic Seas) (WKRRCOD). ICES Scientific Reports. 4:97. 23 pp. <https://doi.org/10.17895/ices.pub.21740090>
- ICES-WKRRCOD2. 2023. Workshops on research needs and a roadmap for further research on cod in the northern shelf seas (including cod in the Celtic Seas) (WKRRCOD2). ICES Scientific Reports. 5:113. 22 pp. <https://doi.org/10.17895/ices.pub.24825543>
- Kraak, S.B.M., Bailey, N., Cardinale, M., Darby, C., De Oliveira, J.A.A., Eero, M., Graham, N., Holmes, S., Jakobsen, T., Kempf, A., Kirkegaard, E., Powell, J., Scott, R.D., Simmonds, E.J., Ulrich, C., Vanhee, W., and M. Vinther. 2013. Lessons for fisheries management from the EU cod recovery plan. *Marine Policy*, 37: 200–213.

4.14 Tables and figures

Table 4.1a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Nominal landings (in tonnes) as officially reported to ICES, and as used by the Working Group.

Subarea 4																			
Country	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023***	2024***
Belgium	1309	1008	894	946	666	653	862	1075	1258	1223	1103	695	817	727	666	544	296	572	274
Denmark	5105	3430	3831	4402	5686	4863	4803	4536	5457	6026	6713	6119	5493	4967	3069	2023	2018	2998	3394
France	354	659	573	950	782	619	369	287	637	517	391	401	583	450	265	284	255	320	286
Germany	2537	1899	1736	2374	2844	2211	2385	1921	2257	2133	2083	2300	1510	824	757	775	716	1401	1069
Ireland	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Netherlands	1585	1523	1896	2649	2657	1928	1955	1344	1242	1403	1365	653	515	719	590	618	594	677	789
Norway	2749	3057	4128	4234	4495	4898	4601	4080	4600	5404	5627	5521	5539	4509	2267	1651	2258	3486	3561
Sweden	309	386	439	378	362	316	471	332	401	415	373	387	274	344	355	396	390	402	373
UK (E/W/Nl)	1491	1587	1546	2384	2553	2170	1630	2129	2962	0	0	0	0	0	0	0	1	0	0
UK (Scotland)	6857	6511	7185	9051	11567	10140	10564	10619	10517	0	0	0	0	0	0	0	0	0	0
UK (combined)	0	0	0	0	0	0	0	0	0	14889	16603	18523	21265	15621	9114	6418	6914	11352	12914
Others	812	18	35	59	32	2	0	0	0	0	2	1	0	0	1	0	2	6	282
BMS landings												1	8	30	21	2	1	2	2
Total Nominal Catch	23108	20079	22261	27425	31645	27799	27640	26322	29331	32011	34260	34599	35997	28160	17083	12709	13054	21214	22942
Unallocated landings	-2029	-1023	-604	208	-665	-1125	-1013	-1006	-782	-767	-1225	-1490	-1553	83	-193	-129	200	-211	-284
WG estimate of total landings^^	21078	19056	21657	27634	30980	26675	26627	25315	28550	31244	33035	33109	34444	28243	16890	12580	13254	21003	22657
Agreed TAC*	23205	19957	22152	28798	33552	26842	26475	26475	27799	29189	33651	39220	43156	29437	14718	13246	13246	21652	19910
Division 6a																			
Country	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023***	2024***
Belgium	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Denmark	0	0	0	0	0	0	0	0	0	0	11	1	0	0	0	0	0	0	0
France	108	92	82	74	60	49	4	3	5	11	86	118	101	142	139	161	124	82	52
Germany	2	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Ireland	18	70	58	24	49	41	18	14	12	17	28	19	12	0	68	98	115	122	124
Netherlands	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Norway	30	30	65	18	20	8	2	24	13	59	39	14	37	47	4	0	2	14	8
Sweden	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
UK (E/W/Nl)	14	21	6	14	4	3	2	1	9	0	0	0	0	0	0	0	1	0	0
UK (Scotland)	318	260	232	104	115	107	135	130	121	0	0	0	0	0	0	0	0	0	0
UK (combined)	0	0	0	0	0	0	0	0	0	168	183	200	217	1224	736	923	915	878	837
Others	1	12	1	0	0	0	0	0	0	0	1	0	0	31	35	27	36	26	3
BMS landings															2				
Total Nominal Catch	492	488	445	235	248	208	162	172	161	256	348	352	367	1443	982	1210	1194	1121	1023
Unallocated landings	13	27	117	50	110	133	65	93	234	272	261	323	624	617	292	55	7	-16	-16
WG estimate of total landings^^	504	515	561	284	358	341	227	266	394	528	609	675	990	2060	1273	1265	1201	1105	1007
Agreed TAC*	613	490	402	302	240	182	0	0	0	0	0	0	0	1735	1279	1279	1279	1210	1114

Division 7d

Country	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023***	2024***
Belgium	80	84	154	73	57	56	40	53	72	78	39	18	8	3	8	12	4	3	1
Denmark	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
France	1124	1743	1326	1779	1606	1078	885	768	1270	1142	279	92	35	16	10	5	20	7	3
Germany	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
Ireland	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Netherlands	9	59	30	35	45	51	40	38	50	52	40	22	10	3	2	1	1	2	1
Norway	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sweden	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
UK (E/W/Nl)	267	174	144	133	127	125	99	100	156	0	0	0	0	0	0	0	0	0	0
UK (Scotland)	1	12	7	3	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
UK (combined)	0	0	0	0	0	0	0	0	0	162	102	48	39	17	20	11	11	19	5
Others	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
BMS landings															0	0	0	0	0
Total Nominal Catch	1480	2073	1662	2023	1836	1311	1064	959	1548	1434	459	180	92	38	40	28	38	31	10
Unallocated landings	-2	74	-33	-135	-128	8	56	-43	-112	-36	-38	-10	-8	-3	-8	9	0	1	0
WG estimate of total landings	1479	2147	1629	1887	1708	1319	1120	916	1436	1398	421	170	84	36	32	37	38	32	10
Agreed TAC^				1678	1955	1564	1543	1543	1620	1701	1961	2059	1733	1715	858	772	772	11261	1158
Subdivision 20 **																			
Country	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023***	2024***
Belgium	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Denmark	2513	2246	2553	3024	3286	3118	3178	3033	3430	3344	3696	3665	4222	2740	1856	1600	1519	2212	2158
France	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Germany	84	67	52	55	56	60	78	69	84	87	94	67	87	53	35	25	35	67	45
Ireland	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
Netherlands	47	19	13	1	26	0	0	28	28	26	25	37	58	39	18	11	8	12	10
Norway	628	681	779	440	375	421	615	575	533	500	551	486	288	151	94	67	123	117	142
Sweden	373	371	366	459	459	518	520	529	570	571	641	557	670	355	224	238	242	377	420
UK (E/W/Nl)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
UK (Scotland)	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0
UK (combined)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Others	326	366	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BMS landings												1	4						0
Total Nominal Catch	3970	3749	3763	3984	4201	4117	4390	4239	4646	4527	5006	4812	5325	3339	2226	1942	1687	2786	2775
Unallocated landings	-712	-730	-370	-190	-144	-161	-63	-85	41	36	-232	-97	159	26	15	31	230	18	2
WG estimate of total landings	3258	3020	3393	3794	4057	3956	4327	4154	4687	4563	4774	4715	5484	3364	2241	1973	1916	2803	2777
Agreed TAC	3315	2851	3165	4114	4793	3835	3783	3783	3972	4171	4807	5744	7995	4205	2103	1893	1893	3095	32846

Subarea 4, Division 6.a, Division 7.d, and Subdivision 20																			
	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023***	2024***
Total Nominal Catch	29050	26389	28131	33667	37930	33436	33256	31692	35686	38228	40073	39943	41781	32981	20331	15889	15973	25151	26749
WG estimate of total landings	26319	24737	27240	33600	37104	32291	32300	30652	35067	37734	38839	38669	41002	33703	20436	15855	16409	24943	26451
BMS landings	0	0	0	0	0	0	0	0	0	0	0	2	12	30	23	2	1	2	2
Agreed TAC	27133	23298	25719	35132	40540	32423	31801	31801	33391	35061	40419	47023	52884	37092	18958	17190	17190	27218	2

* TAC prior to 2010 is for the whole of Subdivision 5.b1 and subareas 6, 12, and 14. TAC from 2010 onwards is for Subdivision 5.b1 and Division 6.a only.

** Skagerrak/Kattegat split derived from national statistics prior to 2016.

*** Official landings are preliminary.

^ No agreed TAC in division 7d prior to 2009.

^^ Includes ICES estimates of area misreporting from 2006–2021.

Table 4.1b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Sampling coverage for discard ratio, landings age composition and discards age composition by area and season (quarter or annual, depending on data stratification) for 2024, calculated as the weight in each area–season–métier stratum covered by the relevant sampling, then summed over métiers and expressed as a proportion of the total for the area–season (note the country dimension is not used). Also provided is the contribution of landings, discards and BMS in each area (by weight) to the total for that catch category (before raising is conducted).

Discard ratio coverage

Area/Season	Q1	Q2	Q3	Q4	annual
27.4	81%	73%	63%	74%	29%
27.3.a.20	70%	84%	77%	70%	
27.6.a	93%	92%	73%	93%	98%
27.7.d	39%	60%	54%	76%	0%

Landings age composition coverage

Area/Season	Q1	Q2	Q3	Q4	annual
27.4	93%	82%	79%	88%	29%
27.3.a.20	97%	97%	93%	98%	
27.6.a	93%	82%	79%	88%	29%
27.7.d	0%	0%	0%	0%	0%

Discards age composition coverage

Area/Season	Q1	Q2	Q3	Q4	annual
27.4	99%	100%	100%	97%	100%
27.3.a.20	100%	99%	99%	99%	
27.6.a	0%	0%	0%	0%	100%
27.7.d	0%	0%	0%	0%	0%

Contribution to total (before raising)

Area/Type	Landings	Discards	BMS
27.4	86%	91%	0%
27.3.a.20	10%	4%	0%
27.6.a	4%	5%	0%
27.7.d	0%	0%	0%

Table 4.2a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Landings numbers-at-age (thousands).

Year/Age	1	2	3	4	5	6	7+
1983	27823	132911	22574	13187	3271	1394	967
1984	66501	68386	33651	4686	3970	1197	858
1985	10150	122453	20157	8870	1509	1425	746
1986	100633	33794	36028	6202	3171	642	890
1987	32690	132610	10749	9594	1665	1127	589
1988	22370	63361	45906	3395	2754	690	483
1989	24089	37165	21914	10500	1107	1097	489
1990	12141	57807	12227	5222	2614	331	452
1991	14806	24873	18359	3454	1637	1110	423
1992	28569	33340	9098	5636	1110	540	463
1993	5016	57767	12183	3257	1729	450	382
1994	16308	26176	22430	3325	916	564	271
1995	16238	65400	13452	5853	860	297	232
1996	5211	38634	24255	3372	2038	424	313
1997	24856	29859	19379	6690	1287	752	244
1998	1554	84266	17086	6084	2435	520	315
1999	5962	10434	32671	4023	1466	638	343
2000	8851	22989	6521	6760	1011	388	211
2001	2260	21550	6199	1176	1104	143	133
2002	4762	8369	14353	2828	408	398	123
2003	491	9380	3645	3193	405	70	85
2004	1515	3611	5544	1539	965	157	57
2005	1046	8299	3081	1754	487	346	78
2006	1304	4498	4488	992	534	189	149
2007	784	6504	2318	1590	401	140	89
2008	338	3288	4236	1166	719	214	113
2009	520	4842	2849	2929	602	239	91
2010	1126	5117	4603	1602	1325	199	94
2011	1099	4569	4097	1426	615	456	84

Year/Age	1	2	3	4	5	6	7+
2012	666	2230	5385	1986	637	251	165
2013	683	2696	3070	2630	875	191	129
2014	2240	4212	4449	1639	1311	334	110
2015	686	6428	4943	1963	766	603	176
2016	168	2052	5726	3202	1029	286	259
2017	351	2252	3284	3541	1707	399	210
2018	172	6015	3628	2117	1827	812	430
2019	871	1869	6139	1165	1071	546	491
2020	845	2899	1649	1530	465	322	215
2021	261	1425	1975	735	626	118	171
2022	169	838	1729	1290	463	274	147
2023	74	1714	1564	2232	1090	293	195
2024	56	984	2789	1545	1478	454	210

Table 4.2b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Discard numbers-at-age (including BMS landings from 2016; thousands).

Year/Age	1	2	3	4	5	6	7+
1983	55792	11155	25	0	0	0	0
1984	538609	12529	5	0	0	0	0
1985	68489	36687	115	0	0	0	0
1986	564476	5775	303	0	0	0	0
1987	38992	61960	0	0	0	0	0
1988	15607	18143	218	0	0	0	0
1989	183163	8691	489	0	0	0	0
1990	38056	48285	78	0	0	0	0
1991	49991	8397	450	0	0	0	0
1992	109795	10024	5	0	0	0	0
1993	33712	28622	12	0	0	0	0
1994	323468	16810	160	0	0	0	0
1995	45381	43490	30	0	0	0	0
1996	14445	23108	764	0	0	0	0

Year/Age	1	2	3	4	5	6	7+
1997	87630	13735	40	0	0	0	0
1998	16248	91231	1500	0	0	0	0
1999	31192	5635	8280	0	0	0	0
2000	39913	5542	0	0	0	0	0
2001	5636	33209	753	0	0	0	0
2002	27318	13435	3181	17	0	0	0
2003	5820	6108	782	55	0	0	0
2004	20574	8964	2007	122	6	0	0
2005	10340	8325	1795	149	66	12	2
2006	27948	35387	1990	53	7	1	1
2007	16453	24368	2703	517	54	27	2
2008	10911	9482	8008	226	15	11	0
2009	10199	9211	2771	796	58	24	12
2010	10277	9960	1403	81	36	0	0
2011	4239	8430	2180	183	7	8	2
2012	12178	5370	2028	192	5	1	1
2013	8363	6402	1076	564	107	40	19
2014	8669	7452	2078	279	104	7	0
2015	4244	9506	2586	252	73	66	22
2016	4320	5318	3299	351	44	1	0
2017	10007	3244	2183	716	138	14	2
2018	2858	9485	944	447	62	18	8
2019	4258	1698	944	22	5	0	0
2020	11192	4387	296	19	0	0	0
2021	2907	2852	419	7	10	0	0
2022	3155	2963	1769	204	19	0	0
2023	2697	4341	1546	413	29	0	0
2024	529	2742	1999	230	58	6	1

Table 4.2c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Catch numbers-at-age (thousands).

Year/Age	1	2	3	4	5	6	7+
1983	83615	144066	22599	13187	3271	1394	967
1984	605110	80915	33656	4686	3970	1197	858
1985	78639	159140	20272	8870	1509	1425	746
1986	665109	39569	36331	6202	3171	642	890
1987	71682	194571	10749	9594	1665	1127	589
1988	37977	81504	46124	3395	2754	690	483
1989	207252	45856	22403	10500	1107	1097	489
1990	50197	106092	12305	5222	2614	331	452
1991	64797	33269	18809	3454	1637	1110	423
1992	138364	43364	9103	5636	1110	540	463
1993	38728	86389	12196	3257	1729	450	382
1994	339777	42986	22590	3325	916	564	271
1995	61620	108890	13482	5853	860	297	232
1996	19656	61741	25019	3372	2038	424	313
1997	112487	43595	19419	6690	1287	752	244
1998	17801	175497	18586	6084	2435	520	315
1999	37155	16069	40950	4023	1466	638	343
2000	48764	28531	6521	6760	1011	388	211
2001	7895	54759	6952	1176	1104	143	133
2002	32080	21805	17534	2845	408	398	123
2003	6311	15488	4427	3247	405	70	85
2004	22089	12574	7550	1661	971	158	57
2005	11385	16624	4876	1903	553	358	80
2006	29253	39885	6479	1045	541	190	150
2007	17237	30872	5021	2107	455	167	91
2008	11249	12770	12244	1392	734	225	113
2009	10719	14053	5620	3724	660	263	103
2010	11403	15077	6006	1683	1362	199	94
2011	5338	12999	6277	1608	621	464	86

Year/Age	1	2	3	4	5	6	7+
2012	12845	7600	7412	2178	641	252	166
2013	9046	9098	4146	3194	982	231	148
2014	10909	11664	6527	1918	1415	342	110
2015	4930	15934	7529	2215	839	669	198
2016	4488	7371	9025	3553	1074	287	259
2017	10358	5496	5467	4257	1845	412	212
2018	3030	15500	4571	2564	1889	830	438
2019	5129	3568	7082	1187	1075	546	491
2020	12037	7286	1944	1549	465	322	215
2021	3168	4277	2394	743	637	118	171
2022	3324	3801	3498	1494	482	274	147
2023	2771	6055	3110	2645	1119	293	195
2024	586	3726	4788	1775	1536	461	211

Table 4.3. Cod in Subarea 4, Divisions 6.a and 7.d, and Subdivision 20. Reported landings, estimated discards (including BMS landings from 2016) and total catch (landings + discards) as SOP values in tonnes. Note any differences in values between Table 4.3 and those given in the report and advice are due to SOP correction.

Year	Landings	Discards	Catch
1963	129642	12347	141988
1964	148555	4922	153477
1965	203152	29526	232677
1966	237301	37986	275287
1967	274661	23377	298037
1968	311283	17563	328846
1969	221348	4795	226143
1970	237648	17737	255385
1971	337142	84089	421230
1972	366854	33810	400664
1973	250139	30224	280362
1974	226843	39759	266602
1975	217412	37149	254561

Year	Landings	Discards	Catch
1976	250417	72669	323086
1977	220942	139341	360284
1978	308168	32625	340793
1979	282104	162517	444620
1980	311634	294529	606163
1981	357480	57379	414859
1982	323875	54579	378454
1983	278939	21628	300567
1984	248342	151333	399675
1985	232961	32261	265221
1986	213099	138867	351966
1987	235011	30094	265105
1988	203615	10873	214487
1989	156747	63732	220479
1990	137010	27318	164328
1991	112369	18821	131190
1992	121826	37972	159798
1993	130261	21550	151811
1994	118843	98869	217712
1995	145836	31848	177684
1996	134148	14093	148240
1997	129468	33683	163151
1998	150351	40641	190992
1999	98309	13711	112019
2000	72544	14181	86725
2001	50787	13611	64398
2002	55395	12380	67775
2003	31712	4065	35777
2004	28318	8796	37114

Year	Landings	Discards	Catch
2005	28679	9973	38652
2006	26167	12380	38547
2007	24730	32070	56800
2008	27376	26021	53396
2009	33469	22131	55600
2010	37115	13899	51014
2011	32323	12515	44839
2012	32363	10554	42916
2013	30744	12073	42817
2014	35048	13906	48954
2015	37938	15038	52976
2016	38754	13702	52456
2017	38355	12202	50557
2018	41147	10170	51317
2019	33995	3677	37672
2020	20185	4905	25090
2021	15754	4405	20159
2022	16150	6761	22911
2023	24634	7783	32418
2024	26232	6167	32399

Table 4.4a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Landings weights-at-age (kg).

Year/Age	1	2	3	4	5	6	7+
1983	0.628	0.931	1.892	3.935	6.464	8.387	10.667
1984	0.599	1.021	2.233	4.106	6.247	8.400	11.063
1985	0.595	0.942	2.170	4.254	6.375	8.420	11.319
1986	0.584	0.872	1.888	3.580	6.236	8.149	10.829
1987	0.610	0.988	2.031	3.737	6.061	8.291	11.700
1988	0.595	0.921	2.025	3.347	6.068	7.930	11.303
1989	0.676	1.059	1.948	3.663	5.452	7.896	10.766

Year/Age	1	2	3	4	5	6	7+
1990	0.732	0.997	2.199	3.884	6.002	7.970	11.092
1991	0.667	1.079	2.095	3.991	6.146	8.039	10.442
1992	0.698	1.154	2.550	4.229	6.243	8.414	10.566
1993	0.704	1.081	2.519	4.555	6.509	8.078	10.194
1994	0.675	1.085	2.249	4.491	7.119	8.435	10.216
1995	0.716	1.026	2.244	4.364	7.183	8.958	10.859
1996	0.697	1.123	2.176	4.109	6.662	8.464	10.566
1997	0.653	0.969	2.143	3.862	6.265	8.347	10.688
1998	0.555	0.927	1.733	3.524	5.404	7.577	10.104
1999	0.641	0.944	1.680	3.327	5.751	7.435	9.318
2000	0.618	1.027	1.770	3.247	4.930	7.502	9.918
2001	0.722	1.003	2.312	3.711	5.870	7.341	9.837
2002	0.629	1.002	1.862	3.751	5.558	7.978	10.101
2003	0.582	1.077	1.905	3.365	5.760	6.731	9.751
2004	0.722	1.071	2.091	3.261	5.208	7.451	9.714
2005	0.747	1.159	1.959	3.654	5.237	7.214	10.170
2006	0.791	1.199	2.239	3.892	5.688	7.239	9.575
2007	0.829	1.181	2.370	4.055	6.055	8.239	9.939
2008	1.067	1.388	2.471	4.077	6.225	7.396	10.330
2009	0.788	1.413	2.675	4.154	6.118	7.495	10.515
2010	0.716	1.295	2.672	4.221	6.047	8.302	10.194
2011	0.862	1.331	2.529	4.595	6.476	7.827	9.902
2012	0.938	1.369	2.357	4.189	6.391	8.110	9.554
2013	0.883	1.240	2.462	4.163	6.186	8.335	9.907
2014	0.699	1.213	2.385	4.157	5.659	7.421	9.524
2015	0.596	1.214	2.296	4.108	5.922	6.877	9.221
2016	0.800	1.316	2.342	3.863	5.734	7.319	8.284
2017	0.753	1.120	2.374	3.914	5.399	6.836	9.388
2018	0.609	1.066	1.949	3.850	5.627	6.813	8.358

Year/Age	1	2	3	4	5	6	7+
2019	0.769	1.120	2.150	3.721	5.482	7.123	8.011
2020	0.756	1.358	1.928	3.842	5.448	6.609	8.777
2021	0.746	1.328	2.570	3.902	5.389	7.110	8.780
2022	0.692	1.315	2.583	3.904	4.862	6.744	9.058
2023	0.639	1.199	2.207	4.124	5.566	6.659	9.544
2024	0.713	1.226	2.143	3.714	5.582	7.072	8.610

Table 4.4b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Discard weights-at-age (includes BMS landings from 2016; kg).

Year/Age	1	2	3	4	5	6	7+
1983	0.296	0.458	0.534	0.000	0.000	0.000	0.000
1984	0.270	0.469	0.509	0.000	0.000	0.000	0.000
1985	0.269	0.376	0.652	0.000	0.000	0.000	0.000
1986	0.242	0.365	0.437	0.000	0.000	0.000	0.000
1987	0.211	0.353	0.000	0.000	0.000	0.000	0.000
1988	0.300	0.336	0.459	0.000	0.000	0.000	0.000
1989	0.326	0.431	0.484	0.000	0.000	0.000	0.000
1990	0.246	0.371	0.526	0.000	0.000	0.000	0.000
1991	0.311	0.366	0.393	0.000	0.000	0.000	0.000
1992	0.308	0.417	2.353	0.000	0.000	0.000	0.000
1993	0.274	0.430	0.947	0.000	0.000	0.000	0.000
1994	0.287	0.362	0.483	0.000	0.000	0.000	0.000
1995	0.314	0.404	0.553	0.000	0.000	0.000	0.000
1996	0.340	0.380	0.515	0.000	0.000	0.000	0.000
1997	0.313	0.453	0.616	0.000	0.000	0.000	0.000
1998	0.352	0.375	0.481	0.000	0.000	0.000	0.000
1999	0.257	0.389	0.422	0.000	0.000	0.000	0.000
2000	0.297	0.422	0.000	0.000	0.000	0.000	0.000
2001	0.232	0.361	0.406	0.000	0.000	0.000	0.000
2002	0.244	0.324	0.413	2.205	0.000	0.000	0.000
2003	0.263	0.346	0.497	0.528	0.000	0.000	0.000

Year/Age	1	2	3	4	5	6	7+
2004	0.236	0.271	0.686	0.864	3.852	11.300	0.000
2005	0.301	0.565	0.814	2.223	4.255	6.509	8.100
2006	0.226	0.118	0.852	2.593	5.093	5.527	11.061
2007	0.286	0.821	1.713	3.984	6.540	10.245	15.507
2008	0.403	0.739	1.717	3.014	5.303	9.341	5.128
2009	0.378	0.990	2.025	3.549	6.498	8.084	13.047
2010	0.291	0.825	1.611	3.153	4.933	8.369	6.728
2011	0.276	0.704	2.147	3.483	5.511	5.047	7.102
2012	0.226	0.575	1.960	3.569	6.456	8.579	11.610
2013	0.280	0.815	1.619	3.279	4.267	8.419	7.065
2014	0.294	0.802	1.964	3.102	3.901	4.561	4.423
2015	0.331	0.777	1.788	3.236	4.091	5.328	6.896
2016	0.318	0.950	1.793	3.270	4.613	5.439	0.000
2017	0.232	0.808	2.007	3.149	3.865	6.008	5.036
2018	0.279	0.628	1.587	3.325	5.078	4.980	4.359
2019	0.322	0.554	1.381	2.295	2.805	0.000	0.000
2020	0.219	0.477	0.976	3.709	0.000	0.000	0.000
2021	0.290	0.928	1.972	5.079	4.754	0.000	0.000
2022	0.278	0.775	1.645	2.906	4.110	0.000	11.371
2023	0.249	0.802	1.613	2.565	2.593	10.516	11.562
2024	0.322	0.727	1.479	2.406	3.065	4.788	4.039

Table 4.4c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Catch weights-at-age (kg).

Year/Age	1	2	3	4	5	6	7+
1983	0.406	0.894	1.891	3.935	6.464	8.387	10.667
1984	0.307	0.935	2.233	4.106	6.247	8.400	11.063
1985	0.310	0.812	2.161	4.254	6.375	8.420	11.319
1986	0.293	0.798	1.876	3.580	6.236	8.149	10.829
1987	0.393	0.786	2.031	3.737	6.061	8.291	11.700
1988	0.474	0.791	2.018	3.347	6.068	7.930	11.303

Year/Age	1	2	3	4	5	6	7+
1989	0.366	0.941	1.916	3.663	5.452	7.896	10.766
1990	0.364	0.712	2.188	3.884	6.002	7.970	11.092
1991	0.393	0.899	2.055	3.991	6.146	8.039	10.442
1992	0.389	0.984	2.550	4.229	6.243	8.414	10.566
1993	0.329	0.866	2.517	4.555	6.509	8.078	10.194
1994	0.305	0.802	2.237	4.491	7.119	8.435	10.216
1995	0.420	0.777	2.240	4.364	7.183	8.958	10.859
1996	0.435	0.845	2.126	4.109	6.662	8.464	10.566
1997	0.388	0.807	2.140	3.862	6.265	8.347	10.688
1998	0.370	0.641	1.632	3.524	5.404	7.577	10.104
1999	0.319	0.749	1.425	3.327	5.751	7.435	9.318
2000	0.355	0.909	1.770	3.247	4.930	7.502	9.918
2001	0.373	0.614	2.106	3.711	5.870	7.341	9.837
2002	0.301	0.585	1.598	3.742	5.558	7.978	10.101
2003	0.289	0.789	1.656	3.317	5.760	6.731	9.751
2004	0.270	0.500	1.718	3.084	5.199	7.462	9.714
2005	0.342	0.861	1.537	3.542	5.119	7.191	10.123
2006	0.251	0.240	1.813	3.827	5.681	7.226	9.589
2007	0.311	0.897	2.017	4.037	6.113	8.565	10.061
2008	0.423	0.906	1.978	3.904	6.207	7.493	10.325
2009	0.398	1.136	2.355	4.024	6.151	7.549	10.803
2010	0.333	0.984	2.424	4.169	6.017	8.302	10.194
2011	0.397	0.924	2.396	4.469	6.466	7.780	9.827
2012	0.263	0.808	2.248	4.134	6.392	8.112	9.565
2013	0.325	0.941	2.243	4.007	5.977	8.350	9.547
2014	0.377	0.950	2.251	4.004	5.531	7.361	9.523
2015	0.368	0.953	2.122	4.009	5.763	6.724	8.960
2016	0.336	1.052	2.142	3.804	5.687	7.310	8.284
2017	0.249	0.935	2.227	3.785	5.283	6.809	9.339

Year/Age	1	2	3	4	5	6	7+
2018	0.298	0.798	1.875	3.758	5.609	6.774	8.284
2019	0.397	0.850	2.048	3.695	5.471	7.123	8.011
2020	0.257	0.827	1.784	3.841	5.448	6.609	8.777
2021	0.328	1.062	2.465	3.913	5.379	7.110	8.780
2022	0.300	0.895	2.109	3.768	4.832	6.744	9.059
2023	0.259	0.914	1.912	3.880	5.488	6.663	9.546
2024	0.359	0.859	1.866	3.545	5.486	7.041	8.588

Table 4.5a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Stock weights-at-age (kg) by substock.

Northwestern

Year/Age	1	2	3	4	5	6	7+
1983	0.071	0.567	1.466	4.184	7.201	11.633	11.911
1984	0.047	0.433	1.803	4.314	6.958	9.500	13.265
1985	0.074	0.360	1.840	4.350	5.777	9.384	13.581
1986	0.051	0.554	1.562	3.489	6.647	8.892	10.836
1987	0.093	0.340	1.393	4.403	7.239	9.508	12.624
1988	0.088	0.530	1.555	3.500	7.306	8.656	12.835
1989	0.074	0.413	1.431	3.594	5.761	9.290	12.523
1990	0.048	0.382	1.413	4.232	6.950	8.466	11.711
1991	0.084	0.523	1.665	4.031	7.375	10.536	13.642
1992	0.051	0.621	2.071	4.731	6.838	10.120	12.466
1993	0.039	0.468	2.167	5.054	7.677	9.879	15.264
1994	0.041	0.402	1.727	5.396	8.546	9.440	12.805
1995	0.061	0.269	1.223	4.384	6.451	10.229	15.943
1996	0.089	0.552	1.442	4.306	6.762	8.077	12.755
1997	0.023	0.375	1.688	4.160	7.396	9.581	11.919
1998	0.110	0.307	1.830	4.522	7.196	9.747	10.863
1999	0.099	0.464	0.657	2.519	7.310	9.314	12.127
2000	0.079	0.546	1.983	2.551	5.165	8.473	13.458
2001	0.099	0.383	1.489	4.401	5.688	6.640	11.018

Year/Age	1	2	3	4	5	6	7+
2002	0.089	0.370	1.338	4.173	5.390	9.246	11.270
2003	0.094	0.495	0.905	2.896	5.847	8.926	13.490
2004	0.071	0.343	1.389	3.975	6.485	11.962	#N/A
2005	0.086	0.391	1.283	2.452	5.596	9.850	13.198
2006	0.067	0.484	1.377	2.696	5.341	4.622	10.934
2007	0.065	0.396	1.683	4.005	6.212	8.406	#N/A
2008	0.075	0.572	1.793	5.081	7.015	#N/A	8.908
2009	0.070	0.493	1.533	3.546	6.699	#N/A	10.909
2010	0.089	0.625	1.994	3.512	6.005	7.177	11.639
2011	0.093	0.512	1.999	4.211	5.789	8.388	10.098
2012	0.113	0.679	1.545	3.659	6.153	8.216	10.998
2013	0.098	0.773	2.004	3.645	5.599	6.548	8.825
2014	0.065	0.607	1.881	3.604	5.188	6.951	8.912
2015	0.081	0.577	1.685	3.727	5.934	6.683	7.718
2016	0.072	0.722	1.751	3.507	5.803	7.618	9.560
2017	0.065	0.627	1.937	4.032	5.816	6.723	8.662
2018	0.071	0.394	1.752	3.319	5.361	5.867	7.933
2019	0.064	0.442	1.289	3.818	5.346	7.754	9.007
2020	0.077	0.512	1.876	3.383	5.028	8.198	12.796
2021	0.098	0.485	2.003	4.015	5.783	6.874	9.521
2022	0.126	0.430	1.315	4.686	6.057	8.330	11.334
2023	0.108	0.532	1.493	3.188	6.227	7.808	8.916
2024	0.107	0.437	1.332	2.817	4.833	7.183	9.827
2025	0.100	0.483	1.138	3.231	5.014	7.459	7.848

Southern

Year/Age	1	2	3	4	5	6	7+
1983	0.078	0.700	2.209	4.786	7.350	10.698	11.623
1984	0.084	0.590	1.847	3.458	7.455	9.949	12.599
1985	0.049	0.558	2.295	4.977	7.164	10.634	12.040

Year/Age	1	2	3	4	5	6	7+
1986	0.068	0.319	1.768	3.849	7.282	10.302	11.912
1987	0.083	0.455	1.100	5.043	7.478	11.120	12.349
1988	0.058	0.716	2.131	3.489	7.682	10.829	12.951
1989	0.062	0.795	2.238	3.900	6.995	10.078	13.240
1990	0.076	0.626	2.477	4.388	7.209	10.961	12.423
1991	0.059	0.737	1.940	4.866	7.591	10.320	14.458
1992	0.066	0.279	1.938	5.542	7.556	11.225	12.021
1993	0.043	0.607	2.398	5.371	8.338	12.761	13.195
1994	0.039	0.410	1.986	5.250	8.633	9.960	13.022
1995	0.045	0.336	1.323	4.475	6.996	10.303	12.699
1996	0.044	0.500	1.494	2.859	6.704	10.545	15.837
1997	0.029	0.198	2.200	3.556	7.025	7.576	10.411
1998	0.040	0.378	0.919	3.468	8.764	9.427	14.976
1999	0.145	0.612	1.566	3.524	7.860	9.430	11.875
2000	0.060	0.524	1.650	3.728	4.588	9.251	12.892
2001	0.146	0.327	1.206	2.996	7.411	8.985	10.495
2002	0.087	0.430	1.763	5.129	8.198	10.682	12.220
2003	0.091	0.641	1.207	4.666	7.276	11.853	12.927
2004	0.052	0.185	2.023	3.519	7.810	8.787	15.263
2005	0.055	0.545	1.619	3.524	6.572	11.271	13.692
2006	0.070	0.742	2.374	4.137	8.845	7.495	11.939
2007	0.078	0.364	2.245	4.464	6.662	12.414	11.649
2008	0.110	1.032	2.043	3.816	6.242	9.212	13.879
2009	0.075	0.924	2.713	4.490	6.629	9.142	14.330
2010	0.078	0.682	2.862	5.280	6.635	10.434	13.406
2011	0.071	0.566	2.139	5.517	7.959	9.249	11.537
2012	0.067	1.073	2.369	4.007	8.653	10.913	9.145
2013	0.063	0.559	2.599	3.758	5.693	8.540	13.043
2014	0.068	0.857	2.414	4.243	7.234	5.533	8.552

Year/Age	1	2	3	4	5	6	7+
2015	0.113	0.871	1.868	3.857	6.698	10.979	16.229
2016	0.100	0.878	2.609	3.831	7.424	7.998	10.019
2017	0.060	0.559	1.599	3.346	4.863	10.242	13.070
2018	0.106	0.498	1.155	2.693	6.533	8.330	12.606
2019	0.060	0.510	1.790	2.765	4.525	8.870	10.378
2020	0.070	0.508	1.621	3.591	3.650	#N/A	9.765
2021	0.087	0.649	1.549	2.377	8.281	10.747	7.185
2022	0.079	0.721	1.975	3.701	6.441	9.451	#N/A
2023	0.097	0.573	1.535	3.700	6.477	9.432	#N/A
2024	0.116	0.497	1.654	2.638	6.750	9.359	11.439
2025	0.091	0.674	1.530	3.305	#N/A	6.683	#N/A

Viking

Year/Age	1	2	3	4	5	6	7+
1983	0.059	0.463	1.580	3.276	5.215	8.273	9.396
1984	0.040	0.374	1.561	2.673	4.760	7.621	14.965
1985	0.047	0.311	1.189	3.162	4.717	7.513	7.156
1986	0.045	0.296	1.042	2.630	4.079	5.141	10.310
1987	0.064	0.341	1.103	3.146	4.452	6.571	11.091
1988	0.076	0.190	1.076	2.718	4.524	6.269	6.781
1989	0.060	0.317	0.799	2.914	4.017	7.512	10.086
1990	0.060	0.347	1.190	3.119	5.062	7.390	10.107
1991	0.052	0.422	1.380	2.817	5.422	6.791	9.603
1992	0.054	0.540	1.631	3.884	4.861	7.930	9.325
1993	0.054	0.333	1.340	3.360	4.529	7.917	11.076
1994	0.036	0.359	1.262	3.961	6.210	8.839	11.078
1995	0.043	0.345	1.115	3.177	5.473	7.840	12.136
1996	0.031	0.317	1.230	3.642	5.243	7.711	12.405
1997	0.040	0.312	1.185	3.327	5.073	7.633	11.883
1998	0.064	0.241	1.180	3.423	5.409	7.374	12.027

Year/Age	1	2	3	4	5	6	7+
1999	0.048	0.270	0.815	2.690	4.961	6.883	9.611
2000	0.040	0.380	1.172	2.739	4.840	5.931	9.383
2001	0.070	0.349	1.496	3.034	4.325	5.282	10.367
2002	0.051	0.299	0.830	3.054	6.139	9.108	9.279
2003	0.051	0.384	0.946	2.423	5.307	5.370	11.599
2004	0.058	0.367	1.373	3.238	4.653	6.246	14.712
2005	0.060	0.449	1.273	2.950	5.009	7.427	9.342
2006	0.054	0.439	1.464	3.132	4.378	7.136	10.172
2007	0.048	0.445	0.943	3.456	5.348	5.928	10.304
2008	0.050	0.529	1.501	3.231	5.516	7.085	11.410
2009	0.051	0.477	1.589	3.684	4.613	6.169	9.986
2010	0.049	0.524	1.584	3.413	4.696	5.928	10.105
2011	0.045	0.403	1.448	3.672	4.390	7.234	11.510
2012	0.059	0.470	1.376	3.247	5.412	7.168	9.414
2013	0.053	0.420	1.277	3.177	4.493	6.298	8.855
2014	0.045	0.436	1.458	3.466	5.640	8.569	10.675
2015	0.058	0.463	1.568	3.306	4.545	7.816	7.376
2016	0.064	0.517	1.566	3.133	4.349	6.570	9.695
2017	0.050	0.545	1.596	3.317	4.579	6.617	10.254
2018	0.053	0.484	1.071	3.061	5.520	6.941	10.008
2019	0.061	0.424	1.229	2.722	6.247	7.516	10.597
2020	0.057	0.516	1.393	3.668	5.762	5.951	9.822
2021	0.049	0.429	1.367	2.722	5.455	6.270	9.067
2022	0.054	0.371	1.425	3.227	4.997	8.202	10.466
2023	0.054	0.427	1.103	2.889	5.541	7.581	15.101
2024	0.054	0.299	1.280	2.819	4.756	7.219	9.182
2025	0.061	0.460	0.985	2.696	4.222	6.596	10.079

Table 4.5b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Proportion mature by age group and substock.**Northwestern**

Year/Age	1	2	3	4	5	6	7+
1983	0.003	0.039	0.272	0.304	0.724	0.807	0.906
1984	0.002	0.021	0.133	0.437	0.648	0.683	0.878
1985	0.002	0.028	0.144	0.457	0.557	0.786	0.872
1986	0.002	0.112	0.099	0.185	0.463	0.720	0.853
1987	0.003	0.019	0.261	0.539	0.741	0.781	0.896
1988	0.002	0.039	0.169	0.458	0.563	0.759	0.890
1989	0.130	0.270	0.335	0.641	0.895	0.880	0.973
1990	0.003	0.082	0.226	0.547	0.749	0.806	0.789
1991	0.005	0.091	0.403	0.556	0.829	0.892	0.943
1992	0.016	0.338	0.657	0.895	0.945	0.966	0.983
1993	0.014	0.115	0.762	0.874	0.943	0.960	0.979
1994	0.017	0.304	0.688	0.884	0.952	0.968	0.984
1995	0.008	0.079	0.438	0.806	0.889	0.923	0.960
1996	0.013	0.181	0.583	0.854	0.938	0.954	0.977
1997	0.017	0.268	0.821	0.931	0.964	0.976	0.988
1998	0.013	0.152	0.648	0.834	0.932	0.950	0.972
1999	0.031	0.576	0.537	0.896	0.960	0.974	0.987
2000	0.030	0.350	0.814	0.898	0.963	0.976	0.988
2001	0.015	0.248	0.633	0.869	0.914	0.957	0.979
2002	0.021	0.263	0.695	0.887	0.947	0.967	0.983
2003	0.023	0.287	0.649	0.877	0.949	0.966	0.983
2004	0.018	0.389	0.729	0.898	0.955	0.971	0.986
2005	0.021	0.380	0.730	0.879	0.950	0.970	0.985
2006	0.035	0.532	0.848	0.937	0.973	0.983	0.992
2007	0.017	0.238	0.701	0.883	0.942	0.964	0.982
2008	0.023	0.469	0.689	0.916	0.961	0.975	0.988
2009	0.016	0.269	0.708	0.885	0.936	0.964	0.982
2010	0.013	0.382	0.812	0.904	0.959	0.973	0.986

Year/Age	1	2	3	4	5	6	7+
2011	0.027	0.268	0.779	0.934	0.968	0.979	0.988
2012	0.028	0.533	0.722	0.898	0.968	0.978	0.989
2013	0.037	0.637	0.794	0.971	0.978	0.987	0.993
2014	0.028	0.194	0.609	0.901	0.937	0.966	0.982
2015	0.022	0.367	0.617	0.864	0.930	0.960	0.983
2016	0.010	0.298	0.464	0.813	0.931	0.954	0.970
2017	0.016	0.500	0.696	0.862	0.895	0.969	0.975
2018	0.012	0.239	0.688	0.786	0.942	0.945	0.980
2019	0.026	0.450	0.716	0.912	0.956	0.976	0.988
2020	0.035	0.251	0.819	0.921	0.964	0.977	0.988
2021	0.022	0.279	0.509	0.884	0.899	0.960	0.979
2022	0.041	0.555	0.789	0.938	0.972	0.977	0.991
2023	0.023	0.398	0.690	0.841	0.945	0.967	0.985
2024	0.014	0.177	0.637	0.806	0.951	0.961	0.979
2025	0.017	0.383	0.718	0.814	0.937	0.962	0.983

Southern

Year/Age	1	2	3	4	5	6	7+
1983	0.001	0.021	0.218	0.468	0.692	0.629	0.973
1984	0.000	0.033	0.264	0.581	0.383	0.852	0.978
1985	0.001	0.023	0.319	0.698	0.767	0.775	0.975
1986	0.001	0.067	0.342	0.561	0.653	0.521	0.976
1987	0.000	0.010	0.263	0.567	0.790	0.647	0.977
1988	0.001	0.065	0.195	0.602	0.623	0.738	0.966
1989	0.038	0.158	0.376	0.585	0.908	0.747	0.993
1990	0.002	0.207	0.279	0.559	0.739	0.841	0.979
1991	0.002	0.154	0.627	0.751	0.920	0.937	0.994
1992	0.002	0.224	0.709	0.909	0.943	0.945	0.996
1993	0.002	0.079	0.442	0.808	0.899	0.900	0.991
1994	0.002	0.098	0.614	0.827	0.880	0.899	0.992

Year/Age	1	2	3	4	5	6	7+
1995	0.003	0.093	0.457	0.834	0.894	0.903	0.992
1996	0.002	0.065	0.460	0.766	0.883	0.892	0.990
1997	0.002	0.139	0.582	0.691	0.860	0.861	0.991
1998	0.002	0.054	0.405	0.719	0.842	0.861	0.988
1999	0.004	0.328	0.650	0.838	0.948	0.948	0.995
2000	0.010	0.285	0.646	0.928	0.962	0.964	0.997
2001	0.003	0.240	0.635	0.909	0.948	0.947	0.996
2002	0.009	0.514	0.836	0.949	0.974	0.977	0.998
2003	0.007	0.342	0.669	0.915	0.960	0.961	0.997
2004	0.004	0.160	0.872	0.931	0.951	0.960	0.997
2005	0.015	0.544	0.826	0.932	0.976	0.977	0.998
2006	0.009	0.342	0.856	0.937	0.970	0.971	0.998
2007	0.007	0.272	0.844	0.943	0.967	0.969	0.998
2008	0.041	0.565	0.933	0.969	0.986	0.988	0.999
2009	0.011	0.327	0.815	0.952	0.973	0.974	0.998
2010	0.007	0.234	0.785	0.939	0.954	0.963	0.997
2011	0.021	0.140	0.840	0.948	0.971	0.971	0.998
2012	0.017	0.560	0.884	0.964	0.983	0.984	0.999
2013	0.017	0.397	0.877	0.958	0.978	0.980	0.999
2014	0.009	0.548	0.911	0.965	0.981	0.982	0.999
2015	0.057	0.818	0.951	0.983	0.993	0.994	1.000
2016	0.019	0.503	0.903	0.966	0.983	0.984	0.999
2017	0.014	0.432	0.877	0.958	0.978	0.979	0.999
2018	0.013	0.488	0.850	0.955	0.977	0.978	0.998
2019	0.009	0.331	0.829	0.936	0.967	0.969	0.998
2020	0.006	0.272	0.742	0.913	0.954	0.957	0.997
2021	0.025	0.596	0.912	0.972	0.986	0.987	0.999
2022	0.027	0.579	0.912	0.970	0.986	0.987	0.999
2023	0.015	0.556	0.883	0.961	0.981	0.982	0.999

Year/Age	1	2	3	4	5	6	7+
2024	0.017	0.557	0.890	0.965	0.982	0.984	0.999
2025	0.031	0.763	0.940	0.982	0.991	0.992	0.999

Viking

Year/Age	1	2	3	4	5	6	7+
1983	0.003	0.025	0.066	0.230	0.560	0.848	0.917
1984	0.005	0.033	0.204	0.440	0.701	0.916	0.954
1985	0.003	0.012	0.102	0.332	0.631	0.883	0.913
1986	0.002	0.024	0.042	0.200	0.547	0.845	0.910
1987	0.004	0.027	0.146	0.378	0.710	0.897	0.946
1988	0.002	0.021	0.077	0.327	0.343	0.810	0.897
1989	0.140	0.040	0.082	0.288	0.704	0.906	0.966
1990	0.003	0.027	0.129	0.295	0.535	0.871	0.910
1991	0.003	0.017	0.153	0.327	0.706	0.884	0.936
1992	0.045	0.090	0.416	0.734	0.851	0.972	0.983
1993	0.003	0.027	0.121	0.432	0.594	0.889	0.922
1994	0.007	0.045	0.260	0.643	0.811	0.950	0.971
1995	0.004	0.027	0.231	0.614	0.731	0.871	0.946
1996	0.012	0.129	0.407	0.830	0.852	0.971	0.984
1997	0.010	0.121	0.440	0.644	0.826	0.965	0.980
1998	0.026	0.103	0.478	0.824	0.914	0.978	0.988
1999	0.016	0.273	0.327	0.752	0.928	0.977	0.987
2000	0.010	0.218	0.597	0.696	0.839	0.968	0.978
2001	0.019	0.124	0.349	0.646	0.894	0.969	0.983
2002	0.018	0.233	0.402	0.755	0.925	0.977	0.987
2003	0.008	0.052	0.289	0.540	0.851	0.949	0.972
2004	0.011	0.127	0.652	0.642	0.850	0.955	0.982
2005	0.012	0.091	0.517	0.823	0.913	0.972	0.983
2006	0.007	0.109	0.286	0.684	0.816	0.962	0.978
2007	0.037	0.186	0.381	0.718	0.933	0.973	0.988

Year/Age	1	2	3	4	5	6	7+
2008	0.017	0.280	0.527	0.720	0.909	0.978	0.988
2009	0.015	0.086	0.414	0.752	0.883	0.965	0.983
2010	0.011	0.082	0.428	0.677	0.908	0.954	0.981
2011	0.014	0.077	0.471	0.716	0.884	0.968	0.982
2012	0.008	0.072	0.210	0.649	0.856	0.954	0.974
2013	0.008	0.069	0.261	0.693	0.816	0.961	0.976
2014	0.010	0.055	0.300	0.709	0.847	0.960	0.976
2015	0.009	0.120	0.285	0.700	0.833	0.957	0.978
2016	0.006	0.068	0.173	0.580	0.766	0.931	0.969
2017	0.008	0.062	0.394	0.697	0.796	0.958	0.973
2018	0.010	0.060	0.275	0.689	0.894	0.961	0.977
2019	0.015	0.213	0.427	0.757	0.908	0.975	0.984
2020	0.009	0.077	0.459	0.791	0.874	0.967	0.982
2021	0.006	0.065	0.241	0.453	0.794	0.939	0.968
2022	0.006	0.043	0.175	0.532	0.835	0.937	0.966
2023	0.006	0.059	0.233	0.485	0.773	0.939	0.965
2024	0.006	0.052	0.232	0.461	0.719	0.926	0.962
2025	0.008	0.098	0.334	0.562	0.784	0.938	0.974

Table 4.5c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Natural mortality by age group. Values are assumed to apply to all substocks.

Year/Age	1	2	3	4	5	6	7+
1983	1.112	0.770	0.277	0.210	0.2	0.2	0.2
1984	0.982	0.711	0.257	0.201	0.2	0.2	0.2
1985	1.090	0.731	0.265	0.201	0.2	0.2	0.2
1986	0.944	0.694	0.268	0.211	0.2	0.2	0.2
1987	0.989	0.696	0.268	0.212	0.2	0.2	0.2
1988	1.047	0.721	0.269	0.229	0.2	0.2	0.2
1989	0.953	0.753	0.286	0.215	0.2	0.2	0.2
1990	1.009	0.769	0.285	0.220	0.2	0.2	0.2
1991	1.006	0.771	0.290	0.201	0.2	0.2	0.2

Year/Age	1	2	3	4	5	6	7+
1992	0.916	0.742	0.254	0.200	0.2	0.2	0.2
1993	0.990	0.729	0.249	0.200	0.2	0.2	0.2
1994	0.985	0.726	0.269	0.200	0.2	0.2	0.2
1995	0.980	0.733	0.277	0.200	0.2	0.2	0.2
1996	1.074	0.764	0.280	0.200	0.2	0.2	0.2
1997	0.965	0.721	0.263	0.211	0.2	0.2	0.2
1998	1.083	0.757	0.296	0.218	0.2	0.2	0.2
1999	1.049	0.829	0.311	0.225	0.2	0.2	0.2
2000	0.899	0.775	0.296	0.236	0.2	0.2	0.2
2001	0.942	0.759	0.322	0.227	0.2	0.2	0.2
2002	1.029	0.857	0.390	0.233	0.2	0.2	0.2
2003	1.115	0.961	0.450	0.258	0.2	0.2	0.2
2004	1.205	1.033	0.484	0.312	0.2	0.2	0.2
2005	1.273	1.085	0.508	0.272	0.2	0.2	0.2
2006	1.251	1.069	0.469	0.262	0.2	0.2	0.2
2007	1.303	1.070	0.466	0.200	0.2	0.2	0.2
2008	1.356	1.121	0.489	0.260	0.2	0.2	0.2
2009	1.291	1.117	0.468	0.201	0.2	0.2	0.2
2010	1.178	0.981	0.437	0.200	0.2	0.2	0.2
2011	1.283	1.054	0.386	0.201	0.2	0.2	0.2
2012	1.350	1.120	0.541	0.201	0.2	0.2	0.2
2013	1.350	1.116	0.502	0.204	0.2	0.2	0.2
2014	1.252	1.046	0.472	0.204	0.202	0.2	0.2
2015	1.289	1.070	0.510	0.207	0.205	0.202	0.2
2016	1.299	1.069	0.474	0.285	0.2	0.2	0.2
2017	1.201	0.988	0.406	0.241	0.204	0.2	0.2
2018	1.350	1.088	0.479	0.260	0.2	0.2	0.2
2019	1.266	1.030	0.454	0.272	0.2	0.2	0.2
2020	1.143	0.926	0.431	0.247	0.2	0.2	0.2

Year/Age	1	2	3	4	5	6	7+
2021	1.223	0.956	0.481	0.205	0.2	0.2	0.2
2022	1.354	1.050	0.546	0.277	0.2	0.2	0.2
2023*	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
2024*	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
2025*	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

*A new SMS run was performed for cod in 2024 with data up to 2022. Natural mortality is modelled as a GMRF process in multistock SAM, allowing for estimation of missing values in 2023–2025.

Table 4.6a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM tuning indices and standard deviations by substock in Q1.

Northwestern

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
1983	482.283	6722.989	878.629	529.545	204.260	174.736	35.646	0.258	0.477	0.349	0.320	0.325	0.328	0.278
1984	1555.500	2323.544	1124.848	216.737	171.233	87.155	45.647	0.603	0.313	0.346	0.332	0.307	0.346	0.389
1985	90.976	5593.020	748.864	737.072	92.941	88.147	9.922	0.274	0.161	0.159	0.206	0.191	0.198	0.229
1986	1779.969	1203.986	1916.734	377.815	195.481	62.129	34.516	0.188	0.176	0.159	0.153	0.172	0.169	0.205
1987	825.336	5505.848	633.156	528.356	91.217	66.677	22.194	0.241	0.149	0.222	0.224	0.174	0.180	0.187
1988	105.233	3044.181	1627.989	212.795	128.122	52.348	27.706	0.197	0.249	0.153	0.213	0.161	0.180	0.190
1989	2548.518	1160.417	1641.775	452.992	60.770	74.524	29.256	0.169	0.157	0.184	0.169	0.177	0.178	0.224
1990	360.423	5139.126	356.594	311.087	135.956	25.665	34.306	0.207	0.171	0.167	0.246	0.208	0.200	0.191
1991	258.090	988.368	1222.098	164.319	126.529	51.005	18.482	0.247	0.168	0.178	0.215	0.271	0.233	0.209
1992	1452.986	1573.250	392.388	311.999	51.462	31.720	23.337	0.162	0.191	0.182	0.235	0.244	0.273	0.238
1993	778.531	5744.552	770.813	131.809	58.664	21.565	7.530	0.199	0.175	0.186	0.194	0.189	0.240	0.276
1994	1121.356	1563.859	1249.825	283.528	60.069	27.737	15.294	0.171	0.178	0.198	0.200	0.173	0.176	0.219
1995	849.207	4243.619	1532.448	525.213	103.909	15.431	6.308	0.194	0.168	0.221	0.269	0.239	0.233	0.617
1996	287.504	2314.765	1180.431	174.244	102.659	20.744	7.007	0.208	0.184	0.192	0.224	0.200	0.213	0.288
1997	3762.598	1238.214	654.337	296.476	88.632	51.886	12.372	0.169	0.179	0.212	0.223	0.249	0.263	0.241
1998	110.688	4153.664	408.121	168.244	78.478	17.415	14.031	0.180	0.150	0.170	0.217	0.196	0.202	0.259

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
1999	139.391	475.332	1500.881	105.525	65.372	16.380	18.485	0.238	0.205	0.142	0.191	0.209	0.243	0.247
2000	902.616	758.664	318.351	290.792	34.811	22.608	14.322	0.198	0.169	0.252	0.180	0.195	0.252	0.273
2001	203.562	1765.712	321.424	113.074	101.096	14.810	11.621	0.499	0.163	0.179	0.321	0.298	0.340	0.431
2002	498.757	594.766	970.468	109.739	22.754	18.207	2.279	0.222	0.187	0.187	0.212	0.288	0.351	0.340
2003	114.967	1204.768	204.375	212.435	59.658	13.863	13.505	0.355	0.196	0.139	0.180	0.215	0.247	0.261
2004	506.267	437.598	293.611	93.692	76.211	9.745	2.652	0.223	0.194	0.205	0.296	0.273	0.249	0.417
2005	132.767	558.242	177.511	84.595	17.436	23.453	3.880	0.287	0.178	0.195	0.292	0.304	0.307	0.358
2006	901.681	287.595	226.987	79.775	22.009	11.988	15.374	0.185	0.233	0.212	0.310	0.301	0.318	0.329
2007	224.385	1684.591	279.870	105.919	18.525	10.882	5.836	0.221	0.182	0.174	0.245	0.276	0.279	0.374
2008	187.072	407.169	959.593	96.675	36.091	6.680	2.455	0.256	0.232	0.215	0.231	0.239	0.248	0.307
2009	247.326	573.062	239.273	162.584	18.437	9.829	6.085	0.293	0.198	0.232	0.275	0.262	0.249	0.357
2010	771.274	1197.731	324.650	57.855	49.238	12.256	2.530	0.217	0.200	0.200	0.260	0.226	0.243	0.330
2011	120.288	1832.464	659.803	129.712	69.768	42.191	8.204	0.218	0.166	0.185	0.202	0.210	0.239	0.269
2012	359.398	747.006	1471.675	356.821	105.260	29.513	31.010	0.226	0.159	0.181	0.194	0.221	0.256	0.395
2013	679.850	794.418	509.717	588.757	198.745	22.287	9.364	0.216	0.186	0.190	0.222	0.190	0.175	0.253
2014	592.240	1061.973	487.174	207.527	175.909	37.489	10.135	0.207	0.204	0.221	0.233	0.213	0.238	0.327
2015	580.146	1691.959	1097.225	407.141	181.265	112.592	20.176	0.186	0.166	0.193	0.216	0.245	0.266	0.314
2016	401.991	1109.641	1481.266	680.109	124.663	40.705	30.870	0.249	0.219	0.196	0.214	0.188	0.199	0.232

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
2017	2301.245	710.491	1268.077	1182.897	897.910	116.166	77.265	0.171	0.175	0.195	0.198	0.229	0.264	0.330
2018	169.953	1750.821	402.718	416.822	188.379	105.488	43.840	0.249	0.177	0.184	0.212	0.196	0.210	0.213
2019	557.502	319.514	601.304	81.467	65.643	23.921	10.027	0.237	0.192	0.184	0.221	0.241	0.256	0.309
2020	635.817	957.405	226.046	223.033	32.171	13.430	11.283	0.212	0.201	0.188	0.207	0.230	0.298	0.367
2021	502.775	971.030	691.874	147.608	75.272	35.360	23.189	0.208	0.162	0.180	0.202	0.203	0.225	0.269
2022	260.838	995.717	814.402	271.901	72.461	52.880	18.992	0.434	0.368	0.326	0.416	0.494	0.494	0.734
2023	234.876	1231.112	765.796	507.694	155.616	49.813	24.531	0.220	0.166	0.163	0.192	0.190	0.204	0.248
2024	128.295	767.966	669.488	322.677	216.662	80.306	18.167	0.257	0.177	0.162	0.201	0.180	0.194	0.230
2025	315.714	302.245	614.659	236.366	84.782	55.026	26.665	0.270	0.178	0.180	0.218	0.220	0.209	0.235

Southern

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
1983	2071.341	3747.109	271.154	216.204	90.522	146.542	53.771	0.160	0.161	0.146	0.150	0.142	0.167	0.179
1984	8303.090	1128.333	450.731	83.234	111.041	64.356	48.490	0.161	0.157	0.161	0.152	0.149	0.178	0.208
1985	280.850	4080.247	260.549	146.314	52.032	100.151	25.707	0.180	0.134	0.157	0.171	0.146	0.161	0.186
1986	6993.418	539.529	448.662	171.839	77.416	54.584	24.281	0.129	0.139	0.146	0.159	0.151	0.169	0.190
1987	4431.373	5111.988	120.032	233.092	38.312	73.518	38.324	0.164	0.166	0.142	0.165	0.141	0.162	0.175
1988	1532.961	1337.348	1024.720	25.837	95.709	87.181	26.908	0.190	0.201	0.190	0.189	0.168	0.189	0.206
1989	2696.951	1169.912	516.195	285.086	32.815	73.003	29.569	0.153	0.194	0.163	0.149	0.151	0.158	0.201

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
1990	543.683	1358.948	347.536	147.444	174.014	41.921	48.504	0.201	0.194	0.188	0.179	0.160	0.175	0.190
1991	1148.940	444.401	298.871	120.040	52.600	107.065	28.978	0.219	0.166	0.155	0.156	0.150	0.161	0.193
1992	3175.805	842.646	139.864	115.674	52.203	26.451	36.078	0.157	0.176	0.174	0.178	0.178	0.190	0.194
1993	1053.580	1203.022	131.540	68.685	69.478	28.859	13.475	0.193	0.164	0.164	0.171	0.167	0.207	0.243
1994	2816.836	406.537	291.630	79.065	61.145	57.517	14.730	0.160	0.186	0.181	0.186	0.194	0.204	0.249
1995	1384.063	1471.728	423.394	96.218	53.171	33.976	21.064	0.191	0.158	0.161	0.185	0.193	0.218	0.412
1996	768.498	540.233	572.178	102.747	63.759	29.290	26.288	0.187	0.184	0.164	0.181	0.166	0.221	0.293
1997	4655.222	812.921	249.349	141.816	39.320	41.129	16.661	0.127	0.167	0.179	0.176	0.190	0.201	0.241
1998	120.541	2368.796	391.112	138.268	94.577	33.217	29.363	0.192	0.139	0.168	0.171	0.154	0.198	0.249
1999	413.108	279.865	704.408	64.640	58.040	28.566	53.056	0.230	0.207	0.148	0.165	0.184	0.214	0.246
2000	666.768	463.080	104.475	235.376	45.956	39.370	40.972	0.185	0.198	0.166	0.158	0.166	0.174	0.212
2001	406.659	758.061	171.779	44.792	37.938	10.975	17.201	0.207	0.178	0.163	0.192	0.196	0.211	0.258
2002	849.249	309.007	147.257	63.985	11.490	30.695	16.130	0.186	0.154	0.163	0.195	0.219	0.227	0.269
2003	104.343	632.375	102.961	68.960	23.144	10.300	15.950	0.235	0.165	0.157	0.188	0.236	0.234	0.258
2004	687.734	283.819	344.700	65.174	40.851	14.628	8.040	0.197	0.158	0.160	0.170	0.190	0.255	0.340
2005	447.255	267.126	72.380	131.810	11.433	29.826	3.746	0.197	0.186	0.167	0.195	0.212	0.229	0.365
2006	874.026	245.908	125.155	35.335	12.070	10.051	19.513	0.168	0.204	0.189	0.243	0.262	0.296	0.326
2007	729.347	296.659	146.716	42.302	12.860	20.862	7.305	0.206	0.172	0.191	0.234	0.248	0.282	0.371

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
2008	645.277	300.383	147.803	94.290	52.873	18.326	14.400	0.211	0.240	0.175	0.192	0.178	0.208	0.251
2009	140.924	464.735	228.256	79.817	31.806	25.818	26.625	0.255	0.193	0.177	0.172	0.183	0.217	0.272
2010	492.471	144.082	257.052	82.215	40.050	28.262	15.572	0.229	0.213	0.201	0.205	0.225	0.227	0.269
2011	225.207	369.913	61.646	81.060	46.977	24.491	9.610	0.296	0.225	0.204	0.219	0.220	0.232	0.291
2012	199.957	338.639	228.820	41.946	49.925	26.604	10.662	0.202	0.190	0.169	0.206	0.202	0.231	0.273
2013	235.903	190.743	131.393	100.085	63.303	18.423	14.953	0.309	0.211	0.199	0.203	0.186	0.241	0.273
2014	633.101	210.942	110.210	45.953	62.580	14.208	2.858	0.203	0.240	0.212	0.225	0.229	0.290	0.491
2015	136.804	549.689	145.526	62.750	32.463	20.473	6.207	0.207	0.201	0.194	0.233	0.253	0.281	0.305
2016	68.305	66.984	176.569	46.127	51.113	30.480	26.435	0.314	0.251	0.208	0.205	0.204	0.224	0.268
2017	854.109	77.609	86.990	53.701	41.162	21.380	19.825	0.166	0.233	0.265	0.245	0.260	0.303	0.335
2018	64.183	213.661	43.747	23.590	32.232	14.974	11.413	0.311	0.212	0.262	0.283	0.289	0.293	0.327
2019	228.842	48.243	85.268	8.355	8.934	8.012	5.990	0.248	0.226	0.268	0.309	0.322	0.401	0.483
2020	527.413	79.652	44.021	18.664	3.913	2.378	5.300	0.241	0.231	0.309	0.303	0.419	0.492	0.595
2021	345.938	192.633	61.674	14.625	8.739	6.981	6.905	0.259	0.222	0.244	0.283	0.355	0.397	0.429
2022	378.060	140.828	80.348	31.234	10.866	10.011	4.893	0.264	0.237	0.283	0.319	0.424	0.470	0.516
2023	262.894	487.025	86.444	34.656	24.981	11.351	5.647	0.333	0.193	0.192	0.247	0.328	0.328	0.358
2024	99.131	153.816	123.677	25.970	34.171	21.072	7.268	0.280	0.229	0.207	0.235	0.221	0.273	0.366
2025	403.211	157.653	92.999	33.829	9.089	11.022	7.682	0.325	0.293	0.226	0.266	0.279	0.311	0.403

Viking

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
1983	1228.976	7266.398	1358.561	491.782	133.911	87.499	37.532	0.275	0.193	0.186	0.199	0.180	0.194	0.240
1984	1451.060	2883.956	972.993	240.455	145.823	41.251	22.054	0.201	0.171	0.160	0.156	0.154	0.203	0.264
1985	217.918	6841.935	1334.989	494.414	117.700	73.039	17.945	0.287	0.185	0.168	0.165	0.164	0.176	0.232
1986	2294.472	1447.022	1805.341	719.450	241.033	101.117	45.376	0.250	0.191	0.149	0.148	0.149	0.160	0.173
1987	500.307	6424.346	264.139	286.316	108.778	46.116	14.130	0.202	0.192	0.146	0.152	0.154	0.163	0.204
1988	1678.249	1605.397	1207.352	117.107	136.012	56.869	41.051	0.302	0.161	0.161	0.170	0.156	0.171	0.195
1989	2233.638	1768.513	1301.041	495.266	88.707	87.975	33.271	0.223	0.183	0.154	0.151	0.150	0.160	0.202
1990	799.251	3317.980	497.483	195.976	117.655	20.973	41.602	0.239	0.163	0.158	0.182	0.166	0.184	0.206
1991	1182.999	1368.613	795.675	214.023	76.902	50.422	29.705	0.306	0.187	0.154	0.171	0.165	0.176	0.197
1992	2863.298	1321.222	325.524	207.950	51.457	20.471	36.183	0.255	0.178	0.176	0.181	0.176	0.220	0.226
1993	1724.915	5655.621	497.144	160.717	112.811	32.739	14.600	0.273	0.169	0.153	0.165	0.159	0.192	0.253
1994	2271.888	1094.374	707.508	195.547	82.470	29.713	19.915	0.233	0.172	0.170	0.166	0.191	0.217	0.268
1995	3033.933	7604.117	745.446	276.743	55.031	19.808	19.596	0.236	0.173	0.157	0.162	0.167	0.206	0.444
1996	835.146	4123.168	1226.183	191.777	103.320	26.078	17.296	0.283	0.187	0.177	0.156	0.180	0.199	0.269
1997	2897.187	2209.344	567.285	207.832	76.625	39.597	12.078	0.191	0.182	0.166	0.159	0.179	0.188	0.228
1998	391.539	7317.847	439.770	194.095	93.151	33.141	23.447	0.268	0.176	0.157	0.175	0.172	0.189	0.262
1999	1123.041	414.937	1520.762	167.694	78.222	31.898	31.608	0.294	0.216	0.161	0.157	0.167	0.195	0.225
2000	1181.665	1699.712	148.201	427.410	94.237	50.524	32.410	0.210	0.192	0.165	0.167	0.177	0.176	0.222

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
2001	745.716	2756.238	427.764	74.011	72.860	19.813	15.096	0.270	0.170	0.146	0.170	0.175	0.195	0.252
2002	2031.043	1025.688	1037.622	125.377	24.179	19.618	9.011	0.301	0.222	0.155	0.162	0.209	0.247	0.282
2003	221.208	966.405	255.995	280.870	86.507	17.011	20.746	0.284	0.220	0.169	0.174	0.177	0.200	0.246
2004	1968.186	628.198	328.595	90.073	96.952	40.844	11.166	0.271	0.207	0.187	0.194	0.204	0.206	0.285
2005	523.603	1270.999	245.067	119.067	45.303	51.004	22.782	0.262	0.219	0.181	0.170	0.190	0.210	0.266
2006	1750.896	638.105	377.223	71.845	40.888	25.543	35.416	0.269	0.233	0.194	0.188	0.203	0.224	0.261
2007	691.190	2370.892	338.766	106.243	56.491	31.781	32.701	0.282	0.173	0.167	0.193	0.220	0.225	0.296
2008	663.450	606.722	636.827	125.185	72.997	16.945	6.959	0.273	0.213	0.172	0.185	0.178	0.202	0.262
2009	563.575	777.688	208.136	220.166	64.420	29.129	18.550	0.306	0.186	0.174	0.192	0.188	0.192	0.233
2010	1019.588	1230.012	305.314	111.128	96.610	28.507	6.590	0.281	0.174	0.175	0.198	0.173	0.198	0.297
2011	252.369	1603.561	311.166	125.533	46.455	51.917	10.577	0.355	0.184	0.182	0.206	0.195	0.218	0.267
2012	1257.319	789.882	859.239	257.809	83.091	21.815	14.578	0.237	0.166	0.164	0.184	0.181	0.216	0.242
2013	491.900	1538.849	446.532	344.541	153.803	44.281	19.386	0.232	0.196	0.168	0.179	0.156	0.170	0.210
2014	584.411	1360.581	456.680	197.211	181.859	64.935	19.951	0.219	0.215	0.186	0.216	0.219	0.239	0.296
2015	538.939	2502.962	775.984	256.806	76.653	45.481	9.461	0.230	0.181	0.159	0.172	0.168	0.203	0.256
2016	262.625	834.714	1311.349	443.924	163.126	55.698	49.878	0.260	0.176	0.173	0.156	0.146	0.160	0.215
2017	1116.719	630.003	569.050	527.564	255.564	45.318	24.300	0.184	0.161	0.158	0.157	0.159	0.181	0.224
2018	163.033	1935.099	285.658	150.452	101.935	63.039	24.817	0.320	0.161	0.154	0.180	0.175	0.181	0.195

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
2019	194.173	304.136	491.261	63.241	52.907	29.600	13.760	0.273	0.160	0.156	0.177	0.222	0.231	0.297
2020	1194.125	615.856	148.936	171.249	36.280	10.482	24.409	0.273	0.186	0.172	0.189	0.201	0.234	0.303
2021	308.454	2656.157	632.531	153.638	70.606	34.974	30.222	0.247	0.184	0.170	0.178	0.185	0.198	0.239
2022	423.665	817.474	1279.411	354.733	67.178	39.874	22.806	0.319	0.211	0.203	0.215	0.243	0.264	0.383
2023	478.058	1942.927	621.040	470.734	110.907	30.270	15.423	0.315	0.215	0.187	0.193	0.189	0.229	0.274
2024	54.365	690.386	642.288	248.669	188.036	69.674	17.021	0.344	0.180	0.169	0.198	0.177	0.178	0.222
2025	424.396	240.745	449.852	172.244	74.890	49.999	28.062	0.331	0.216	0.173	0.168	0.187	0.190	0.225

Table 4.6b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM tuning indices and standard deviations for the Northern Shelf cod stock (all substocks) in Q3+Q4.

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
1992	14492.686	1814.096	472.825	227.252	77.609	54.158	37.000	0.108	0.140	0.135	0.146	0.200	0.237	0.426
1993	3099.240	3697.428	738.292	198.922	91.534	28.137	46.555	0.121	0.131	0.186	0.199	0.215	0.227	0.368
1994	17045.174	2334.550	1125.386	188.348	47.388	34.064	22.524	0.112	0.125	0.147	0.223	0.224	0.349	0.858
1995	8675.213	6079.828	1400.263	652.258	91.762	22.996	36.457	0.106	0.126	0.243	0.300	0.348	0.299	0.506
1996	4184.278	2187.519	807.755	246.420	148.428	26.790	12.853	0.121	0.130	0.138	0.170	0.235	0.267	1.050
1997	24482.536	2067.676	810.240	304.111	68.903	39.357	21.856	0.133	0.150	0.174	0.197	0.227	0.258	0.646
1998	841.145	7274.882	565.881	145.926	79.416	28.463	20.404	0.142	0.129	0.142	0.187	0.235	0.239	0.434
1999	3428.524	463.381	1556.281	141.501	37.085	13.632	7.216	0.171	0.141	0.133	0.159	0.267	0.308	0.618
2000	5610.031	1001.505	106.050	256.260	28.807	28.580	30.343	0.235	0.226	0.186	0.245	0.327	0.540	1.055

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
2001	1622.510	1890.031	340.377	75.773	43.375	26.480	37.714	0.182	0.133	0.134	0.199	0.248	0.274	0.667
2002	4487.211	884.430	747.302	209.546	50.038	18.022	5.516	0.180	0.146	0.140	0.158	0.197	0.260	0.873
2003	718.304	1115.606	254.893	185.831	83.485	64.011	28.294	0.155	0.146	0.147	0.179	0.220	0.273	0.578
2004	3274.991	696.068	448.715	80.598	68.887	19.020	#N/A	0.186	0.134	0.155	0.195	0.222	0.271	#N/A
2005	1137.107	745.113	204.226	113.819	27.394	31.851	13.407	0.165	0.137	0.141	0.156	0.215	0.288	0.673
2006	4362.304	736.957	505.235	98.010	14.746	14.174	11.472	0.130	0.157	0.176	0.176	0.272	0.877	0.987
2007	2375.520	2662.417	419.456	163.220	61.912	35.915	26.487	0.176	0.159	0.158	0.162	0.191	0.213	0.531
2008	2560.627	1114.432	1047.844	203.979	75.295	24.127	11.795	0.185	0.140	0.146	0.147	0.185	0.224	0.796
2009	2137.958	833.921	276.402	250.195	48.167	21.962	28.905	0.197	0.158	0.176	0.195	0.231	0.240	0.505
2010	2658.880	1676.044	540.640	185.126	113.181	23.408	3.888	0.125	0.147	0.156	0.179	0.229	0.257	0.817
2011	1142.711	3019.413	1452.900	342.350	145.796	118.411	14.339	0.153	0.118	0.175	0.148	0.255	0.263	0.653
2012	2448.086	947.178	1330.736	429.498	95.480	19.206	26.559	0.157	0.137	0.144	0.140	0.186	0.244	0.586
2013	2200.152	1088.279	516.650	622.406	173.192	66.636	34.493	0.154	0.153	0.141	0.156	0.194	0.212	0.550
2014	2642.887	1568.905	733.328	309.778	219.277	88.138	53.308	0.132	0.138	0.150	0.146	0.154	0.166	0.363
2015	1458.054	2996.982	1288.348	474.504	150.097	140.680	57.302	0.158	0.119	0.129	0.137	0.164	0.198	0.464
2016	1067.273	1203.305	1694.843	1160.355	207.141	123.630	83.555	0.158	0.131	0.126	0.154	0.158	0.169	0.391
2017	4742.234	617.114	692.764	529.411	304.912	51.067	54.642	0.122	0.126	0.155	0.149	0.171	0.178	0.485
2018	753.610	1919.201	365.682	455.864	292.503	158.499	20.458	0.150	0.132	0.126	0.186	0.202	0.213	0.523

Year/Age	1	2	3	4	5	6	7+	1	2	3	4	5	6	7+
2019	1673.544	386.783	534.012	138.446	77.863	44.231	30.208	0.126	0.145	0.145	0.174	0.190	0.217	0.559
2020	3028.299	1144.633	204.393	241.787	70.043	29.049	30.716	0.157	0.135	0.148	0.166	0.193	0.214	0.515
2021	1411.237	2467.280	727.423	164.528	112.433	18.199	22.750	0.128	0.141	0.140	0.159	0.171	0.233	0.625
2022	2883.317	980.271	965.312	419.878	107.662	82.399	23.400	0.152	0.129	0.137	0.155	0.205	0.211	0.540
2023	1497.677	1507.832	650.825	507.815	182.623	38.938	25.889	0.159	0.122	0.135	0.141	0.162	0.215	0.613
2024	376.829	790.260	425.316	202.315	221.013	100.579	9.228	0.189	0.151	0.148	0.154	0.170	0.199	0.734

Table 4.6c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM recruitment indices and standard deviations by substock. These indices consist of age 0 cod from the Q3+Q4 surveys forward shifted to represent age 1 on 1st January.

	Northwestern	Southern	Viking	Northwestern	Southern	Viking
1993	1792.151	1553.782	3755.020	0.321	0.303	0.316
1994	1046.439	3789.362	2111.705	0.432	0.320	0.354
1995	2625.390	4743.717	5768.490	0.278	0.390	0.268
1996	436.630	3719.212	3416.523	0.605	0.359	0.401
1997	3796.705	7112.292	5994.004	0.272	0.340	0.347
1998	31.899	47.643	140.237	0.496	0.472	0.490
1999	340.952	2793.756	4763.996	0.473	0.568	0.420
2000	350.872	413.220	1324.656	0.394	0.585	0.432
2001	70.853	327.603	904.008	0.870	0.675	1.057
2002	77.861	10564.433	2951.932	0.611	0.592	0.454
2003	25.622	225.749	224.806	0.618	1.012	0.478
2004	45.477	1284.331	2966.673	0.639	0.881	0.462
2005	32.530	1264.619	396.202	0.773	0.782	0.436
2006	1511.869	379.508	1764.610	0.448	0.516	0.455
2007	69.379	1338.719	957.638	0.572	0.653	0.540
2008	63.448	5366.873	1212.565	0.677	0.561	0.451
2009	59.876	54.018	508.175	0.692	1.309	0.557
2010	109.474	318.239	924.962	0.434	0.990	0.547
2011	3.268	26.820	144.133	0.604	0.934	0.665
2012	10.727	221.027	4048.491	0.870	1.005	0.452
2013	48.014	507.069	285.092	0.671	0.955	0.480
2014	74.515	74.287	103.028	0.717	0.785	0.513
2015	154.965	49.899	316.727	0.395	0.716	0.528
2016	37.535	13.894	18.743	0.743	0.918	0.635
2017	2437.655	255.375	447.124	0.305	0.560	0.376
2018	89.574	23.915	160.519	0.646	0.982	0.509
2019	97.573	20.416	76.699	0.588	1.105	0.668

	Northwestern	Southern	Viking	Northwestern	Southern	Viking
2020	500.531	237.313	1494.653	0.322	0.638	0.462
2021	437.498	98.720	452.352	0.324	0.669	0.470
2022	126.506	139.200	2054.125	0.447	0.667	0.657
2023	89.055	97.165	3380.004	0.571	0.952	0.675
2024	12.771	28.611	39.162	0.606	0.850	0.572
2025	11.970	12.222	282.868	1.030	1.415	0.707

Table 4.7a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM final run stock-wise configuration. All three substocks were given the same stock-wise configuration.

```
# Configuration saved: Sat Apr 26 13:26:15 2025
#
# Where a matrix is specified rows corresponds to fleets and columns to ages.
# Same number indicates same parameter used
# Numbers (integers) starts from zero and must be consecutive
# Negative numbers indicate that the parameter is not included in the model
#
$minAge
# The minimum age class in the assessment
1

$maxAge
# The maximum age class in the assessment
7

$maxAgePlusGroup
# Is last age group considered a plus group for each fleet (1 yes, or 0 no).
1 1 1 1 1 0 0 0 0

$keyLogFsta
# Coupling of the fishing mortality states processes for each age (normally only
# the first row (= fleet) is used).
# Sequential numbers indicate that the fishing mortality is estimated individually
# for those ages; if the same number is used for two or more ages, F is bound for
# those ages (assumed to be the same). Binding fully selected ages will result in a
# flat selection pattern for those ages.
0 1 2 3 4 5 6
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyLogFmu
#
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
```

```

-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyLogFrho
#
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$corFlag
# Correlation of fishing mortality across ages (0 independent, 1 compound symmetry,
# 2 AR(1), 3 separable AR(1).
# 0: independent means there is no correlation between F across age
# 1: compound symmetry means that all ages are equally correlated;
# 2: AR(1) first order autoregressive - similar ages are more highly correlated than
# ages that are further apart, so similar ages have similar F patterns over time.
# if the estimated correlation is high, then the F pattern over time for each age
# varies in a similar way. E.g if almost one, then they are parallel (like a
# separable model) and if almost zero then they are independent.
# 3: Separable AR - Included for historic reasons . . . more later
2

$keyLogFpar
# Coupling of the survey catchability parameters (nomally first row is
# not used, as that is covered by fishing mortality).
-1 -1 -1 -1 -1 -1 -1
0 1 2 3 3 4 4
5 6 7 7 7 8 9
10 11 12 13 14 15 15
16 17 17 17 17 17 17
18 -1 -1 -1 -1 -1 -1 -1
19 -1 -1 -1 -1 -1 -1 -1
20 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyQpow
# Density dependent catchability power parameters (if any).
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyVarF
# Coupling of process variance parameters for log(F)-process (Fishing mortality
# normally applies to the first (fishing) fleet; therefore only first row is used)

```

```

0 0 1 1 1 1 1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyVarLogN
# Coupling of the recruitment and survival process variance parameters for the
# log(N)-process at the different ages. It is advisable to have at least the first age
# class (recruitment) separate, because recruitment is a different process than
# survival.
0 1 1 1 1 1 1

$keyVarLogP
#

$keyVarObs
# Coupling of the variance parameters for the observations.
# First row refers to the coupling of the variance parameters for the catch data
# observations by age
# Second and further rows refers to coupling of the variance parameters for the
# index data observations by age
0 1 2 2 2 2 2
3 4 4 4 5 5 6
7 8 8 8 9 10 10
11 12 13 13 13 13 13
14 14 15 15 15 16 16
17 -1 -1 -1 -1 -1 -1
18 -1 -1 -1 -1 -1 -1
19 -1 -1 -1 -1 -1 -1
20 20 20 -1 -1 -1 -1
21 21 -1 -1 -1 -1 -1

$sobsCorConstruct
# Covariance structure for each fleet ("ID" independent, "AR" AR(1), or "US" for unstruc-
tured). | Possible values are: "ID" "AR" "US"
"AR" "AR" "AR" "AR" "AR" "ID" "ID" "ID" "US" "US"

$keyCorObs
# Coupling of correlation parameters can only be specified if the AR(1) structure is
chosen above.
# NA's indicate where correlation parameters can be specified (-1 where they cannot).
#1-2 2-3 3-4 4-5 5-6 6-7
0 1 2 2 2 2
3 4 4 5 5 5
6 6 7 7 8 8
9 9 9 9 9 9
10 11 11 11 11 11
-1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1

$stockRecruitmentModelCode

```

```
# Stock recruitment code (0 for plain random walk, 1 for Ricker, 2 for Beverton-Holt, 3
piece-wise constant, 61 for segmented regression/hockey stick, 62 for AR(1), 63 for bent
hyperbola / smooth hockey stick, 64 for power function with degree < 1, 65 for power
function with degree > 1, 66 for Shepherd, 67 for Deriso, 68 for Saila-Lorda, 69 for
sigmoidal Beverton-Holt, 90 for CMP spline, 91 for more flexible spline, and 92 for most
flexible spline).
```

```
0
```

```
$noScaledYears
```

```
# Number of years where catch scaling is applied.
```

```
0
```

```
$keyScaledYears
```

```
# A vector of the years where catch scaling is applied.
```

```
$keyParScaledYA
```

```
# A matrix specifying the couplings of scale parameters (nrow = no scaled years, ncol =
no ages).
```

```
$fbarRange
```

```
# lowest and highest age included in Fbar
```

```
2 4
```

```
$keyBiomassTreat
```

```
# To be defined only if a biomass survey is used (0 SSB index, 1 catch index, 2 FSB index,
3 total catch, 4 total landings, 5 TSB index, 6 TSN index, and 10 Fbar idx).
```

```
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1
```

```
$obsLikelihoodFlag
```

```
# Option for observational likelihood | Possible values are: "LN" "ALN" "Dirichlet"
```

```
"LN" "LN" "LN" "LN" "LN" "LN" "LN" "LN" "ALN" "ALN"
```

```
$fixVarToWeight
```

```
# If weight attribute is supplied for observations this option sets the treatment (0
relative weight, 1 fix variance to weight). Can be specified fleetwise.
```

```
0 0 0 0 0 0 0 0 0 0
```

```
$fracMixF
```

```
# The fraction of t(3) distribution used in logF increment distribution
```

```
0
```

```
$fracMixN
```

```
# The fraction of t(3) distribution used in logN increment distribution (for each age
group)
```

```
0 0 0 0 0 0 0
```

```
$fracMixObs
```

```
# A vector with same length as number of fleets, where each element is the fraction of
t(3) distribution used in the distribution of that fleet
```

```
0 0 0 0 0 0 0 0 0 0
```

```
$constRecBreaks
```

```
# For stock-recruitment code 3: Vector of break years between which recruitment is at
constant level. The break year is included in the left interval. For spline stock-
recruitment: Vector of log-ssb knots. (This option is only used in combination with stock-
recruitment code 3, 90-92, and 290)
```

```
$predVarObsLink
```

```
# Coupling of parameters used in a prediction-variance link for observations.
```

```
-1 -1 -1 -1 -1 -1 -1
```

```

-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
NA NA NA NA NA NA NA
NA NA NA NA NA NA NA
NA NA NA NA NA NA NA
NA NA NA NA NA NA NA
NA NA NA NA NA NA NA

```

\$stockweightModel

Integer code describing the treatment of stock weights in the model (0 use as known, 1 use as observations to inform stock weight process (GMRF with cohort and within year correlations)), 2 to add extra correlation to plusgroup
2

\$keyStockweightMean

Coupling of stock-weight process mean parameters (not used if stockweightModel==0)
0 1 2 3 4 5 6

\$keyStockweightObsvar

Coupling of stock-weight observation variance parameters (not used if stockweightModel==0)
0 0 0 0 0 0 0

\$catchweightModel

Integer code describing the treatment of catch weights in the model (0 use as known, 1 use as observations to inform catch weight process (GMRF with cohort and within year correlations)), 2 to add extra correlation to plusgroup
0

\$keyCatchweightMean

Coupling of catch-weight process mean parameters (not used if catchweightModel==0)
NA NA NA NA NA NA NA

\$keyCatchweightObsvar

Coupling of catch-weight observation variance parameters (not used if catchweightModel==0)
NA NA NA NA NA NA NA

\$matureModel

Integer code describing the treatment of proportion mature in the model (0 use as known, 1 use as observations to inform proportion mature process (GMRF with cohort and within year correlations on logit(proportion mature))), 2 to add extra correlation to plusgroup
2

\$keyMatureMean

Coupling of mature process mean parameters (not used if matureModel==0)
0 1 2 3 4 5 6

\$mortalityModel

Integer code describing the treatment of natural mortality in the model (0 use as known, 1 use as observations to inform natural mortality process (GMRF with cohort and within year correlations)), 2 to add extra correlation to plusgroup
1

\$keyMortalityMean

0 1 2 3 3 3 3

\$keyMortalityObsvar

```
# Coupling of natural mortality observation variance parameters (not used if mortalityModel==0)
0 0 0 0 0 0 0
```

```
$keyXtraSd
```

```
# An integer matrix with 4 columns (fleet year age coupling), which allows additional uncertainty to be estimated for the specified observations
```

```
$logNMeanAssumption
```

```
# Flags indicating what the population model should correspond to. 0: Median, 1: Mean, 2: Mode. Two values are given to differentiate recruitment and other ages.
0 0
```

```
$initState
```

```
# Flag indicating whether initial parameters should be added for the latent processes.
0
```

```
$recruitmentAutocorrelation
```

```
# Number of auto-correlation parameters for recruitment. The auto-regressive process is forced to be stationary with real characteristic roots.
0
```

```
$keyLogFseason
```

```
# Coupling of the seasonality parameters.
```

```
0 0 0 0 0 0 0
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
```

```
$seasonTimes
```

```
# Vector of time endpoints for seasons. Should start with 0 and end with 1. Default is c(0,1).
0 0.25 0.5 0.75 1
```

```
$isFishingSeason
```

```
# Vector of 0/1 indicating if there is fishing in the season. Length should be one less than seasonTimes
1 1 1 1
```

```
$seasonFirstYear
```

```
# First year where seasons are used. Default is -Inf.
-Inf
```

```
$seasonFixedEffect
```

```
# 0: Seasonality is modelled by random effects. 1: Seasonality is modelled by fixed effects.
1
```

```
$keyMatureObsVar
```

```
#
```

```
$keyLogFbound_kappa
```

```
#
```

```
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
```

```

-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyLogFbound_alpha
#
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyLogFbound_tau
#
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$boundF
# (Soft) lower and upper bound for the average fishing mortality. Default is NA for no
bound.
NA NA

$fecundityScaling
# Allometric scaling of fecundity. Use NA to estimate.
1

Table 4.7b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM final run stock-complex configuration arguments.

$newtonsteps
0

$rm.unidentified
FALSE

$lower
list(initLogN = 10, shared_lfssd = -3, logSdLogObs = -5, logitArea = -5, sigmaObsParUS =
-3, transfIRARDist = -3, seasonMu = -8)

$upper
list(initLogN = 15, transfIRARDist = 3, logSdMO = 5, logSdLogObs = 2, logitArea = 5,
sigmaObsParUS = 3)

$starting
list()

$shared_keys
c("keyLogFpar", "keyQpow", "keyVarF", "keyVarLogN", "keyVarObs", "keyCorObs", "sea-
sonRho", "seasonSd", "keyLogFmu", "keyLogFrho", "keyStockWeightObsVar", "procSdSw",

```



```
"phiSw", "procSdMO", "phiMO", "sdMO", "keyMortalityObsVar", "procSdNM", "phiNM", "keyMor-
talityMean", "sigmaObsParUS", "rec_transphi", "predVarObsLink", "keyXtraSd")

$shared_selectivity
4L

$shared_seasonality
0L

$shared_stockrecruitment
FALSE

$shared_oneFScalePars
TRUE

$shared_initN
FALSE

$shared_proportionalHazard
~poly(Age, 3)

$skip_stock_observations
TRUE

$initN
2L

$initF
FALSE

$inner.control
list(maxit = 2000)

Finally, the following fleet-to-stock key was passed to the fitting procedure with the
shared observations:
```

Table 4.7c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM final run fleet-to-stock key.

	North- west	South	Viking
Catch	1	1	1
Survey_Q34	1	1	1
Survey_Q1_NW	1	0	0
Survey_Q1_SO	0	1	0
Survey_Q1_VI	0	0	1
Survey_Rec_NW	1	0	0
Survey_Rec_SO	0	0.75	0
Survey_Rec_VI	0	0.25	1
AUX_SeasonProp	1	1	1
AUX_SubNew	1	1	1

Table 4.7d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM final run model fitting diagnostics.

Model fitting		
log(L)	#par	AIC
1405.673	146	-2519.35

Table 4.8. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20: Multistock SAM final run estimated fishing mortality-at-age for each substock.**Northwestern**

Year/Age	1	2	3	4	5	6	7+	Fbar
1983	0.3	0.928	0.767	0.745	0.628	0.673	1.017	0.813
1984	0.345	1.067	0.895	0.86	0.748	0.81	1.179	0.941
1985	0.343	1.057	0.908	0.86	0.744	0.801	1.13	0.942
1986	0.342	1.067	0.963	0.907	0.782	0.818	1.094	0.979
1987	0.315	0.998	0.934	0.871	0.749	0.786	0.989	0.934
1988	0.286	0.911	0.899	0.825	0.719	0.738	0.866	0.878
1989	0.304	0.977	0.97	0.903	0.786	0.788	0.86	0.95
1990	0.322	1.053	1.03	0.98	0.874	0.86	0.878	1.021
1991	0.304	1.003	1.009	0.973	0.901	0.904	0.889	0.995
1992	0.279	0.931	0.944	0.913	0.855	0.869	0.833	0.93
1993	0.275	0.941	0.962	0.897	0.843	0.853	0.789	0.933
1994	0.265	0.919	0.992	0.902	0.843	0.849	0.759	0.938
1995	0.282	1.033	1.121	0.987	0.908	0.895	0.772	1.047
1996	0.275	1.047	1.183	1.026	0.953	0.93	0.788	1.085
1997	0.242	0.929	1.103	0.987	0.926	0.893	0.735	1.006
1998	0.25	0.981	1.195	1.094	1.04	0.971	0.765	1.09
1999	0.24	0.957	1.224	1.156	1.113	1.048	0.806	1.112
2000	0.221	0.891	1.15	1.16	1.16	1.089	0.82	1.067
2001	0.189	0.768	0.997	1.018	1.037	0.996	0.753	0.928
2002	0.181	0.741	0.968	0.988	0.987	0.972	0.729	0.899
2003	0.161	0.665	0.897	0.933	0.929	0.917	0.681	0.832
2004	0.154	0.637	0.879	0.908	0.914	0.903	0.655	0.808
2005	0.135	0.562	0.795	0.817	0.856	0.857	0.613	0.725
2006	0.124	0.518	0.731	0.737	0.804	0.831	0.592	0.662
2007	0.113	0.471	0.695	0.7	0.773	0.811	0.577	0.622
2008	0.114	0.486	0.751	0.754	0.866	0.917	0.647	0.664
2009	0.099	0.422	0.69	0.716	0.85	0.923	0.658	0.609

Year/Age	1	2	3	4	5	6	7+	Fbar
2010	0.087	0.37	0.635	0.669	0.822	0.896	0.653	0.558
2011	0.073	0.309	0.556	0.603	0.763	0.839	0.619	0.489
2012	0.072	0.301	0.551	0.605	0.781	0.851	0.634	0.486
2013	0.066	0.28	0.526	0.581	0.753	0.807	0.617	0.463
2014	0.059	0.249	0.479	0.525	0.676	0.713	0.555	0.418
2015	0.059	0.25	0.484	0.527	0.679	0.699	0.555	0.421
2016	0.064	0.278	0.544	0.589	0.756	0.75	0.603	0.47
2017	0.075	0.335	0.655	0.719	0.915	0.889	0.719	0.57
2018	0.084	0.381	0.751	0.816	1.063	1.029	0.864	0.649
2019	0.071	0.319	0.653	0.717	0.949	0.937	0.816	0.563
2020	0.055	0.245	0.513	0.581	0.772	0.771	0.686	0.446
2021	0.043	0.19	0.416	0.487	0.655	0.656	0.611	0.364
2022	0.04	0.181	0.403	0.478	0.64	0.645	0.614	0.354
2023	0.043	0.197	0.447	0.533	0.705	0.701	0.668	0.392
2024	0.043	0.204	0.477	0.576	0.765	0.743	0.713	0.419

Southern

Year/Age	1	2	3	4	5	6	7+	Fbar
1983	0.37	1.263	1.097	1.057	0.828	0.763	0.913	1.139
1984	0.432	1.473	1.298	1.237	1	0.932	1.073	1.336
1985	0.427	1.455	1.313	1.234	0.992	0.919	1.026	1.334
1986	0.422	1.451	1.376	1.286	1.03	0.927	0.981	1.371
1987	0.382	1.338	1.316	1.218	0.973	0.879	0.875	1.291
1988	0.344	1.21	1.255	1.143	0.925	0.817	0.759	1.203
1989	0.365	1.292	1.35	1.246	1.008	0.869	0.751	1.296
1990	0.389	1.404	1.442	1.363	1.128	0.956	0.772	1.403
1991	0.363	1.322	1.396	1.336	1.149	0.993	0.773	1.351
1992	0.326	1.198	1.276	1.225	1.064	0.932	0.707	1.233
1993	0.312	1.175	1.264	1.168	1.019	0.889	0.65	1.202
1994	0.294	1.125	1.276	1.152	1	0.867	0.613	1.184

Year/Age	1	2	3	4	5	6	7+	Fbar
1995	0.31	1.251	1.427	1.247	1.065	0.904	0.617	1.309
1996	0.302	1.269	1.508	1.297	1.119	0.94	0.631	1.358
1997	0.264	1.118	1.395	1.239	1.079	0.896	0.584	1.251
1998	0.28	1.215	1.554	1.412	1.247	1.002	0.625	1.394
1999	0.277	1.218	1.637	1.534	1.371	1.112	0.677	1.463
2000	0.261	1.164	1.578	1.58	1.467	1.185	0.707	1.44
2001	0.224	1.009	1.376	1.394	1.319	1.091	0.653	1.26
2002	0.218	0.984	1.351	1.368	1.269	1.076	0.639	1.235
2003	0.194	0.884	1.252	1.293	1.195	1.015	0.597	1.143
2004	0.189	0.863	1.251	1.283	1.199	1.02	0.586	1.132
2005	0.165	0.757	1.126	1.148	1.117	0.963	0.545	1.01
2006	0.152	0.7	1.038	1.038	1.052	0.936	0.528	0.925
2007	0.139	0.644	0.998	0.996	1.022	0.923	0.52	0.879
2008	0.146	0.686	1.115	1.111	1.184	1.08	0.602	0.971
2009	0.13	0.614	1.055	1.087	1.199	1.121	0.632	0.919
2010	0.119	0.559	1.007	1.053	1.202	1.128	0.651	0.873
2011	0.102	0.473	0.897	0.965	1.133	1.073	0.626	0.778
2012	0.101	0.468	0.901	0.982	1.175	1.103	0.65	0.783
2013	0.095	0.44	0.869	0.953	1.147	1.058	0.64	0.754
2014	0.084	0.391	0.792	0.86	1.029	0.935	0.576	0.681
2015	0.084	0.395	0.804	0.869	1.039	0.921	0.579	0.689
2016	0.092	0.437	0.896	0.963	1.149	0.981	0.624	0.765
2017	0.105	0.513	1.055	1.149	1.358	1.135	0.726	0.905
2018	0.114	0.571	1.185	1.277	1.545	1.288	0.856	1.011
2019	0.093	0.463	0.995	1.084	1.332	1.133	0.78	0.847
2020	0.071	0.346	0.763	0.856	1.057	0.909	0.64	0.655
2021	0.055	0.266	0.609	0.708	0.884	0.762	0.562	0.528
2022	0.05	0.248	0.58	0.684	0.85	0.737	0.555	0.504
2023	0.053	0.272	0.646	0.765	0.941	0.805	0.607	0.561

Year/Age	1	2	3	4	5	6	7+	Fbar
2024	0.054	0.278	0.683	0.818	1.008	0.844	0.641	0.593
Viking								
Year/Age	1	2	3	4	5	6	7+	Fbar
1983	0.076	0.549	0.655	0.686	0.56	0.629	1.309	0.63
1984	0.086	0.619	0.749	0.775	0.653	0.743	1.487	0.714
1985	0.084	0.607	0.753	0.768	0.643	0.727	1.411	0.709
1986	0.084	0.612	0.798	0.809	0.675	0.742	1.365	0.739
1987	0.076	0.562	0.76	0.763	0.635	0.7	1.213	0.695
1988	0.068	0.506	0.721	0.713	0.601	0.648	1.047	0.646
1989	0.072	0.543	0.779	0.781	0.658	0.693	1.041	0.701
1990	0.076	0.582	0.822	0.842	0.727	0.752	1.056	0.748
1991	0.072	0.554	0.805	0.836	0.75	0.79	1.07	0.732
1992	0.066	0.513	0.752	0.783	0.709	0.758	0.999	0.683
1993	0.065	0.52	0.768	0.77	0.701	0.746	0.949	0.686
1994	0.063	0.51	0.795	0.778	0.704	0.745	0.916	0.694
1995	0.067	0.572	0.897	0.85	0.757	0.784	0.931	0.773
1996	0.064	0.569	0.929	0.867	0.78	0.8	0.933	0.788
1997	0.055	0.491	0.843	0.812	0.738	0.747	0.847	0.716
1998	0.055	0.508	0.894	0.881	0.812	0.796	0.862	0.761
1999	0.054	0.499	0.921	0.937	0.873	0.864	0.914	0.785
2000	0.049	0.464	0.864	0.939	0.909	0.896	0.929	0.755
2001	0.042	0.398	0.746	0.821	0.809	0.817	0.85	0.655
2002	0.039	0.377	0.711	0.781	0.755	0.781	0.807	0.623
2003	0.036	0.347	0.676	0.757	0.73	0.757	0.773	0.593
2004	0.035	0.334	0.666	0.742	0.723	0.75	0.749	0.581
2005	0.031	0.3	0.614	0.68	0.69	0.726	0.714	0.532
2006	0.028	0.277	0.566	0.614	0.649	0.704	0.691	0.486
2007	0.025	0.249	0.531	0.576	0.616	0.679	0.664	0.452
2008	0.026	0.257	0.575	0.622	0.692	0.77	0.746	0.485

Year/Age	1	2	3	4	5	6	7+	Fbar
2009	0.023	0.227	0.537	0.6	0.69	0.787	0.772	0.455
2010	0.02	0.198	0.491	0.557	0.663	0.759	0.761	0.415
2011	0.017	0.167	0.434	0.507	0.621	0.717	0.728	0.369
2012	0.017	0.163	0.431	0.51	0.636	0.729	0.747	0.368
2013	0.015	0.151	0.411	0.488	0.613	0.69	0.725	0.35
2014	0.013	0.132	0.367	0.433	0.541	0.599	0.642	0.311
2015	0.013	0.133	0.372	0.436	0.544	0.589	0.643	0.314
2016	0.015	0.152	0.43	0.501	0.623	0.649	0.718	0.361
2017	0.018	0.192	0.544	0.643	0.792	0.808	0.898	0.459
2018	0.021	0.223	0.635	0.743	0.938	0.953	1.101	0.534
2019	0.018	0.186	0.55	0.651	0.834	0.865	1.036	0.462
2020	0.013	0.139	0.421	0.512	0.659	0.692	0.848	0.357
2021	0.01	0.105	0.333	0.419	0.546	0.574	0.737	0.286
2022	0.01	0.1	0.323	0.413	0.535	0.567	0.742	0.279
2023	0.01	0.109	0.355	0.457	0.586	0.611	0.801	0.307
2024	0.011	0.116	0.392	0.51	0.655	0.67	0.884	0.339

Table 4.9. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM final run estimated population numbers-at-age (start of year; thousands) for each substock.

Northwestern

Year/Age	1	2	3	4	5	6	7+	Total
1983	485860	122291	22759	8066	3110	1359	568	644013
1984	705249	101092	30038	5542	4128	1234	717	848000
1985	217360	182976	15195	10726	1822	1732	545	430356
1986	617300	53268	30131	5343	3445	746	792	711024
1987	368304	167800	10341	8658	1737	1197	498	558535
1988	141601	105680	27487	3392	2691	745	556	282153
1989	501989	40266	22253	7866	1150	1113	493	575130
1990	154300	138215	7410	5968	2645	420	595	309554
1991	221082	39482	18835	2401	1752	892	363	284808
1992	530724	61803	6919	4486	865	560	423	605781

Year/Age	1	2	3	4	5	6	7+	Total
1993	247560	148073	12336	2284	1282	344	336	412216
1994	528984	69563	23827	3784	791	432	260	627641
1995	317602	143065	14856	6928	1197	278	257	484183
1996	158151	82734	21144	3474	2173	389	206	268272
1997	597300	42331	12626	4776	1046	674	214	658966
1998	68584	160212	7686	3119	1394	335	309	241638
1999	96646	19166	23816	1750	889	346	247	142860
2000	200838	27145	4007	4251	463	244	187	237135
2001	65633	63439	4954	1104	1001	115	132	136378
2002	96320	20602	13321	1551	322	292	79	132488
2003	56697	27659	3797	3142	452	98	120	91965
2004	86786	16642	4813	1160	942	138	74	110556
2005	71474	21414	3273	1190	364	311	74	98101
2006	270473	17774	4330	963	369	130	140	294180
2007	87319	71292	3976	1441	359	138	102	164627
2008	97619	20276	15136	1299	554	143	90	135117
2009	171864	23294	4302	3861	491	177	88	204077
2010	212578	46139	5497	1397	1463	187	88	267348
2011	95756	63316	11349	1793	665	525	101	173506
2012	147506	27097	17854	4130	797	260	235	197880
2013	202760	35837	7472	6400	1767	287	197	254721
2014	249065	52004	10101	3123	2755	630	199	317877
2015	154181	71458	16702	4167	1460	1140	338	249446
2016	100262	41358	20242	8042	1823	592	592	172909
2017	271091	24541	12141	7451	3329	690	496	319741
2018	64727	62493	6269	4498	2339	1025	485	141836
2019	138905	15268	12324	1887	1526	589	458	170957
2020	176537	37790	4335	3612	754	490	332	223851
2021	128521	46755	10728	2105	1456	285	334	190184

Year/Age	1	2	3	4	5	6	7+	Total
2022	152396	34814	14485	4613	1137	607	267	208319
2023	106626	40181	10402	6299	2206	539	358	166611
2024	39135	29183	10289	3868	2692	905	355	86427
2025	64887	10480	9142	3829	1634	960	511	91443

Southern

Year/Age	1	2	3	4	5	6	7+	Total
1983	318785	74586	9866	2511	709	254	97	406808
1984	686071	71219	11194	1812	776	218	126	771417
1985	135121	157153	8177	2335	420	237	95	303537
1986	831514	29750	16005	1806	571	135	98	879879
1987	325646	207162	3390	3158	383	172	80	539990
1988	243076	77899	26170	703	725	139	81	348793
1989	300883	63762	11028	5130	218	227	81	381330
1990	133075	77693	8892	1964	1167	76	113	222981
1991	137714	33354	8825	1625	368	293	68	182248
1992	245280	34236	4453	1728	341	87	105	286230
1993	169948	63406	4568	1074	454	90	63	239604
1994	420627	46378	8871	1046	304	141	54	477421
1995	273149	108916	8068	1837	283	94	73	392419
1996	220064	69046	14678	1468	451	79	66	305852
1997	585945	57619	9195	2455	330	124	54	655722
1998	57442	166276	9440	1860	594	97	72	235782
1999	122522	16620	21759	1435	393	132	71	162932
2000	115164	33015	2627	2806	259	87	71	154029
2001	69625	35507	4637	462	400	44	51	110727
2002	143043	21767	5697	890	92	86	35	171609
2003	50920	39359	3449	1100	165	22	38	95053
2004	89465	14741	6728	680	256	39	23	111932
2005	92569	21473	2296	1100	128	68	21	117654

Year/Age	1	2	3	4	5	6	7+	Total
2006	112309	23779	3800	466	229	34	32	140649
2007	131576	27508	4051	904	135	67	26	164268
2008	121415	30850	5105	950	291	44	37	158692
2009	48710	27814	5212	1075	240	72	32	83154
2010	77389	11819	5155	1144	294	60	34	95894
2011	53391	21472	2451	1092	322	72	29	78828
2012	55460	14110	4934	701	305	84	32	75626
2013	53190	13619	3019	1211	238	68	36	71381
2014	64955	12992	3113	752	373	57	32	82274
2015	21528	17882	3159	893	258	105	32	43856
2016	14824	5506	3937	814	302	75	51	25509
2017	30841	3712	1291	888	235	74	45	37087
2018	12145	7444	816	285	210	48	37	20985
2019	16251	2882	1312	150	60	35	23	20714
2020	22633	4169	673	278	39	13	18	27824
2021	20384	6084	1142	200	88	11	12	27923
2022	38976	5350	1593	376	78	29	10	46412
2023	28374	10300	1423	505	148	28	16	40793
2024	15952	7344	2531	382	177	47	17	26451
2025	22210	4226	2037	729	121	50	24	29397

Viking

Year/Age	1	2	3	4	5	6	7+	Total
1983	100334	31535	8530	3370	1377	645	281	146073
1984	112122	33551	9595	2640	1518	581	334	160339
1985	32054	40208	9697	3660	981	666	279	87545
1986	109079	10058	10807	3693	1396	444	322	135799
1987	55326	40143	2728	3344	1182	577	228	103528
1988	50358	20379	10749	1127	1275	489	296	84673
1989	77062	15988	7106	3456	535	606	285	105038

Year/Age	1	2	3	4	5	6	7+	Total
1990	40102	26669	4639	2080	1224	227	349	75289
1991	43440	13217	6658	1646	661	484	208	66314
1992	82984	14326	3545	2170	573	243	256	104097
1993	55726	29086	3968	1364	837	243	171	91395
1994	103407	18269	7459	1463	570	326	156	131649
1995	74721	35654	5501	2523	522	230	184	119334
1996	49845	23318	8309	1669	878	201	156	84376
1997	120273	15862	5540	2138	636	332	123	144904
1998	16962	44142	4323	1781	778	266	172	68424
1999	41764	5121	12087	1475	640	278	171	61536
2000	60216	13639	1425	2878	528	240	153	79079
2001	26664	23097	3657	571	695	168	131	54982
2002	38179	9108	7126	1238	220	225	103	56199
2003	24756	12448	2660	2348	521	90	116	42940
2004	51790	8037	3234	920	807	217	74	65079
2005	35749	14415	2219	973	370	323	111	54161
2006	71289	9811	3593	733	353	163	169	86112
2007	31849	19926	2672	1068	356	149	128	56149
2008	38827	8017	5452	975	502	166	104	54043
2009	45917	9927	2098	1910	450	193	104	60601
2010	68327	12568	2715	837	827	187	97	85557
2011	35294	21032	3808	1172	404	375	97	62182
2012	59499	9733	6515	1806	603	186	172	78513
2013	62775	14848	3160	2365	917	267	141	84472
2014	77546	18051	4260	1301	1098	371	158	102785
2015	39769	25270	6445	1946	667	508	205	74808
2016	24330	11248	9122	3271	1051	325	304	49650
2017	54148	6723	4034	3676	1868	441	242	71132
2018	18086	16051	2292	1481	1298	698	248	40154

Year/Age	1	2	3	4	5	6	7+	Total
2019	33049	4730	4536	747	535	397	285	44280
2020	70615	9944	1565	1403	337	175	218	84257
2021	34006	21808	3765	796	531	147	155	61210
2022	48755	10065	7690	1864	430	245	128	69177
2023	32673	13458	3715	3043	919	213	153	54174
2024	7800	9036	4592	1654	1486	470	148	25186
2025	14658	2027	3335	1668	757	578	254	23277

Table 4.10. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20: Multistock SAM final run estimated catches-at-age (thousands) for each substock.

Northwestern

Year/Age	1	2	3	4	5	6	7+
1983	68807	49796	10490	3778	1296	595	326
1984	116898	45590	15409	2861	1946	614	447
1985	35205	82110	7856	5538	856	855	332
1986	103725	24301	16131	2843	1674	373	473
1987	57060	73320	5428	4481	820	583	281
1988	19894	43159	14064	1687	1235	348	289
1989	76091	16978	11876	4159	561	543	255
1990	24300	60588	4095	3321	1382	217	312
1991	33273	16812	10291	1338	933	476	192
1992	75876	25328	3674	2408	446	292	214
1993	34318	61372	6644	1213	654	177	164
1994	70552	28479	12989	2017	404	222	124
1995	44646	62822	8663	3903	641	147	124
1996	21215	36349	12686	2000	1199	211	101
1997	72907	17373	7304	2676	567	357	100
1998	8392	67241	4599	1848	810	187	148
1999	11592	7767	14338	1066	537	202	123
2000	23325	10591	2333	2584	286	146	94
2001	6521	22323	2620	624	581	65	63

Year/Age	1	2	3	4	5	6	7+
2002	8875	6799	6722	857	182	163	37
2003	4494	8044	1775	1657	245	53	53
2004	6304	4532	2180	592	506	74	32
2005	4452	5174	1375	573	188	160	30
2006	15460	4023	1734	436	183	66	56
2007	4462	14885	1539	642	173	69	40
2008	4972	4277	6155	603	288	77	38
2009	7829	4404	1662	1757	252	96	38
2010	8828	8100	2025	608	735	99	38
2011	3279	9333	3826	725	318	267	42
2012	4808	3819	5720	1674	387	134	99
2013	6145	4743	2317	2513	837	142	81
2014	6930	6322	2934	1133	1209	288	76
2015	4242	8701	4862	1509	643	513	129
2016	3011	5562	6534	3091	865	279	239
2017	9672	3950	4598	3329	1788	365	227
2018	2476	10993	2572	2178	1376	591	252
2019	4639	2362	4596	835	839	321	229
2020	4804	4774	1354	1386	363	236	148
2021	2676	4663	2777	714	625	123	136
2022	2842	3204	3569	1511	480	258	110
2023	2121	3997	2807	2245	996	243	156
2024	790	3005	2945	1463	1283	424	162

Southern

Year/Age	1	2	3	4	5	6	7+
1983	54546	36901	5699	1469	358	121	52
1984	138397	38898	7123	1159	441	119	75
1985	26595	85278	5228	1491	237	128	55
1986	167897	16278	10462	1175	330	73	55

Year/Age	1	2	3	4	5	6	7+
1987	59998	108524	2165	1993	214	90	42
1988	40422	38222	16286	427	393	70	39
1989	53709	32115	7087	3271	125	118	39
1990	24833	40670	5900	1311	711	42	54
1991	24303	16900	5770	1081	226	166	33
1992	40294	16526	2812	1101	201	47	48
1993	26442	30439	2876	667	261	48	27
1994	61858	21746	5583	645	173	73	22
1995	41936	54155	5349	1181	167	50	30
1996	32315	34366	9975	963	274	43	27
1997	77854	26813	6037	1565	196	66	21
1998	7853	80327	6456	1262	382	55	30
1999	16809	7896	15131	1008	265	80	31
2000	15644	15393	1798	1990	180	55	32
2001	8153	15131	2938	310	265	27	22
2002	15716	8816	3484	589	60	51	15
2003	4830	14205	1974	700	104	13	15
2004	7938	5069	3783	425	161	23	9
2005	7029	6584	1206	652	78	38	8
2006	7838	6863	1921	264	134	19	12
2007	8318	7424	2002	507	78	36	9
2008	7894	8621	2685	562	182	26	15
2009	2913	7184	2674	638	151	44	14
2010	4374	2935	2602	669	186	36	14
2011	2532	4583	1161	606	197	42	12
2012	2544	2920	2249	394	190	50	14
2013	2292	2681	1348	667	146	40	15
2014	2571	2358	1314	387	215	31	13
2015	847	3264	1340	460	150	57	12

Year/Age	1	2	3	4	5	6	7+
2016	633	1099	1818	438	186	42	21
2017	1524	861	672	533	158	45	21
2018	632	1847	448	180	150	31	19
2019	712	615	652	87	40	21	11
2020	790	718	282	140	23	7	8
2021	538	823	400	90	46	5	5
2022	907	660	525	161	40	14	4
2023	707	1373	511	234	81	14	7
2024	400	1001	951	185	101	24	7

Viking

Year/Age	1	2	3	4	5	6	7+
1983	4406	9519	3613	1519	537	274	189
1984	5691	11270	4491	1299	663	278	239
1985	1577	13305	4547	1788	424	314	195
1986	5521	3378	5266	1863	625	212	221
1987	2520	12614	1285	1620	506	265	147
1988	2038	5850	4877	519	524	213	176
1989	3373	4803	3386	1697	235	276	169
1990	1828	8405	2290	1075	577	109	209
1991	1879	4012	3245	851	318	242	126
1992	3357	4119	1664	1075	265	118	148
1993	2198	8484	1894	669	385	117	96
1994	3928	5256	3626	722	263	156	86
1995	3008	11202	2887	1321	253	114	102
1996	1886	7249	4454	885	435	101	86
1997	3976	4417	2795	1082	303	160	64
1998	554	12504	2246	947	395	133	91
1999	1338	1405	6367	813	341	147	94
2000	1836	3554	721	1583	289	130	85

Year/Age	1	2	3	4	5	6	7+
2001	687	5297	1656	288	352	86	68
2002	900	1934	3045	603	106	111	52
2003	518	2375	1072	1109	246	44	57
2004	1007	1446	1270	423	379	105	35
2005	608	2317	815	424	168	152	51
2006	1111	1468	1249	299	154	75	77
2007	439	2700	886	421	149	67	56
2008	537	1107	1916	404	228	81	50
2009	570	1229	704	781	205	96	51
2010	760	1411	858	325	365	91	47
2011	328	1982	1099	423	170	175	46
2012	531	883	1802	654	259	88	83
2013	520	1256	842	828	382	121	66
2014	574	1366	1041	413	416	152	68
2015	292	1922	1584	619	254	205	88
2016	201	976	2554	1138	443	141	142
2017	557	736	1385	1552	932	223	131
2018	205	1981	874	691	723	393	152
2019	323	503	1556	317	277	210	169
2020	539	823	435	500	148	80	114
2021	194	1379	845	244	203	58	74
2022	252	594	1652	555	162	96	61
2023	179	861	872	984	370	89	77
2024	45	618	1176	584	648	209	79

Table 4.11. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM final run estimated stock weights-at-age (kg) for each substock.

Northwestern

Year/Age	1	2	3	4	5	6	7+
1983	0.07	0.548	1.543	4.177	7.14	10.824	12.525
1984	0.054	0.453	1.77	4.076	6.882	9.53	12.621

Year/Age	1	2	3	4	5	6	7+
1985	0.072	0.387	1.694	4.234	6.209	9.291	12.556
1986	0.056	0.491	1.534	3.804	6.731	8.887	12.362
1987	0.086	0.379	1.468	4.15	6.871	9.295	12.404
1988	0.082	0.504	1.485	3.594	7.013	8.947	12.459
1989	0.072	0.445	1.53	3.723	6.04	9.259	12.478
1990	0.058	0.426	1.489	4.103	6.804	8.698	12.532
1991	0.081	0.496	1.651	4.002	7.167	9.905	12.728
1992	0.057	0.579	1.939	4.5	6.893	9.893	12.88
1993	0.045	0.452	2.041	4.892	7.469	9.693	13.091
1994	0.045	0.376	1.679	4.935	8.028	9.631	13.038
1995	0.064	0.315	1.341	4.283	6.83	10.027	13.039
1996	0.076	0.487	1.401	4.063	6.858	8.659	12.71
1997	0.033	0.411	1.656	4.011	7.117	9.335	12.364
1998	0.091	0.291	1.528	4.22	6.974	9.499	12.143
1999	0.092	0.486	0.875	2.948	6.902	9.22	12.104
2000	0.077	0.527	1.738	2.703	5.271	8.651	12.045
2001	0.086	0.415	1.568	4.054	5.549	7.313	11.802
2002	0.084	0.401	1.346	3.964	5.878	8.64	11.697
2003	0.083	0.474	1.119	3.212	6.285	8.715	11.77
2004	0.071	0.395	1.358	3.461	6.195	10.127	11.646
2005	0.081	0.416	1.297	2.838	5.406	8.985	11.532
2006	0.068	0.487	1.43	3.009	5.341	6.019	11.125
2007	0.07	0.433	1.683	3.86	5.86	7.879	10.72
2008	0.076	0.514	1.649	4.489	6.6	7.914	10.491
2009	0.076	0.507	1.579	3.709	6.567	8.205	10.501
2010	0.086	0.567	1.832	3.626	6.049	7.676	10.455
2011	0.095	0.521	1.848	4.078	5.867	8.097	10.235
2012	0.108	0.63	1.603	3.782	6.021	7.888	10.038
2013	0.095	0.689	1.889	3.635	5.665	7.037	9.686

Year/Age	1	2	3	4	5	6	7+
2014	0.073	0.585	1.873	3.782	5.396	7.086	9.423
2015	0.084	0.544	1.705	3.825	5.875	6.921	9.238
2016	0.076	0.626	1.732	3.647	5.773	7.468	9.268
2017	0.066	0.56	1.835	3.89	5.655	7.006	9.256
2018	0.071	0.416	1.708	3.582	5.636	6.581	9.29
2019	0.069	0.46	1.378	3.73	5.559	7.592	9.523
2020	0.077	0.499	1.722	3.454	5.363	7.807	9.875
2021	0.09	0.473	1.83	3.96	5.779	7.156	9.869
2022	0.109	0.464	1.403	4.305	6.073	7.915	9.883
2023	0.098	0.527	1.447	3.272	6.16	7.748	9.729
2024	0.098	0.466	1.415	3.057	5.143	7.418	9.65
2025	0.094	0.506	1.292	3.285	5.118	7.294	9.512

Southern

Year/Age	1	2	3	4	5	6	7+
1983	0.078	0.661	2.085	4.562	7.323	10.421	12.258
1984	0.078	0.601	1.999	3.85	7.533	10.025	12.35
1985	0.052	0.553	2.124	4.549	7.157	10.397	12.379
1986	0.066	0.368	1.839	4.103	7.488	10.202	12.452
1987	0.081	0.493	1.301	4.496	7.453	10.695	12.58
1988	0.067	0.668	1.932	3.492	7.601	10.571	12.739
1989	0.067	0.694	2.194	3.97	6.971	10.251	12.844
1990	0.076	0.605	2.358	4.41	7.211	10.504	12.885
1991	0.058	0.649	2.063	4.773	7.654	10.295	13.003
1992	0.065	0.39	2.069	5.021	7.892	10.894	12.933
1993	0.047	0.553	2.066	4.961	8.212	11.647	13.005
1994	0.042	0.409	1.929	4.757	8.318	10.413	13.029
1995	0.048	0.366	1.411	4.193	7.368	10.537	12.986
1996	0.041	0.457	1.495	3.159	6.766	10.272	12.99
1997	0.035	0.262	1.847	3.599	6.703	8.609	12.643

Year/Age	1	2	3	4	5	6	7+
1998	0.049	0.368	1.081	3.652	7.679	9.475	12.635
1999	0.106	0.517	1.503	3.174	7.197	9.75	12.444
2000	0.063	0.546	1.648	3.62	5.364	9.505	12.373
2001	0.108	0.406	1.508	3.413	7.08	8.942	12.247
2002	0.084	0.496	1.693	4.337	7.554	10.312	12.407
2003	0.072	0.595	1.435	4.274	7.287	11.026	12.669
2004	0.057	0.312	1.912	3.514	7.606	9.514	12.895
2005	0.061	0.527	1.571	3.816	6.551	10.617	12.788
2006	0.068	0.63	2.109	3.901	7.977	8.591	12.618
2007	0.085	0.464	2.122	4.251	6.807	11.246	12.518
2008	0.105	0.843	1.97	3.993	6.627	9.527	12.649
2009	0.079	0.851	2.613	4.304	6.822	9.35	12.556
2010	0.078	0.661	2.708	5.081	6.916	9.941	12.28
2011	0.078	0.606	2.135	5.215	7.97	9.417	11.899
2012	0.07	0.832	2.193	4.028	8.076	10.399	11.557
2013	0.071	0.593	2.459	4.004	6.012	9.116	11.58
2014	0.076	0.708	2.199	4.298	7.091	7.056	11.349
2015	0.103	0.753	1.979	4.048	6.814	9.987	11.56
2016	0.09	0.748	2.299	3.759	7.132	8.682	11.399
2017	0.065	0.554	1.723	3.641	5.594	9.716	11.387
2018	0.089	0.507	1.369	2.948	6.313	8.383	11.19
2019	0.064	0.534	1.725	2.824	4.97	8.676	10.77
2020	0.072	0.508	1.612	3.601	4.604	7.838	10.432
2021	0.084	0.605	1.634	2.89	7.196	9.043	10.171
2022	0.078	0.64	1.902	3.597	6.227	9.322	10.417
2023	0.088	0.567	1.648	3.697	6.443	9.098	10.619
2024	0.103	0.54	1.694	3.013	6.389	9.033	10.77
2025	0.088	0.667	1.651	3.393	5.821	7.662	10.799

Viking

Year/Age	1	2	3	4	5	6	7+
1983	0.055	0.451	1.453	3.199	5.189	7.849	10.248
1984	0.042	0.38	1.447	2.932	4.888	7.357	10.371
1985	0.046	0.32	1.191	3.123	4.584	7.115	9.923
1986	0.045	0.316	1.082	2.777	4.385	5.827	9.884
1987	0.054	0.333	1.088	2.97	4.474	6.427	9.757
1988	0.065	0.254	1.087	2.749	4.642	6.397	9.524
1989	0.058	0.352	0.905	2.901	4.339	7.067	9.741
1990	0.059	0.38	1.17	2.939	4.905	6.973	9.93
1991	0.055	0.433	1.351	2.937	5.199	6.898	10.072
1992	0.053	0.473	1.508	3.55	4.947	7.577	10.272
1993	0.052	0.363	1.404	3.514	5.033	7.601	10.622
1994	0.04	0.365	1.255	3.695	5.834	8.076	10.895
1995	0.042	0.341	1.185	3.226	5.537	7.851	11.136
1996	0.035	0.327	1.215	3.373	5.242	7.677	11.214
1997	0.04	0.306	1.178	3.253	5.16	7.514	11.124
1998	0.055	0.267	1.116	3.209	5.226	7.295	10.929
1999	0.049	0.315	0.908	2.782	4.896	6.995	10.594
2000	0.043	0.376	1.145	2.693	4.638	6.378	10.401
2001	0.059	0.334	1.333	3.044	4.602	6.016	10.363
2002	0.052	0.334	0.951	3.1	5.363	7.796	10.378
2003	0.051	0.384	1.04	2.576	5.074	6.288	10.643
2004	0.058	0.379	1.288	3.025	4.651	6.623	10.744
2005	0.058	0.435	1.271	2.999	4.918	7.108	10.471
2006	0.055	0.418	1.399	3.165	4.549	7.025	10.406
2007	0.052	0.437	1.118	3.389	5.157	6.341	10.37
2008	0.053	0.477	1.439	3.109	5.246	7.017	10.346
2009	0.054	0.456	1.518	3.487	4.703	6.584	10.238
2010	0.051	0.47	1.517	3.4	4.903	6.331	10.161
2011	0.049	0.405	1.431	3.572	4.747	7.002	10.115

Year/Age	1	2	3	4	5	6	7+
2012	0.057	0.429	1.349	3.286	5.248	6.934	9.935
2013	0.054	0.424	1.318	3.261	4.882	6.67	9.809
2014	0.05	0.431	1.401	3.309	5.377	7.624	9.799
2015	0.059	0.442	1.461	3.252	4.763	7.449	9.671
2016	0.063	0.483	1.485	3.218	4.618	6.684	9.827
2017	0.053	0.482	1.499	3.387	4.783	6.652	9.946
2018	0.054	0.445	1.211	3.276	5.379	6.889	10.027
2019	0.06	0.421	1.301	2.958	5.595	7.305	10.123
2020	0.057	0.473	1.341	3.445	5.313	6.588	10.153
2021	0.05	0.424	1.381	2.984	5.425	6.738	10.199
2022	0.054	0.373	1.352	3.259	5.027	7.756	10.438
2023	0.052	0.404	1.152	3.017	5.305	7.365	10.708
2024	0.055	0.333	1.237	2.837	4.778	7.202	10.445
2025	0.059	0.427	1.063	2.827	4.423	6.729	10.351

Table 4.12. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM final run estimated maturity-at-age for each substock.

Northwestern

Year/Age	1	2	3	4	5	6	7+
1983	0.002	0.039	0.271	0.305	0.722	0.805	0.905
1984	0.002	0.024	0.135	0.437	0.647	0.686	0.878
1985	0.003	0.029	0.143	0.454	0.558	0.784	0.871
1986	0.002	0.108	0.102	0.191	0.466	0.721	0.854
1987	0.003	0.023	0.26	0.535	0.736	0.779	0.894
1988	0.005	0.042	0.171	0.461	0.568	0.762	0.893
1989	0.121	0.263	0.333	0.639	0.889	0.875	0.963
1990	0.005	0.088	0.23	0.549	0.751	0.809	0.801
1991	0.007	0.095	0.405	0.56	0.83	0.892	0.941
1992	0.014	0.335	0.655	0.891	0.942	0.963	0.979
1993	0.015	0.12	0.76	0.874	0.943	0.96	0.979
1994	0.015	0.3	0.685	0.882	0.949	0.965	0.981

Year/Age	1	2	3	4	5	6	7+
1995	0.009	0.084	0.442	0.808	0.891	0.926	0.964
1996	0.014	0.183	0.583	0.854	0.937	0.954	0.976
1997	0.017	0.267	0.816	0.927	0.961	0.974	0.984
1998	0.015	0.155	0.649	0.838	0.934	0.953	0.976
1999	0.03	0.571	0.539	0.895	0.958	0.973	0.985
2000	0.029	0.351	0.812	0.896	0.961	0.975	0.986
2001	0.016	0.25	0.635	0.87	0.916	0.958	0.98
2002	0.021	0.263	0.694	0.886	0.946	0.965	0.982
2003	0.023	0.288	0.65	0.877	0.948	0.966	0.983
2004	0.019	0.388	0.727	0.897	0.954	0.97	0.985
2005	0.023	0.38	0.731	0.88	0.951	0.971	0.986
2006	0.033	0.529	0.844	0.934	0.97	0.981	0.989
2007	0.019	0.241	0.703	0.885	0.943	0.965	0.984
2008	0.023	0.466	0.688	0.914	0.959	0.973	0.986
2009	0.017	0.272	0.709	0.885	0.937	0.964	0.983
2010	0.014	0.381	0.81	0.904	0.958	0.972	0.986
2011	0.028	0.27	0.778	0.933	0.967	0.978	0.988
2012	0.029	0.531	0.723	0.9	0.968	0.978	0.989
2013	0.036	0.631	0.792	0.966	0.975	0.984	0.991
2014	0.028	0.199	0.612	0.901	0.939	0.966	0.984
2015	0.022	0.364	0.615	0.864	0.931	0.96	0.981
2016	0.012	0.299	0.468	0.813	0.931	0.954	0.972
2017	0.016	0.496	0.694	0.86	0.896	0.967	0.975
2018	0.014	0.242	0.689	0.789	0.941	0.946	0.98
2019	0.026	0.447	0.715	0.91	0.955	0.974	0.986
2020	0.034	0.252	0.816	0.918	0.962	0.975	0.987
2021	0.024	0.281	0.513	0.886	0.903	0.961	0.982
2022	0.04	0.551	0.785	0.934	0.969	0.975	0.988
2023	0.023	0.397	0.691	0.842	0.946	0.967	0.984

Year/Age	1	2	3	4	5	6	7+
2024	0.016	0.181	0.638	0.807	0.95	0.961	0.98
2025	0.018	0.381	0.716	0.815	0.936	0.962	0.982

Southern

Year/Age	1	2	3	4	5	6	7+
1983	0.001	0.022	0.219	0.467	0.692	0.63	0.973
1984	0.001	0.034	0.264	0.58	0.389	0.848	0.977
1985	0.001	0.026	0.318	0.695	0.764	0.772	0.976
1986	0.001	0.064	0.34	0.562	0.655	0.525	0.976
1987	0.001	0.014	0.264	0.566	0.788	0.647	0.975
1988	0.003	0.065	0.196	0.603	0.627	0.738	0.969
1989	0.032	0.156	0.374	0.584	0.903	0.747	0.989
1990	0.003	0.208	0.284	0.562	0.743	0.843	0.982
1991	0.003	0.156	0.627	0.751	0.918	0.934	0.992
1992	0.002	0.221	0.706	0.905	0.94	0.943	0.994
1993	0.002	0.082	0.445	0.809	0.9	0.901	0.992
1994	0.002	0.099	0.612	0.826	0.88	0.899	0.992
1995	0.002	0.094	0.458	0.833	0.894	0.902	0.991
1996	0.002	0.068	0.46	0.766	0.882	0.891	0.99
1997	0.002	0.137	0.579	0.692	0.86	0.861	0.99
1998	0.003	0.059	0.408	0.721	0.844	0.862	0.989
1999	0.005	0.324	0.648	0.838	0.946	0.946	0.993
2000	0.009	0.284	0.648	0.926	0.96	0.962	0.996
2001	0.006	0.243	0.638	0.909	0.949	0.948	0.996
2002	0.009	0.509	0.833	0.946	0.972	0.974	0.997
2003	0.007	0.342	0.673	0.916	0.96	0.961	0.997
2004	0.006	0.165	0.869	0.93	0.952	0.961	0.997
2005	0.014	0.539	0.825	0.932	0.974	0.975	0.997
2006	0.009	0.344	0.856	0.936	0.969	0.971	0.998
2007	0.01	0.276	0.844	0.943	0.968	0.969	0.998

Year/Age	1	2	3	4	5	6	7+
2008	0.037	0.561	0.928	0.967	0.983	0.985	0.998
2009	0.011	0.329	0.816	0.952	0.973	0.974	0.998
2010	0.008	0.236	0.786	0.939	0.956	0.963	0.998
2011	0.02	0.146	0.839	0.947	0.971	0.971	0.998
2012	0.018	0.557	0.881	0.963	0.981	0.982	0.998
2013	0.017	0.399	0.878	0.959	0.979	0.98	0.998
2014	0.012	0.549	0.91	0.965	0.982	0.982	0.998
2015	0.054	0.812	0.948	0.981	0.99	0.991	0.999
2016	0.02	0.505	0.904	0.966	0.983	0.984	0.999
2017	0.015	0.432	0.877	0.958	0.978	0.979	0.998
2018	0.013	0.487	0.849	0.954	0.976	0.978	0.998
2019	0.009	0.331	0.828	0.936	0.966	0.969	0.998
2020	0.009	0.275	0.745	0.916	0.956	0.959	0.998
2021	0.024	0.593	0.909	0.97	0.984	0.984	0.998
2022	0.027	0.578	0.911	0.969	0.985	0.986	0.998
2023	0.016	0.555	0.884	0.961	0.981	0.982	0.999
2024	0.019	0.557	0.891	0.966	0.983	0.984	0.999
2025	0.031	0.759	0.939	0.98	0.989	0.99	0.999

Viking

Year/Age	1	2	3	4	5	6	7+
1983	0.003	0.026	0.068	0.232	0.561	0.849	0.919
1984	0.004	0.032	0.201	0.437	0.698	0.913	0.95
1985	0.003	0.014	0.103	0.333	0.631	0.883	0.915
1986	0.003	0.024	0.046	0.204	0.549	0.846	0.911
1987	0.004	0.027	0.144	0.375	0.705	0.894	0.942
1988	0.005	0.023	0.079	0.328	0.349	0.813	0.902
1989	0.129	0.039	0.083	0.288	0.701	0.903	0.96
1990	0.005	0.032	0.13	0.296	0.537	0.872	0.914
1991	0.006	0.022	0.156	0.331	0.707	0.886	0.938

Year/Age	1	2	3	4	5	6	7+
1992	0.039	0.085	0.409	0.727	0.845	0.967	0.976
1993	0.005	0.032	0.126	0.437	0.599	0.892	0.93
1994	0.007	0.045	0.258	0.64	0.808	0.946	0.966
1995	0.006	0.032	0.234	0.615	0.734	0.875	0.951
1996	0.012	0.127	0.404	0.825	0.851	0.968	0.98
1997	0.012	0.121	0.441	0.647	0.829	0.965	0.98
1998	0.026	0.104	0.477	0.822	0.912	0.976	0.986
1999	0.017	0.271	0.329	0.751	0.926	0.976	0.986
2000	0.011	0.216	0.594	0.696	0.841	0.968	0.98
2001	0.019	0.125	0.352	0.649	0.893	0.968	0.982
2002	0.017	0.229	0.401	0.753	0.922	0.975	0.985
2003	0.009	0.058	0.293	0.543	0.851	0.951	0.975
2004	0.011	0.126	0.647	0.643	0.85	0.955	0.981
2005	0.012	0.091	0.515	0.821	0.91	0.97	0.981
2006	0.01	0.109	0.288	0.687	0.82	0.963	0.98
2007	0.036	0.185	0.382	0.718	0.93	0.972	0.986
2008	0.017	0.277	0.525	0.72	0.908	0.977	0.987
2009	0.015	0.089	0.415	0.752	0.883	0.965	0.983
2010	0.012	0.083	0.427	0.678	0.906	0.955	0.981
2011	0.013	0.077	0.468	0.715	0.883	0.967	0.981
2012	0.009	0.072	0.213	0.649	0.856	0.954	0.975
2013	0.008	0.069	0.262	0.691	0.817	0.96	0.975
2014	0.01	0.057	0.301	0.708	0.847	0.959	0.976
2015	0.009	0.117	0.284	0.698	0.832	0.956	0.977
2016	0.007	0.07	0.177	0.581	0.768	0.932	0.97
2017	0.008	0.062	0.392	0.695	0.796	0.957	0.973
2018	0.011	0.063	0.277	0.689	0.893	0.961	0.977
2019	0.015	0.209	0.426	0.756	0.906	0.974	0.983
2020	0.01	0.078	0.457	0.788	0.873	0.966	0.981

Year/Age	1	2	3	4	5	6	7+
2021	0.007	0.065	0.242	0.457	0.796	0.94	0.97
2022	0.007	0.045	0.177	0.531	0.833	0.937	0.966
2023	0.007	0.059	0.232	0.485	0.773	0.939	0.964
2024	0.007	0.054	0.233	0.462	0.72	0.926	0.963
2025	0.008	0.097	0.333	0.562	0.783	0.937	0.972

Table 4.13. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM final run estimated natural mortality-at-age for each substock.

Northwestern

Year/Age	1	2	3	4	5	6	7+
1983	1.081	0.753	0.273	0.207	0.2	0.2	0.2
1984	1.017	0.725	0.262	0.203	0.2	0.2	0.2
1985	1.047	0.723	0.265	0.203	0.2	0.2	0.2
1986	0.976	0.702	0.268	0.21	0.2	0.2	0.2
1987	0.991	0.704	0.269	0.214	0.2	0.2	0.2
1988	1.018	0.722	0.272	0.222	0.2	0.2	0.2
1989	0.977	0.748	0.283	0.216	0.2	0.2	0.2
1990	0.999	0.765	0.284	0.215	0.2	0.2	0.2
1991	0.99	0.763	0.282	0.204	0.2	0.2	0.2
1992	0.946	0.744	0.261	0.201	0.2	0.2	0.2
1993	0.981	0.735	0.256	0.2	0.2	0.2	0.2
1994	0.986	0.729	0.268	0.2	0.2	0.2	0.2
1995	0.995	0.738	0.275	0.2	0.2	0.2	0.2
1996	1.04	0.753	0.278	0.202	0.2	0.2	0.2
1997	1	0.738	0.273	0.21	0.2	0.2	0.2
1998	1.051	0.764	0.293	0.217	0.2	0.2	0.2
1999	1.024	0.802	0.307	0.225	0.2	0.2	0.2
2000	0.939	0.784	0.306	0.232	0.2	0.2	0.2
2001	0.962	0.787	0.332	0.23	0.2	0.2	0.2
2002	1.032	0.863	0.387	0.238	0.2	0.2	0.2
2003	1.113	0.955	0.44	0.26	0.2	0.2	0.2

Year/Age	1	2	3	4	5	6	7+
2004	1.196	1.021	0.475	0.29	0.2	0.2	0.2
2005	1.254	1.065	0.491	0.272	0.2	0.2	0.2
2006	1.26	1.067	0.473	0.254	0.2	0.2	0.2
2007	1.3	1.075	0.47	0.219	0.2	0.2	0.2
2008	1.326	1.103	0.479	0.239	0.2	0.2	0.2
2009	1.28	1.089	0.463	0.209	0.2	0.2	0.2
2010	1.22	1.019	0.439	0.202	0.2	0.2	0.2
2011	1.277	1.053	0.421	0.202	0.2	0.2	0.2
2012	1.328	1.097	0.501	0.202	0.2	0.2	0.2
2013	1.327	1.096	0.495	0.205	0.201	0.2	0.2
2014	1.273	1.056	0.48	0.207	0.202	0.2	0.2
2015	1.285	1.06	0.492	0.218	0.204	0.201	0.2
2016	1.287	1.055	0.467	0.26	0.202	0.2	0.2
2017	1.247	1.021	0.431	0.249	0.203	0.2	0.2
2018	1.308	1.061	0.462	0.258	0.201	0.2	0.2
2019	1.257	1.018	0.454	0.262	0.2	0.2	0.2
2020	1.19	0.956	0.446	0.245	0.2	0.2	0.2
2021	1.237	0.972	0.481	0.225	0.2	0.2	0.2
2022	1.315	1.03	0.522	0.26	0.201	0.2	0.2
2023	1.307	1.036	0.511	0.26	0.205	0.201	0.201
2024	1.308	1.03	0.5	0.259	0.208	0.202	0.201
2025	1.307	1.033	0.49	0.257	0.211	0.203	0.202

Southern

Year/Age	1	2	3	4	5	6	7+
1983	1.083	0.759	0.273	0.208	0.2	0.2	0.2
1984	1.018	0.724	0.263	0.203	0.2	0.2	0.2
1985	1.048	0.725	0.265	0.203	0.2	0.2	0.2
1986	0.976	0.704	0.268	0.21	0.2	0.2	0.2
1987	0.994	0.704	0.269	0.214	0.2	0.2	0.2

Year/Age	1	2	3	4	5	6	7+
1988	1.019	0.724	0.273	0.222	0.2	0.2	0.2
1989	0.977	0.748	0.283	0.217	0.2	0.2	0.2
1990	0.998	0.763	0.284	0.215	0.2	0.2	0.2
1991	0.992	0.762	0.282	0.204	0.2	0.2	0.2
1992	0.947	0.745	0.261	0.201	0.2	0.2	0.2
1993	0.981	0.733	0.256	0.2	0.2	0.2	0.2
1994	0.986	0.729	0.268	0.2	0.2	0.2	0.2
1995	0.996	0.736	0.275	0.2	0.2	0.2	0.2
1996	1.04	0.751	0.278	0.202	0.2	0.2	0.2
1997	0.998	0.736	0.273	0.21	0.2	0.2	0.2
1998	1.049	0.762	0.293	0.217	0.2	0.2	0.2
1999	1.023	0.802	0.306	0.225	0.2	0.2	0.2
2000	0.939	0.783	0.306	0.232	0.2	0.2	0.2
2001	0.961	0.786	0.332	0.23	0.2	0.2	0.2
2002	1.032	0.86	0.386	0.238	0.2	0.2	0.2
2003	1.113	0.951	0.44	0.26	0.2	0.2	0.2
2004	1.195	1.02	0.475	0.29	0.2	0.2	0.2
2005	1.251	1.064	0.492	0.272	0.2	0.2	0.2
2006	1.261	1.068	0.473	0.254	0.2	0.2	0.2
2007	1.299	1.076	0.47	0.219	0.2	0.2	0.2
2008	1.327	1.105	0.478	0.239	0.2	0.2	0.2
2009	1.283	1.092	0.463	0.209	0.2	0.2	0.2
2010	1.221	1.021	0.439	0.203	0.2	0.2	0.2
2011	1.278	1.055	0.421	0.202	0.2	0.2	0.2
2012	1.327	1.1	0.501	0.202	0.2	0.2	0.2
2013	1.327	1.098	0.497	0.204	0.201	0.2	0.2
2014	1.273	1.061	0.481	0.207	0.202	0.2	0.2
2015	1.284	1.065	0.494	0.218	0.204	0.201	0.2
2016	1.284	1.056	0.469	0.26	0.202	0.2	0.2

Year/Age	1	2	3	4	5	6	7+
2017	1.244	1.018	0.432	0.249	0.203	0.2	0.2
2018	1.306	1.059	0.463	0.258	0.201	0.2	0.2
2019	1.257	1.019	0.455	0.262	0.2	0.2	0.2
2020	1.19	0.957	0.447	0.245	0.2	0.2	0.2
2021	1.238	0.975	0.482	0.225	0.2	0.2	0.2
2022	1.316	1.033	0.524	0.261	0.201	0.2	0.2
2023	1.311	1.041	0.516	0.26	0.205	0.201	0.201
2024	1.308	1.041	0.505	0.259	0.209	0.202	0.201
2025	1.308	1.043	0.495	0.257	0.211	0.203	0.202

Viking

Year/Age	1	2	3	4	5	6	7+
1983	1.082	0.758	0.274	0.208	0.2	0.2	0.2
1984	1.016	0.723	0.263	0.203	0.2	0.2	0.2
1985	1.047	0.722	0.265	0.203	0.2	0.2	0.2
1986	0.975	0.702	0.268	0.21	0.2	0.2	0.2
1987	0.991	0.703	0.269	0.214	0.2	0.2	0.2
1988	1.02	0.721	0.273	0.222	0.2	0.2	0.2
1989	0.978	0.747	0.283	0.217	0.2	0.2	0.2
1990	0.999	0.763	0.284	0.216	0.2	0.2	0.2
1991	0.992	0.763	0.282	0.204	0.2	0.2	0.2
1992	0.947	0.745	0.261	0.201	0.2	0.2	0.2
1993	0.981	0.734	0.256	0.2	0.2	0.2	0.2
1994	0.987	0.73	0.268	0.2	0.2	0.2	0.2
1995	0.997	0.739	0.275	0.2	0.2	0.2	0.2
1996	1.04	0.754	0.278	0.202	0.2	0.2	0.2
1997	0.999	0.738	0.273	0.21	0.2	0.2	0.2
1998	1.053	0.763	0.293	0.217	0.2	0.2	0.2
1999	1.025	0.804	0.306	0.224	0.2	0.2	0.2
2000	0.939	0.785	0.306	0.232	0.2	0.2	0.2

Year/Age	1	2	3	4	5	6	7+
2001	0.963	0.786	0.332	0.23	0.2	0.2	0.2
2002	1.033	0.861	0.387	0.238	0.2	0.2	0.2
2003	1.114	0.952	0.44	0.26	0.2	0.2	0.2
2004	1.196	1.02	0.476	0.29	0.2	0.2	0.2
2005	1.254	1.067	0.492	0.272	0.2	0.2	0.2
2006	1.262	1.069	0.475	0.254	0.2	0.2	0.2
2007	1.302	1.076	0.47	0.219	0.2	0.2	0.2
2008	1.328	1.104	0.478	0.238	0.2	0.2	0.2
2009	1.283	1.091	0.463	0.209	0.2	0.2	0.2
2010	1.221	1.02	0.438	0.203	0.2	0.2	0.2
2011	1.28	1.054	0.42	0.202	0.2	0.2	0.2
2012	1.33	1.097	0.501	0.202	0.2	0.2	0.2
2013	1.323	1.099	0.496	0.205	0.201	0.2	0.2
2014	1.269	1.056	0.481	0.207	0.202	0.2	0.2
2015	1.282	1.057	0.492	0.218	0.204	0.201	0.2
2016	1.281	1.049	0.468	0.26	0.202	0.2	0.2
2017	1.239	1.013	0.431	0.249	0.203	0.2	0.2
2018	1.303	1.054	0.462	0.258	0.201	0.2	0.2
2019	1.253	1.015	0.455	0.262	0.2	0.2	0.2
2020	1.186	0.952	0.446	0.245	0.2	0.2	0.2
2021	1.234	0.968	0.481	0.225	0.2	0.2	0.2
2022	1.315	1.02	0.523	0.26	0.201	0.2	0.2
2023	1.31	1.011	0.509	0.259	0.204	0.201	0.201
2024	1.313	1.005	0.5	0.258	0.208	0.202	0.201
2025	1.31	1.012	0.489	0.256	0.211	0.203	0.202

Table 4.14. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multi-stock SAM final run estimated stock and management metrics for each sub-stock, together with the lower and upper bounds of the pointwise 95% confidence intervals. Estimated recruitment, total stock biomass (TSB), spawning stock biomass (SSB), catches and average fishing mortality for ages 2 to 4 ($F_{bar\ 2-4}$).

Northwestern

Year	R(age 1) (‘000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
1983	485860	323206	730369	213911	155684	293914	56820	40744	79240	124015	85932	178975	0.813	0.714	0.927
1984	705249	419273	1186282	208614	157590	276159	52638	37888	73129	146872	102891	209654	0.941	0.787	1.124
1985	217360	134176	352117	191879	149309	246586	51346	38031	69324	134546	100901	179409	0.942	0.798	1.112
1986	617300	379906	1003035	167019	131144	212707	35453	26794	46911	108884	79800	148567	0.979	0.832	1.152
1987	368304	233169	581758	175505	137341	224274	47717	35989	63267	120948	88989	164385	0.934	0.793	1.101
1988	141601	89905	223022	150370	117361	192663	36909	28212	48287	91109	69330	119729	0.878	0.743	1.039
1989	501989	319239	789355	140596	110644	178655	60205	47011	77104	91924	67541	125110	0.95	0.806	1.12
1990	154300	98139	242601	132463	103216	169998	43660	33281	57276	87320	65793	115892	1.021	0.865	1.204
1991	221082	141542	345321	104233	82216	132145	42593	32470	55871	66249	50880	86261	0.995	0.848	1.167
1992	530724	350625	803332	116442	91393	148357	55451	42951	71590	81465	58306	113824	0.93	0.794	1.088
1993	247560	167213	366515	131756	102773	168913	53664	41570	69276	94026	70389	125600	0.933	0.791	1.1
1994	528984	332144	842480	122581	96446	155797	65462	50598	84693	88511	66163	118408	0.938	0.8	1.099
1995	317602	207739	485567	129433	101568	164944	49869	38150	65187	111288	82909	149381	1.047	0.896	1.224
1996	158151	103184	242400	116918	91952	148663	56598	44051	72719	85989	66244	111619	1.085	0.927	1.27
1997	597300	388674	917907	93404	73646	118462	55682	42728	72564	75848	54494	105571	1.006	0.859	1.179

Year	R(age 1) ('000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
1998	68584	45089	104321	94456	73624	121181	41750	32326	53920	67481	50630	89939	1.09	0.94	1.263
1999	96646	61569	151706	56575	44870	71333	33372	26137	42608	39227	30910	49781	1.112	0.96	1.288
2000	200838	133015	303242	54978	43235	69911	28028	21925	35831	33858	25591	44796	1.067	0.919	1.239
2001	65633	42515	101321	52219	40746	66923	22924	17979	29229	28476	21910	37009	0.928	0.797	1.08
2002	96320	61303	151340	45819	36024	58279	25377	19625	32814	23277	18124	29895	0.899	0.767	1.054
2003	56697	36897	87123	37256	29249	47454	20401	15862	26238	18364	14205	23739	0.832	0.704	0.983
2004	86786	55340	136102	31422	24512	40279	18797	14483	24395	13026	10053	16878	0.808	0.683	0.956
2005	71474	44958	113631	27967	21504	36374	15020	11507	19604	12544	9471	16614	0.725	0.611	0.86
2006	270473	182421	401026	40573	30489	53993	17350	13205	22796	11712	8517	16107	0.662	0.549	0.798
2007	87319	55300	137879	53507	40806	70163	21303	16399	27673	22476	16597	30438	0.622	0.517	0.748
2008	97619	63227	150718	54377	42185	70091	33076	25352	43153	23267	17726	30540	0.664	0.553	0.797
2009	171864	115944	254753	51514	40317	65820	26261	20073	34358	21786	16778	28289	0.609	0.51	0.727
2010	212578	147094	307214	70847	55731	90063	33733	26341	43200	23983	18638	30862	0.558	0.466	0.668
2011	95756	64607	141924	79553	62885	100641	41260	32338	52643	26881	20976	34447	0.489	0.406	0.59
2012	147506	97531	223090	86521	68899	108650	53279	42123	67389	28630	22414	36570	0.486	0.403	0.586
2013	202760	137956	298008	95221	75764	119675	63564	50269	80375	28694	22575	36473	0.463	0.384	0.557
2014	249065	170981	362811	100448	79850	126360	48910	38470	62182	29288	22941	37392	0.418	0.345	0.505
2015	154181	107580	220967	115800	92419	145095	64353	51074	81085	34524	27114	43957	0.421	0.346	0.511

Year	R(age 1) (’000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
2016	100262	69397	144854	118314	94370	148333	67447	53064	85727	41563	32711	52811	0.47	0.39	0.567
2017	271091	189262	388300	111208	88915	139091	73516	57669	93718	43004	33779	54749	0.57	0.474	0.684
2018	64727	44526	94093	81842	65628	102061	49653	39100	63053	36324	28553	46211	0.649	0.535	0.787
2019	138905	97321	198257	57916	46511	72117	38688	30493	49086	25063	19925	31527	0.563	0.471	0.674
2020	176537	120094	259509	63494	50972	79092	33617	26742	42260	17752	14042	22442	0.446	0.37	0.538
2021	128521	89414	184732	75446	60732	93725	36764	29399	45975	20901	16240	26900	0.364	0.296	0.449
2022	152396	102362	226885	87241	70090	108590	58054	46326	72751	21992	17121	28248	0.354	0.287	0.436
2023	106626	70773	160641	88504	71286	109879	56714	45266	71058	26856	20949	34430	0.392	0.315	0.489
2024	39135	24292	63050	67800	54138	84909	44308	34769	56465	24957	19228	32391	0.419	0.317	0.554
2025	64887	20866	201786	56027	41815	75068	40171	29899	53973						

Southern

Year	R(age 1) (’000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
1983	318785	198367	512301	115229	80388	165170	17379	10871	27782	75591	48489	117838	1.139	0.935	1.387
1984	686071	385711	1220325	134977	93892	194040	17100	10981	26626	104037	64797	167038	1.336	1.073	1.664
1985	135121	77294	236211	128524	87495	188793	20485	13293	31569	98343	64452	150054	1.334	1.08	1.648
1986	831514	490710	1409011	109762	76593	157293	19619	12923	29785	89346	54103	147547	1.371	1.108	1.697
1987	325646	196824	538782	152681	104917	222189	15083	9310	24436	123303	80671	188465	1.291	1.046	1.593

Year	R(age 1) ('000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
1988	243076	151562	389845	129289	92605	180505	20382	13606	30533	87058	61471	123295	1.203	0.973	1.487
1989	300883	174361	519214	113810	83174	155729	32640	22701	46931	77472	54490	110147	1.296	1.055	1.592
1990	133075	78186	226498	97359	69066	137243	28992	19872	42298	61194	42588	87927	1.403	1.134	1.736
1991	137714	79070	239855	62343	44493	87355	26916	18046	40146	43988	30730	62966	1.351	1.096	1.666
1992	245280	140134	429321	52246	36591	74601	22113	14518	33683	45904	29481	71478	1.233	0.999	1.522
1993	169948	107473	268742	63406	43900	91579	16533	10767	25389	47689	31793	71532	1.202	0.972	1.487
1994	420627	252824	699800	63551	45854	88078	20732	13807	31130	53790	35966	80448	1.184	0.959	1.463
1995	273149	168494	442806	76062	54433	106285	19083	12943	28135	78824	54411	114191	1.309	1.073	1.596
1996	220064	143767	336851	71989	51893	99866	20089	13677	29507	70755	51035	98094	1.358	1.118	1.651
1997	585945	374339	917169	65575	47816	89930	21568	14456	32180	72792	49705	106602	1.251	1.034	1.513
1998	57442	35908	91890	87433	61482	124337	18226	12238	27143	72132	50707	102611	1.394	1.16	1.674
1999	122522	75014	200118	63847	46928	86865	32635	22765	46784	38604	28814	51721	1.463	1.217	1.759
2000	115164	69386	191144	42848	31228	58794	20394	14148	29397	30806	22185	42778	1.44	1.187	1.748
2001	69625	44530	108862	34348	24875	47428	13126	9061	19015	21633	15572	30053	1.26	1.027	1.545
2002	143043	92547	221090	38389	28458	51785	19252	13625	27202	18539	13527	25409	1.235	1.009	1.511
2003	50920	31968	81105	38675	28534	52421	17546	12754	24138	19024	14048	25763	1.143	0.925	1.412
2004	89465	54218	147626	27603	20297	37539	16696	11786	23653	13578	10166	18135	1.132	0.903	1.42
2005	92569	58136	147394	26611	18993	37284	14843	10310	21368	12988	9193	18349	1.01	0.797	1.28

Year	R(age 1) (‘000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
2006	112309	67898	185770	34971	24792	49330	16237	11187	23567	9113	6337	13106	0.925	0.724	1.182
2007	131576	84997	203679	38418	27962	52783	16462	11521	23524	16215	11358	23149	0.879	0.693	1.116
2008	121415	78070	188828	55384	41038	74745	30823	22413	42388	20133	14677	27618	0.971	0.761	1.238
2009	48710	28288	83872	48484	35730	65791	26005	18954	35677	19590	14535	26405	0.919	0.722	1.169
2010	77389	47823	125235	36660	27005	49767	21242	14987	30109	15009	11109	20278	0.873	0.674	1.13
2011	53391	33053	86243	31709	23155	43424	15254	10476	22211	12450	9128	16981	0.778	0.592	1.022
2012	55460	32653	94195	32973	24090	45132	22492	16114	31394	11465	8472	15516	0.783	0.592	1.036
2013	53190	31533	89722	26566	19205	36749	16875	11825	24082	10316	7528	14136	0.754	0.562	1.011
2014	64955	38782	108792	27658	19940	38362	17825	12402	25619	9262	6612	12975	0.681	0.509	0.911
2015	21528	12449	37227	28718	20551	40132	23656	16786	33338	9465	6747	13279	0.689	0.511	0.929
2016	14824	8213	26757	20938	15077	29079	16581	11668	23563	8468	6139	11682	0.765	0.564	1.039
2017	30841	17716	53688	12064	8500	17122	8472	5660	12682	6039	4339	8406	0.905	0.67	1.224
2018	12145	6922	21311	8952	6148	13035	5701	3756	8653	4390	3047	6326	1.011	0.744	1.374
2019	16251	9427	28014	6117	4218	8870	3627	2355	5585	2925	2025	4224	0.847	0.627	1.146
2020	22633	12881	39771	6313	4366	9128	2782	1829	4231	2077	1432	3013	0.655	0.48	0.893
2021	20384	11936	34814	8706	5999	12635	5335	3619	7864	2716	1845	4000	0.528	0.379	0.735
2022	38976	22334	68018	11722	8179	16800	6985	4787	10193	2897	1984	4231	0.504	0.358	0.709
2023	28374	15350	52446	13936	9591	20250	8504	5826	12412	3923	2700	5700	0.561	0.392	0.802

Year	R(age 1) (⁰ 000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
2024	15952	7366	34548	12796	8684	18854	8887	5956	13260	4218	2869	6201	0.593	0.393	0.894
2025	22210	5753	85744	11951	7277	19629	9114	5729	14499						

Viking

Year	R(age 1) (⁰ 000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
1983	100334	57856	174001	58028	37316	90235	14690	9490	22741	30898	19362	49308	0.63	0.524	0.757
1984	112122	60960	206222	54243	35669	82488	18971	12428	28960	36765	23216	58220	0.714	0.58	0.88
1985	32054	17047	60274	49313	32800	74140	14721	9754	22217	36273	23492	56006	0.709	0.58	0.867
1986	109079	58188	204478	41977	28313	62236	11164	7484	16654	28887	19111	43664	0.739	0.603	0.907
1987	55326	30377	100765	40481	26905	60907	13656	9253	20154	26563	17037	41416	0.695	0.57	0.847
1988	50358	27322	92814	35087	23574	52221	9223	6229	13656	24028	15742	36677	0.646	0.53	0.789
1989	77062	42145	140908	35905	24094	53506	12380	8339	18378	23741	15576	36187	0.701	0.573	0.857
1990	40102	21731	74004	35084	23389	52628	10620	7123	15835	22483	14724	34332	0.748	0.614	0.913
1991	43440	23573	80048	30818	20669	45952	10493	6993	15744	19618	12966	29684	0.732	0.602	0.89
1992	82984	44940	153236	31505	21054	47143	15279	10275	22720	18363	11988	28128	0.683	0.562	0.829
1993	55726	31488	98620	31691	21033	47749	9014	5997	13548	20307	13033	31643	0.686	0.561	0.838
1994	103407	57995	184381	33207	22485	49043	13023	8741	19404	20830	13854	31318	0.694	0.571	0.844
1995	74721	41129	135747	36737	24897	54207	12583	8487	18657	26151	17613	38827	0.773	0.641	0.932

Year	R(age 1) (‘000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
1996	49845	28139	88292	32973	22485	48354	16828	11473	24683	24717	16850	36256	0.788	0.657	0.947
1997	120273	68789	210291	30271	20726	44212	14487	9906	21186	19181	13048	28197	0.716	0.599	0.855
1998	16962	9691	29688	31157	21234	45719	15703	10815	22802	19281	13038	28512	0.761	0.641	0.903
1999	41764	23425	74462	25612	17695	37072	13746	9535	19818	17185	11909	24798	0.785	0.659	0.936
2000	60216	34795	104211	22664	15671	32779	12599	8726	18193	13537	9306	19691	0.755	0.634	0.901
2001	26664	14996	47411	21467	14757	31228	9002	6281	12902	11439	7859	16651	0.655	0.546	0.786
2002	38179	21071	69178	19635	13633	28280	10187	7126	14563	10530	7343	15102	0.623	0.514	0.754
2003	24756	13973	43860	19304	13406	27797	8376	5871	11951	9744	6788	13986	0.593	0.488	0.721
2004	51790	28869	92912	18972	13093	27492	10242	7158	14656	7574	5255	10915	0.581	0.477	0.707
2005	35749	20502	62335	19372	13262	28296	9467	6607	13566	7433	5067	10903	0.532	0.434	0.651
2006	71289	40763	124674	19886	13601	29076	7671	5349	11000	6192	4235	9051	0.486	0.39	0.606
2007	31849	17978	56423	21095	14280	31161	9340	6475	13471	8100	5532	11861	0.452	0.365	0.56
2008	38827	22005	68508	21618	14827	31520	11992	8287	17354	9139	6315	13227	0.485	0.393	0.598
2009	45917	25924	81330	21303	14505	31287	10914	7426	16042	8958	6171	13003	0.455	0.369	0.559
2010	68327	39554	118030	22542	15262	33295	9987	6799	14670	8504	5828	12406	0.415	0.337	0.512
2011	35294	20248	61521	25410	17147	37657	11419	7735	16859	9396	6423	13746	0.369	0.299	0.456
2012	59499	33738	104927	28450	19303	41932	11657	7883	17239	10761	7378	15695	0.368	0.297	0.455
2013	62775	36340	108438	29198	19709	43257	13601	9114	20298	10488	7104	15484	0.35	0.282	0.435

Year	R(age 1) (‘000)	Low	High	TSB (tonnes)	Low	High	SSB (tonnes)	Low	High	Catches (tonnes)	Low	High	Fbar (2-4)	Low	High
2014	77546	45020	133570	32196	21738	47686	14540	9733	21720	9584	6476	14183	0.311	0.249	0.388
2015	39769	23050	68614	38191	25757	56628	16608	11225	24571	11414	7745	16820	0.314	0.251	0.392
2016	24330	13947	42441	41046	27824	60550	17558	11867	25977	15623	10572	23086	0.361	0.292	0.447
2017	54148	31145	94138	38901	26585	56924	23510	15995	34557	17457	11812	25800	0.459	0.368	0.573
2018	18086	10328	31672	30031	20561	43863	17856	12178	26183	13854	9508	20185	0.534	0.423	0.673
2019	33049	18631	58625	20887	14284	30543	13009	8819	19191	9278	6413	13425	0.462	0.371	0.577
2020	70615	39973	124748	20799	14056	30776	10015	6786	14779	5848	4014	8518	0.357	0.287	0.445
2021	34006	19561	59120	23982	15979	35993	7717	5199	11454	6726	4481	10095	0.286	0.226	0.362
2022	48755	26945	88218	28244	19051	41873	10126	6814	15049	8173	5434	12292	0.279	0.219	0.355
2023	32673	17484	61058	28672	19419	42333	12596	8509	18645	9673	6421	14572	0.307	0.238	0.395
2024	7800	3444	17668	25839	17553	38037	13395	9047	19832	10520	6878	16091	0.339	0.248	0.464
2025	14658	4351	49378	19861	13184	29918	12750	8435	19271	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

Table 4.15. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Catch scenarios for each substock based on the multistock SAM assessment and assuming $F_{\text{status quo}}$ in the intermediate year. Units are tonnes (SSB, landings, discards and catch) or thousands (recruitment).

Northwestern

Forecast assumptions

variable	value
Fbar(2025)	0.421
SSB(2026)	35837
R(2025)	67359
R(2026)	64705
Catch(2025)	21692
Landings(2025)	17961
Discards(2025)	3731

Catch scenarios

Basis	Catch (2026)	F_{total} (2026)	SSB (2027)	% SSB change	% advice change	Probability of SSB (2027) < B_{lim} (%)
$F_{2026}=F_{2025} \times 0.074^*$	1737	0.031	48221	35	-82	1.00
F_{MSY}	9712	0.196	39170	9.3	-2.1	9.3
$F_{\text{MSY lower}}$	6315	0.121	42997	20.0	-36	3.4
$F = 0$	0	0	50332	40	-100	0.60
F_{pa}	21524	0.543	26058	-27	117	68
$\text{SSB}(2027) = B_{\text{lim}}$	18393	0.43	29378	-18.0	85	50
$\text{SSB}(2027) = B_{\text{trigger}} = B_{\text{pa}}$	8242	0.163	40823	13.9	-16.9	6.1
$F=F_{2025}$	17967	0.421	29901	-16.6	81	47
$\text{SSB}(2027) = \text{SSB}(2026)$	12657	0.269	35837	0.00	28	17.3
MSY approach: $F_{\text{MSY}} \times \text{SSB}(2026) / \text{MSY } B_{\text{trigger}}$	8670	0.172	40379	12.7	-12.6	6.8
$F_{\text{MSY lower}} \times \text{SSB}(2026) / \text{MSY } B_{\text{trigger}}$	5603	0.106	43816	22	-44	2.8
Without precautionary considerations: $F_{2025} \times 0.36^*$	7859	0.154	41261	15.1	-21	5.6

* F multiplier that equates approximately to the cumulative substock advice when following MSY approach for each substock independently and equates to a total catch of 12 675 t. $F_{(2026, \text{Northwestern})} = F_{2025} \times 0.074$ (as $F_{\text{Southern}(2026)} / F_{\text{Southern}(2025)} = 0.074$).

Southern*Forecast assumptions*

variable	value
Fbar(2025)	0.591
SSB(2026)	9259
R(2025)	21822
R(2026)	22280
Catch(2025)	4172
Landings(2025)	2954
Discards(2025)	1218

Catch scenarios

Basis	Catch (2026)	F _{total} (2026)	SSB (2027)	% SSB change	% advice change	Probability of SSB (2027) < B _{lim} (%)
MSY approach: SSB(2027) = B _{lim}	399	0.044	14002	51	-82	50
Without precautionary considerations: F = F ₂₀₂₅ x 0.36***	1824	0.22	11935	29	-18.9	69
F _{MSY} x SSB(2027) / B _{trigger}	933	0.108	13219	43	-58	56
F _{MSY lower} x SSB(2026) / B _{trigger}	595	0.067	13706	48	-74	53
F = 0	0	0	14605	58	-100	44
F _{pa}	3312	0.464	9839	6.3	47	85
SSB(2027) = B _{trigger} = B _{pa} [^]						
F = F ₂₀₂₅	3958	0.591	8905	-3.8	76	89
SSB(2027) = SSB(2026)	3717	0.542	9259	0.00	65	87
F _{MSY}	1866	0.231	11878	28	-17.0	69
F _{MSY lower}	1221	0.144	12769	38	-46	61

[^] B_{pa} and MSY B_{trigger} cannot be achieved in 2027, even with zero catches.

*** F multiplier that equates approximately to the cumulative substock advice when following MSY approach for each substock independently and equates to a total catch of 12 675 t.

Viking*Forecast assumptions*

variable	value
Fbar(2025)	0.342
SSB(2026)	11096
R(2025)	14788
R(2026)	14588
Catch(2025)	8853
Landings(2025)	7838
Discards(2025)	1015

Catch scenarios

Basis	Catch (2026)	F _{total} (2026)	SSB (2027)	% SSB change	% advice change	Probability of SSB (2027) < B _{lim} (%)
$F_{2026} = F_{2025} \times 0.074^*$	676	0.025	14478	30	-80	6.2
F _{MSY}	4322	0.187	10825	-2.4	29	32
F _{MSY lower}	2541	0.102	12596	13.5	-24	15.4
F = 0	0	0	15171	37	-100	-3.9
F _{pa}	6402	0.307	8812	--21	92	62
SSB(2027) = B _{lim}	5522	0.25	9619	-13.3	65	50
SSB(2027) = B _{trigger} = B _{pa}	1406	0.054	13732	24	-58	8.6
F = F ₂₀₂₅	6898	0.340	8327	-25	106	68
SSB(2027) = SSB(2026)	4045	0.173	11096	0	21	30
MSY approach: F _{MSY} x SSB (2026)/MSY B _{trigger}	3610	0.151	11571	4.3	8.0	25
F _{MSY lower} x SSB (2026)/MSY B _{trigger}	2090	0.082	13047	17.6	-37	12.9

* $F_{(2026, \text{Northwestern})} = F_{2025} \times 0.074$ (as $F_{\text{Southern (2026)}} / F_{\text{Southern (2025)}} = 0.074$)

Table 4.16a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Ratio of numbers-at-age from stock assessment and forecast results from WGNSSK 2024 and WGNSSK 2025.

Northwestern

Age	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
1	0.98	0.97	1	0.99	0.95	0.96	0.96	1.06	0.69	0.66
2	0.99	0.97	0.98	1	0.98	0.95	0.94	0.93	1.08	0.72
3	0.97	0.99	0.97	0.98	0.99	0.98	0.96	0.9	0.84	1.17
4	0.98	0.96	0.98	1	1.02	1.01	0.99	0.93	0.76	0.75
5	1	1	0.98	0.99	1.02	1.02	1.01	0.94	0.79	0.66
6	1.02	1	1	1.01	1.02	1.04	1.02	1	0.81	0.65
7+	1.02	1.02	1.01	1	1.01	1.01	0.99	0.97	0.9	0.81

Southern

Age	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
1	1.01	1.07	1.03	1.04	0.98	0.91	0.8	0.91	0.96	0.41
2	1.02	1.02	1.08	1.04	1.04	0.99	0.9	0.79	0.94	0.97
3	1.03	1.02	1.02	1.08	1.05	1.05	0.99	0.86	0.71	0.92
4	0.97	1	0.98	0.98	1.03	1	1	0.89	0.69	0.55
5	0.95	0.94	0.96	0.95	0.94	0.98	0.95	0.91	0.73	0.48
6	0.92	0.93	0.91	0.94	0.92	0.91	0.93	0.87	0.75	0.54
7+	0.91	0.9	0.91	0.91	0.92	0.91	0.89	0.88	0.82	0.64

Viking

Age	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
1	0.96	0.97	1	0.99	0.99	0.99	1.01	1.45	1.04	0.38
2	0.96	0.95	0.97	0.99	0.99	0.99	0.99	1	1.56	1.02
3	0.98	0.97	0.96	0.98	1	1	0.99	1	1	1.76
4	1	0.99	0.98	0.98	1	1.01	1	0.98	0.95	0.8
5	1	0.99	1.01	1.01	1.02	1.01	1.01	0.99	0.97	0.83
6	1.02	0.99	0.99	1.01	1.04	1.04	1	0.98	0.94	0.78
7+	1	0.99	0.99	1	1.02	1.03	1.02	0.99	0.93	0.86

Table 4.16b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Ratio of biomass-at-age from stock assessment and forecast results from WGNSSK 2024 and WGNSSK 2025.**Northwestern**

Age	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
1	0.98	0.98	1	0.99	0.96	0.96	0.96	1.06	0.69	0.75
2	0.99	0.98	0.98	1	0.98	0.95	0.94	0.93	1.04	0.67
3	0.97	0.99	0.97	0.98	0.99	0.98	0.96	0.9	0.82	0.94
4	0.98	0.96	0.98	1	1.02	1.01	0.99	0.93	0.75	0.7
5	1	1	0.98	0.99	1.01	1.02	1.01	0.94	0.79	0.62
6	1.02	1	1	1.01	1.02	1.04	1.02	0.99	0.81	0.65
7+	1.02	1.01	1	0.99	1	1	0.98	0.95	0.87	0.75

Southern

Age	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
1	1.02	1.07	1.03	1.04	0.99	0.91	0.81	0.91	0.96	0.44
2	1.02	1.02	1.08	1.04	1.05	0.99	0.91	0.79	0.93	0.99
3	1.03	1.02	1.02	1.08	1.05	1.05	0.99	0.86	0.69	0.83
4	0.97	1	0.98	0.98	1.02	1	0.99	0.88	0.68	0.51
5	0.95	0.94	0.96	0.94	0.94	0.98	0.95	0.9	0.7	0.47
6	0.91	0.92	0.9	0.93	0.91	0.9	0.92	0.86	0.76	0.44
7+	0.91	0.9	0.9	0.9	0.92	0.91	0.88	0.87	0.8	0.63

Viking

Age	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
1	0.96	0.97	1	0.99	0.99	0.99	1.01	1.45	1.09	0.42
2	0.97	0.95	0.98	1	1	0.99	0.99	1	1.52	1.14
3	0.98	0.96	0.96	0.98	1	1	0.99	0.99	0.98	1.58
4	0.99	0.99	0.98	0.98	1	1.01	1	0.97	0.92	0.73
5	1	0.99	1.01	1.01	1.02	1.01	1	0.99	0.95	0.77
6	1.01	0.99	0.99	1.01	1.04	1.03	1	0.98	0.95	0.74
7+	1	0.99	0.99	1	1.02	1.02	1.02	0.98	0.92	0.85

Table 4.17. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Summary of forecast assumptions from the current (WGNSSK 2025) and previous (WGNSSK 2024) assessments.

Northwestern

	Year	Current assessment (2025)	Previous assessment (2024)
Assumed recruitment	2024	39135	56780
	2025	67359	102098
Catch	2024	24877	26500
SSB	2024	44308	55468
	2025	41270	55567
F	2024	0.417	0.322
Target F	2025	0.421	0.111

Southern

	Year	Current assessment (2025)	Previous assessment (2024)
Assumed recruitment	2024	15952	16556
	2025	21822	53397
Catch	2024	4196	4966
SSB	2024	8887	12131
	2025	9293	12844
F	2024	0.59	0.428
Target F	2025	0.591	0.152

Viking

	Year	Current assessment (2025)	Previous assessment (2024)
Assumed recruitment	2024	7800	7483
	2025	14788	39382
Catch	2024	10501	9060
SSB	2024	13395	14272
	2025	12968	15249
F	2024	0.338	0.256
Target F	2025	0.342	0.088

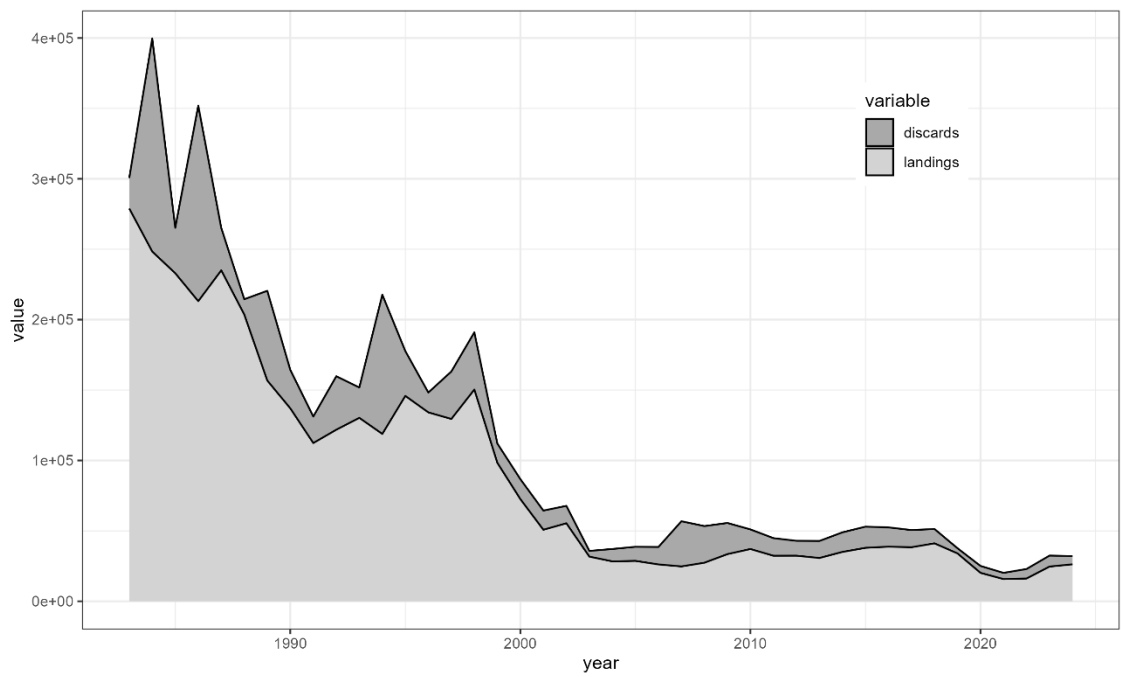


Figure 4.1a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Stacked area plot of reported landings and estimated discards (including BMS landings; in tonnes).

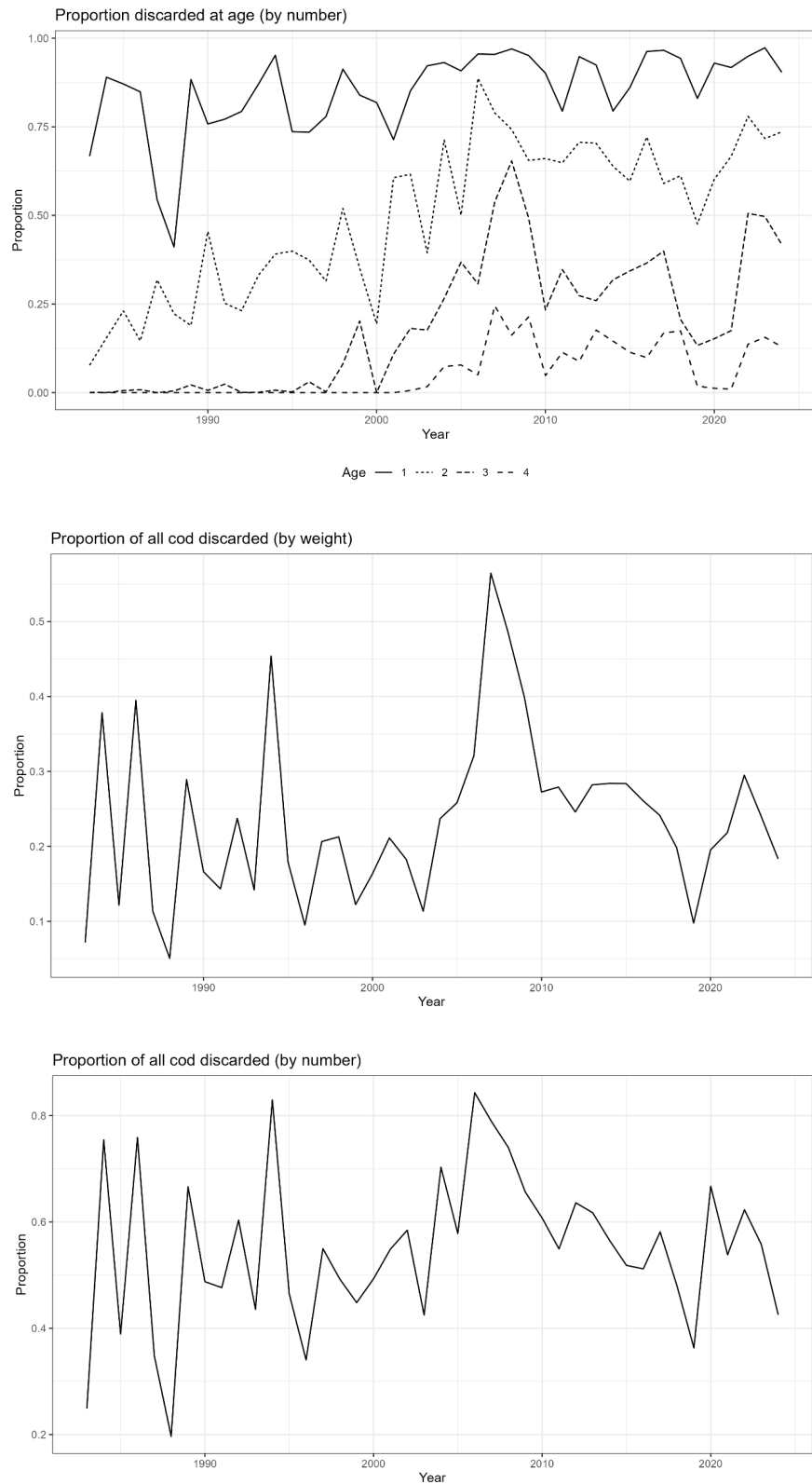


Figure 4.1b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. (Top) Proportion of total numbers caught at age that are discarded; (middle) proportion of total weight caught that is discarded; and (bottom) proportion of the total numbers caught that are discarded.

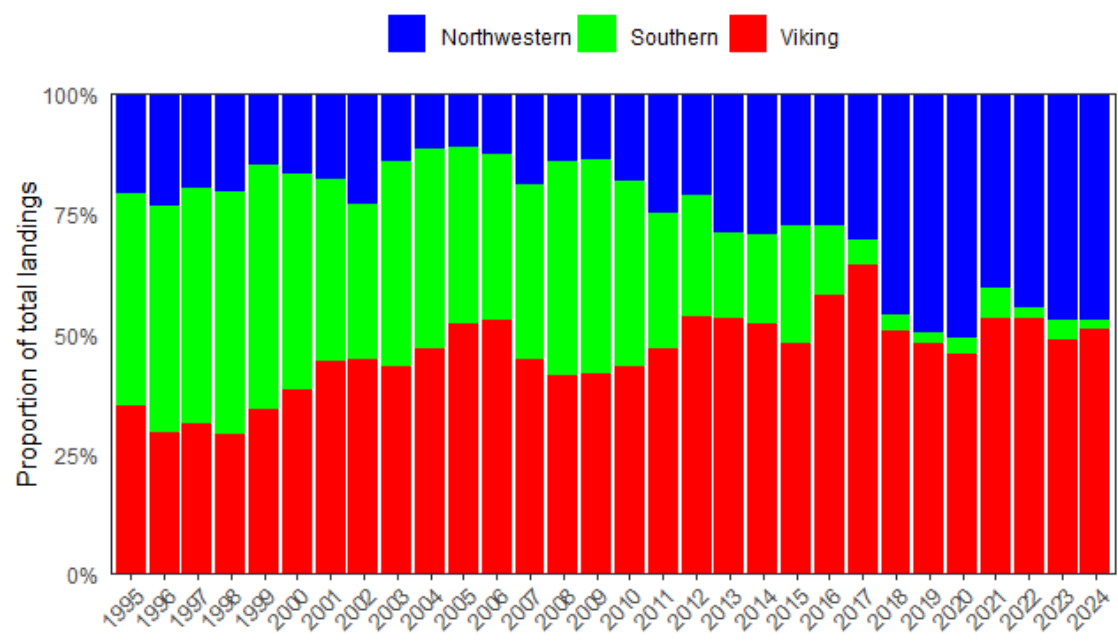


Figure 4.1c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Proportion of commercial landings by substock for the period 1995–2024. Commercial landings data from Q1 only are used to estimate substock compositions as it is assumed substocks do not mix in Q1.

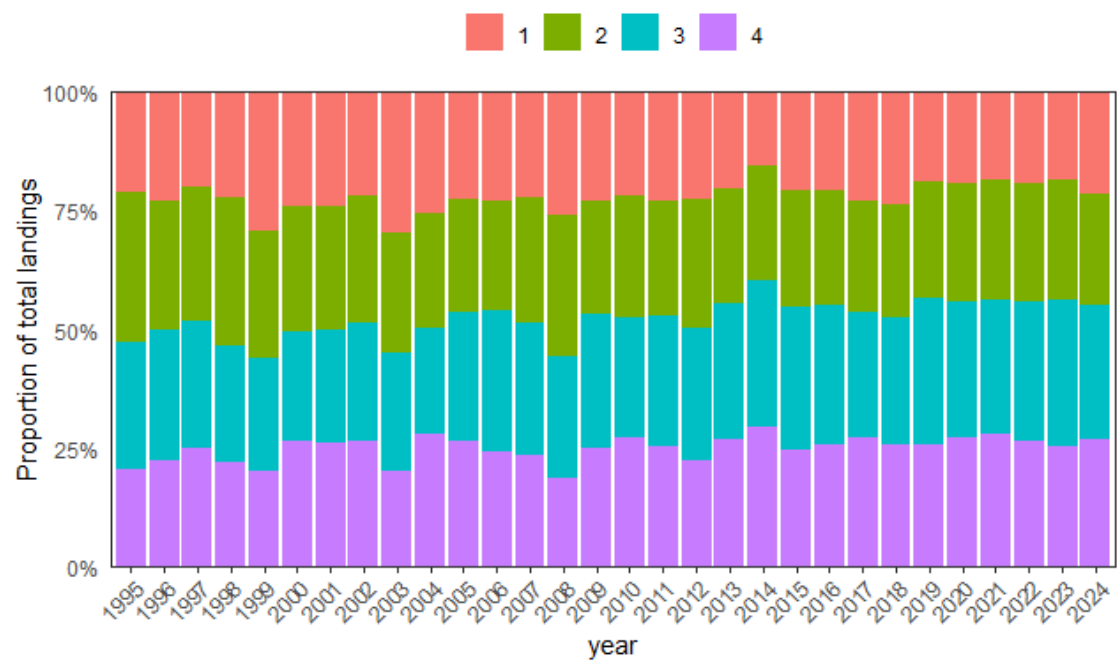


Figure 4.1d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Proportion of commercial landings by quarter for the period 1995–2024.

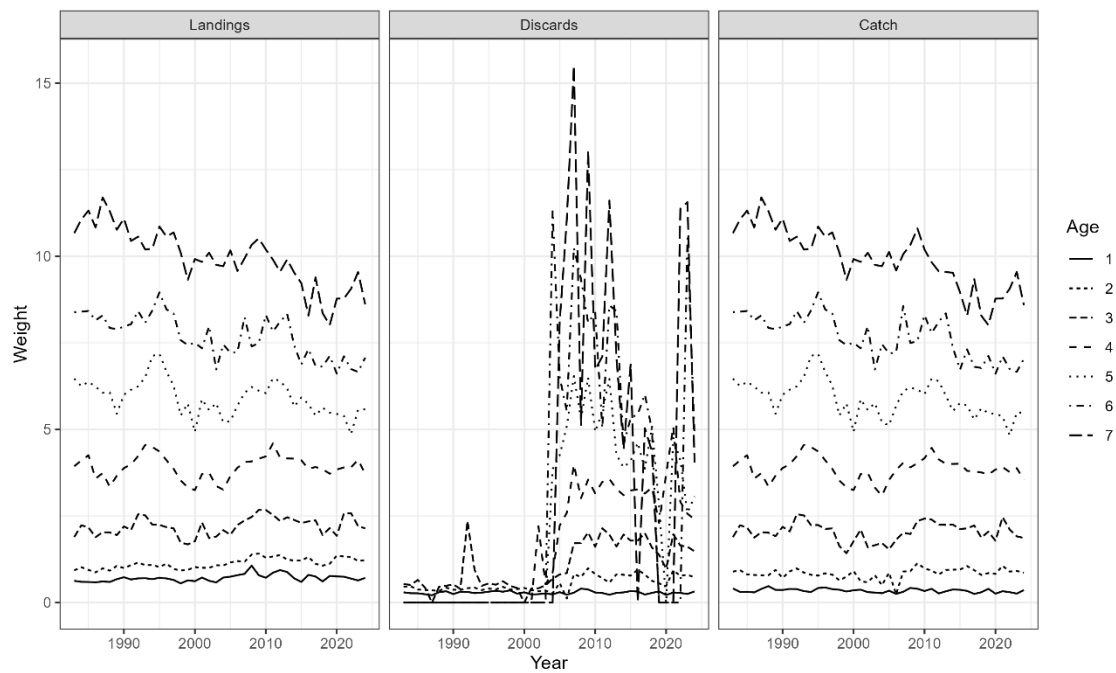


Figure 4.2a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Mean weights-at-age by catch component for ages 1–7+.

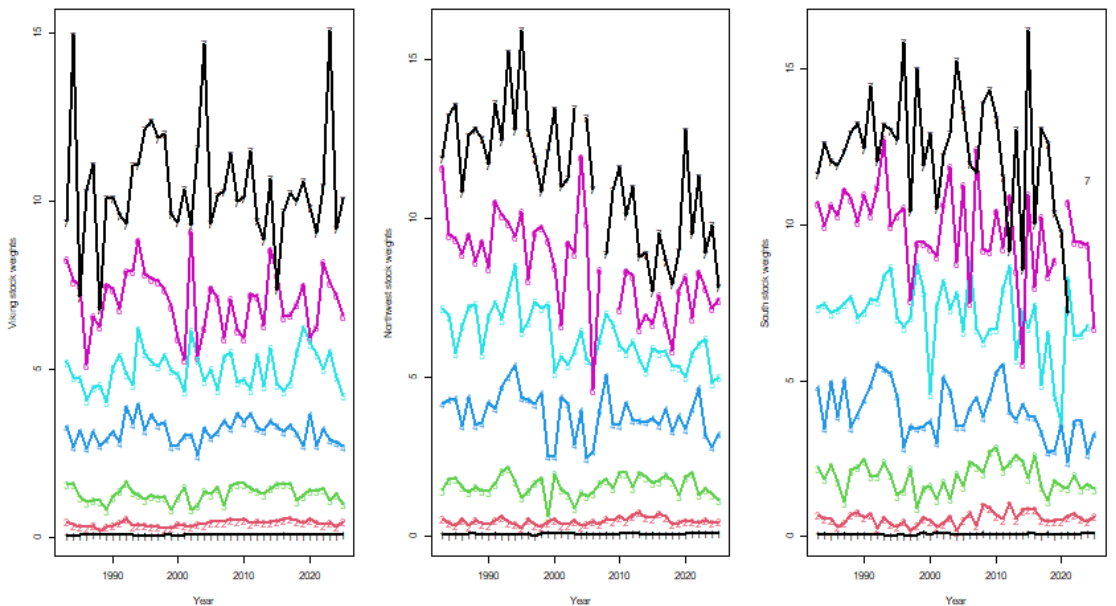


Figure 4.2b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Mean weights-at-age for each substock, as input to the multistock SAM assessment.

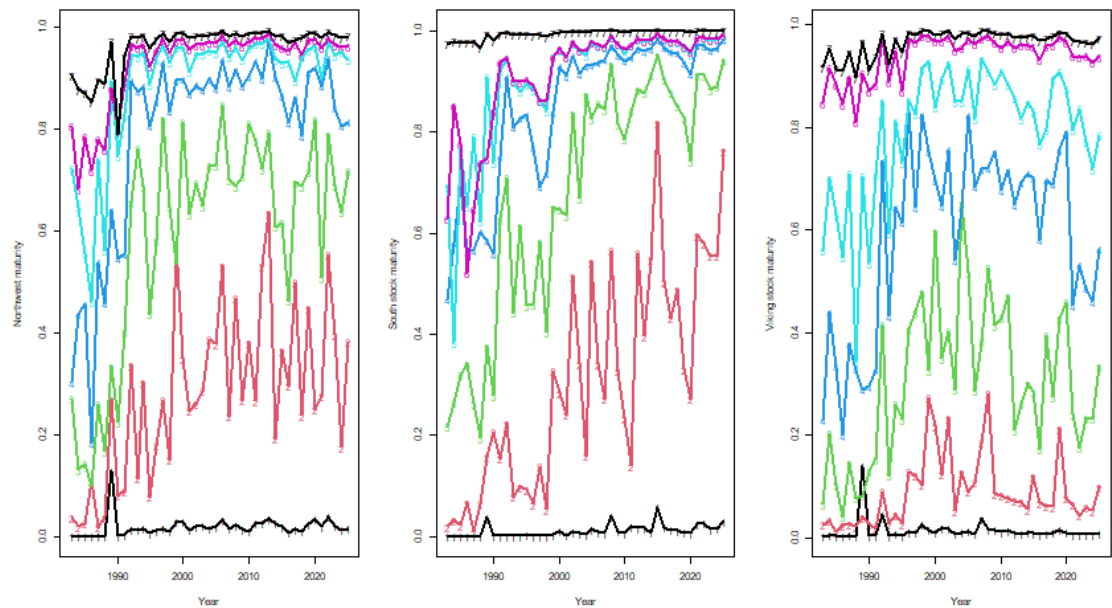


Figure 4.2c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Annually varying maturity-at-age for each sub-stock, as input to the multistock SAM assessment.

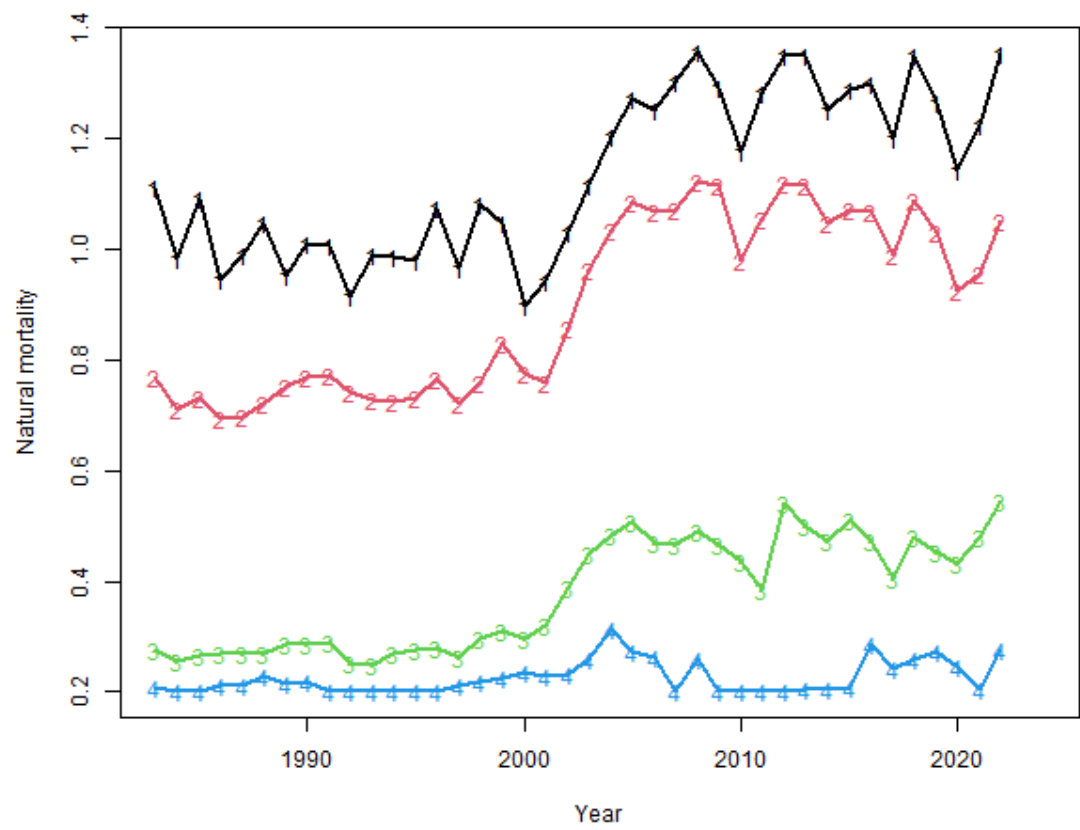


Figure 4.2d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Annually varying natural mortality from the SMS run for Northern Shelf cod, as input to the multistock SAM assessment. Values are assumed to apply to all three sub-stocks.

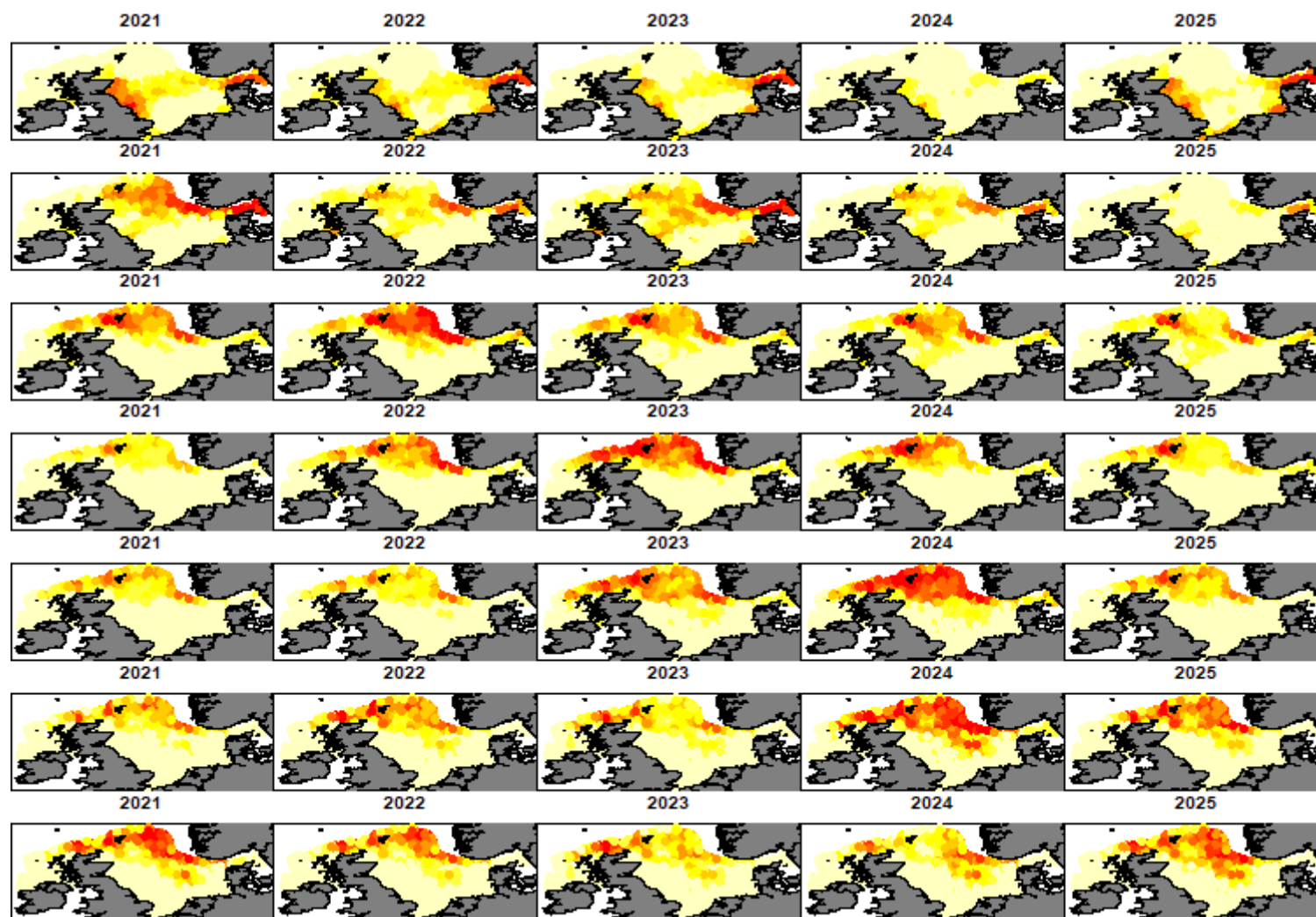


Figure 4.3a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM survey abundances of cod ages 1–7+ (top to bottom) in Q1.

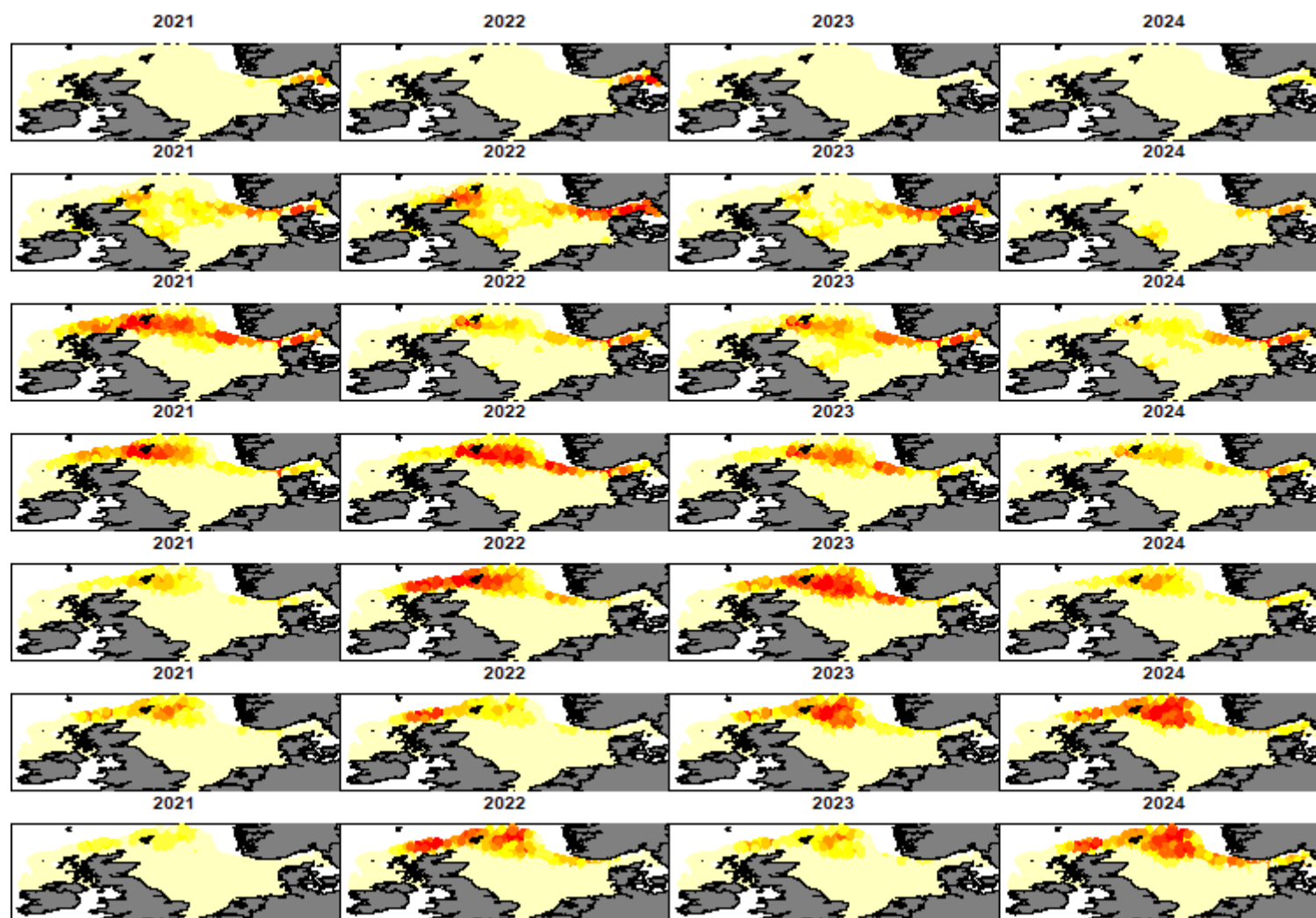


Figure 4.3b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM survey abundances of cod ages 0–6 (top to bottom) in Q3+Q4.

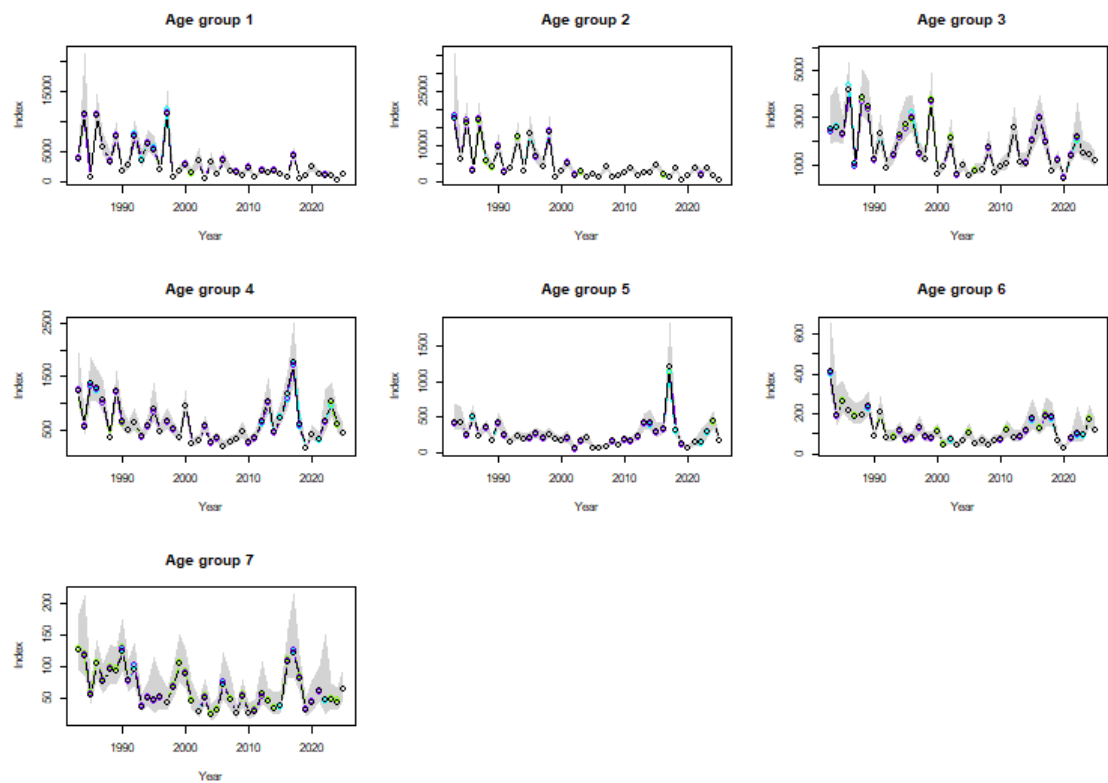


Figure 4.3c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (3 years) from the Delta-GAM model for Q1, together with corresponding pointwise 95% confidence intervals.

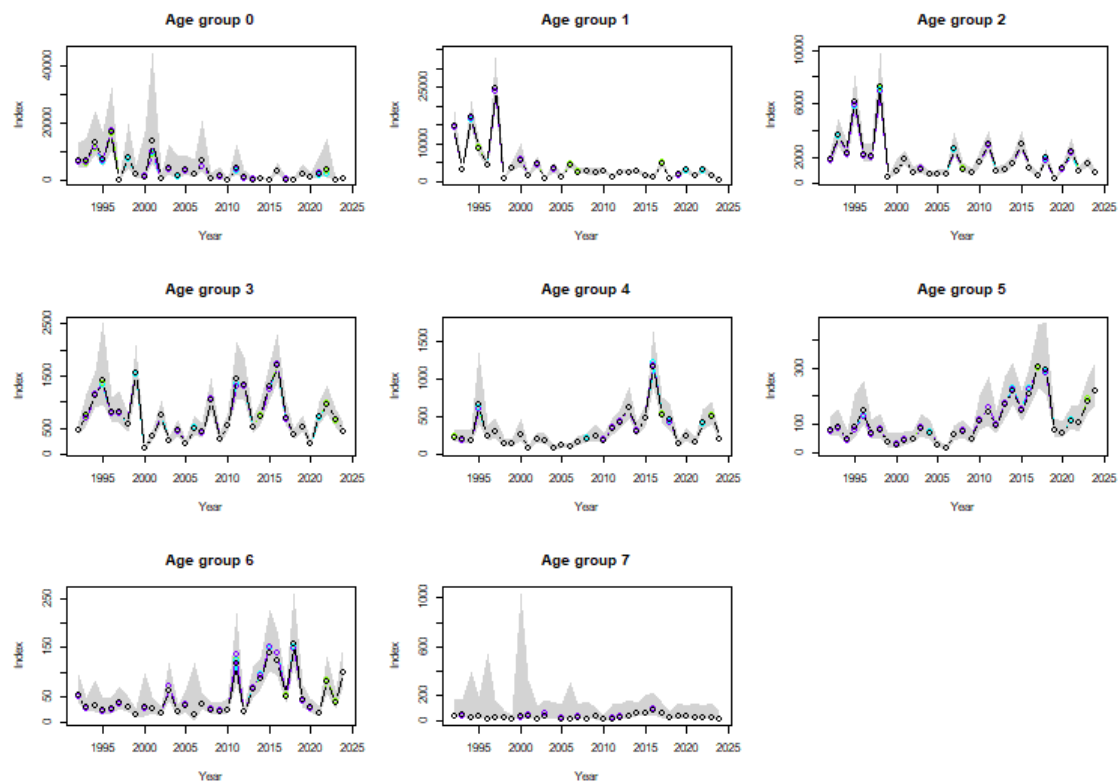


Figure 4.3d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (3 years) from the Delta-GAM model for Q3+Q4, together with corresponding pointwise 95% confidence intervals.

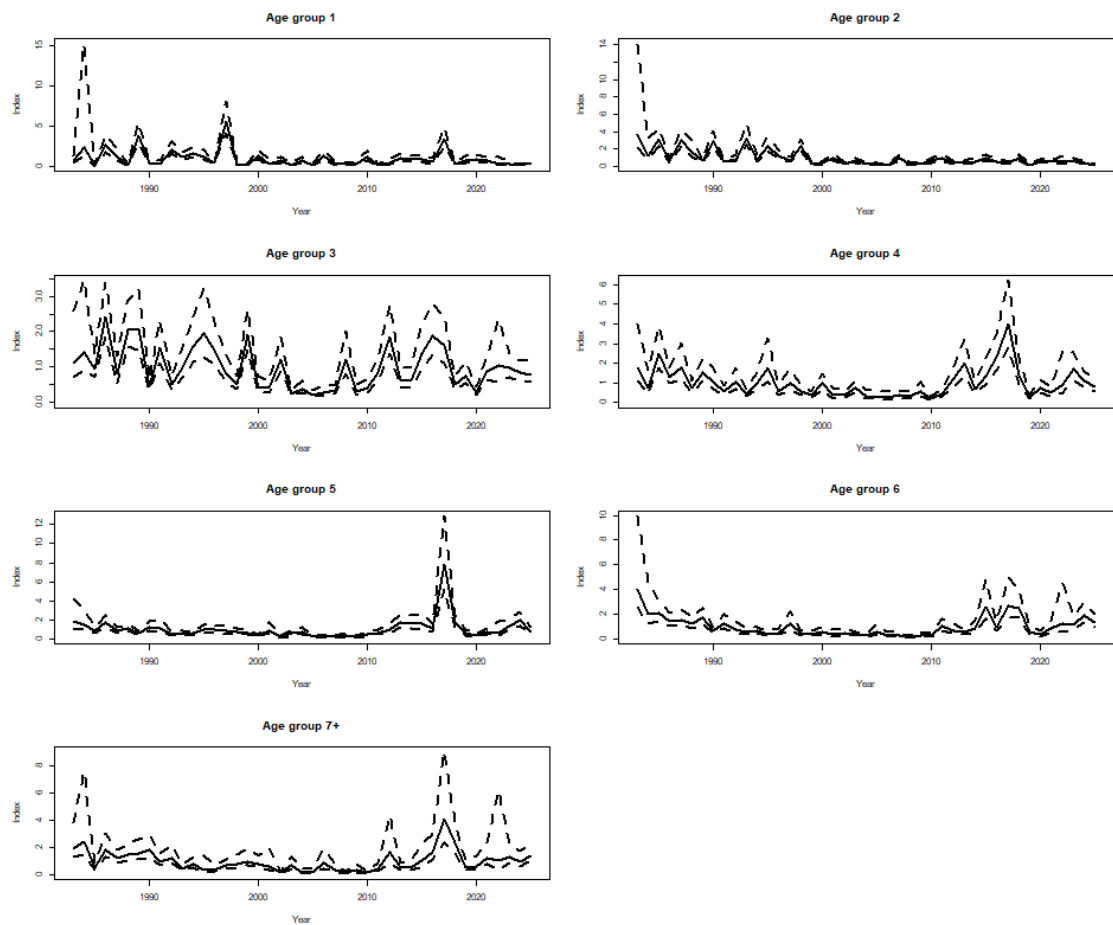


Figure 4.3e. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM survey indices for the Northwestern substock in Q1.

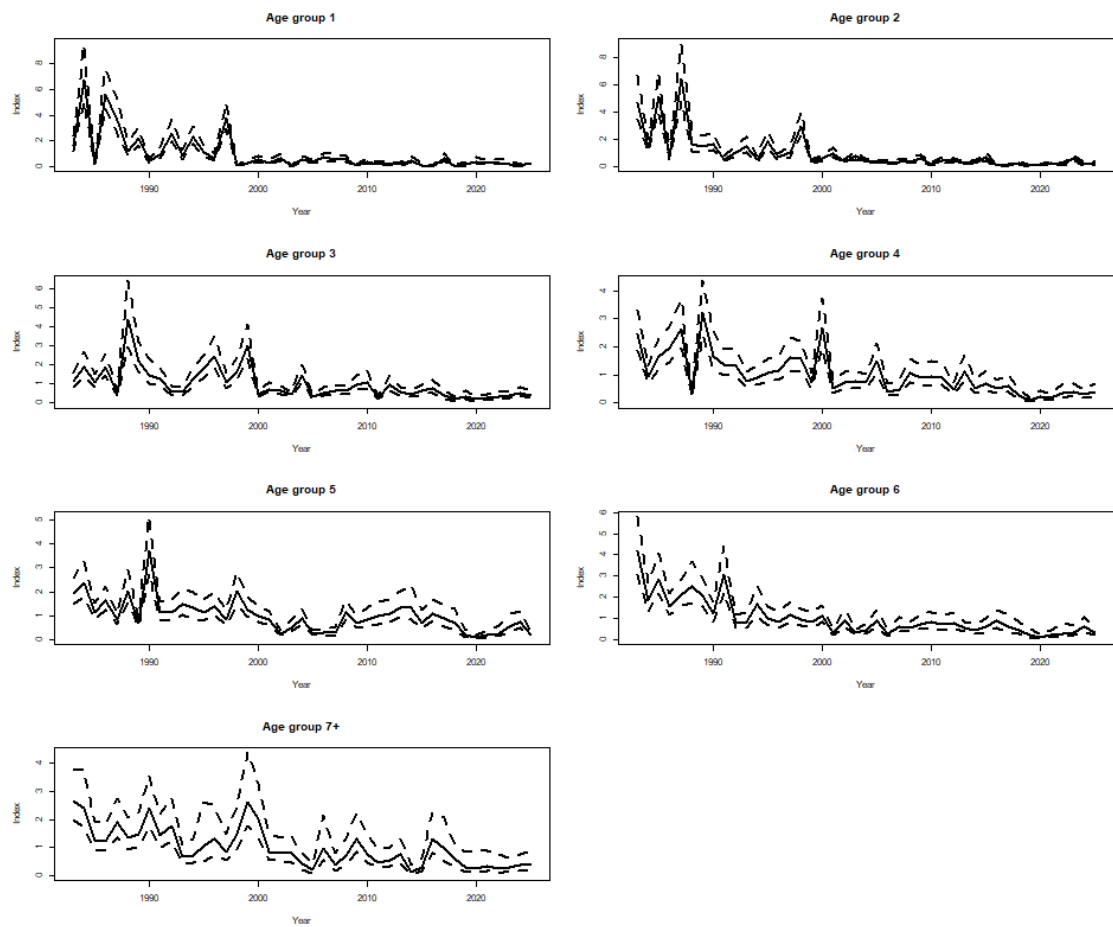


Figure 4.3f. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM survey indices for the Southern sub-stock in Q1.

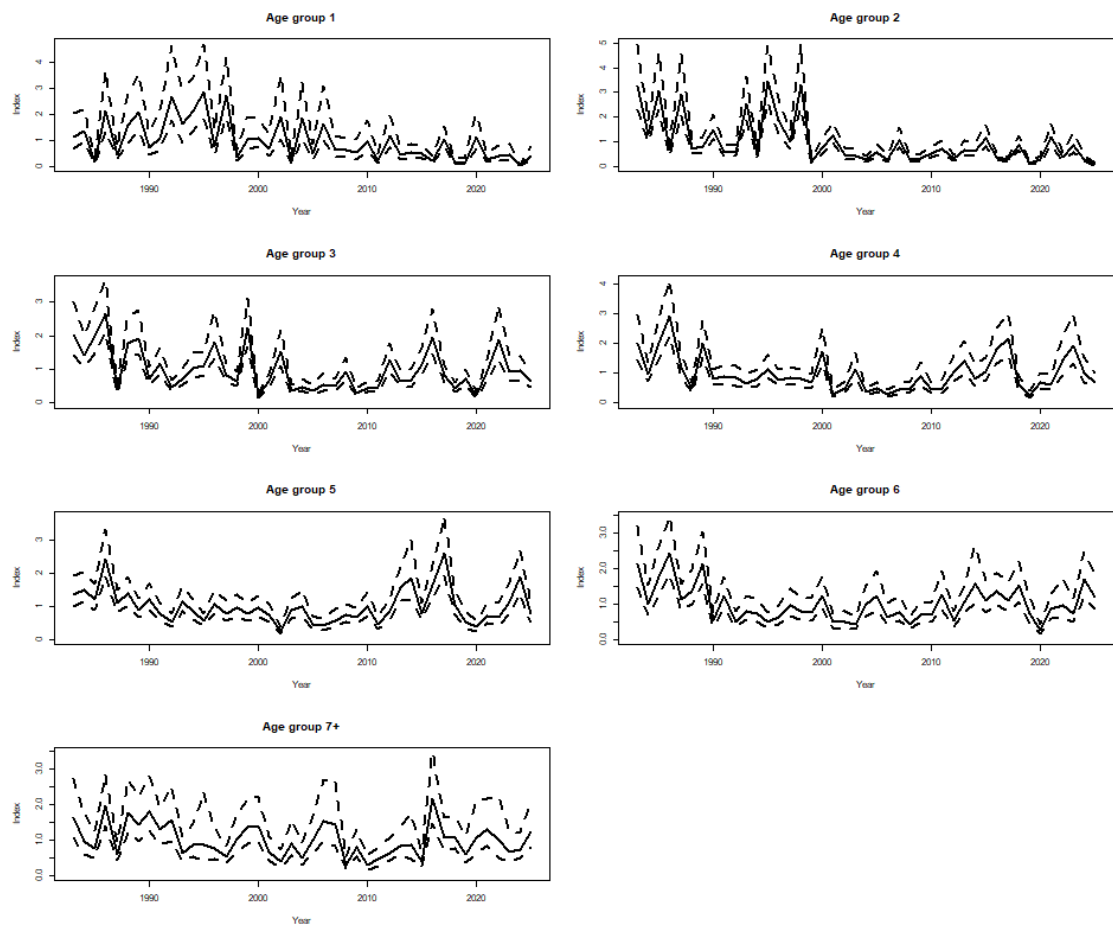


Figure 4.3g. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM survey indices for the Viking substock in Q1.

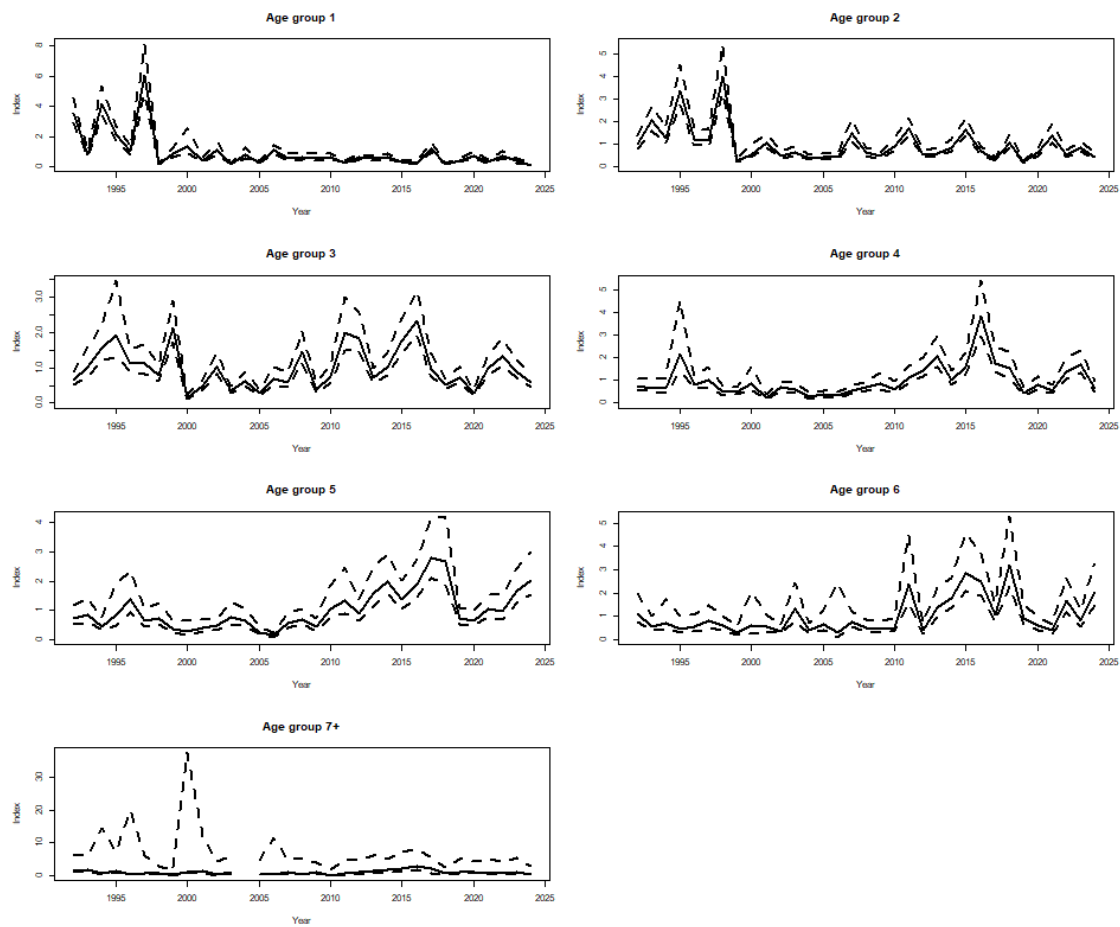


Figure 4.3h. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM survey indices for the Northern Shelf cod stock (all substocks) in Q3+Q4.

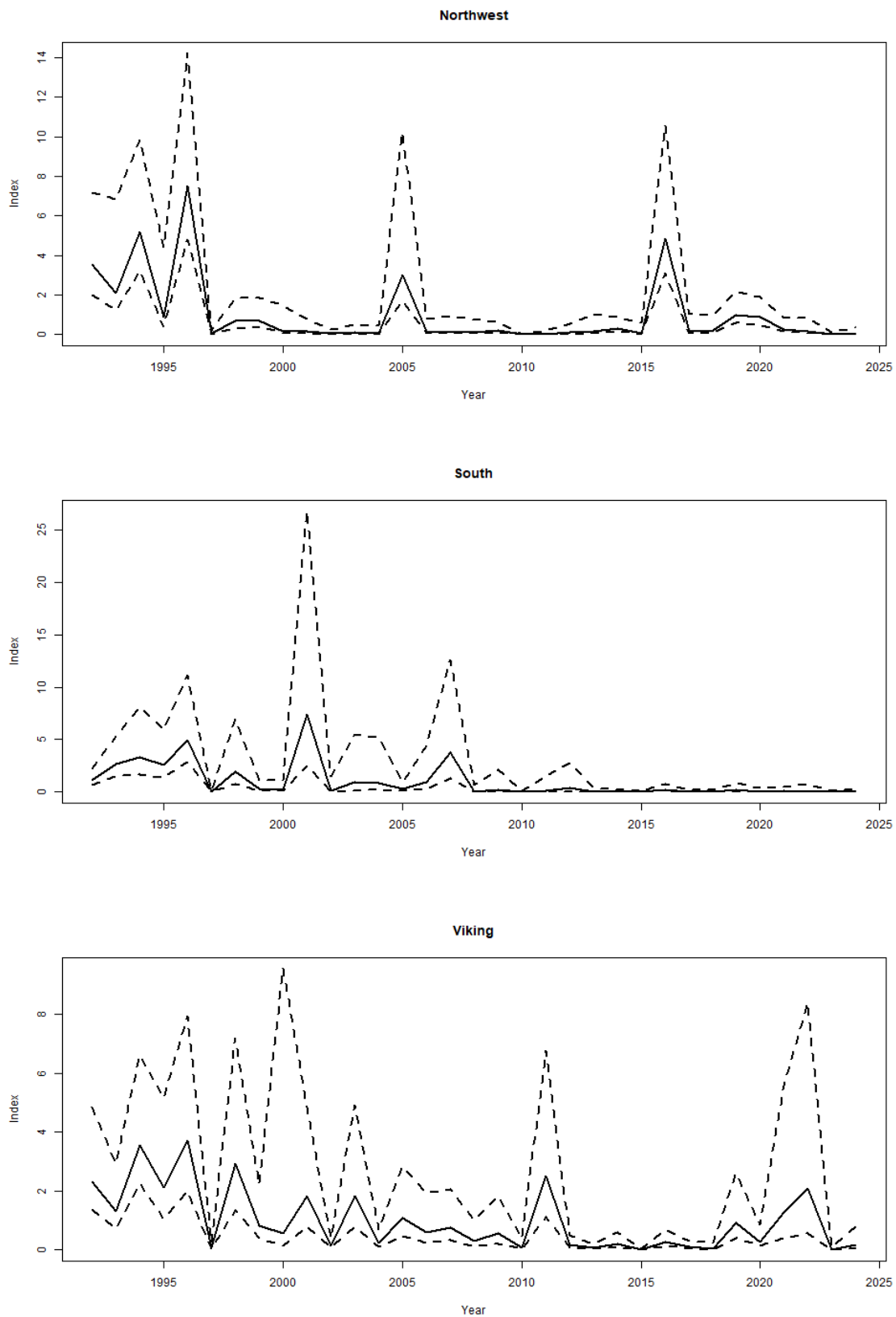


Figure 4.3i. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Delta-GAM indices for substocks of Northern Shelf cod at age 0 in Q3+Q4. These indices are forward shifted in the multistock SAM assessment to represent age 1 cod 1st January the following year.

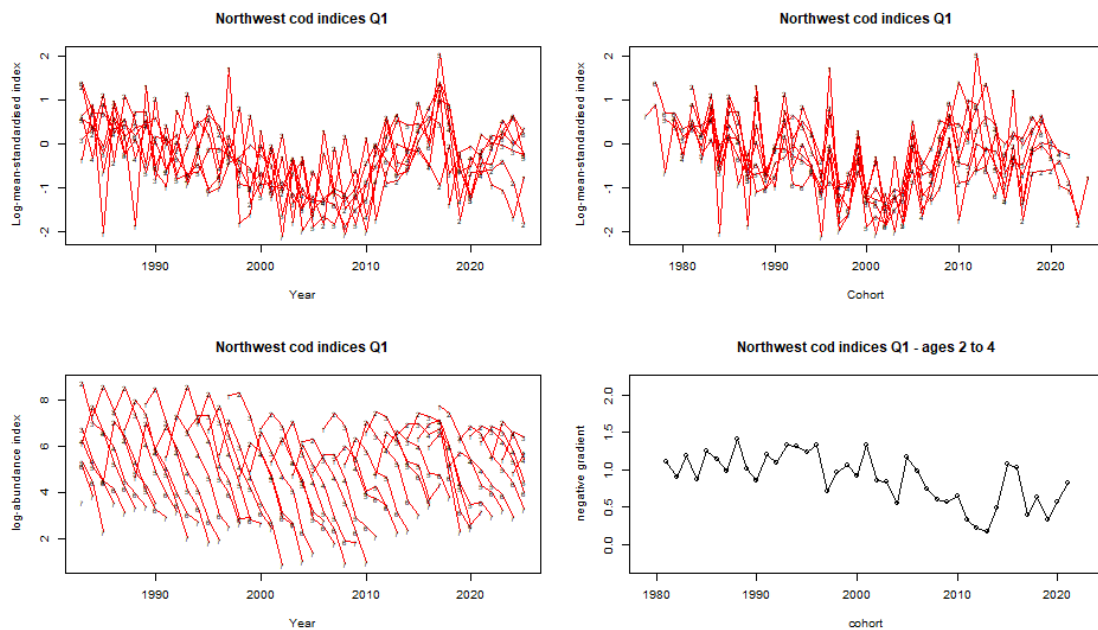


Figure 4.4a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Log mean standardized indices plotted by year (top left) and cohort (top right), log abundance curves (bottom left) and associated negative gradients for each cohort across the reference fishing mortality of age 2–4 (bottom right), for the Northwestern substock in Q1.

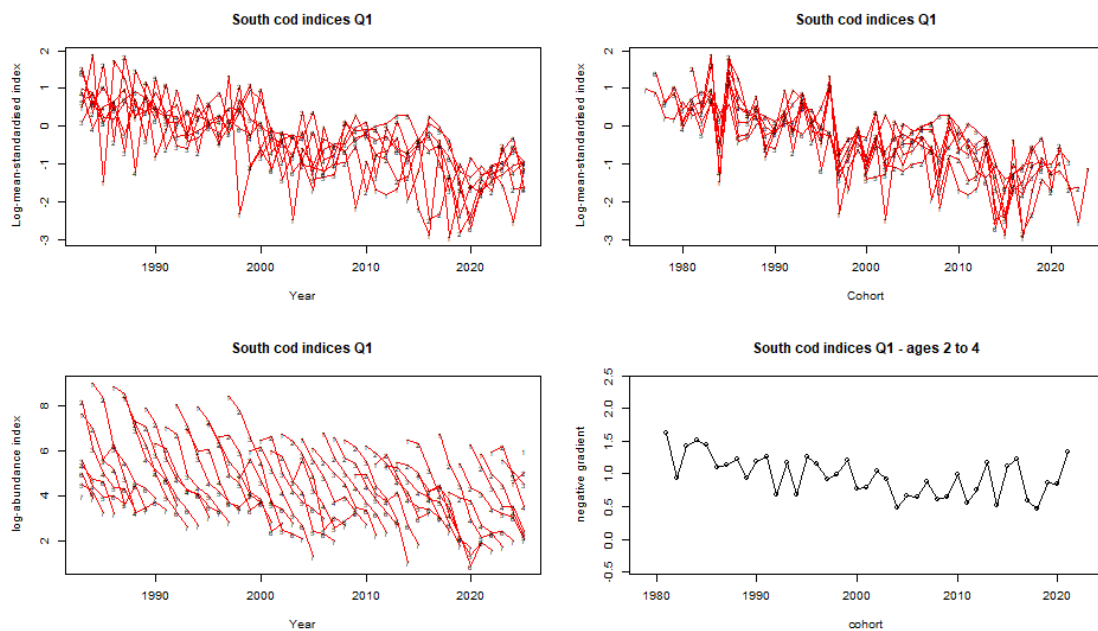


Figure 4.4b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Log mean standardized indices plotted by year (top left) and cohort (top right), log abundance curves (bottom left) and associated negative gradients for each cohort across the reference fishing mortality of age 2–4 (bottom right), for the Southern substock in Q1.

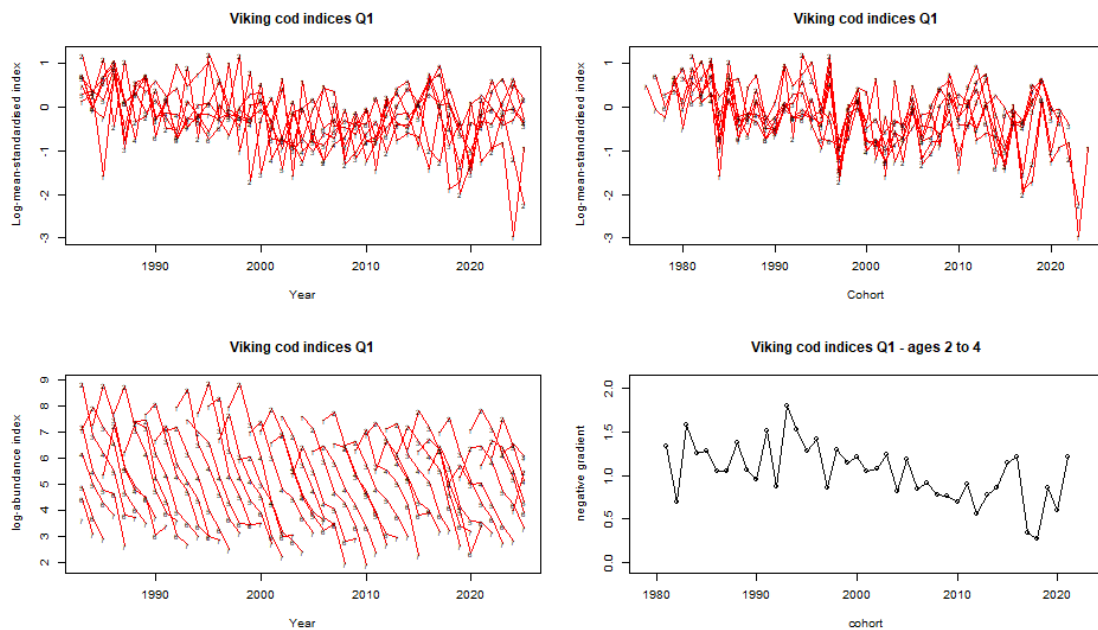


Figure 4.4c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Log mean standardized indices plotted by year (top left) and cohort (top right), log abundance curves (bottom left) and associated negative gradients for each cohort across the reference fishing mortality of age 2–4 (bottom right), for the Viking substock in Q1.

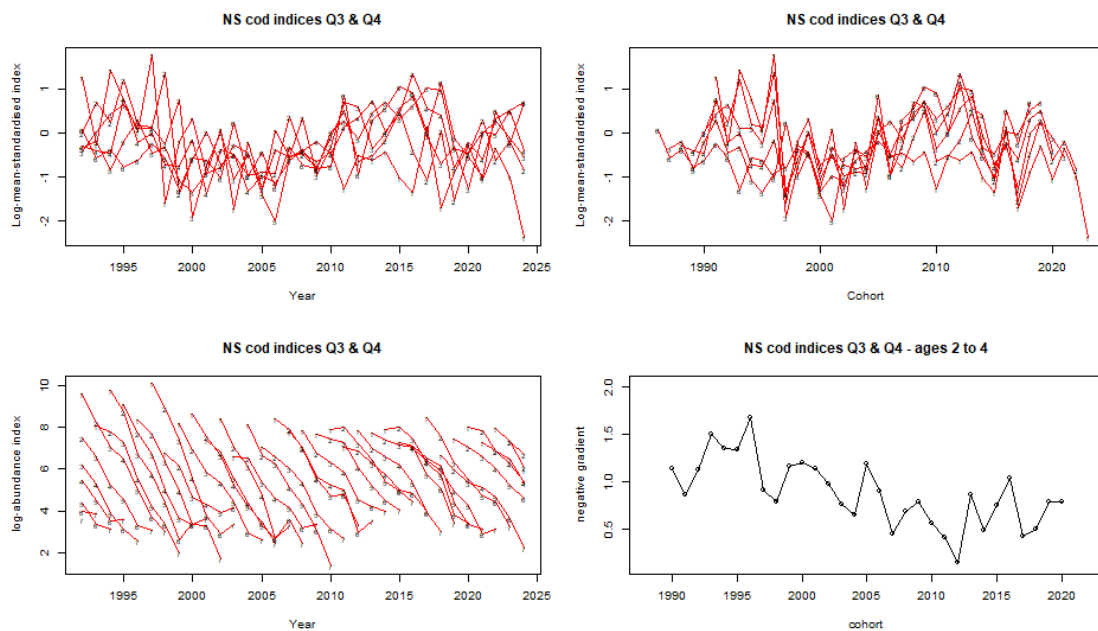


Figure 4.4d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Log mean standardized indices plotted by year (top left) and cohort (top right), log abundance curves (bottom left) and associated negative gradients for each cohort across the reference fishing mortality of age 2–4 (bottom right), in the Q3+Q4 surveys (all substocks).

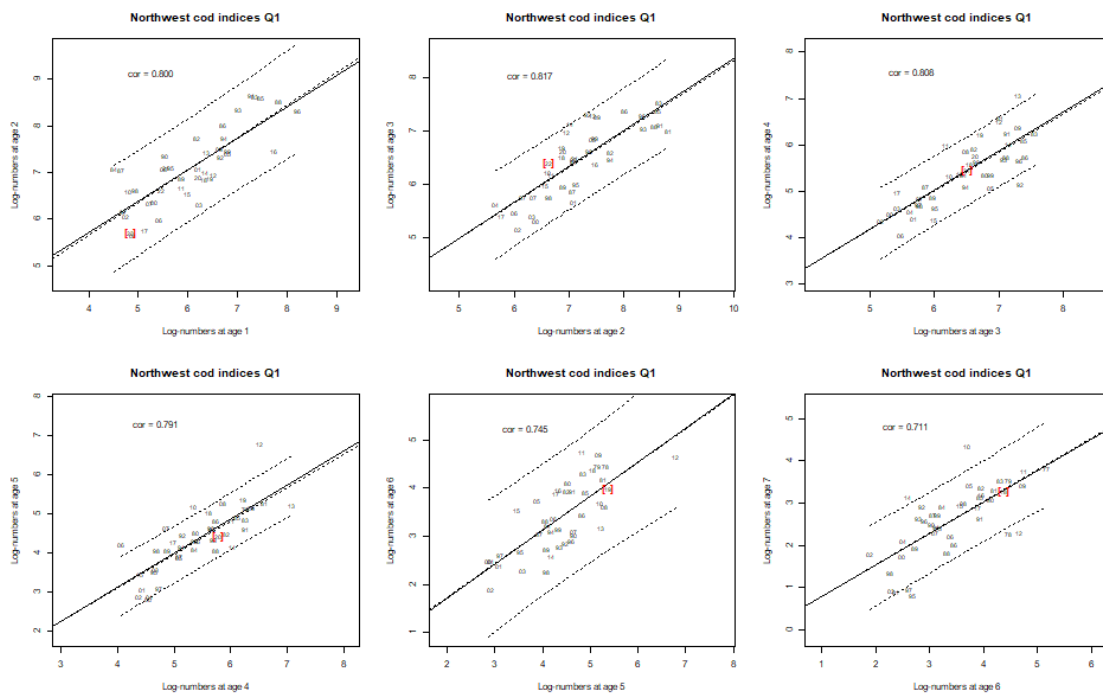


Figure 4.5a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Within survey correlations for the Northwestern substock in Q1 for the period 1983–2025. Individual points are given by cohort (year-class), the solid line is a standard linear regression line, the broken line nearest to it a robust linear regression line, and “cor” denotes the correlation coefficient. The pair of broken lines on either side of the solid line indicate prediction intervals. The most recent data point appears in red square brackets.

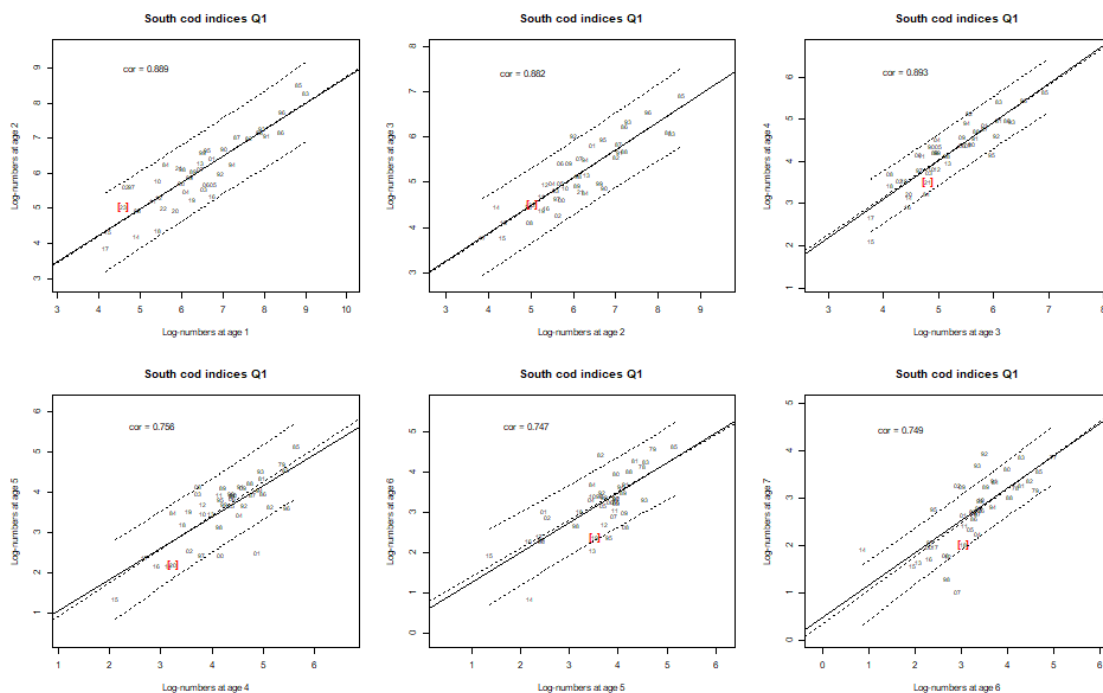


Figure 4.5b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Within survey correlations for the Southern substock in Q1 for the period 1983–2025. Individual points are given by cohort (year-class), the solid line is a standard linear regression line, the broken line nearest to it a robust linear regression line, and “cor” denotes the correlation coefficient. The pair of broken lines on either side of the solid line indicate prediction intervals. The most recent data point appears in red square brackets.

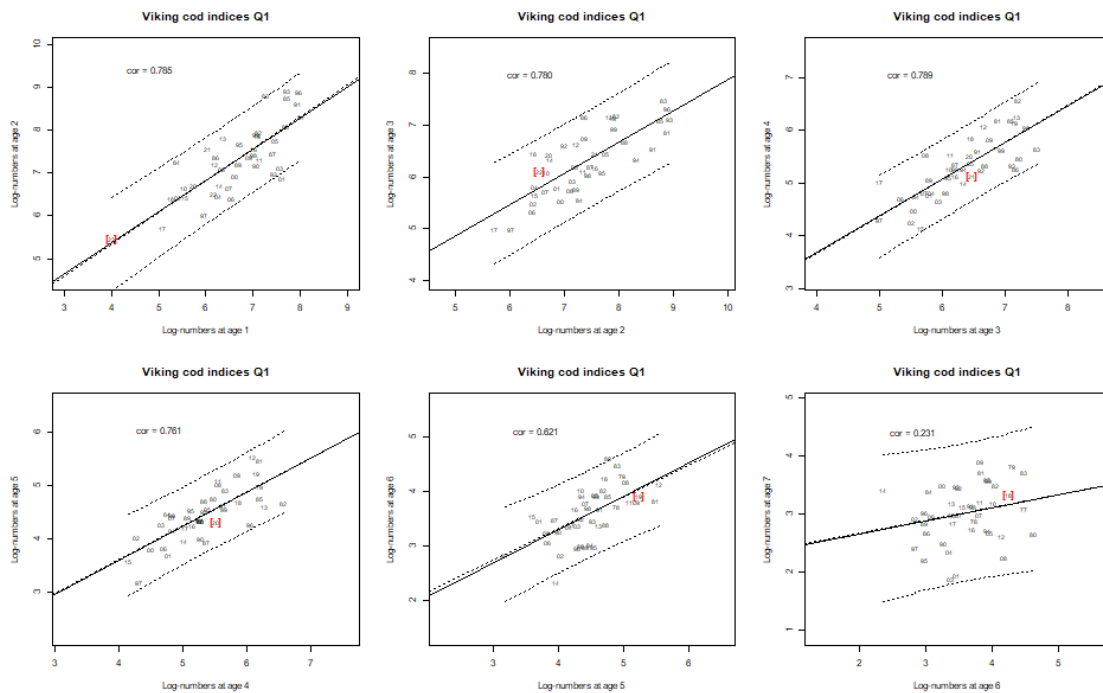


Figure 4.5c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Within survey correlations for the Viking substock in Q1 for the period 1983–2025. Individual points are given by cohort (year-class), the solid line is a standard linear regression line, the broken line nearest to it a robust linear regression line, and “cor” denotes the correlation coefficient. The pair of broken lines on either side of the solid line indicate prediction intervals. The most recent data point appears in red square brackets.

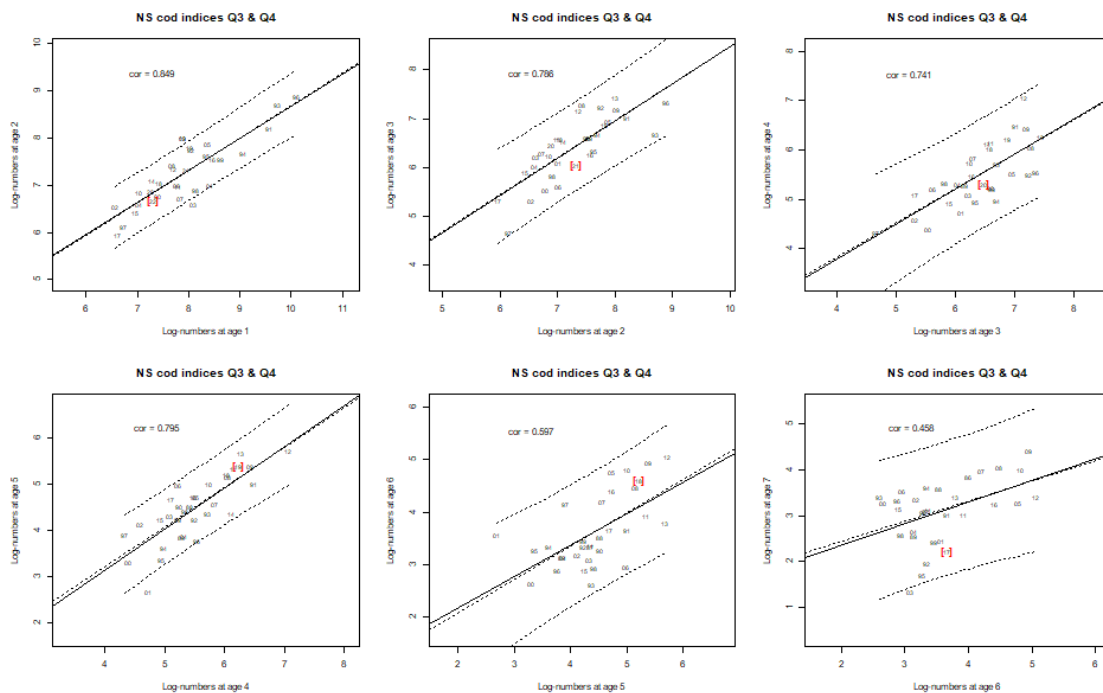


Figure 4.5d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Within survey correlations for the Northern Shelf stock (all substocks) in Q3+Q4 for the period 1992–2024. Individual points are given by cohort (year-class), the solid line is a standard linear regression line, the broken line nearest to it a robust linear regression line, and “cor” denotes the correlation coefficient. The pair of broken lines on either side of the solid line indicate prediction intervals. The most recent data point appears in red square brackets.

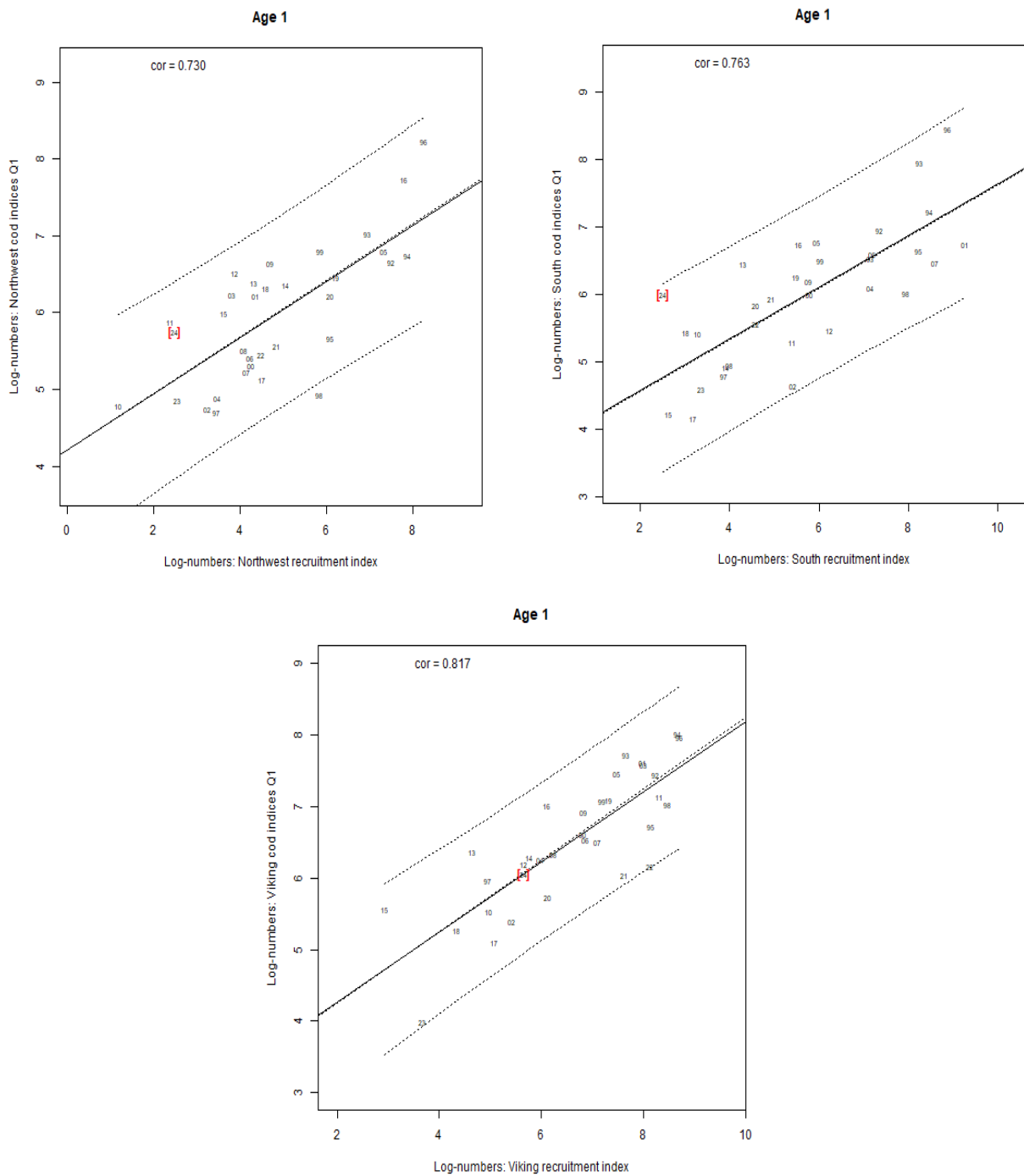


Figure 4.5e. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Between-survey correlations for the Q1 age 1 and Q3+Q4 recruitment indices (age 0 forward shifted to 1st January the following year) by substock for the period 1993–2025. Individual points are given by cohort (year-class), the solid line is a standard linear regression line, and the broken line nearest to it a robust linear regression line. The pair of broken lines on either side of the solid line indicate prediction intervals. The most recent data point appears in red square brackets.

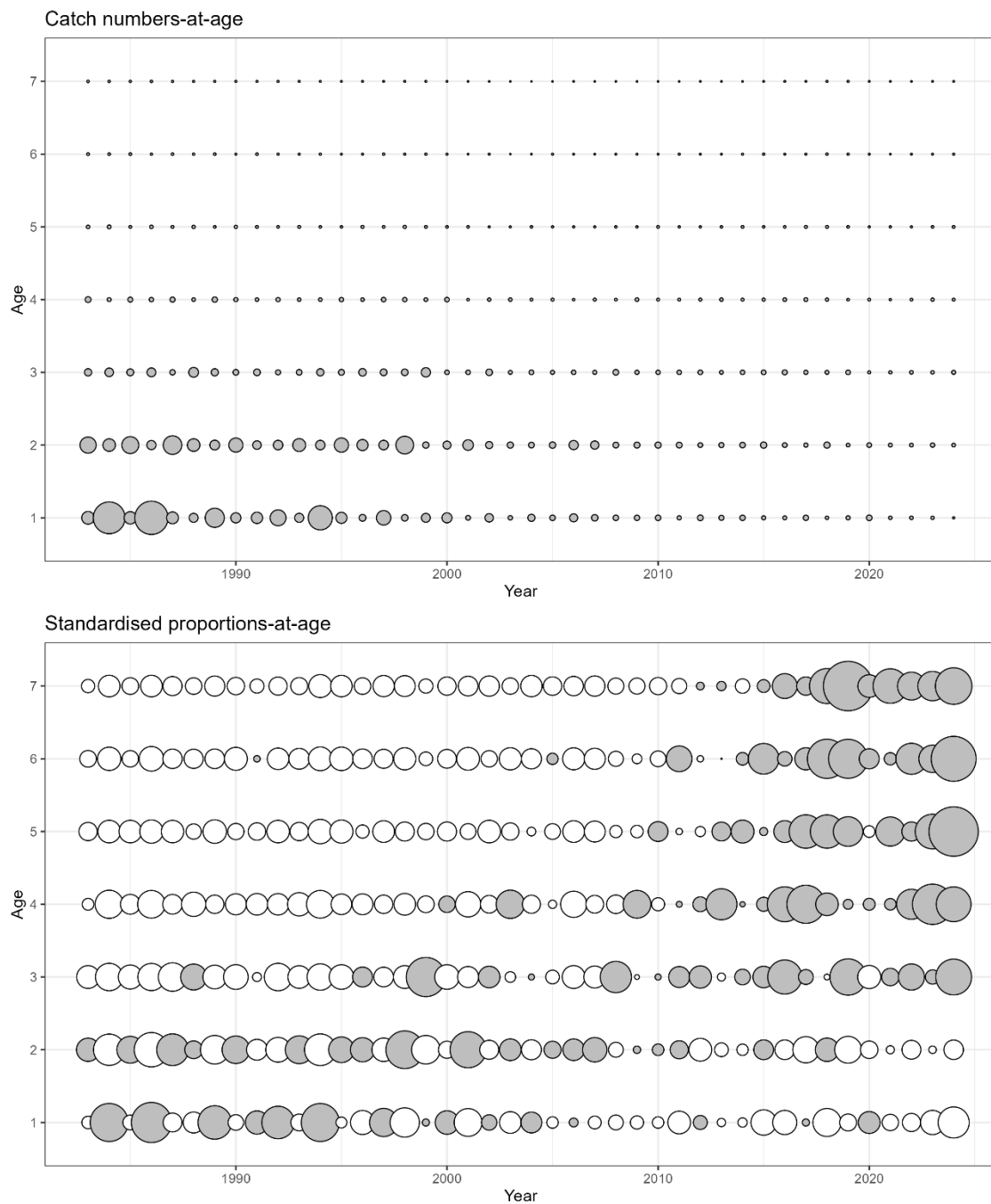


Figure 4.6. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Total catch-at-age matrix expressed for the mixed-stock (all substocks) as (top) numbers-at-age and (bottom) proportions-at-age, which have been standardized over time (for each age, this is achieved by subtracting the mean proportion-at-age over the time series and dividing by the corresponding variance). Grey bubbles indicate proportions above the mean over the time series at each age.

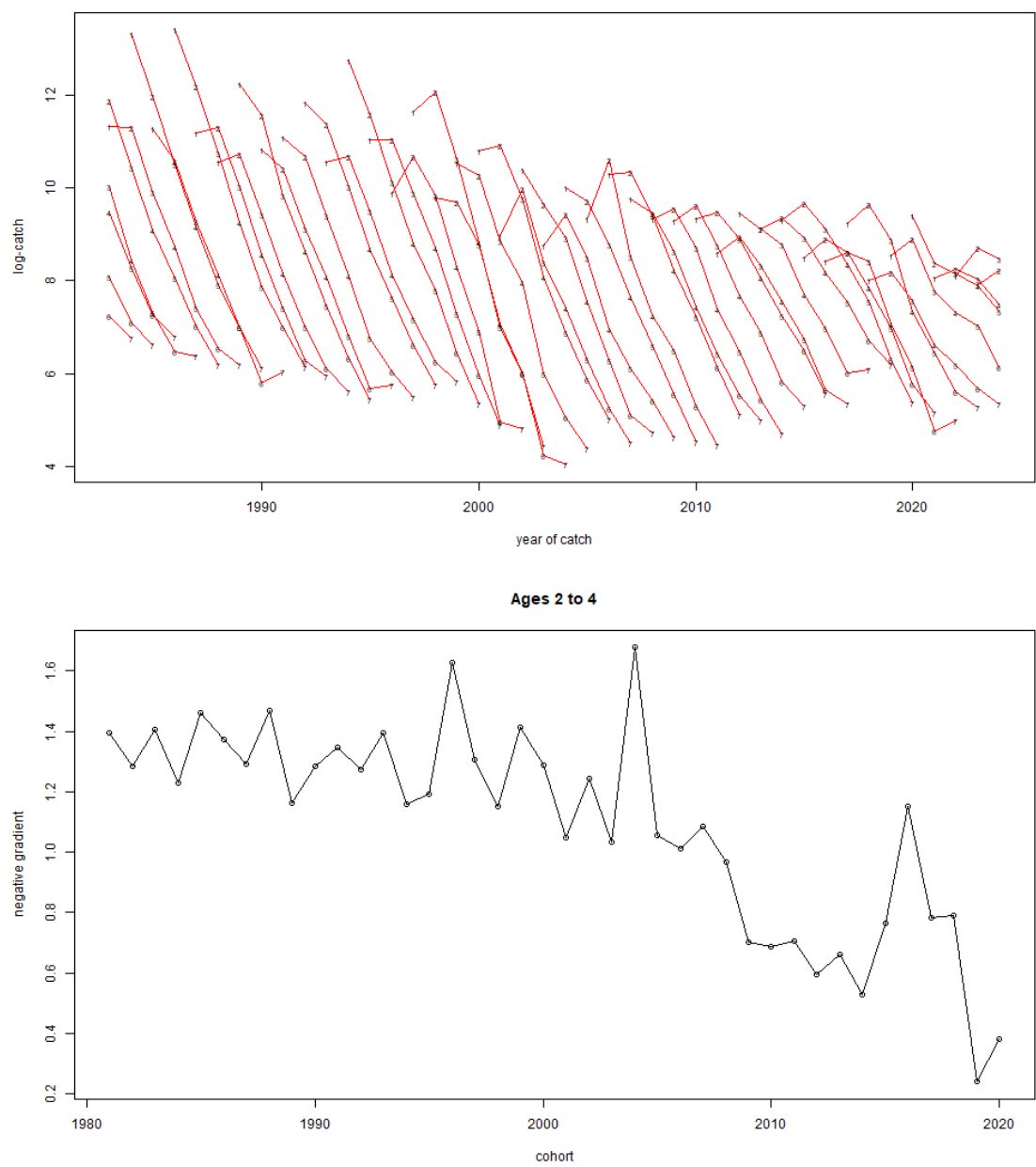


Figure 4.7. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Log-catch cohort curves for the mixed-stock (all substocks; top panel) and the associated negative gradients for each cohort across the reference fishing mortality of age 2–4.

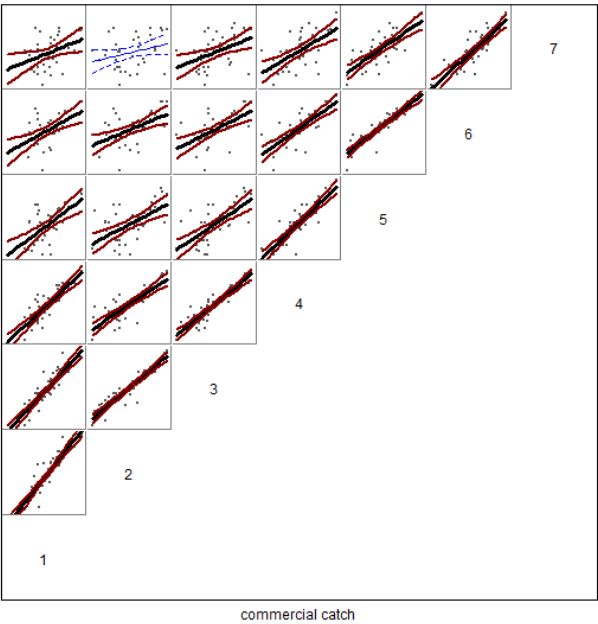


Figure 4.8. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Correlations in the catch-at-age matrix.

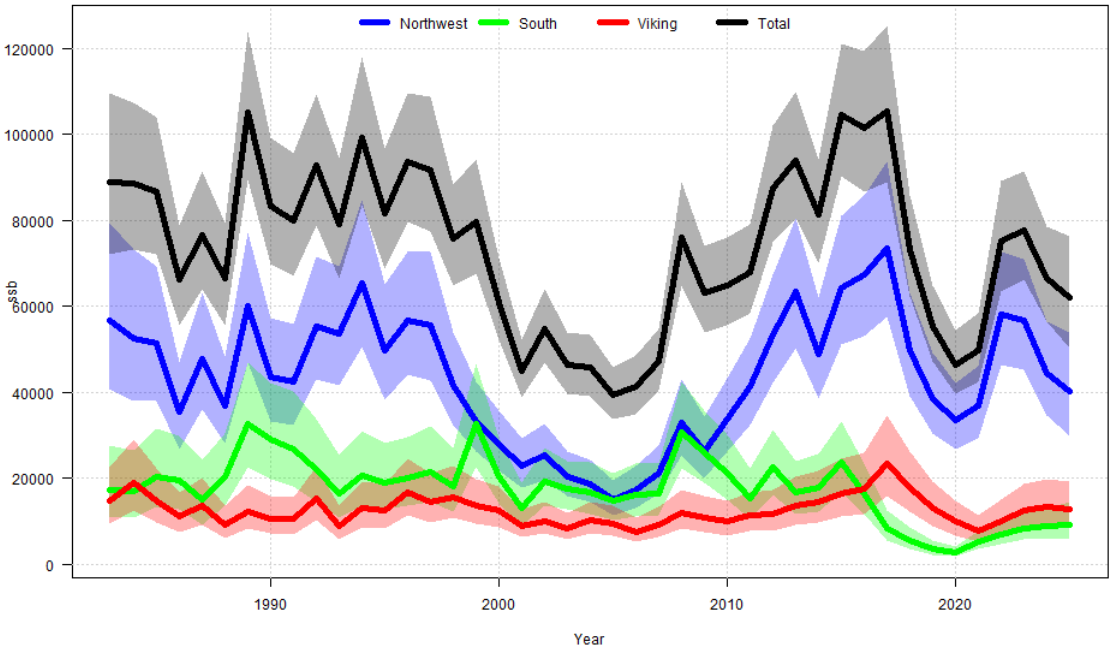


Figure 4.9a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Estimated SSB from the multistock SAM assessment for each substock and the total stock combined, with pointwise 95% confidence intervals.

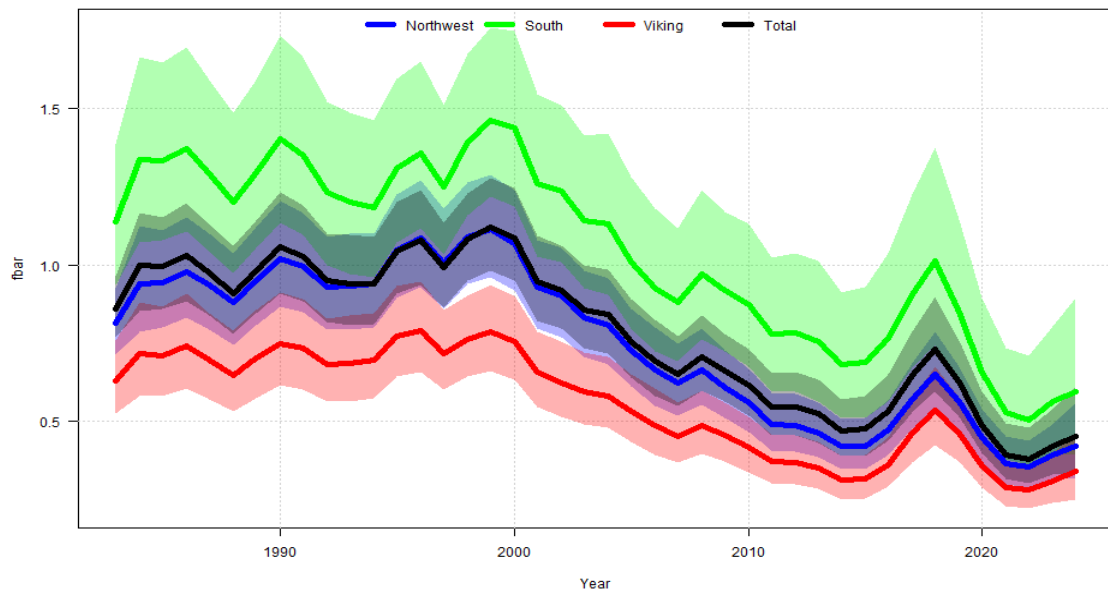


Figure 4.9b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Estimated $F(2-4)$ from the multistock SAM assessment for each substock and the total stock combined, with pointwise 95% confidence intervals.

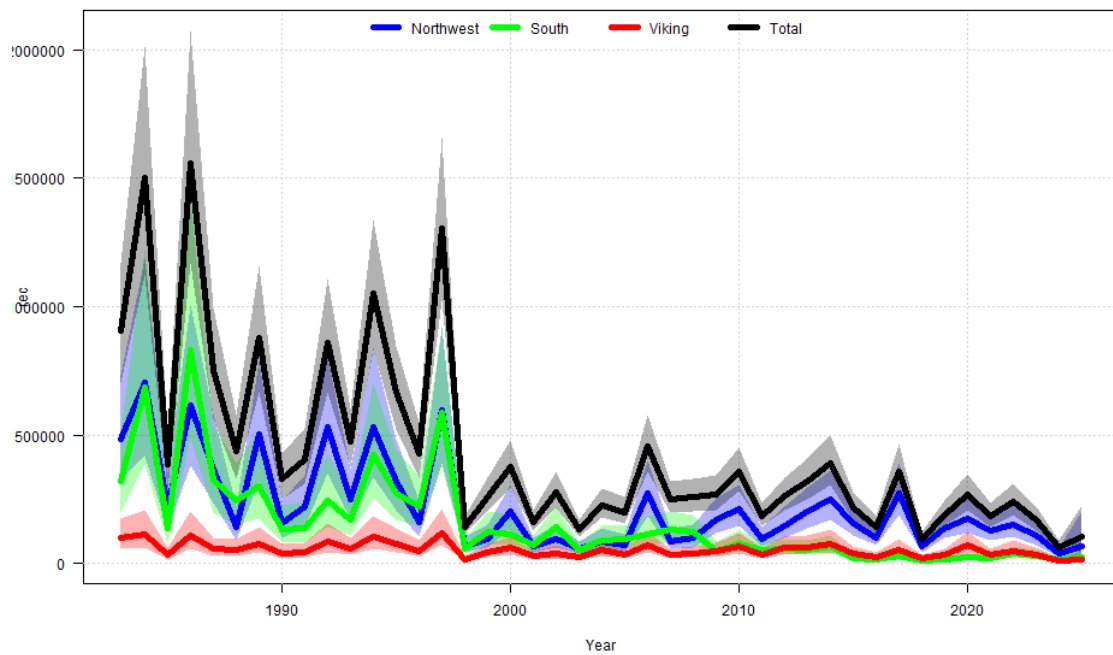


Figure 4.9c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Estimated recruitment (age 1) from the multistock SAM assessment for each substock and the total stock combined, with pointwise 95% confidence intervals.

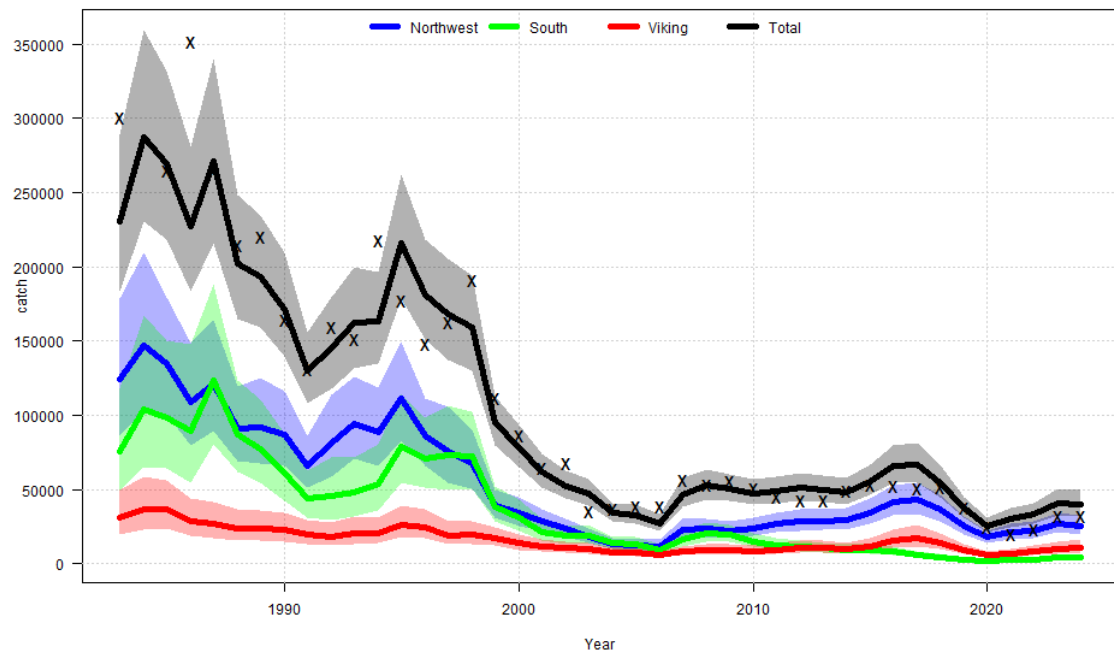


Figure 4.9d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Estimated catches from the multistock SAM assessment for each substock and the total stock combined, with pointwise 95% confidence intervals. Observed catches for the total stock are shown with crosses.

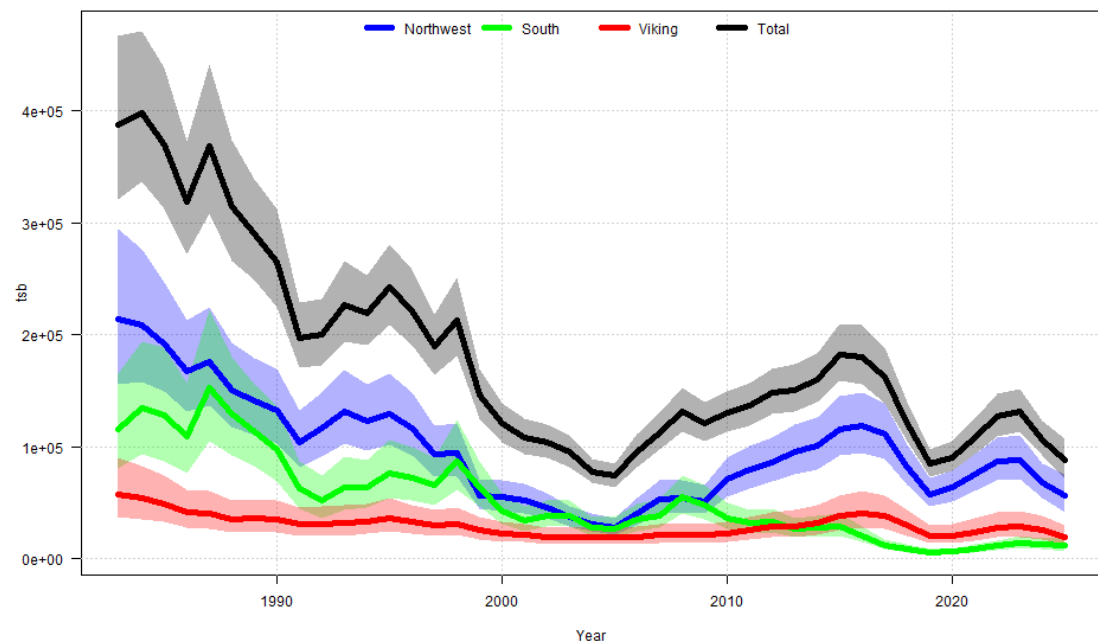


Figure 4.9e. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Estimated TSB from the multistock SAM assessment for each substock and the total stock combined, with pointwise 95% confidence intervals.

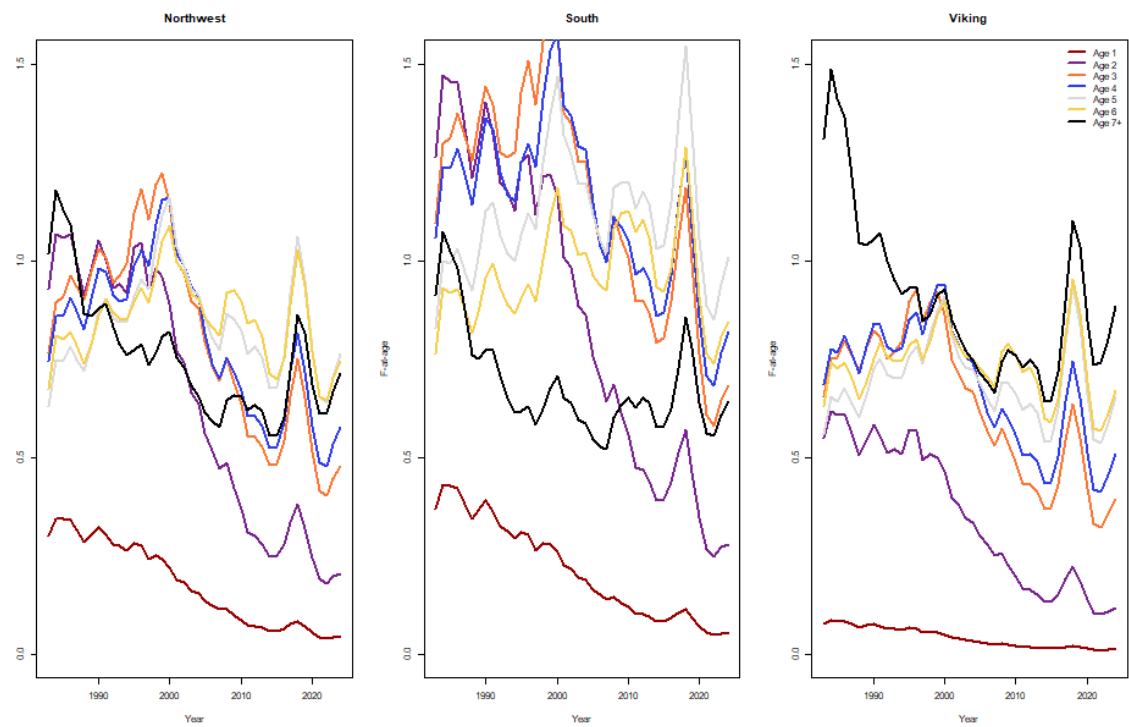


Figure 4.10a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM estimates of fishing mortality-at-age for each substock.

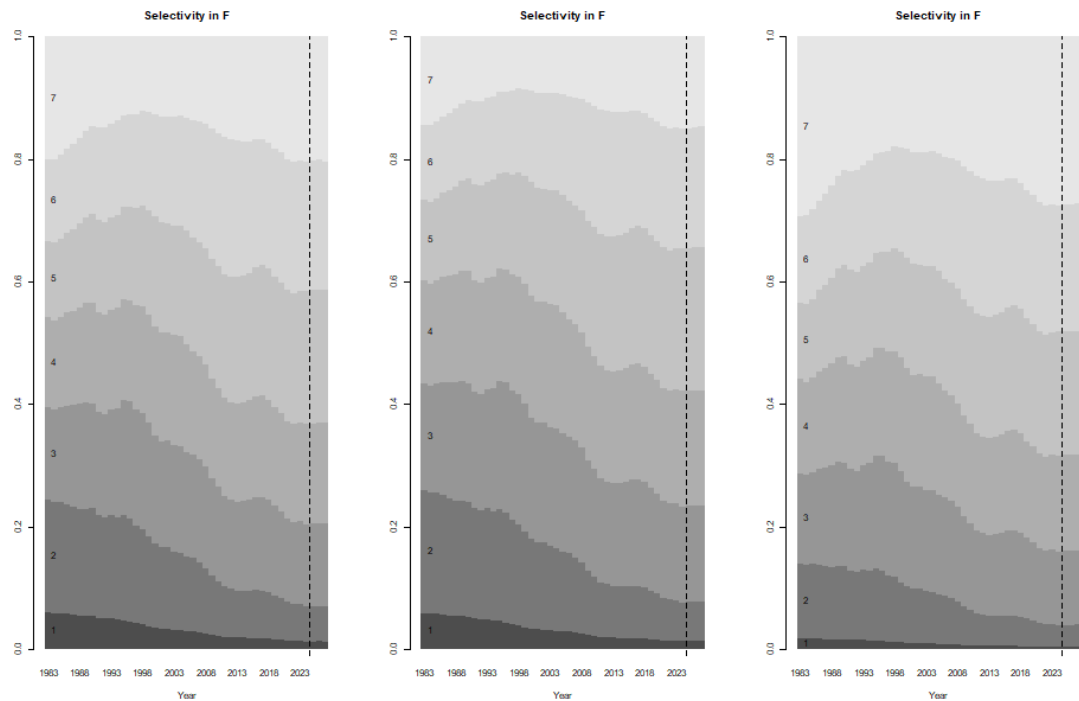


Figure 4.10b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM estimates of selectivity for each substock, derived as the proportions of total fishing mortality-at-age over time. The dashed line represents the beginning of the forecast period.

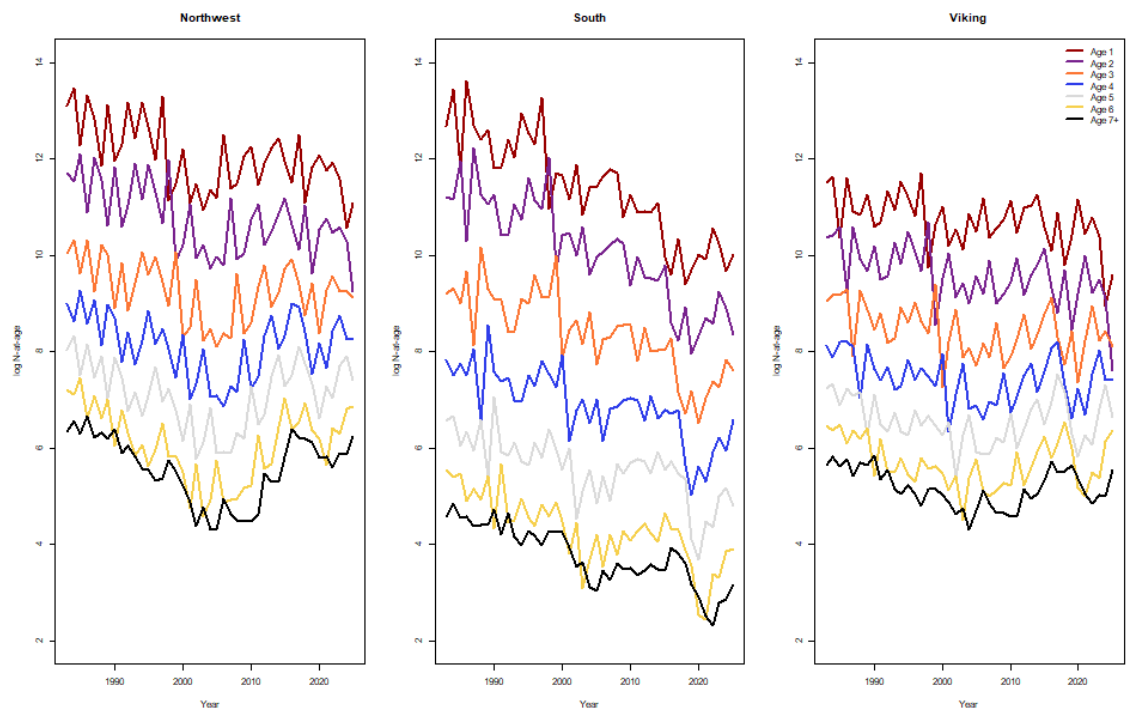


Figure 4.11. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM estimates of log numbers-at-age for each substock.

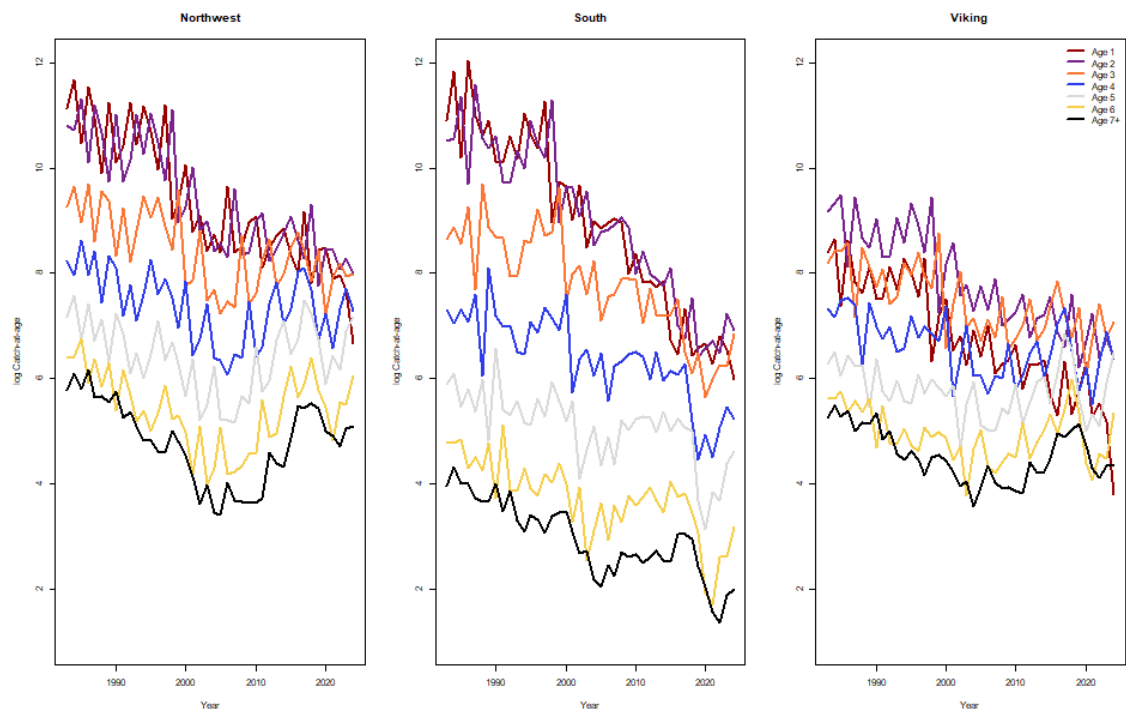


Figure 4.12. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM estimates of log catches-at-age for each substock.

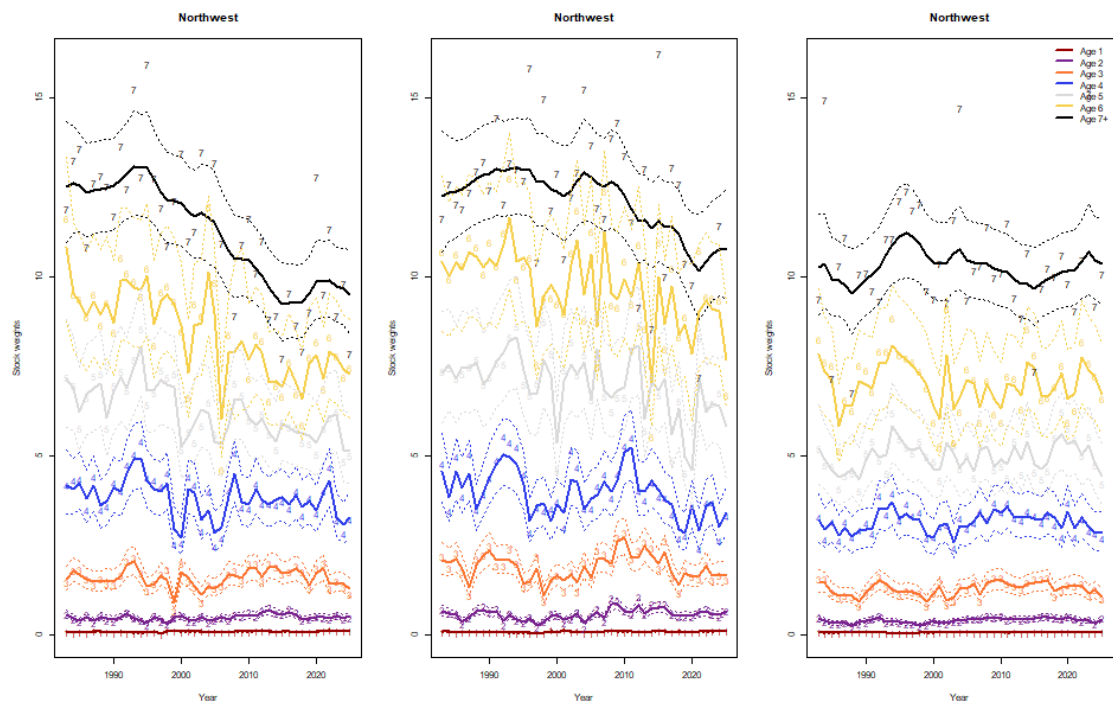


Figure 4.13. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM fits to stock weight data. Numbers are the input data shown in Table 4.5a and Figure 4.2b. The solid lines are the multistock SAM estimates of stock weights-at-age, extending to the forecast period, with the dotted lines showing 95% confidence intervals.

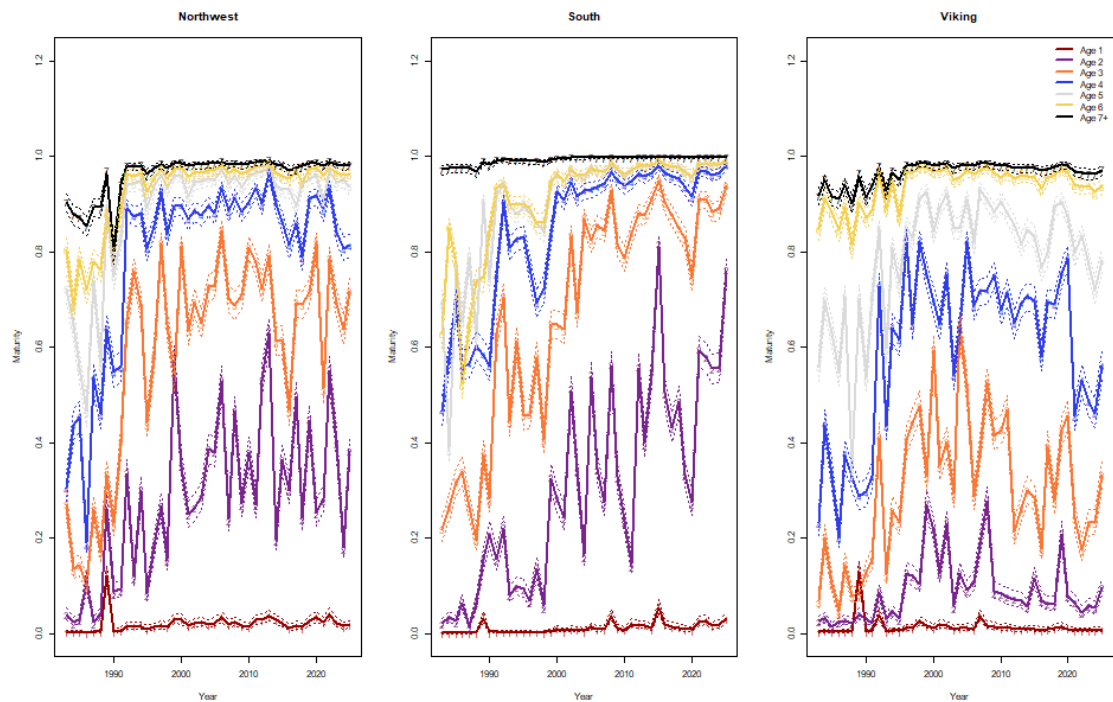


Figure 4.14. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM fits to maturity data. Numbers are the input data shown in Table 4.5b and Figure 4.2c. The solid lines are the multistock SAM estimates of maturity-at-age, extending to the forecast period, with the dotted lines showing 95% confidence intervals.

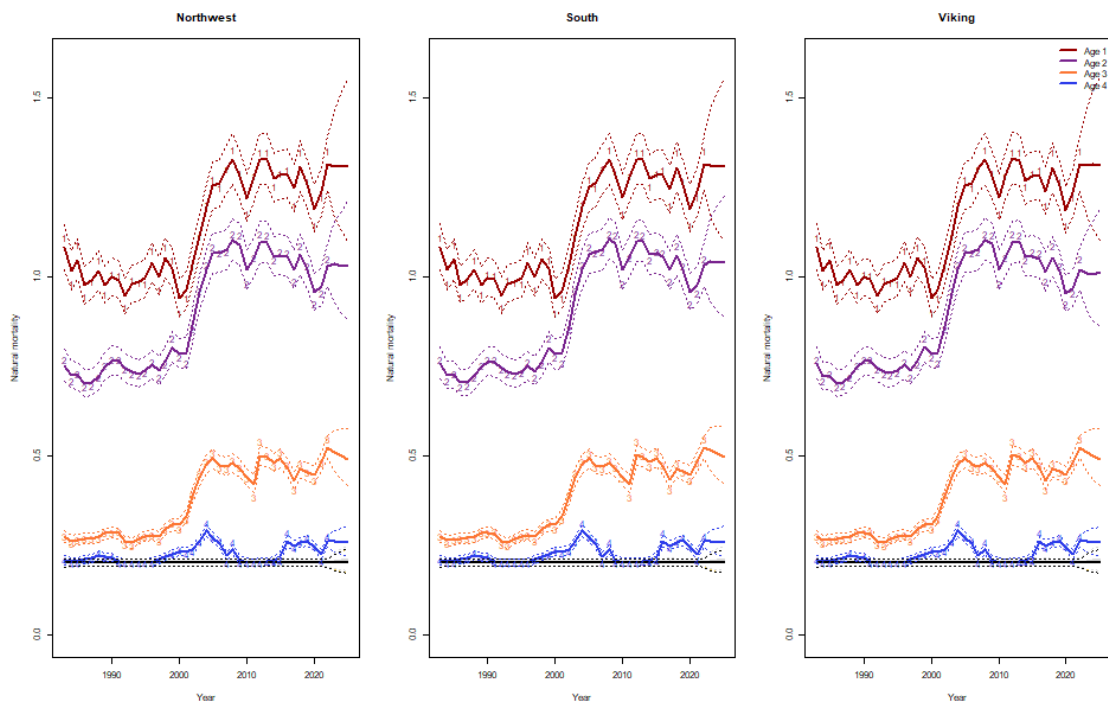


Figure 4.15. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Multistock SAM fits to natural mortality data. Numbers are the input data shown in Table 4.5c and Figure 4.2d. The solid lines are the multistock SAM estimates of natural mortality-at-age, extending to the forecast period, with the dotted lines showing 95% confidence intervals.

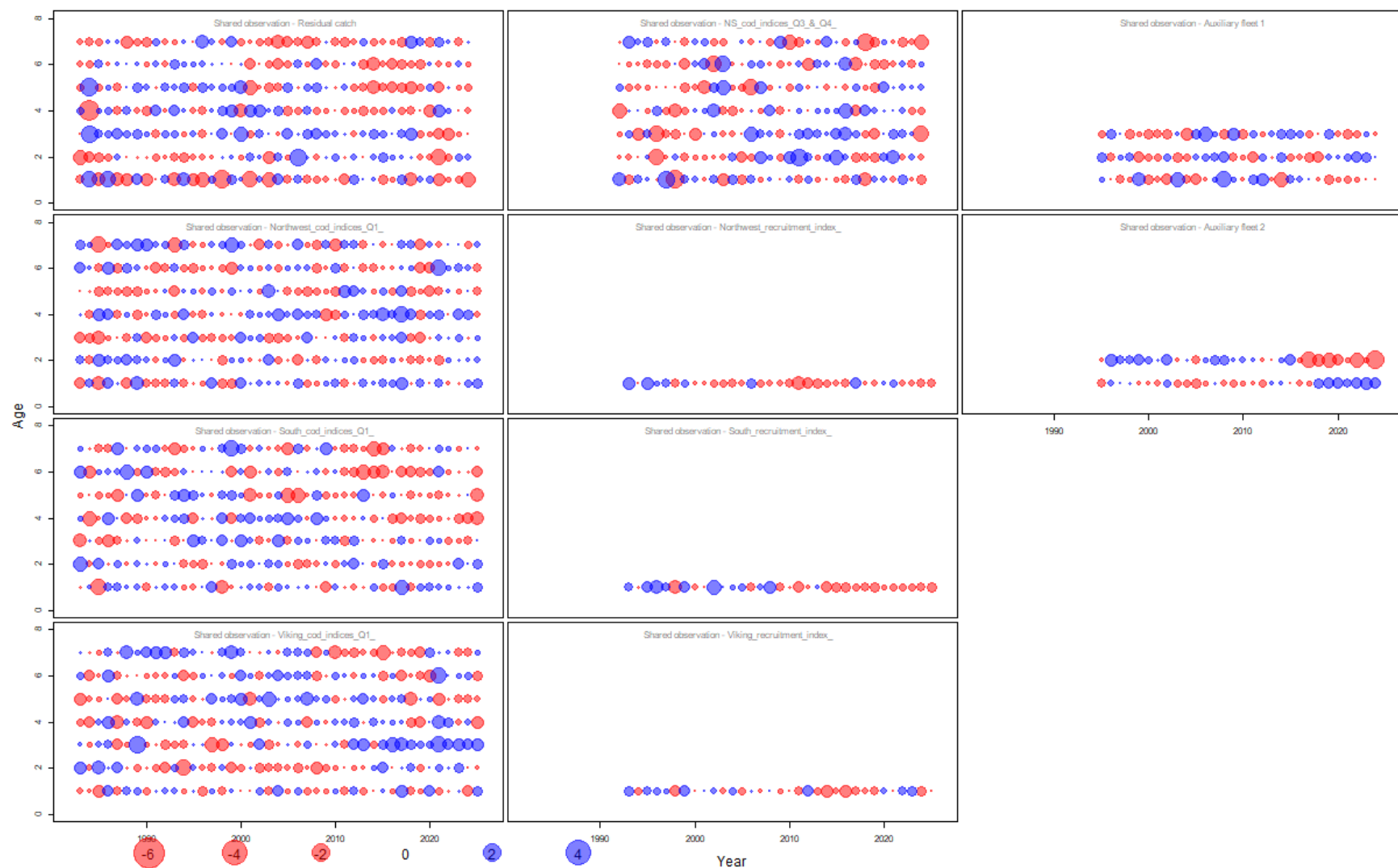


Figure 4.16. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. One step ahead (OSA) residuals for the multistock SAM assessment for total catch, seven survey tuning indices, landings proportions by quarter for all substocks combined (Auxiliary fleet 1) and by substock for Q1 (Auxiliary fleet 2). Blue circles indicate a positive residual and red circles a negative residual.

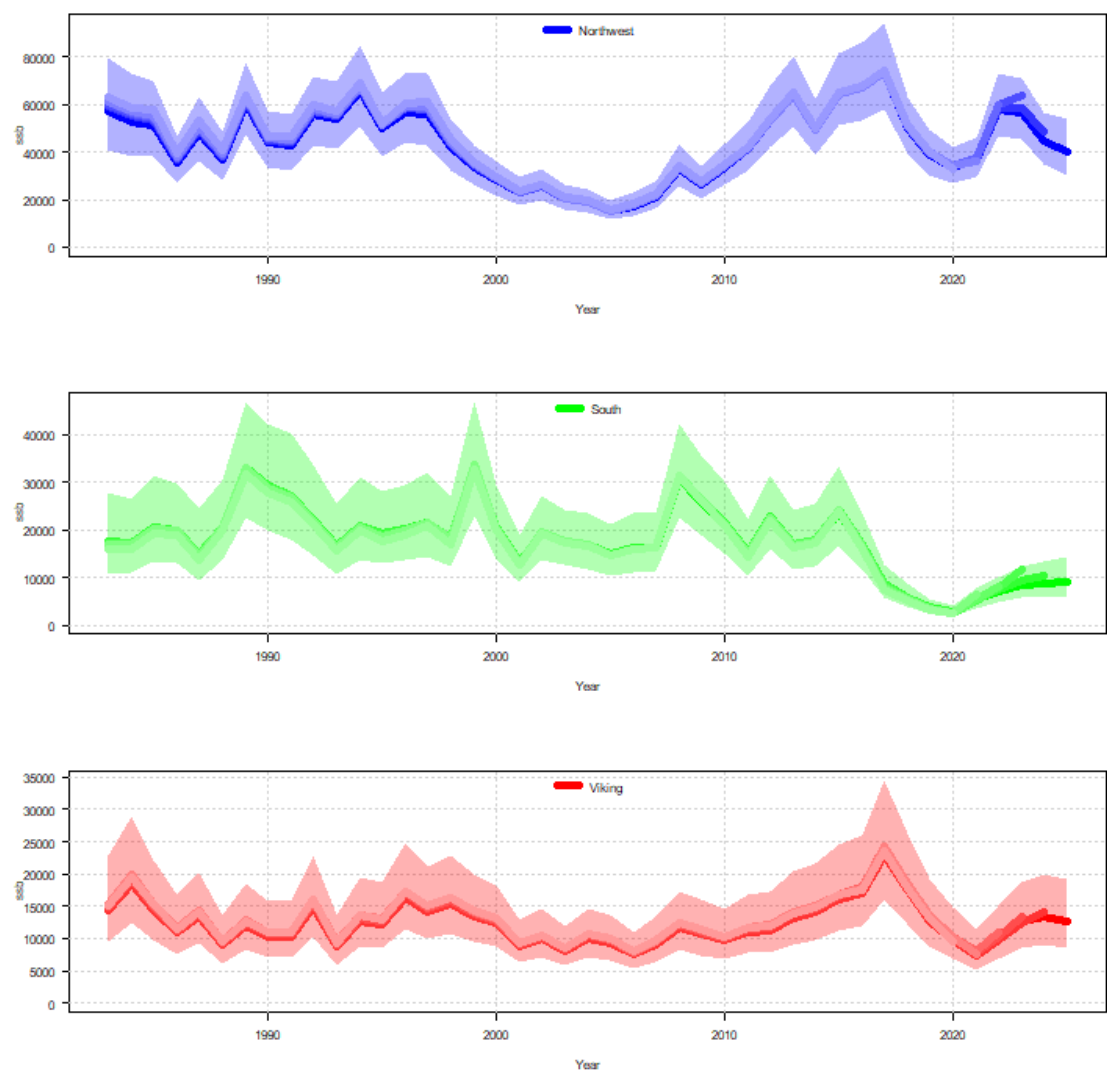


Figure 4.17a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (5 years) of SSB from the multistock SAM assessment for each substock, together with corresponding pointwise 95% confidence intervals.

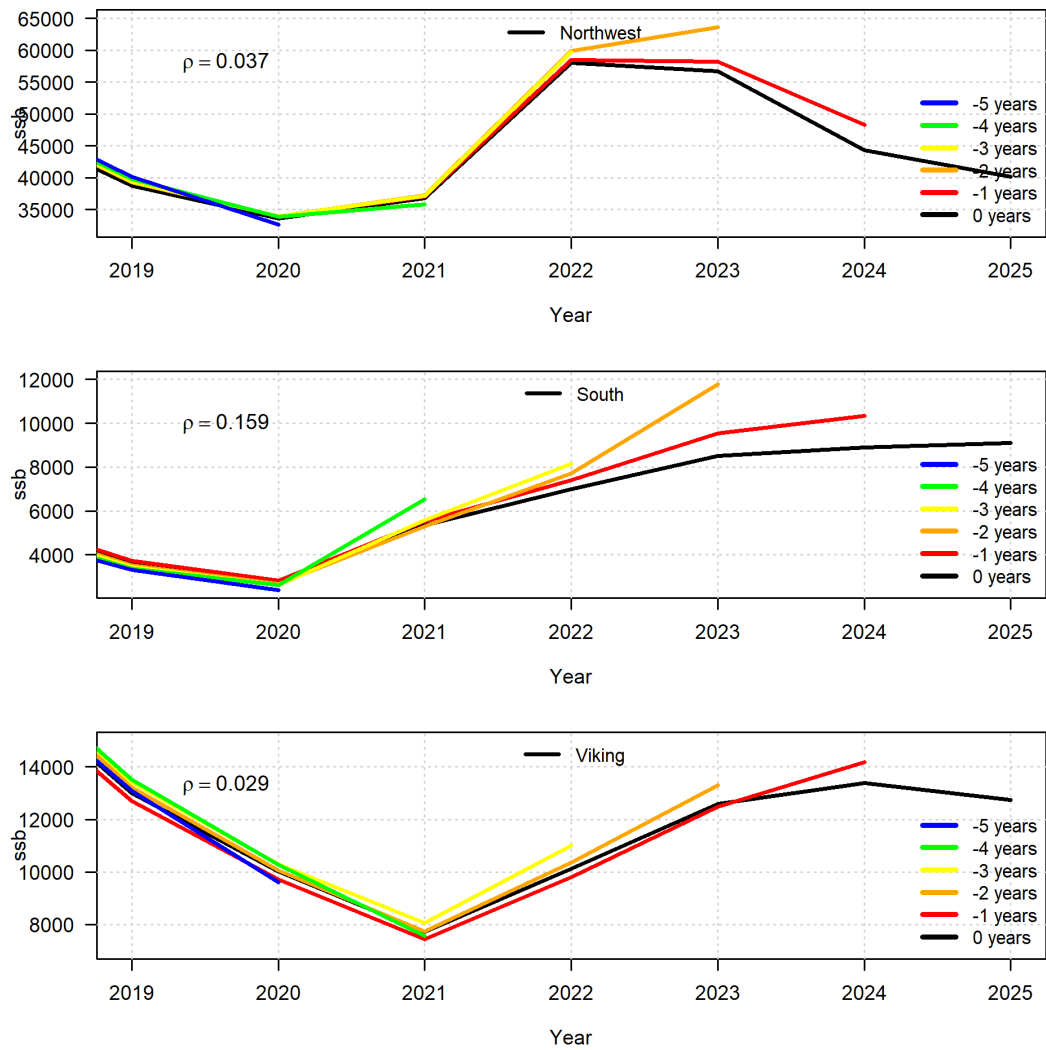


Figure 4.17b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (5 years) of SSB from the multistock SAM assessment for each substock (2019-2025).

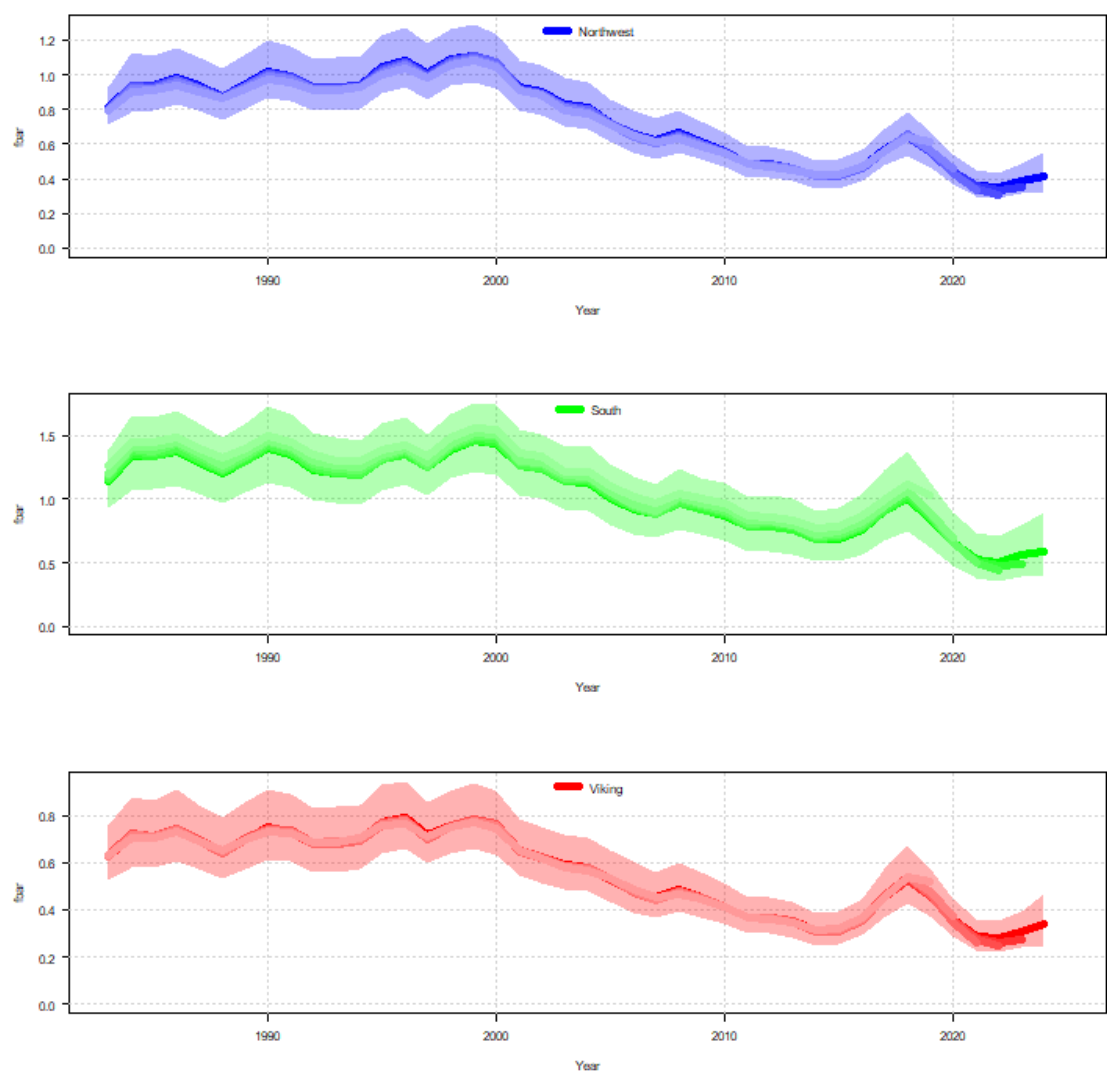


Figure 4.17c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (5 years) of F (2–4) from the multistock SAM assessment for each substock, together with corresponding pointwise 95% confidence intervals.

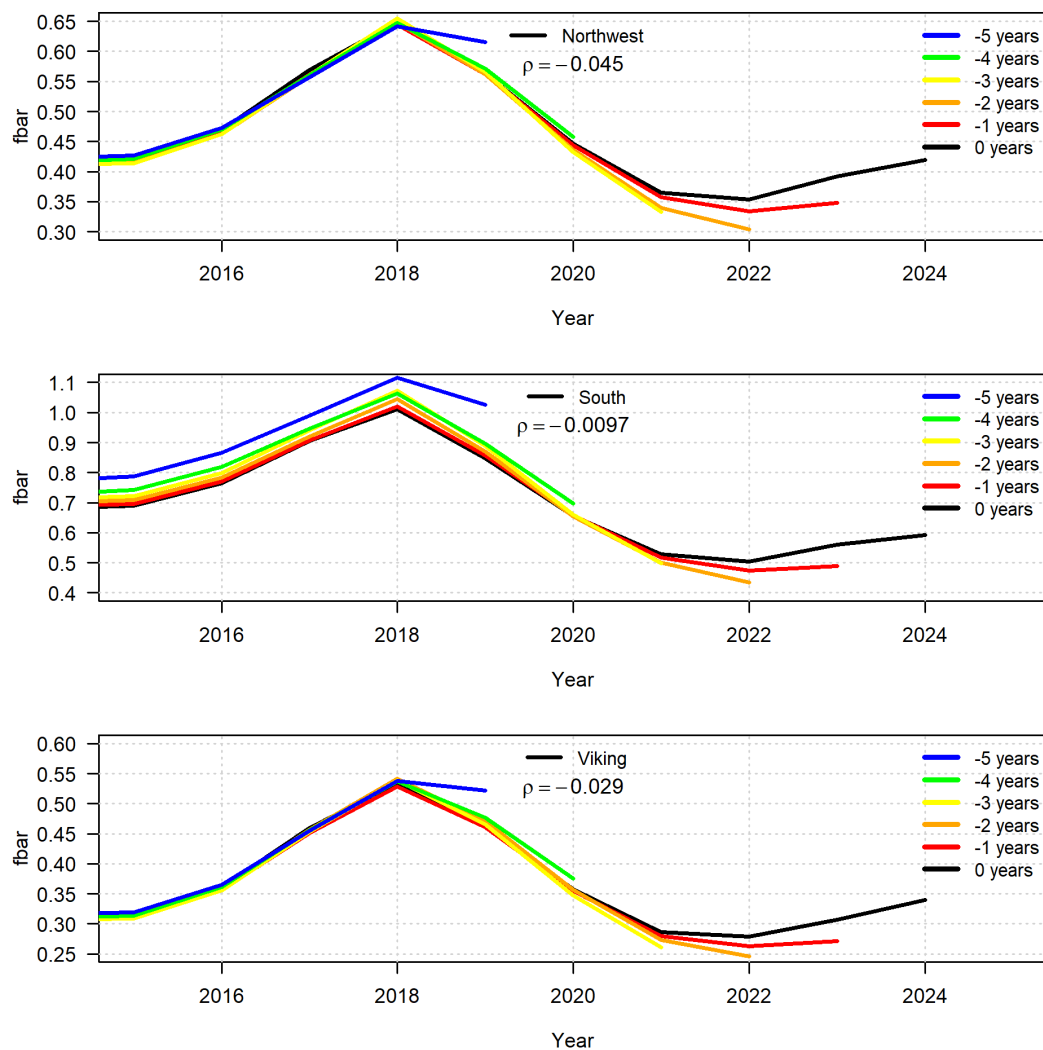


Figure 4.17d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (5 years) of $\bar{F}$ (2–4) from the multistock SAM assessment for each substock (2015–2024).

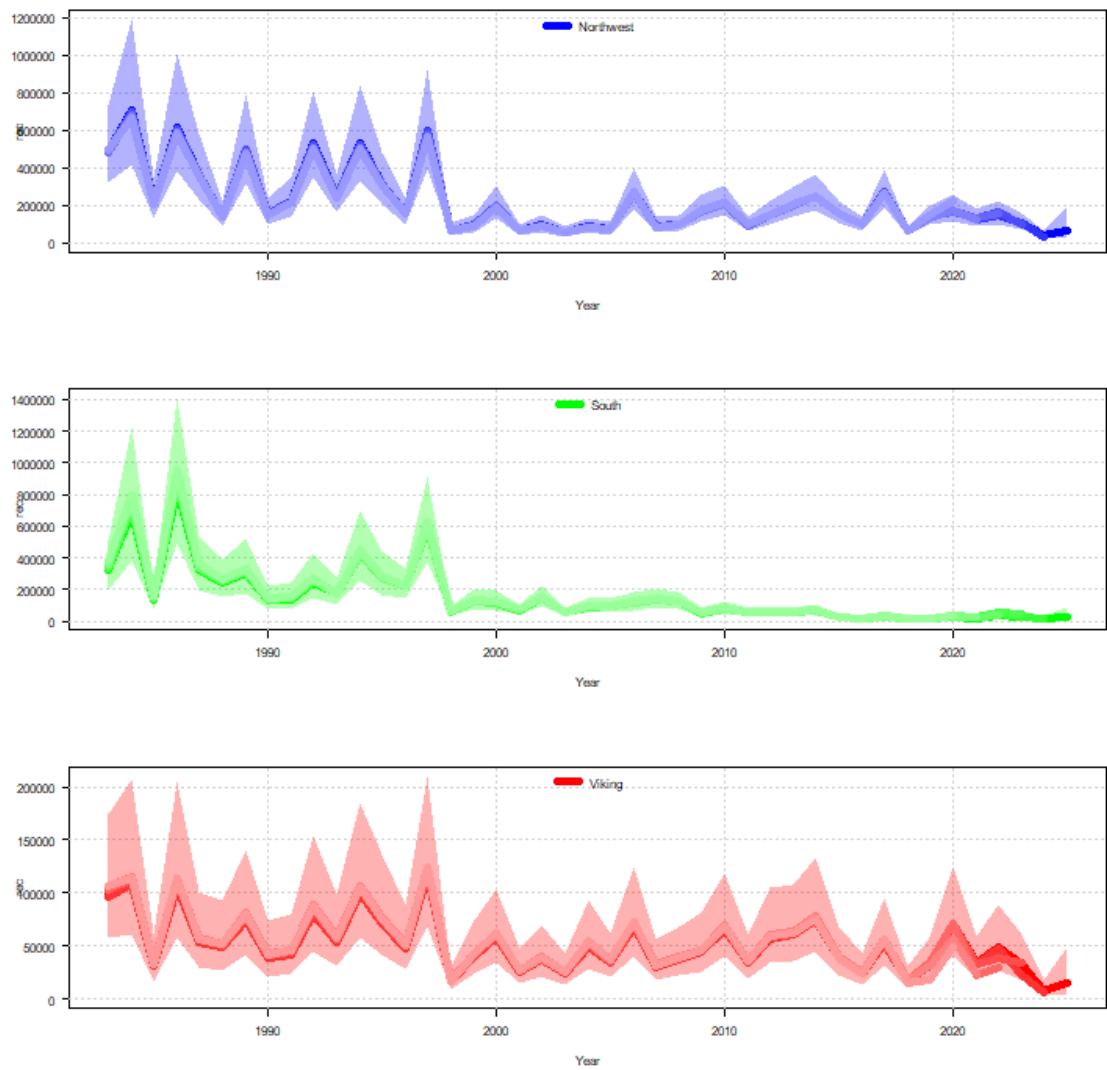


Figure 4.17e. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (5 years) of recruitment from the multistock SAM assessment for each substock, together with corresponding pointwise 95% confidence intervals.

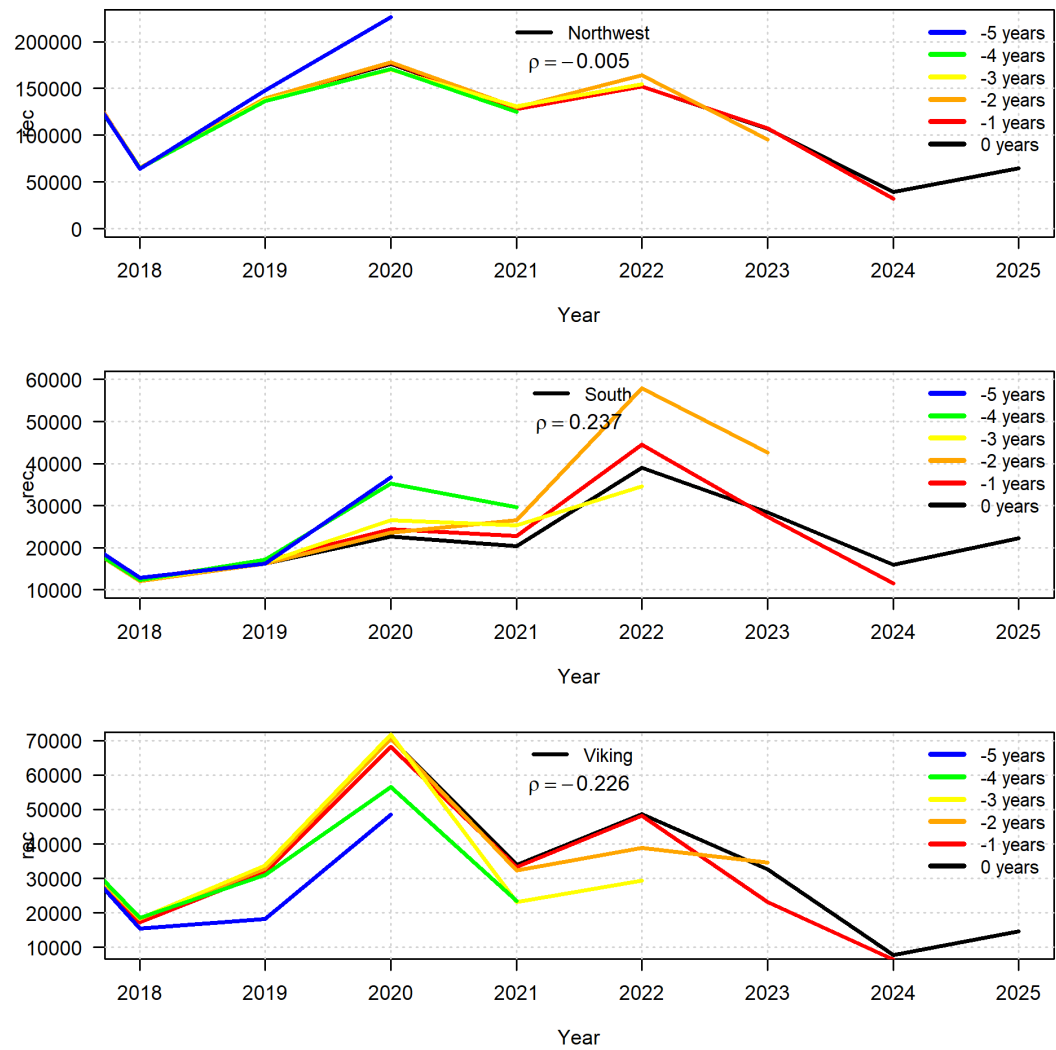


Figure 4.17f. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (5 years) of recruitment from the multistock SAM assessment for each substock (2018-2025).

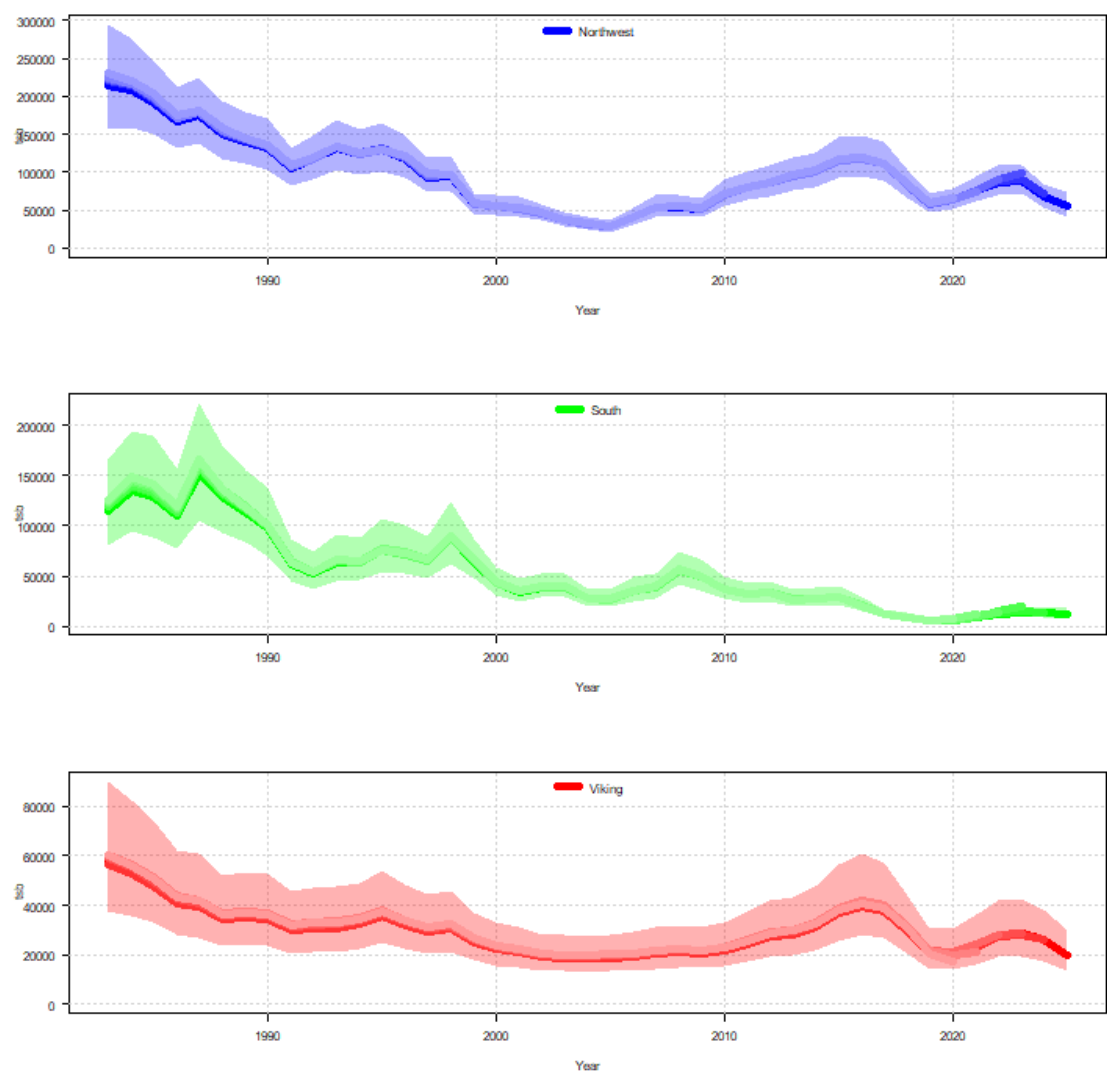


Figure 4.17g. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (5 years) of TSB from the multistock SAM assessment for each substock, together with corresponding pointwise 95% confidence intervals.

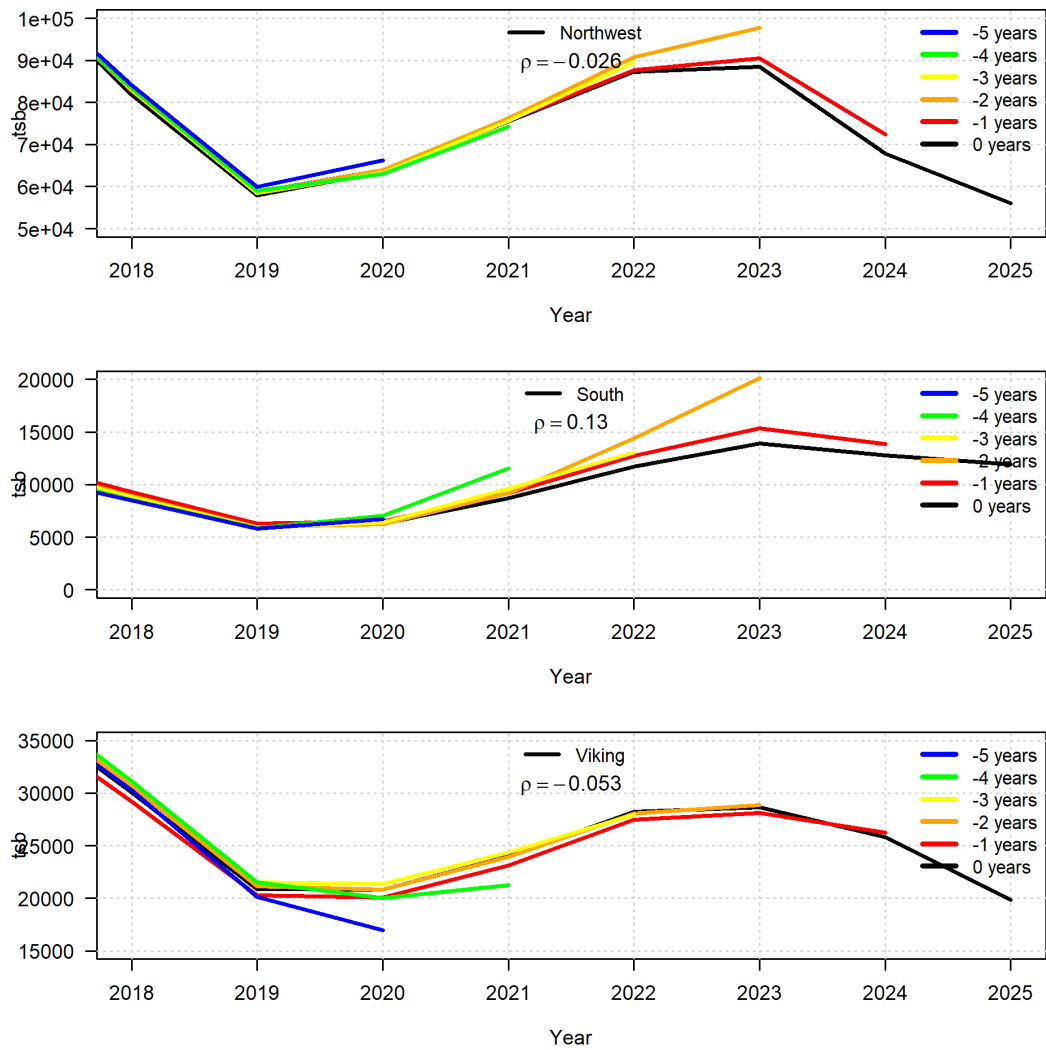


Figure 4.17h. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Retrospective estimates (5 years) of TSB from the multistock SAM assessment for each substock (2018-2025).

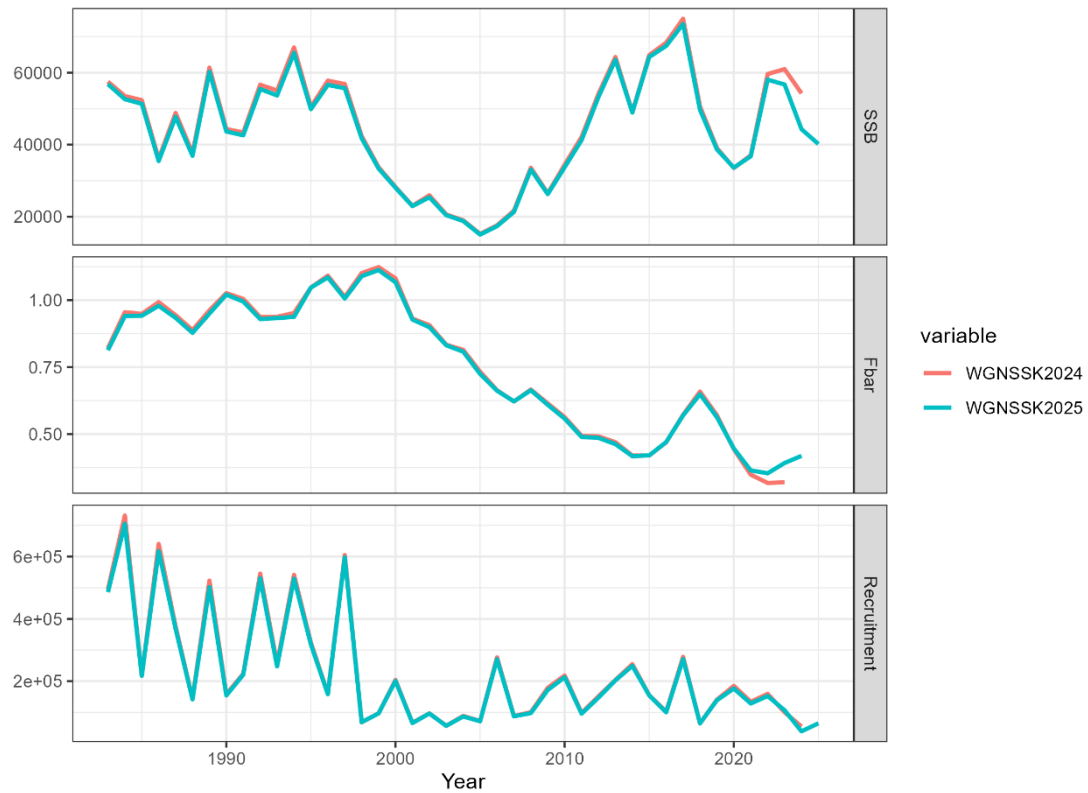


Figure 4.18a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of the final multistock SAM assessment for 2025 with the final multistock SAM assessment from 2024 (ICES WGNSSK 2024) for the Northwestern substock. Estimated yearly SSB (top), average fishing mortality (middle) and recruitment age 1 (bottom).

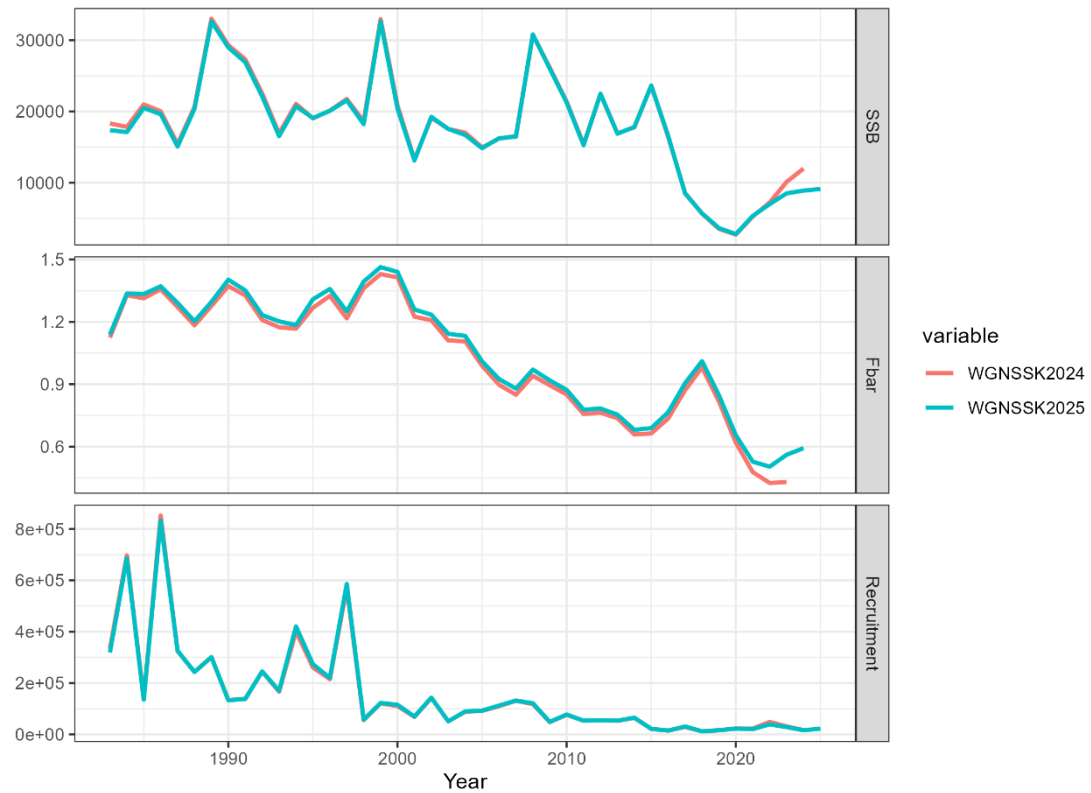


Figure 4.18b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of the final multistock SAM assessment for 2025 with the final multistock SAM assessment from 2024 (ICES WGNSSK 2024) for the Southern substock. Estimated yearly SSB (top), average fishing mortality (middle) and recruitment age 1 (bottom).

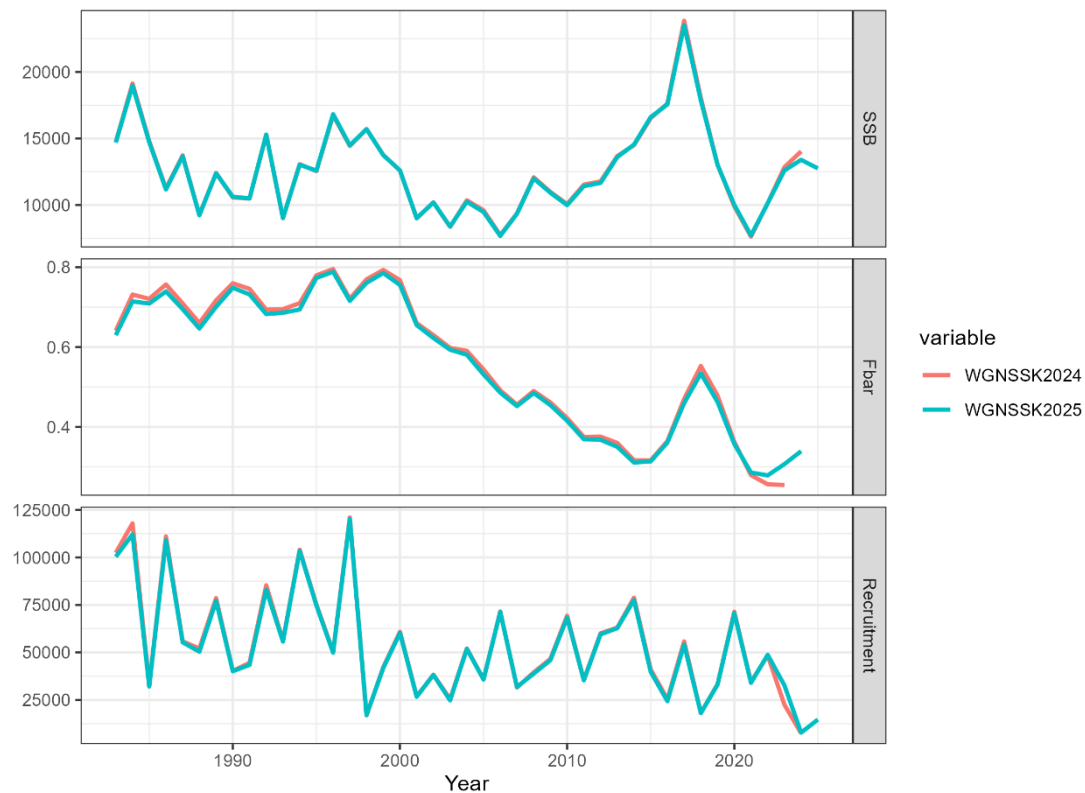


Figure 4.18c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of the final multistock SAM assessment for 2025 with the final multistock SAM assessment from 2024 (ICES WGNSSK 2024) for the Viking substock. Estimated yearly SSB (top), average fishing mortality (middle) and recruitment age 1 (bottom).

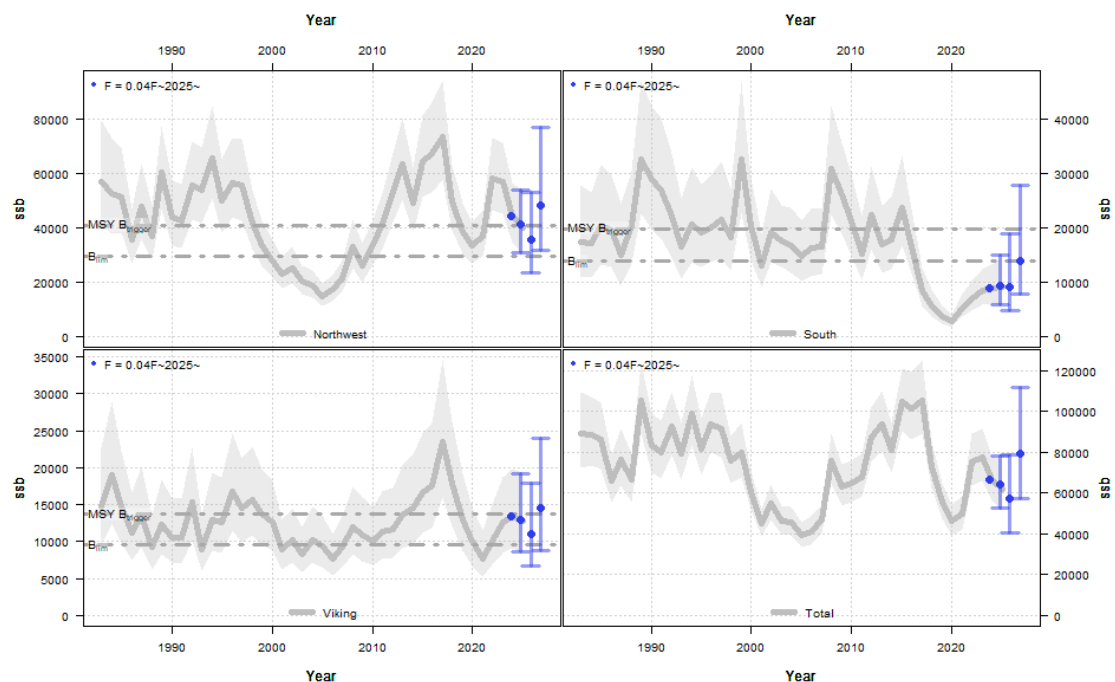


Figure 4.19a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Forecasted SSB for each substock and the total stock combined following the MSY approach for the Southern substock and precautionary considerations to protect the Southern substock for the Northwestern and Viking substocks.

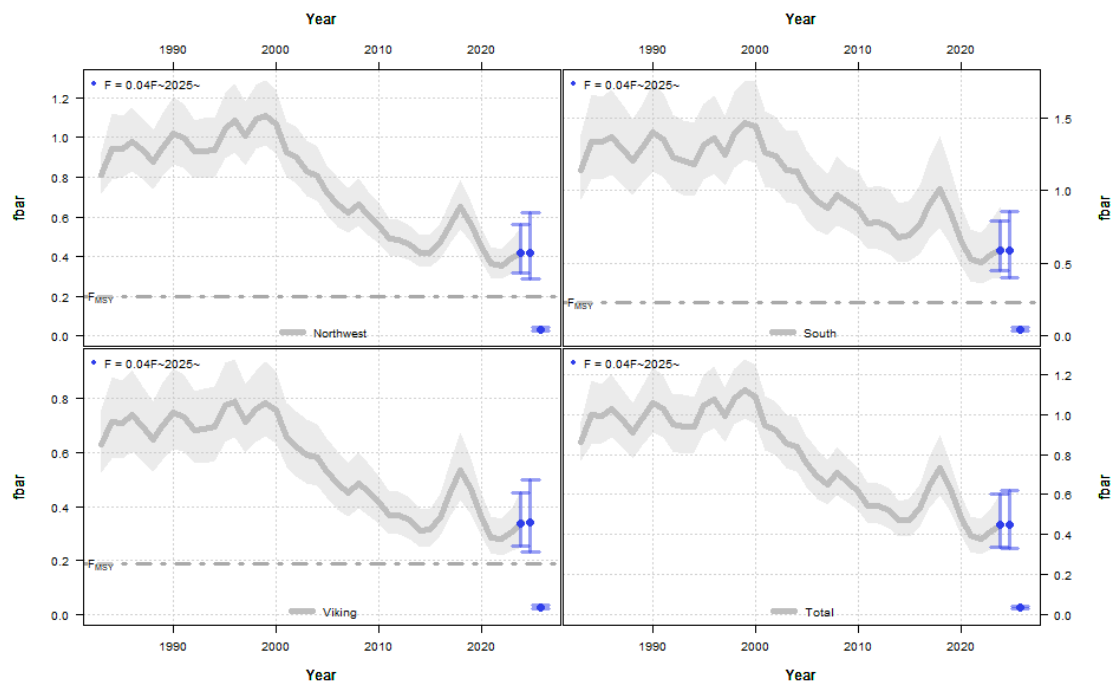


Figure 4.19b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Forecasted $F(2-4)$ for each substock and the total stock combined following the MSY approach for the Southern substock and precautionary considerations to protect the Southern substock for the Northwestern and Viking substocks.

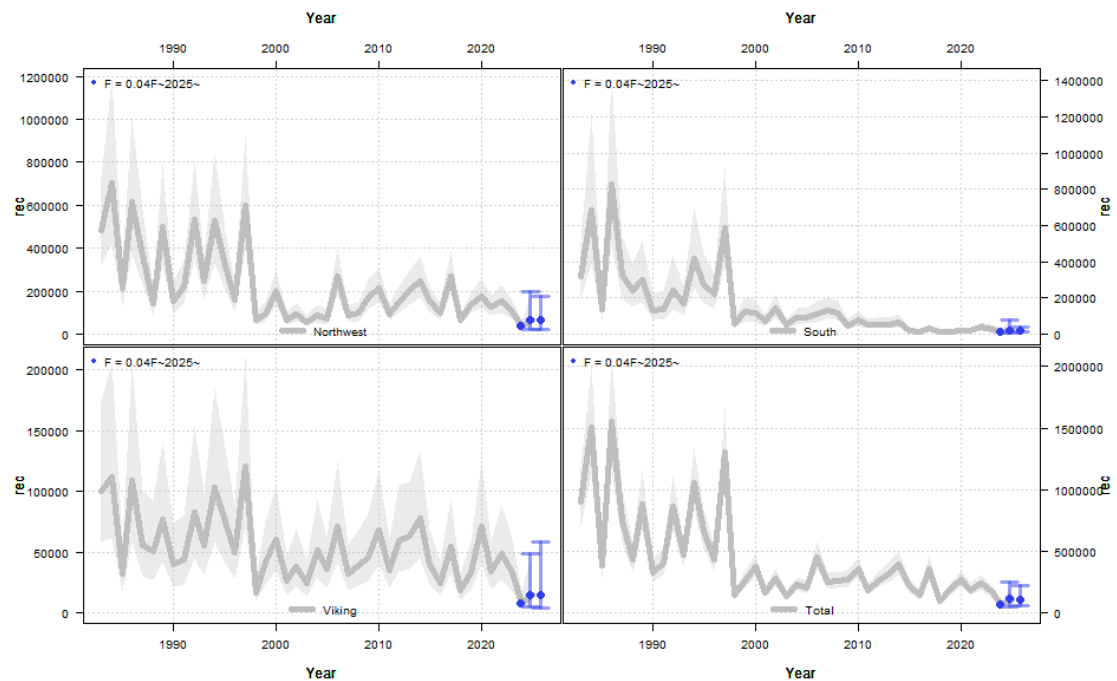


Figure 4.19c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Forecasted recruitment for each substock and the total stock combined following the MSY approach for the Southern substock and precautionary considerations to protect the Southern substock for the Northwestern and Viking substocks.

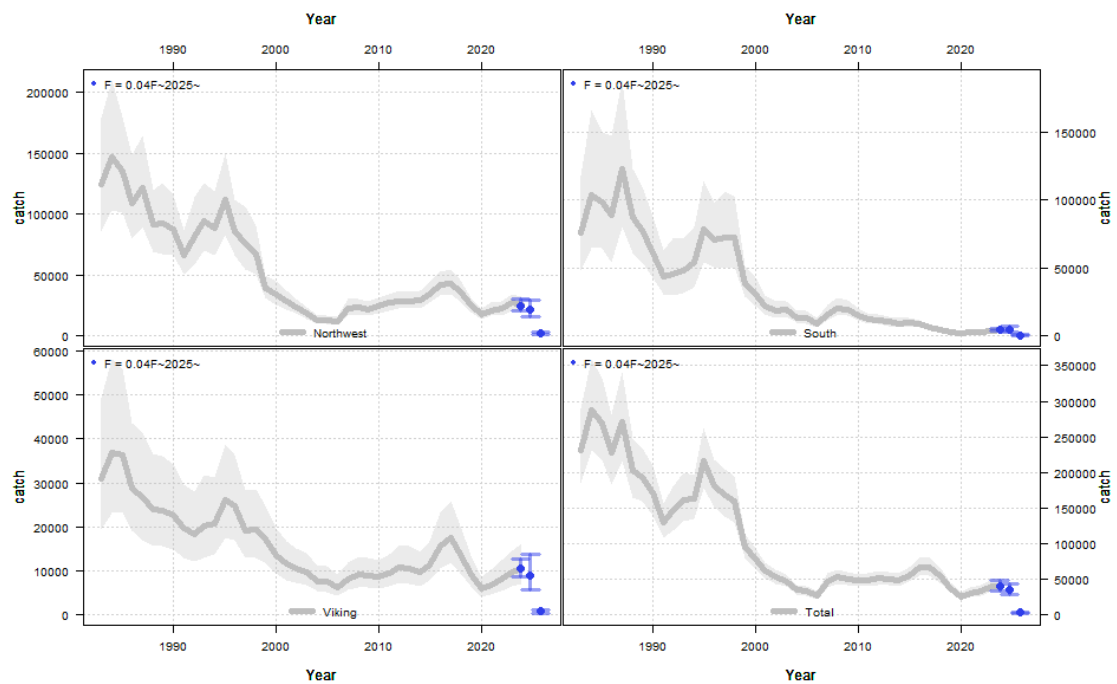


Figure 4.19d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Forecasted catches for each substock and the total stock combined following the MSY approach for the Southern substock and precautionary considerations to protect the Southern substock for the Northwestern and Viking substocks.

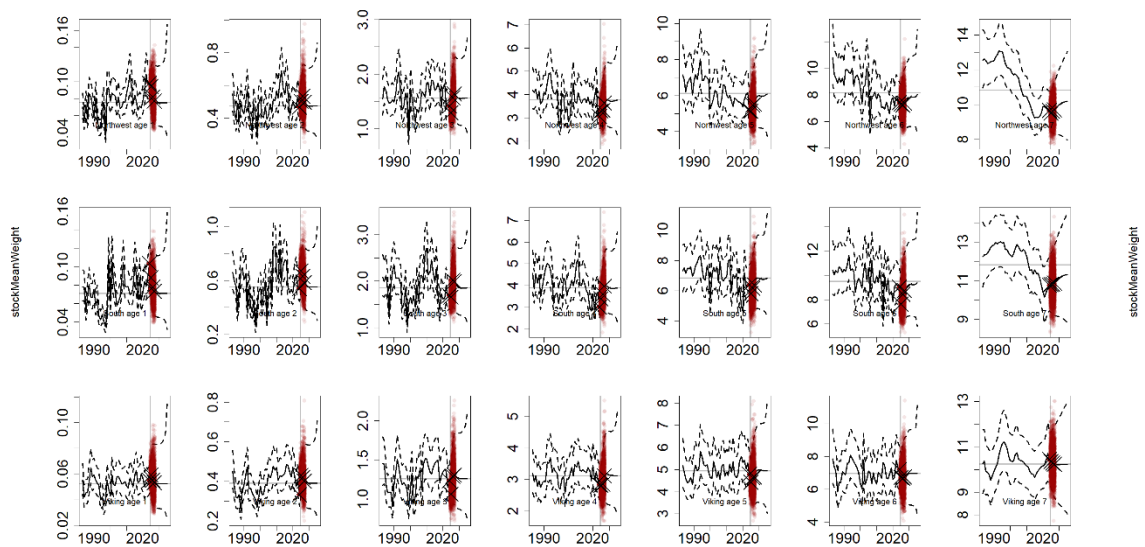


Figure 4.20a. Simulated forecasted stock mean weights for ages 1-7 (red dots, median: black cross) compared to the historical and most likely projected trajectory of the fitted GMRf from the assessment model (Northwest substock top row, south, middle row and Viking lower row, each column represents each age 1-7).

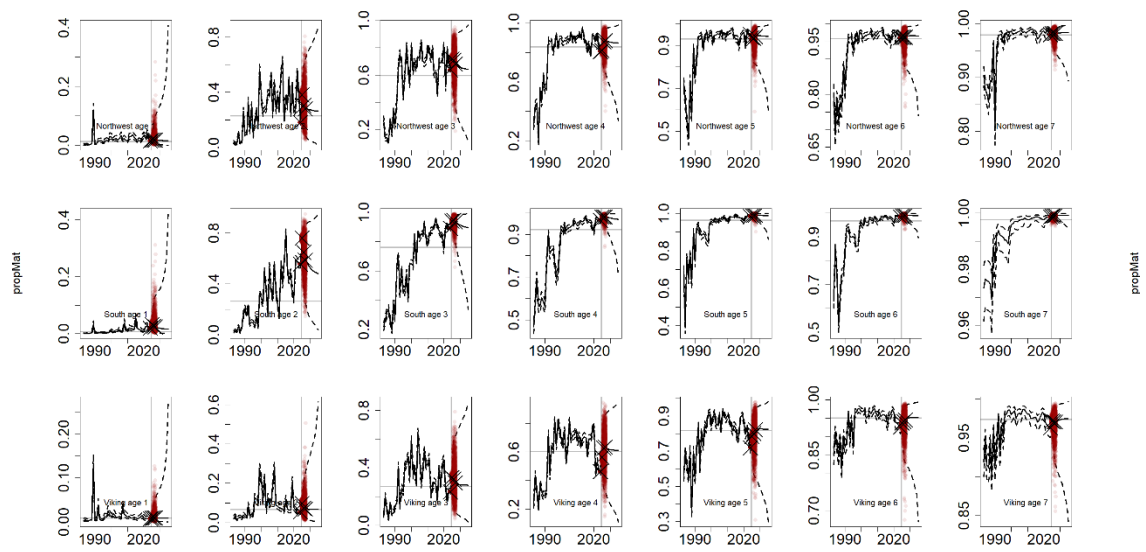


Figure 4.20b. Simulated forecasted proportion mature for ages 1-7 (red dots, median: black cross) compared to the historical and most likely projected trajectory of the fitted GMRF from the assessment model (Northwest substock top row, south, middle row and Viking lower row, each column represents each age 1-7).

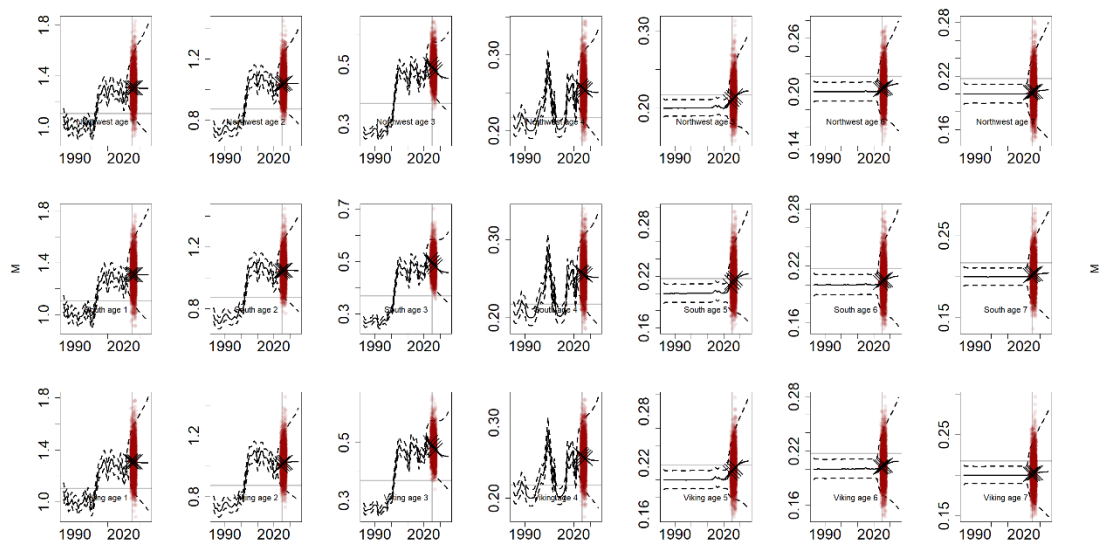


Figure 4.20c. Simulated forecasted natural mortality for ages 1-7 (red dots, median: black cross) compared to the historical and most likely projected trajectory of the fitted GMRF from the assessment model (Northwest substock top row, south middle row and Viking lower row, each column represents each age 1-7).

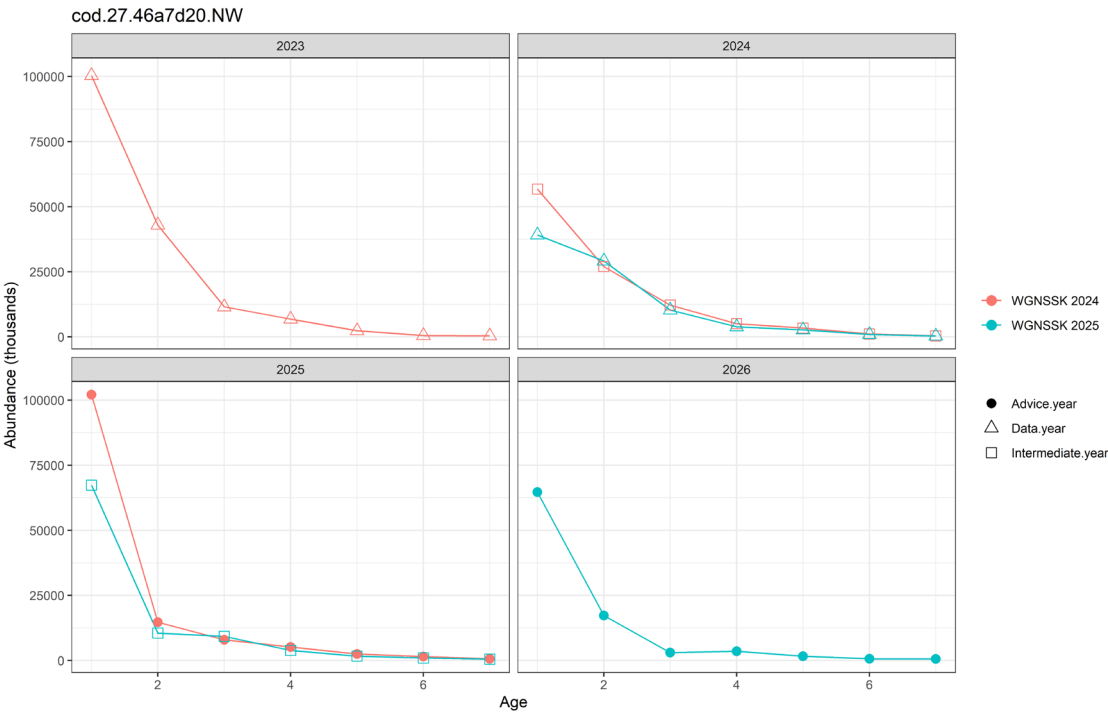


Figure 4.21a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of numbers-at-age from previous stock assessment and forecast results (WGNSSK 2024) and current stock assessment and forecast results (WGNSSK 2025) for the Northwestern substock.

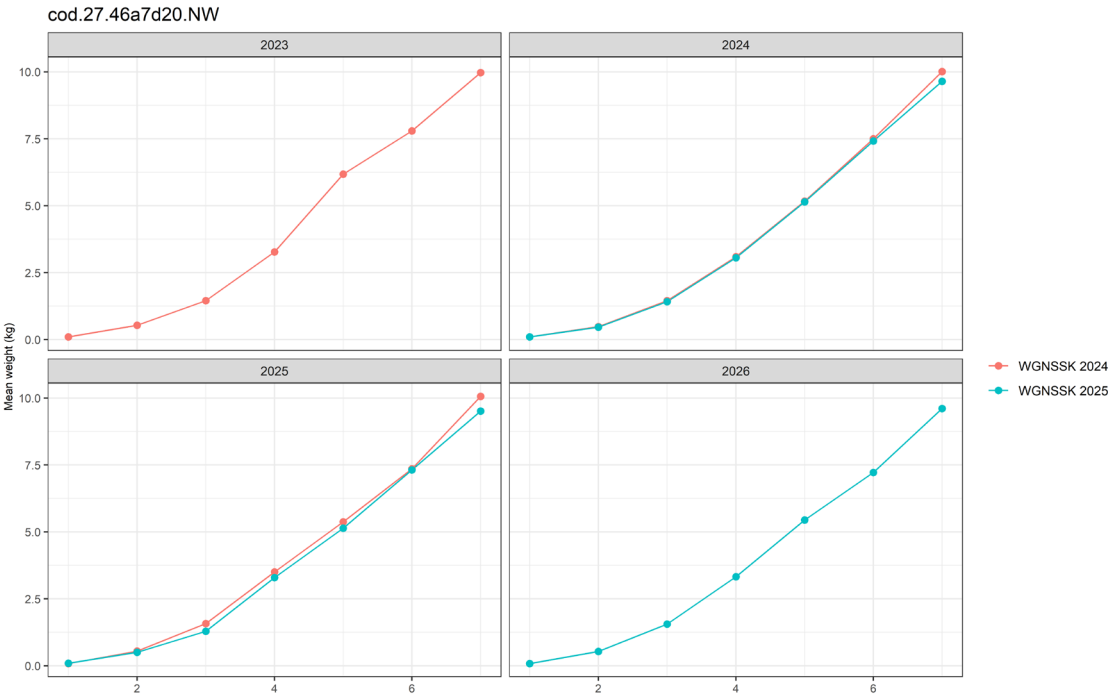


Figure 4.21b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of mean weights-at-age in the stock from the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025) for the Northwestern substock.

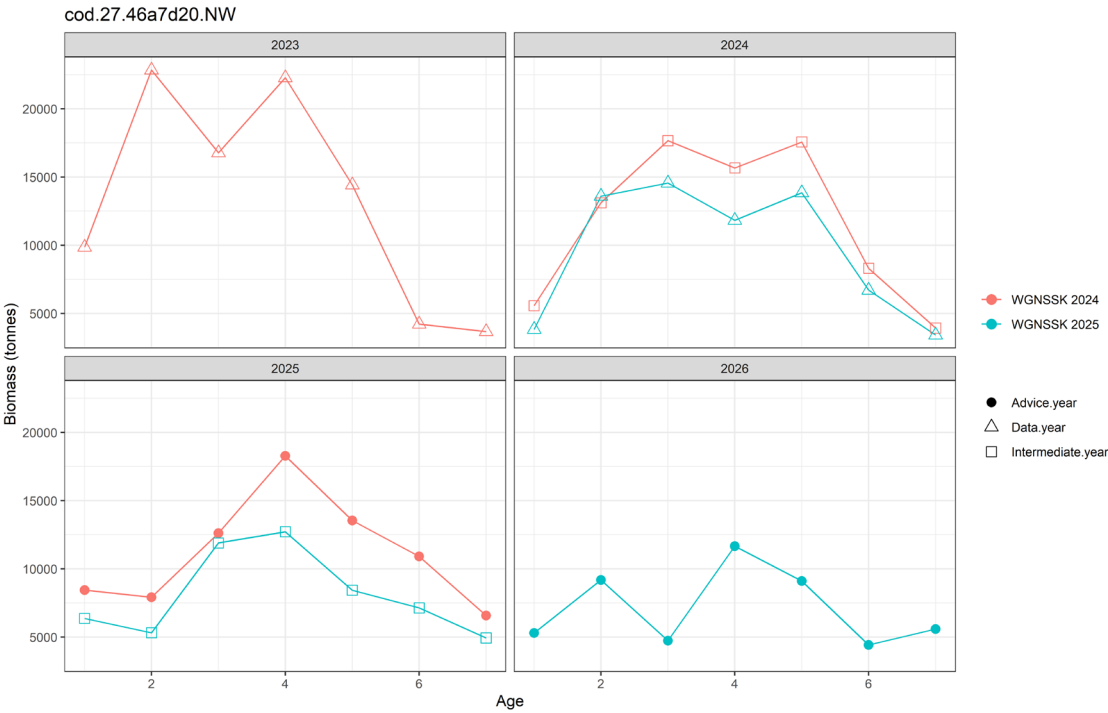


Figure 4.21c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of biomass-at-age from previous stock assessment and forecast results (WGNSK 2024) and current stock assessment and forecast results (WGNSK 2025) for the Northwestern substock.

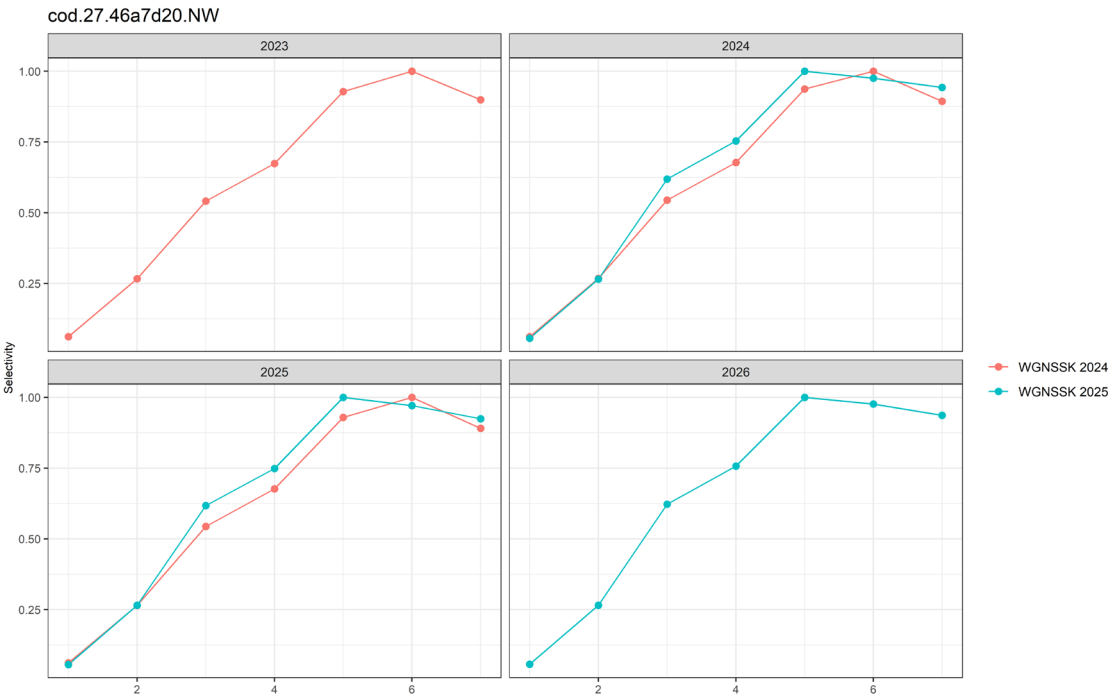


Figure 4.21d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of selectivity used in the previous forecast (WGNSK 2024) and current forecast (WGNSK 2025) for the Northwestern substock.

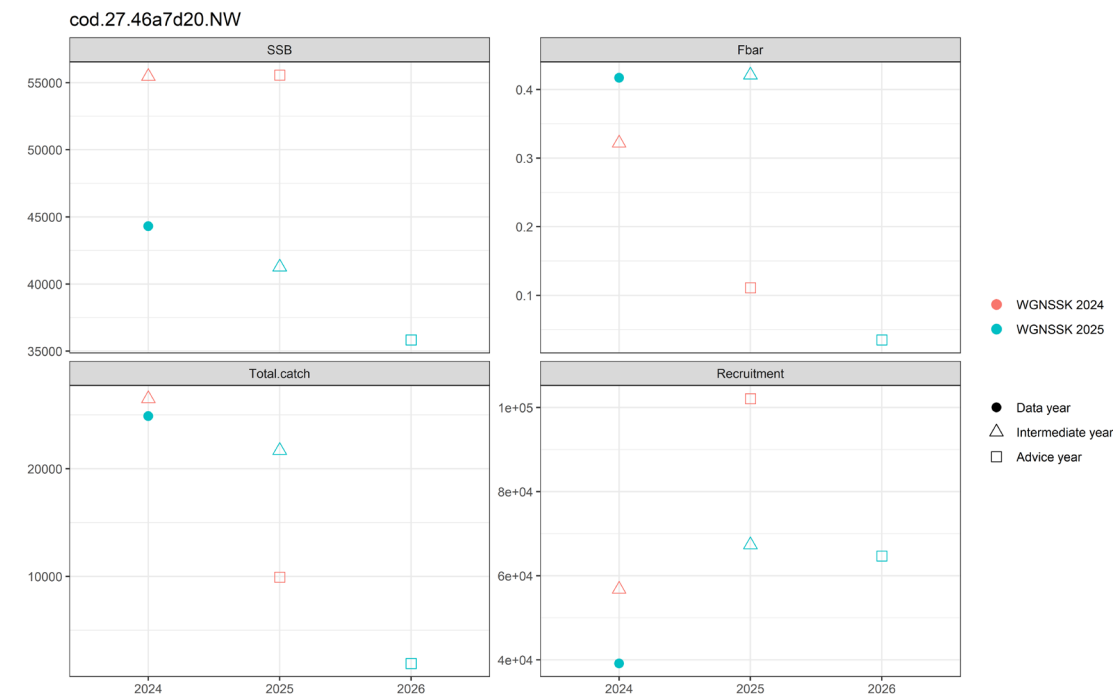


Figure 4.21e. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of SSB, Fbar, total catch and recruitment assumptions used in the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025) for the North-western substock.

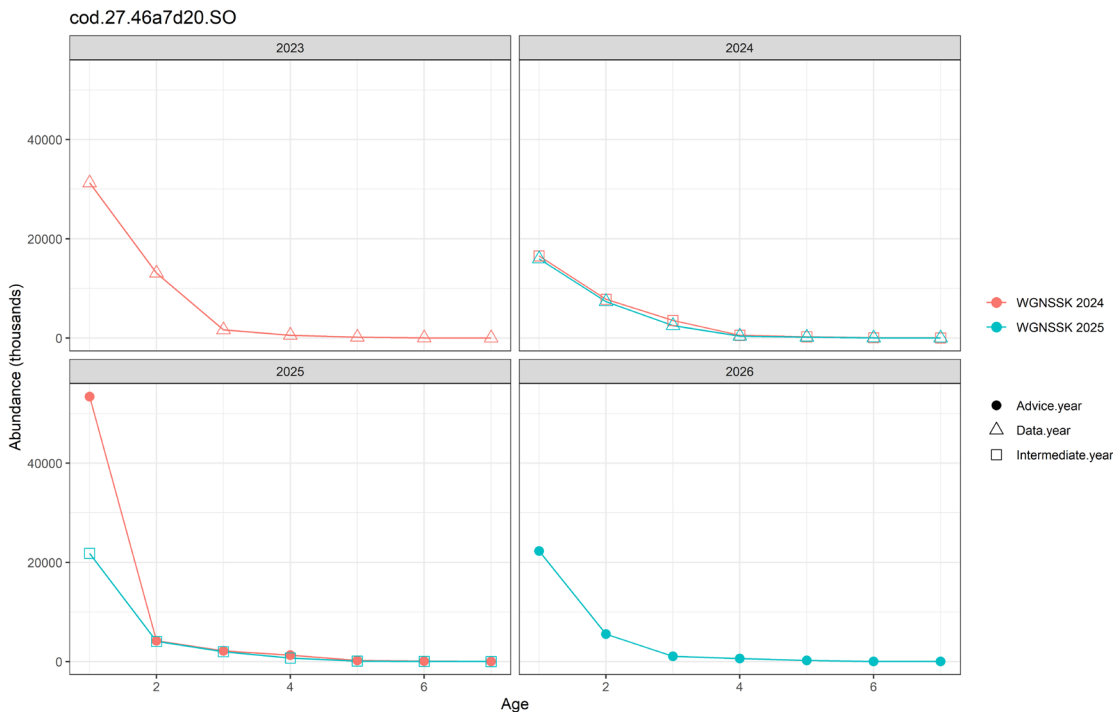


Figure 4.22a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of numbers-at-age from previous stock assessment and forecast results (WGNSSK 2024) and current stock assessment and forecast results (WGNSSK 2025) for the Southern substock.

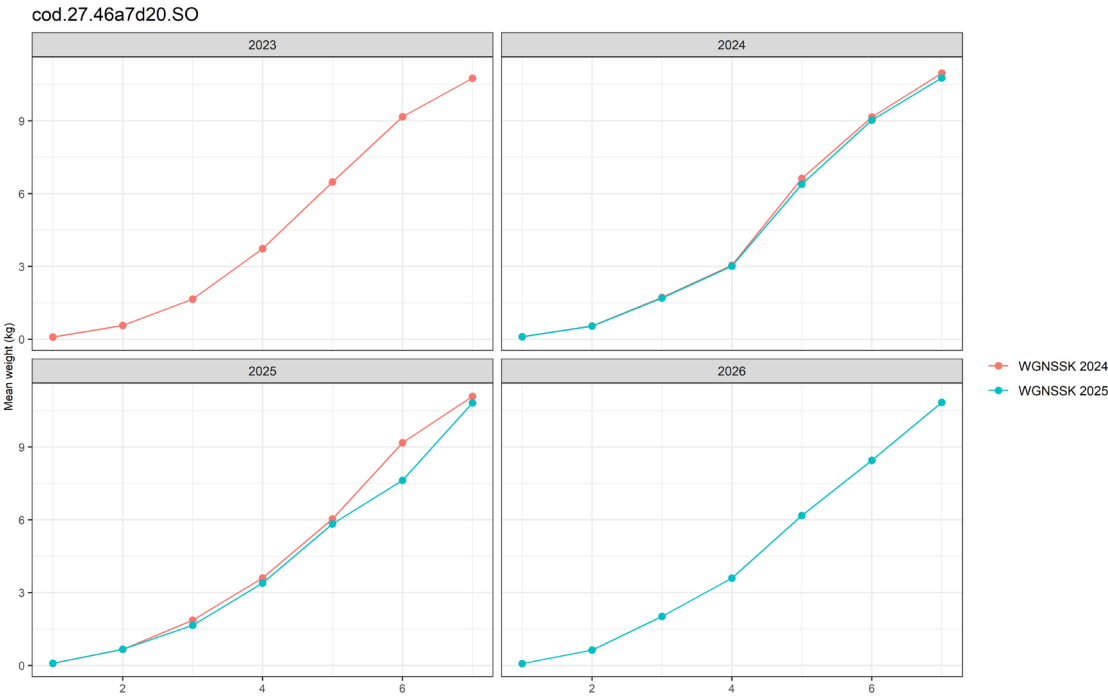


Figure 4.22b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of mean weights-at-age in the stock from the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025) for the Southern substock.

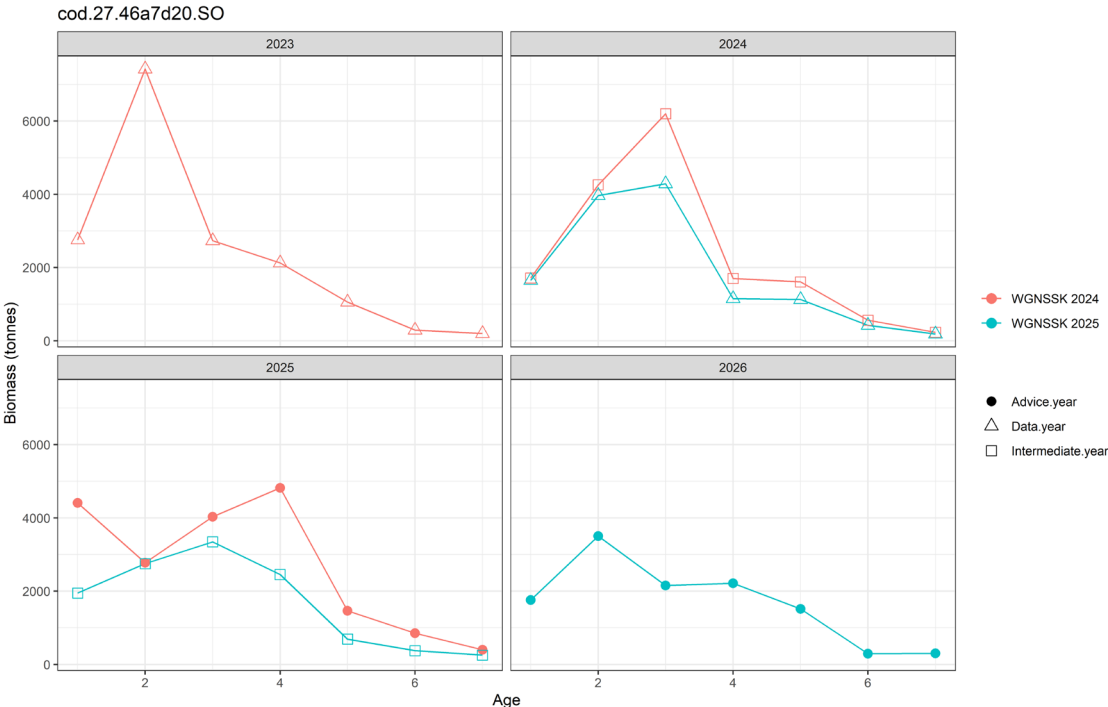


Figure 4.22c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of biomass-at-age from previous stock assessment and forecast results (WGNSSK 2024) and current stock assessment and forecast results (WGNSSK 2025) for the Southern substock.

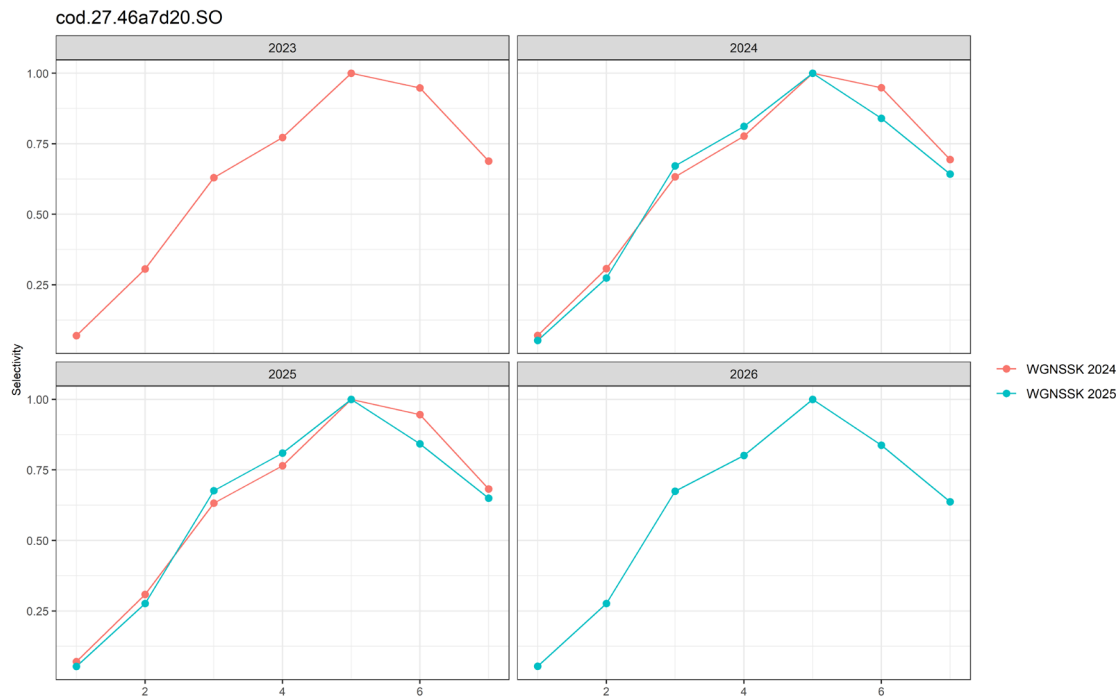


Figure 4.22d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of selectivity used in the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025) for the Southern substock.

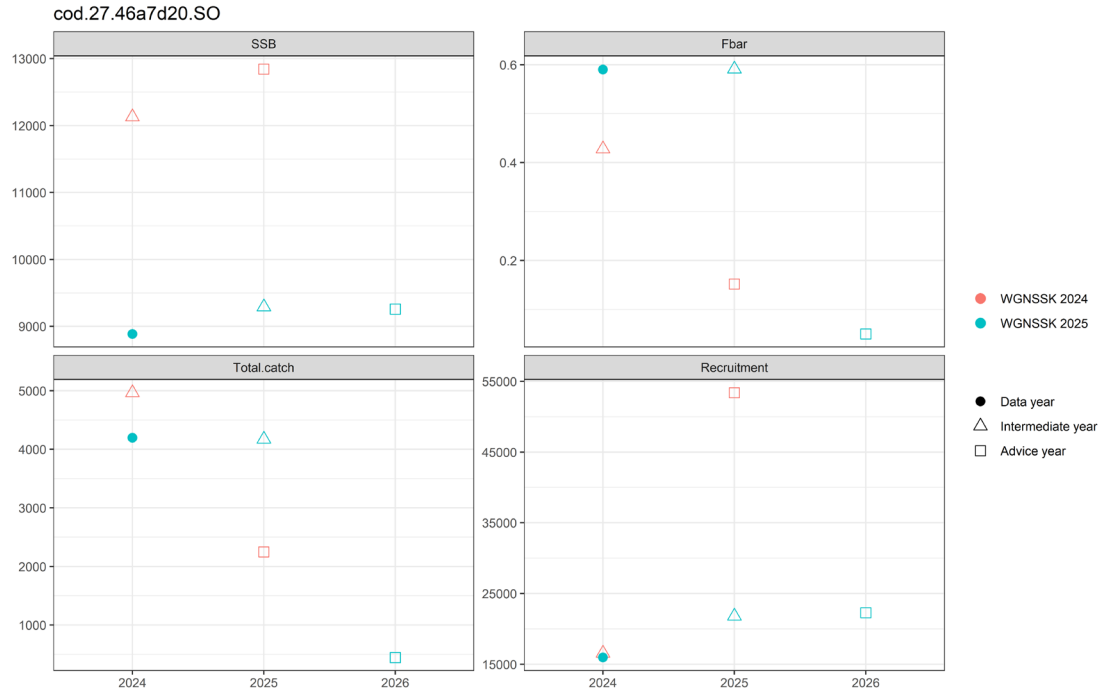


Figure 4.22e. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of SSB, Fbar, total catch and recruitment assumptions used in the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025) for the Southern substock.

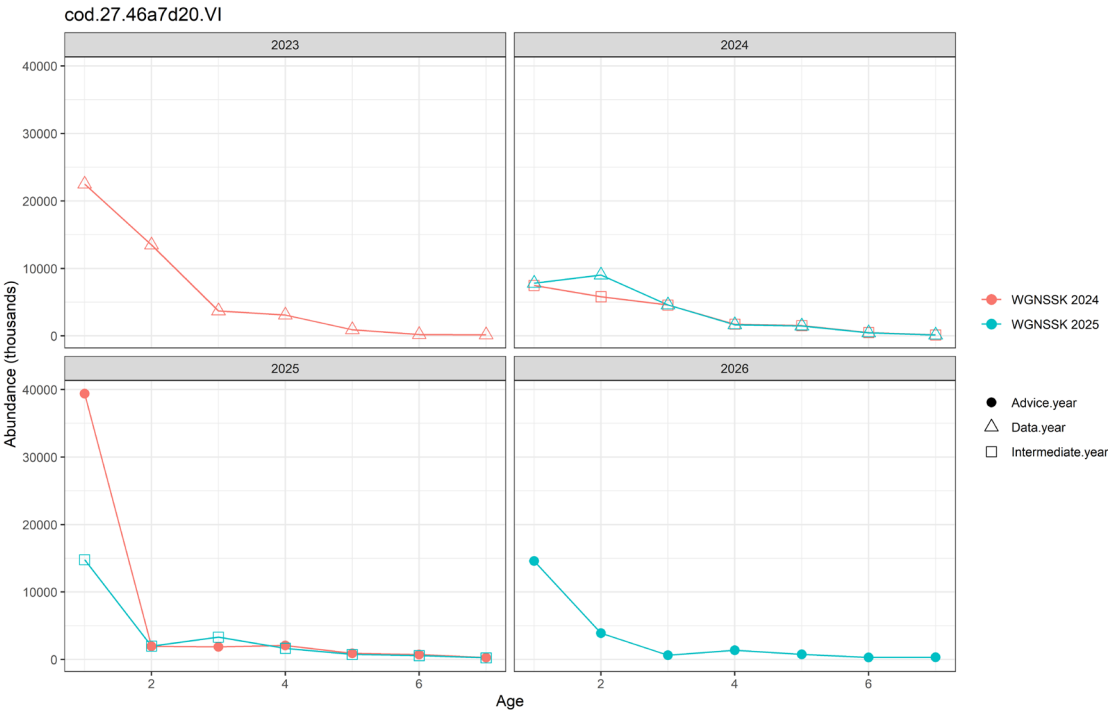


Figure 4.23a. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of numbers-at-age from previous stock assessment and forecast results (WGNSSK 2024) and current stock assessment and forecast results (WGNSSK 2025) for the Viking substock.

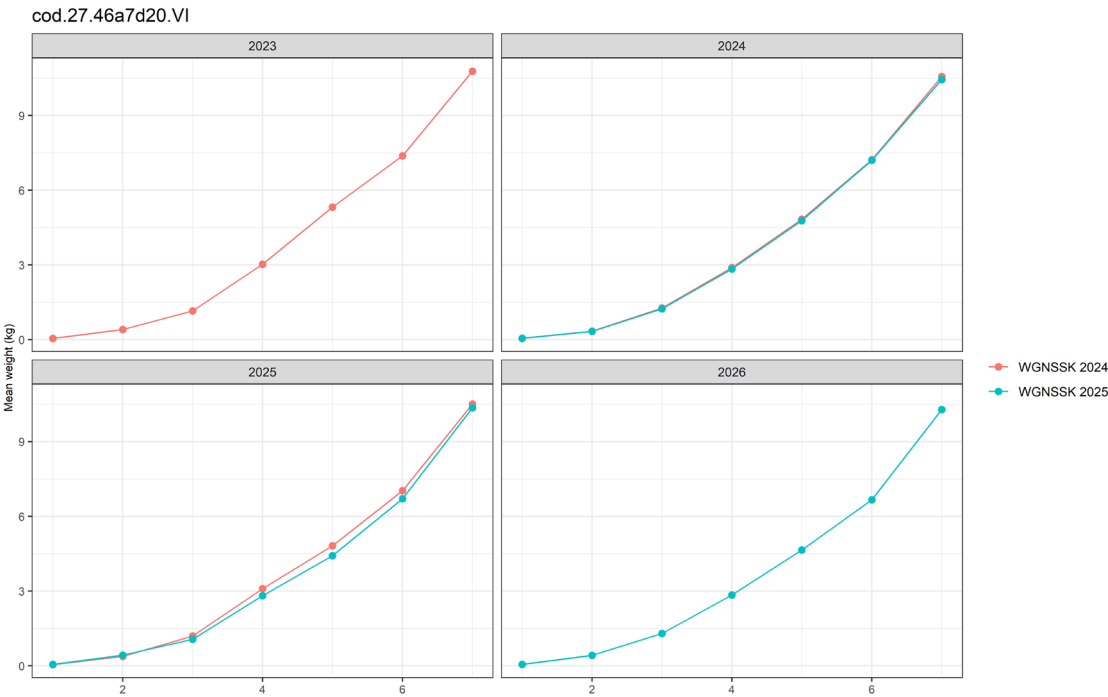


Figure 4.23b. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of mean weights-at-age in the stock from the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025) for the Viking substock.

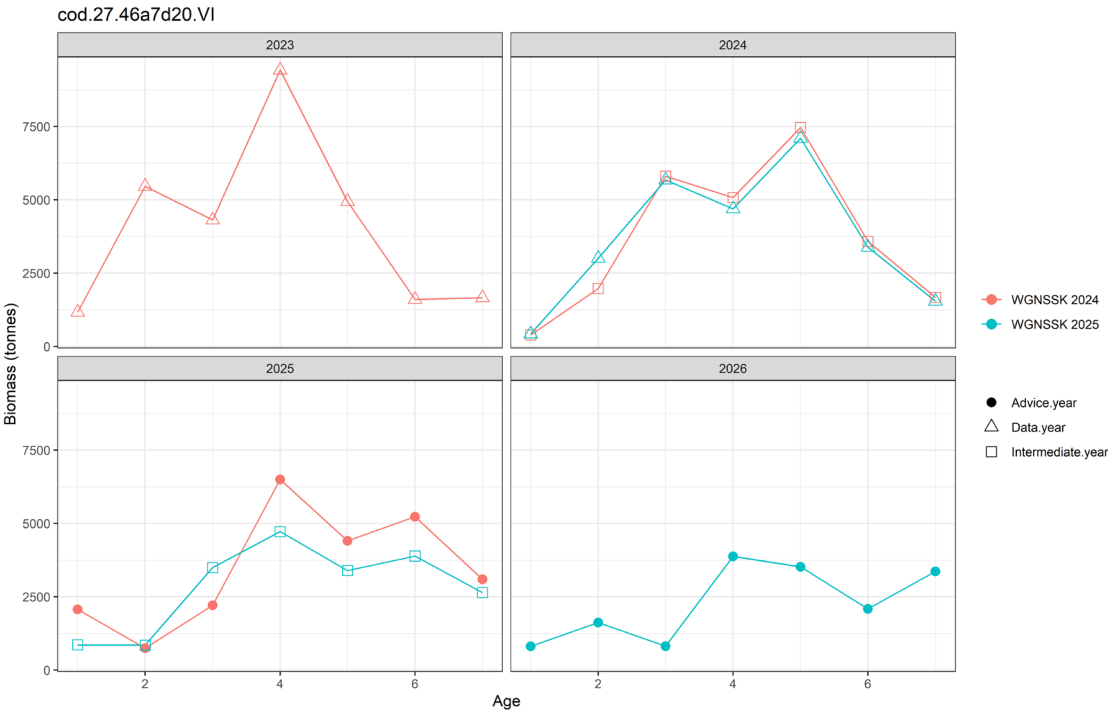


Figure 4.23c. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of biomass-at-age from previous stock assessment and forecast results (WGNSSK 2024) and current stock assessment and forecast results (WGNSSK 2025) for the Viking substock.

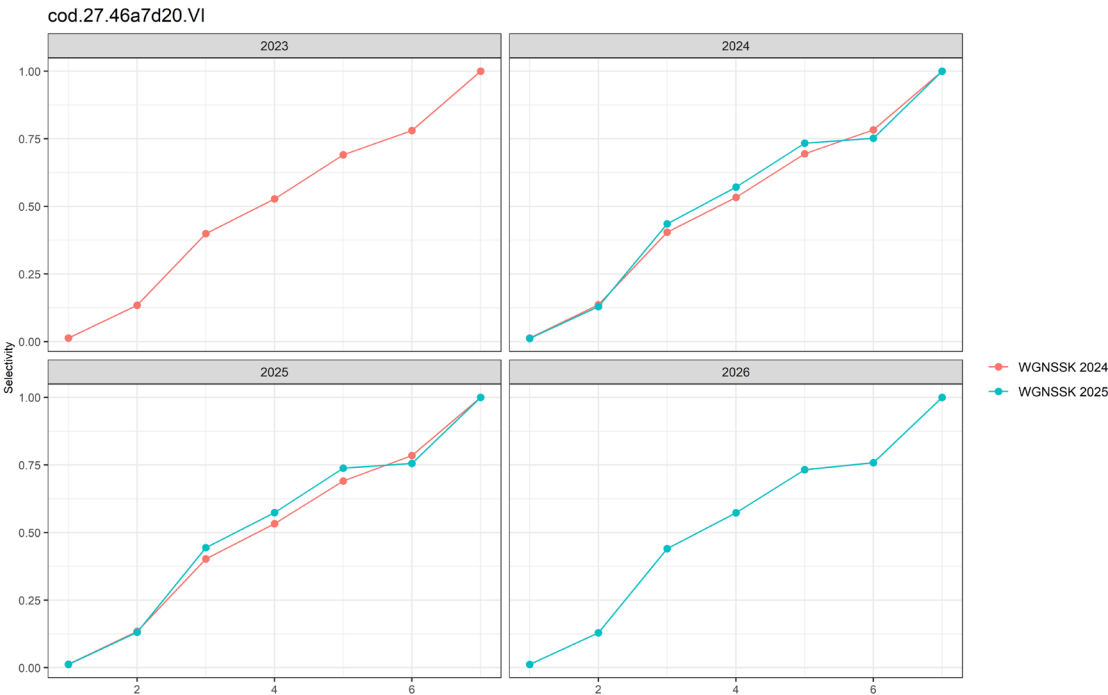


Figure 4.23d. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of selectivity used in the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025) for the Viking substock.

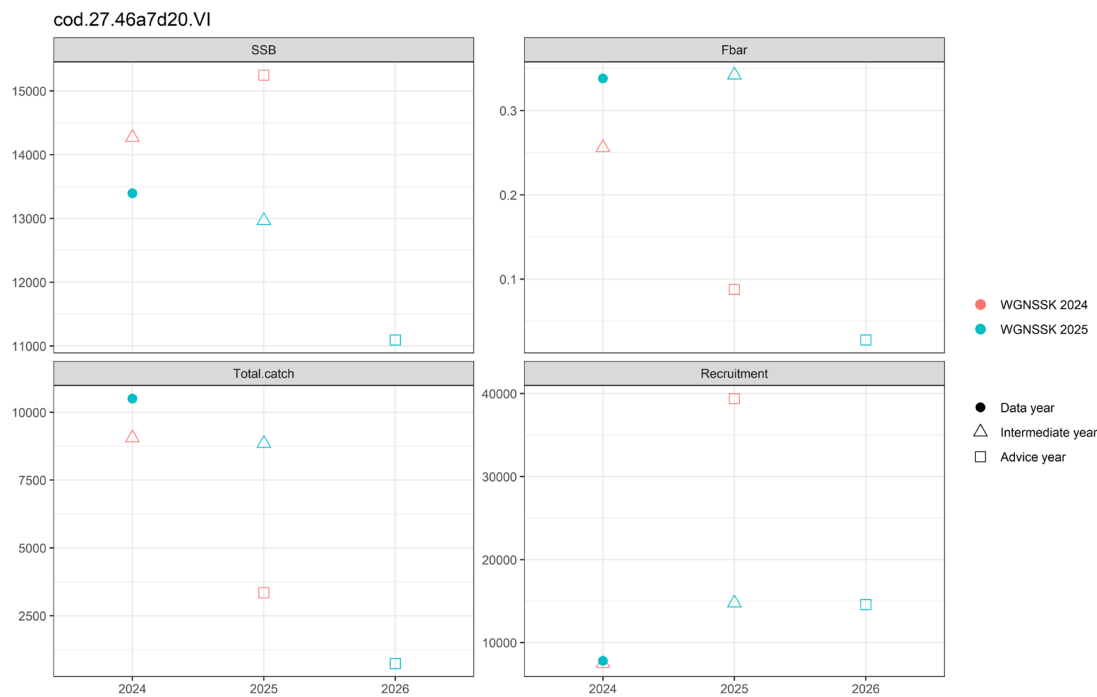


Figure 4.23e. Cod in Subarea 4, divisions 6.a and 7.d, and Subdivision 20. Comparison of SSB, Fbar, total catch and recruitment assumptions used in the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025) for the Viking substock.

Contents

4	Cod in the North Sea, West of Scotland, eastern English Channel, and Skagerrak (Northern Shelf).....	76
4.1	General.....	76
4.1.1	Stock definition	76
4.1.2	Ecosystem aspects	77
4.1.3	Fisheries	77
4.1.4	Management.....	78
4.2	Data available	80
4.2.1	Catch	80
4.2.2	Weight-at-age	88
4.2.3	Maturity and natural mortality	88
4.2.4	Catch, effort and research vessel data	89
4.3	Data analyses	90
4.3.1	Assessment audit.....	90
4.3.2	Exploratory survey-based analyses.....	90
4.3.3	Exploratory catch-at-age-based analyses	91
4.3.4	Final assessment	91
4.4	Historic stock trends	92
4.5	Recruitment estimates.....	93
4.6	Biological reference points	93
4.7	Short-term forecasts	95
4.8	Medium-term forecasts	97
4.9	Quality of the assessment.....	97
4.10	Status of the stock	101
4.11	Management considerations	101
4.12	Issues for future benchmarks	102
4.12.1	Data.....	103
4.12.2	Assessment	104
4.12.3	Forecast.....	104
4.13	References	104
4.14	Tables and figures	106